

卷二·病因、机制与风险因素

猫为什么容易得肾病（AIM 蛋白）、哪些猫风险更高、患病率数字、老龄化共病谱、急性肾损伤如何转成慢性。理解机制才能理解为什么预防的重心在磷、水体重和拦住 AKI。

#	文献（中文标题 / 原文标题 / 要点）	来源	页
1	猫AIM蛋白与猫肾病易感性 Impact of feline AIM on the susceptibility of cats to renal disease www.nature.com/scientificreports 猫 AIM 对猫肾脏易感性的影响 Ryoichi Sugisawa1,* , Emiri Hiramoto1,* , Shigeru Matsuoka1,2, Satomi Iwai3, Ryosuke Takai1, Tomoko Yamazaki1, Nobuko Mori1,4, Yuki Okada5, Naoki Takeda6, Ken-ichi Yamamura6, 肾功能衰竭由于其不可治愈性和患者医疗费用高昂，是最重要的社会问题之一。通过阐明猫对肾脏疾病高度易感的潜在机制，我们在此证明血清AIM蛋白从IgM的有效解离...	—	13
2	猫AIM外显子3变异与自然CKD进展 The protein apoptosis inhibitor of macrophages (AIM) is involved in kidney repair. An AIM (fAIM) genetic variant in 猫巨噬细胞凋亡抑制因子：外显子3变异等位基因与自然进程之间的潜在联系 巨噬细胞凋亡抑制蛋白（AIM）参与肾脏修复。猫的AIM（fAIM）基因变异导致结构域重复，可能会削弱fAIM对肾功能的保护作用。为了确认先前描述的fAIM结构域重复是由基因组水平上的外显子重复引起的，并确定患有慢性肾病（CKD）的猫如果携带变异fAIM等位基因，其肾功能下降的风险是否高于fAIM野生型猫。分析了华盛顿州立大学就诊猫的病历（n=172）和基因组DNA样本（n=100），这些猫均被诊断为CKD。使用测序和PCR来确定fAIM基因型。根据血清肌酐（Scr）浓度，猫按照IRIS CKD分期进行表型分类。使用Fish...	—	17
3	AIM外显子3重复变异患病率 Prevalence of the AIM exon 3 duplication variant, a putative biomarker associated with progression of kidney disease, in 1000 cats 目标 一种称为巨噬细胞凋亡抑制因子（AIM）的蛋白有助于清除肾小管的废物。一些家猫在AIM基因的外显子3中存在重复，产生一种包含额外清道夫受体半胱氨酸富集亚基的四结构域AIM蛋白。这种变异已被认为与慢性肾病风险增加相关。初步研究表明，这种遗传变异的纯合状态在约20%的猫群体中存在。本研究的目的是确定在大量家猫中野生型、杂合型和纯合型AIM外显子3基因型的流行情况。方法 从华盛顿州立大学个体化医学项目DNA银行中存档的1000只由客户拥有的猫的基因组DNA通过滴度数字PCR分析，以确定AIM外显子3的拷贝数变异。样本...	Journal of Feline Medicine and Sur 2026	12
4	猫CKD病理生理与风险因素综述 Feline CKD: Pathophysiology and risk factors — what do we know? - PMC 猫慢性肾病（CKD）的病理生理学及风险因素——我们了解多少？ Brice S Reynolds 1, Hervé P Lefebvre 1,* 实用相关性：慢性肾病（CKD）是猫中最常见的疾病之一，近年来发病率有所增加。虽然其潜在病因很少被确定，但猫CKD的常见最终结果是肾小管间质纤维化。了解CKD的病理生理学对于优化个体化患者管理是必要的，尤其是在诊断和治疗肾外并发症方面。患者群体：CKD在老年猫和高龄猫中最为常见，但在任何患有输尿管结石、甲状腺功能亢进、逆转录病毒感染、系统性高血压、心血管疾病或泌尿道感染的猫患者中也应考虑。证据基础：我们对慢性肾病（CKD）发病机制的了解大多是通过...	—	38
5	随机年龄组猫CKD患病率与分期 Prevalence and classification of chronic kidney disease in cats randomly selected from four age groups and in cats recruited for degenerative joint di 随机从四个年龄组中选择的猫以及为退行性关节疾病研究招募的猫 Christina L Marino 1, B Duncan X Lascelles 2,3, Shelly L Vaden 1,3, Margaret E Gruen 2,4, Steven L Marks 1, ☒ 慢性肾病（CKD）和退行性关节疾病在老年猫中都被认为是常见病。关于这两种疾病的共患率的信息尚缺。这项回顾性研究旨在确定两组猫中慢性肾病（CKD）的患病率：分别为从四个均匀分布的年龄组中随机选出的猫（RS组）以及为退行性关节疾病研究招募的猫（DJD组），并评估这两组猫中CKD与DJD的共存情况。RS组从6个月至20岁的四个...	—	22
6	猫CKD流行病学影响因素分析 Epidemiology of Chronic Kidney Disease (CKD) in Cats: An Analysis of the Factors Involved. 慢性肾病（CKD）是猫最常见且临床上相关的代谢性疾病之一，影响了相当比例的成年猫和老年猫群体。CKD的患病率随年龄增加而升高，据估计，超过10岁的猫中患病率为20%至50%。该疾病的病因涉及遗传易感性、饮食因素以及接触毒素或伴随疾病，这些因素可能加速肾脏损伤的进展。本研究综合了科学文献的数据，并结合对一组120只猫在四年期间进行的临床流行病学分析。结果显示，在研究队列中慢性肾病（CKD）（CKD）的总体患病率较高，九岁以上的猫患病频率明显更高，尤其是纯种猫。男性的患病率略高于女性，而生殖状态显示绝育猫患CKD的风险...	Life (Basel, Switzerland) 2025	42
7	希腊猫CKD真实世界数据临床特征与共病	Veterinary sciences 2026	21

A Real-World Data Analysis on Feline Chronic Kidney Disease in Greece: Clinical Profiles, Comorbidities, and Quality of Life.

猫慢性肾病（fCKD）在老年猫中非常常见，影响高达80%，并且是猫发病率的主要原因之一，其诊断具有挑战性。本研究旨在描述希腊真实临床环境中患有fCKD猫的临床特征、共病情况和生活质量。从208只主人拥有的fCKD猫中收集了真实世界数据进行分析，包括人口学信息、临床发现、身体和肌肉状况评分以及根据国际肾脏学会（IRIS）指南由临床医生评定的分期。此外，宠物主人还完成了VetMetrica™健康相关生活质量（HRQL）问卷。使用适当的统计分析评估了共病、IRIS分期与HRQL各领域之间的关联。在66.8%的猫中至少存在...

8 多囊肾对猫慢性肾功能不全发生的贡献 Veterinary Research Forum 2026 19

Contribution of polycystic kidney disease to the development of chronic renal dysfunction in cats

多囊肾病（PKD）是一种遗传性疾病，其特征是肾囊肿逐渐形成，导致肾功能受损，并增加慢性肾病（CKD）的风险。本前瞻性、为期18个月的纵向研究评估了12只患有PKD的猫的诊断和预后指标，结合超声成像与生化分析进行研究。研究记录了囊肿体积增加14.00%、肾脏长度增大7.00%，以及血浆肌酐、磷和血BUN水平显著升高，强调了这些指标与疾病进展的相关性。血液学分析显示，与晚期CKD阶段相关的进行性贫血，进一步勾勒出PKD的全身性影响。这些研究结果强调，即使宠物主人和兽医提供了细心护理，患有多囊肾病（PKD）的猫咪慢性肾病...

9 老龄化猫的发病共病与死亡模式7年队列 Frontiers in Veterinary Science 2026 18

Patterns of morbidity, multimorbidity, and mortality in aging cats: findings from seven years of the Cat Prospective Ageing and Welfare Study.

引言 随着年龄增长，猫的发病率、多发性疾病和死亡风险增加。然而，在该物种中，纵向队列研究相对有限。关于与年龄相关疾病的流行率和发生率、疾病组合以及与死亡相关的因素的信息，可以提供关于衰老对家猫福祉影响的见解。方法 在一项前瞻性纵向队列研究中，对209只老龄猫的常见年龄相关疾病的流行率和发生率、多发性疾病的模式以及死亡的原因和风险因素进行了调查。猫在中年时入组，并每6个月接受一次兽医检查，直到死亡或审查，最长为7年。结果 共进行了1,241次检查（中位随访时间4.4年；范围0.0–6.9年）。在整个研究期间，牙科疾...

10 慢性肾病急性加重结局差于单纯AKI Frontiers in Veterinary Science 2026 9

Dogs and cats with acute-on-chronic kidney disease have worse outcomes than those with acute kidney injury following renal replacement therapy

目的 比较接受肾脏替代治疗（RRT）的急性加重慢性肾病（ACKD）犬猫与急性肾损伤（AKI）犬猫的出院生存率及长期预后，并识别与预后相关的因素。设计 回顾性观察设计。地点 单一转诊医院。动物 136 只客户拥有的动物（67 只犬和 69 只猫），在 2017 年至 2024 年间接受 RRT 治疗。测量及主要结果 本回顾性观察队列研究回顾了转诊医院的病历。动物被分类为 ACKD（n = 66）或 AKI（n = 70）。短期生存率通过出院生存率、受限平均生存时间（RMST）和逻辑回归模型进行评估。使用Kaplan-...

11 猫CKD肾皮质铁处理与缺氧反应基因表达 Veterinary Sciences 2026 13

Iron-Handling, Lipid-Oxygenation, and Hypoxia-Response Gene Expression in the Renal Cortex of Cats with Chronic Kidney Disease: An Analysis-Plan-Guide

慢性肾病（CKD）（CKD）在老年猫中很常见，但皮质水平的转录关系在缺氧反应、铁处理和脂质氧化方面尚不清楚。本次基于分析计划的、假设生成的二次分析使用了公共猫肾RNA测序数据（GSE303653）。内部计划在审查当前表达结果之前，固定了基因面板、复合指标构建、主要推理检验和质量控制阈值，但未公开注册。经过技术质量控制后，分析了来自对照组、CKD 1/2 期和 CKD 3/4 期猫的21个肾皮质样本。使用似然比检验作为主要的面板级筛查，对23基因面板和全转录组差异表达进行了评估，同时以成对DESeq2对比、斯皮尔曼摘...

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Impact of feline AIM on the susceptibility of cats to renal disease

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Renal failure is one of the most important social problems for its incurability and high costs for patients' health care. Through clarification of the underlying mechanism for the high susceptibility of cats to renal disease, we here demonstrate that the effective dissociation of serum AIM protein from IgM is necessary for the recovery from acute kidney injury (AKI). In cats, the AIM-IgM binding affinity is 1000-fold higher than that in mice, which is caused by the unique positively-charged amino-acid cluster present in feline AIM. Hence, feline AIM does not dissociate from IgM during AKI, abolishing its translocation into urine. This results in inefficient clearance of lumen-obstructing necrotic cell debris at proximal tubules, thereby impairing AKI recovery. Accordingly, mice whose AIM is replaced by feline AIM exhibit higher mortality by AKI than in wild-type mice. Recombinant AIM administration into the mice improves their renal function and survival. As insufficient recovery from AKI predisposes patients to chronic, end-stage renal disease, feline AIM may be involved crucially in the high mortality of cats due to renal disease. Our findings could be the basis of the development of novel AKI therapies targeting AIM-IgM dissociation, and may support renal function in cats and prolong their lives.

The number of pets is increasing markedly worldwide alongside the recently decreasing birth rate and increasingly greater age of the human population, and cats are the most popular pet in the vast majority of areas^{1,2}. It is well known that cats are profoundly more susceptible to and more often die from chronic kidney disease (CKD) than other animals^{3–6}. However, the exact reason for their susceptibility to renal disease, which is one of the most pressing questions in veterinary medicine, remains unclear. Therefore, no effective therapies are available. In this study, we newly identified that feline apoptosis inhibitor of macrophage (AIM, also called CD5-like antigen [CD5L] and encoded by the gene *Cd5l*)⁷ protein remains inactive during acute kidney injury (AKI) in cats, and thus, is crucially involved in the susceptibility of cats to progressive renal disease. Furthermore, we propose that adequate resolution of acute injury by the administration of functional AIM may prevent progression to the lethal, chronic and end-stage renal disease.

AIM is a circulating protein present at a relatively high level in the blood⁸ that was initially identified as a supporter of macrophage survival⁷. AIM is produced by tissue macrophages under transcriptional regulation by nuclear receptor liver X receptor/retinoid X receptor heterodimers^{9–11}. Under healthy states, AIM associates with IgM pentamers in the blood, which protects AIM from renal excretion and maintains high levels of circulating AIM¹². AIM binds to multiple molecules including lipopolysaccharide¹³, cytoplasmic fatty acid synthase¹⁴, IgM^{12,15}, and various regulators of complement activation. In addition, AIM is endocytosed by various cell types, including macrophages, adipocytes, and hepatocytes via scavenger receptors^{14,16–18}. On the basis of

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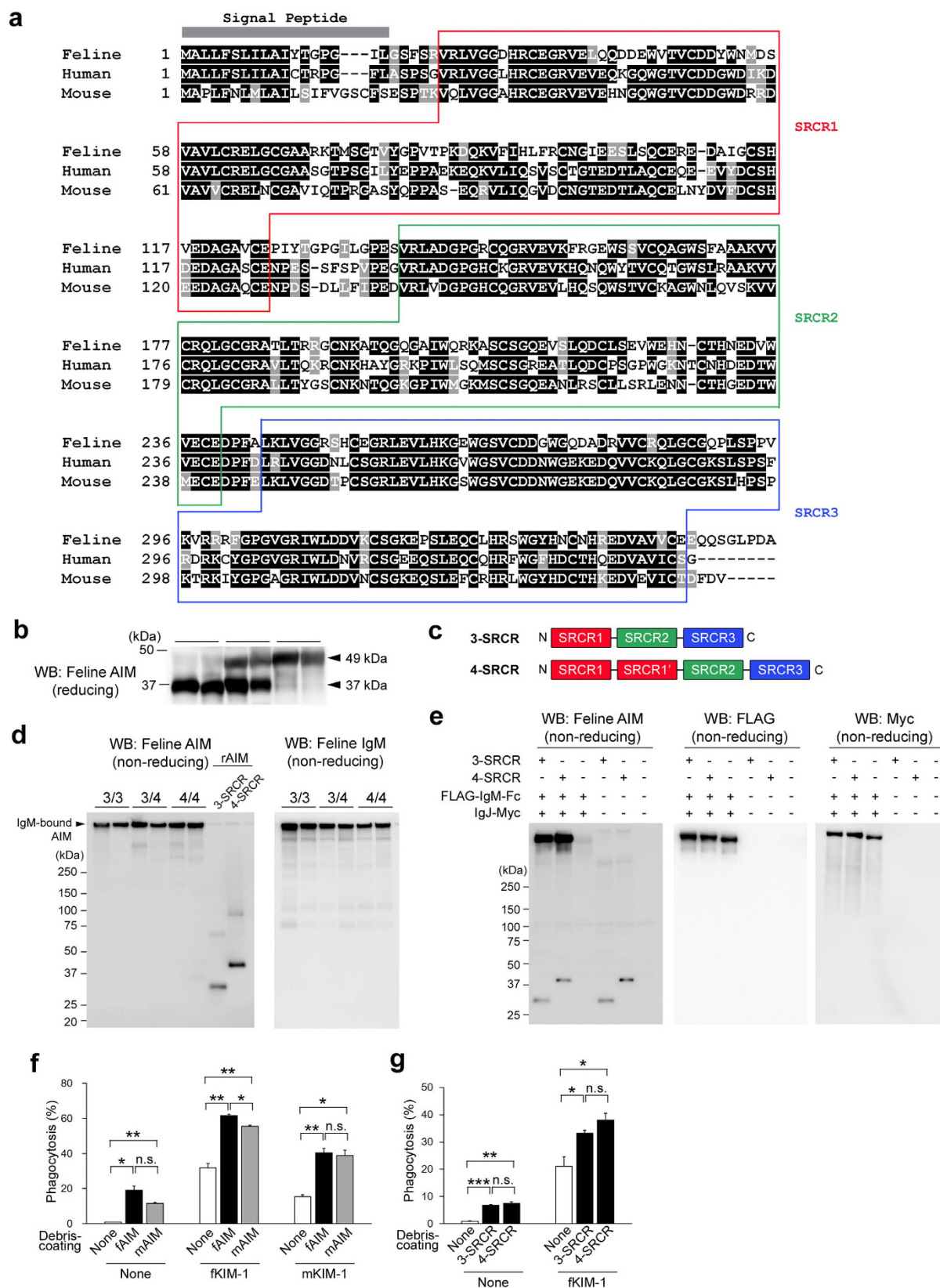


Figure 1. Identification of feline AIM. (a) Amino acid sequence alignment of feline AIM in comparison with human and mouse AIM. Residues with black background indicate identical amino acids and residues with grey background indicate similar amino acids. Three SRCR domains (SRCR1, 2, and 3) are indicated by coloured boxes. (b) Immunoblotting of cat plasma in a reducing condition using a newly generated anti-feline AIM polyclonal antibody. The arrows indicate 2 types of feline AIM. (c) Schema of the 3-SRCR and 4-SRCR

feline AIM. **(d)** Immunoblotting in non-reducing conditions for AIM (left) and IgM (right) using plasma from cats possessing 3-SRCR (indicated by 3/3), 3/4-SRCR (indicated by 3/4), or 4-SRCR (indicated by 4/4) AIM. Recombinant proteins (20 ng each) of 3-SRCR and 4-SRCR AIM were loaded as controls. **(e)** HEK293T cells overexpressing 3- or 4-SRCR feline AIM and those overexpressing FLAG-tagged feline IgM-Fc and Myc-tagged feline IgJ were co-cultured for 16 h, and the supernatant was analysed for AIM by immunoblotting in non-reducing conditions. **(f,g)** *In vitro* phagocytosis assay. Engulfment of debris with rAIM coating by mProx24 cells overexpressing KIM-1 was assessed. The types of rAIM (mouse, 3-SRCR feline, or 4-SRCR feline) and KIM-1 (mouse or feline) are indicated. Engulfment of feline AIM (3-SRCR, indicated by fAIM) or mouse AIM (mAIM) coated debris by feline KIM-1 (fKIM-1) or mouse KIM-1 (mKIM-1) expressing cells **(f)**, and engulfment of 3-SRCR and 4-SRCR feline AIM coated debris by feline KIM-1 expressing cells **(g)**. None: mProx24 cells without KIM-1 expression. The experiment was performed in triplicate, and the percentage (averages $\pm$ s.e.m.) of eFluor[®]780-positive mProx24 cells is shown as “Phagocytosis” on the graph.

these characteristics, we and others have demonstrated recently the involvement of AIM in the pathogenesis of a broad range of diseases including obesity¹⁴, insulin resistance¹⁹, atherosclerosis^{9,20}, autoimmune diseases¹², and non-alcoholic fatty liver diseases^{8,16,21}. Most recently, we found that during AKI in humans and mice, AIM dissociates from IgM pentamers and appears in urine²². Urinary AIM accumulates at the lumen-obstructing necrotic cell debris at proximal tubules, and interacts with kidney injury molecule-1 (KIM-1; also known as TIM-1 or HAVCR1 and encoded by gene *Havcr1*) expressed on the injured tubular epithelial cells^{23–25}. This response promotes the engulfment of intraluminal debris by the epithelial cells, thereby promoting AKI recovery²². Indeed, when subjected to ischemia/reperfusion (IR)-induced AKI, AIM-deficient (AIM[−]) mice exhibit abrogated debris clearance and persistent renal inflammation, resulting in higher mortality than wild-type mice due to progressive renal dysfunction. Treatment with recombinant AIM of mice with IR-induced AKI results in the removal of the debris, thereby ameliorating renal pathology²². Based on such important involvement of AIM in the resolution of AKI, we decided to evaluate feline AIM to assess whether it is related to the susceptibility of cats to renal disease.

Results

Isolation and functional characterization of feline AIM. Feline AIM cDNA was isolated from a cat spleen cDNA pool by 5'-rapid amplification of cDNA ends based on database sequence information of the predicted feline AIM (*cd5l*) mRNA. A comparison of the amino acid sequences of human, feline and mouse AIM is presented (Fig. 1a). The amino acid identity was at similar levels in feline and human AIM and in mouse and human AIM, which was 63.6% and 67.6%, respectively. Having generated several antibodies recognising feline AIM specifically, serum AIM levels were assessed by immunoblotting in >30 cats (strain information and AIM level of each animal are described in Supplementary Table 1). The overall average AIM level in cats was 21.2 μ g/mL [8.48–36.39 μ g/mL range in cats analysed], which is markedly higher than that in mice (2–3 μ g/mL)^{14,16} and humans (approximately 5 μ g/mL)⁸ (Supplementary Table 1). In blood, AIM associates with IgM pentamers; thus, serum levels of AIM and IgM are positively correlated in humans and mice^{8,12}. However, as IgM levels in cats were found to be similar to those in humans and mice (1–2 mg/mL), the increased AIM levels in cats were not caused by higher IgM levels. Interestingly, immunoblotting of serum AIM revealed that cats could be classified into three groups by their AIM pattern: with only 37 kDa AIM, with 37 and 49 kDa AIM, and with only 49 kDa AIM (Fig. 1b). This difference was not due to variable glycosylation, as deglycosylation treatments of serum did not alter the pattern (Supplementary Fig. 1a). To this end, we analysed AIM mRNA from cats showing each serum AIM pattern, and found that 49 kDa AIM possessed 4 cysteine-rich (called scavenger receptor cysteine-rich [SRCR]) domains. Typically, AIM consists of 3 SRCR domains⁷, but feline 49 kDa AIM contained a duplicated first SRCR (SRCR1) domain (Fig. 1c). We also identified a minor variant of 3-SRCR and 4-SRCR feline AIM (one variant for each), in which the hinge region between SRCR1 and SRCR2 was variable (Supplementary Fig. 1b). Thus, blood AIM protein showing the size of 37 kDa only, 37 and 49 kDa, and 49 kDa only in immunoblotting represent 3-SRCR AIM homozygote, 3-/4-SRCR AIM heterozygote, and 4-SRCR AIM homozygote, respectively. Both 3-SRCR and 4-SRCR AIM are associated with IgM pentamers in blood, as shown by immunoblotting of the three types of cat sera in a non-reducing condition (Fig. 1d). This was corroborated by an *in vitro* association experiment using feline recombinant AIM (rAIM) and feline IgM Fc proteins (Fig. 1e). Note that we previously showed that AIM binds to the Fc region of IgM^{12,26}.

The most notable AIM function in facilitating AKI repair is the enhancement of clearance of dead cell debris in the proximal tubules. During AKI, the cell death in the kidney occurs due to apoptosis and necroptosis, particularly in the proximal tubules at the corticomedullary junction, and such dead cells detach from the tubular basement membrane and physically obstruct the tubular lumen. These events reduce glomerular filtration and also induce the production of inflammatory mediators by injured epithelial and infiltrating hematopoietic cells, further exacerbating tubular injury and disturbing the tubular degeneration^{27–29}. KIM-1, the expression of which is highly induced in tubular epithelial cells upon injury and is thus a well-known injury marker^{23–25}, is a ligand for AIM, and induces the engulfment of AIM-deposited necrotic cell debris by tubular epithelial cells^{22,30}. Therefore, we determined whether feline AIM may be deficient in accelerating phagocytosis of debris by feline KIM-1-expressing tubular epithelial cells, by performing an *in vitro* phagocytosis assay²². Debris was prepared from necrotic mProx24 cells, a mouse proximal tubular epithelial cell line, and then coated with feline recombinant AIM (rAIM) by co-incubation. The debris was incubated with living mProx24 cells overexpressing feline KIM-1. The effect of AIM coating on KIM-1-dependent phagocytosis was assessed quantitatively by flow cytometry. Debris engulfment by feline KIM-1-expressing cells increased significantly when debris was coated with feline rAIM and was at similar levels as observed when mouse AIM and mouse KIM-1 were used, suggesting that both

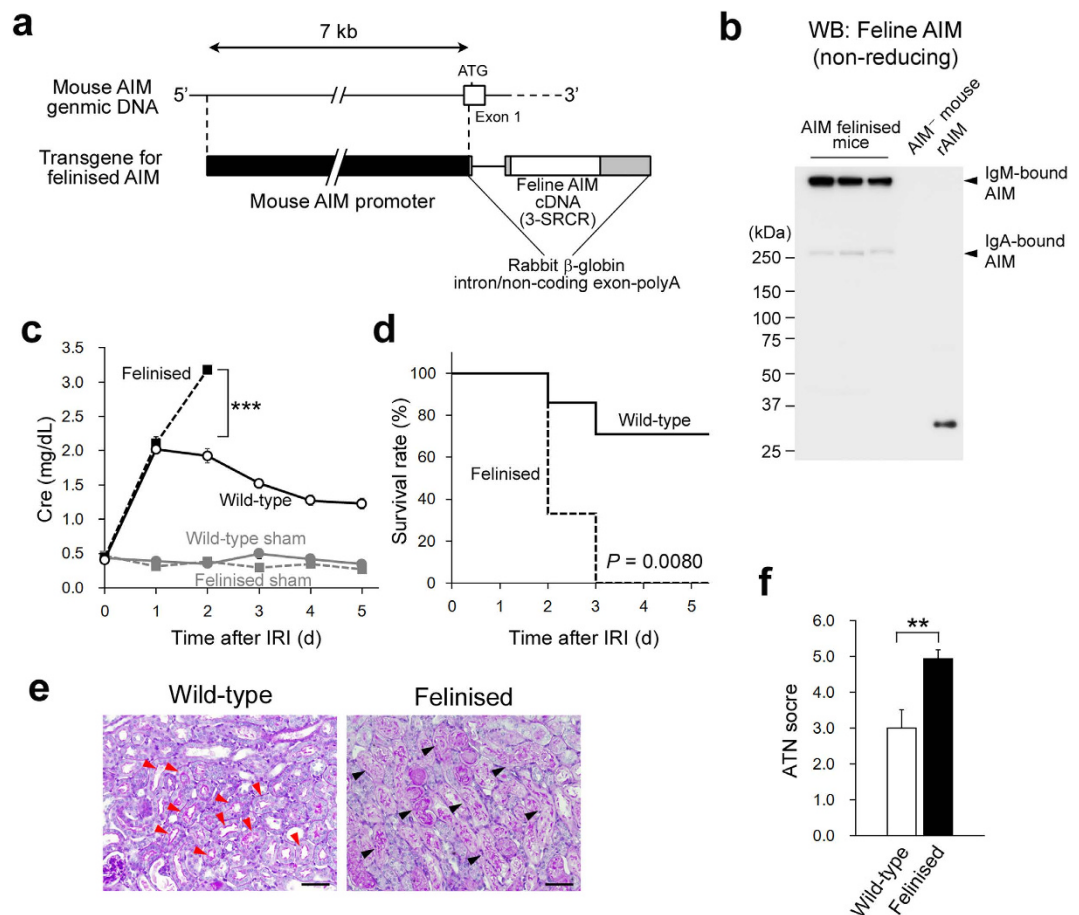


Figure 2. Impaired AKI repair in AIM felinised mice. (a) Schema of the transgene. (b) Immunostaining in a non-reducing condition for feline AIM involving sera from AIM felinised mice ($n = 3$). AIM⁻ mouse serum and 3-SRCR feline rAIM (10 ng) were loaded as a negative and a positive control, respectively. IgM-bound and IgA-bound transgenic feline AIM are indicated. (c) The Cre levels on the indicated days after IR. Open circles, wild-type mice ($n = 7$); closed squares, AIM felinised mice ($n = 6$). Sham control mice ($n = 3$ for each) were also analysed. Grey circles, sham wild-type mice; grey squares, sham AIM felinised mice. Male mice at 8–10 weeks of age were used. *** $P < 0.001$. (d) Survival of wild-type and AIM felinised mice after IR (Kaplan–Meier method). P : statistical significance between each type of mouse was calculated by using the log-rank test. Solid line, wild-type mice ($n = 7$); dotted line, AIM felinised mice ($n = 6$). (e) Representative PAS-stained histologic photomicrographs of the kidney from wild-type and AIM felinised mice ($n = 3$ for each group) on day 3 after IR. Photomicrographs of the corticomedullary junction area are presented. Red arrows: brush border; black arrows: intraluminal debris. Scale bar: 50 μ m. (f) The ATN score in the kidney of IR-treated wild-type mice and AIM felinised mice on day 3 after IR. $n = 3$ each.

feline AIM and feline KIM-1 were functional in promoting phagocytic action for necrotic cell debris (Fig. 1f). This effect was observed equivalently for 3-SRCR and 4-SRCR feline AIM (Fig. 1g). Interestingly, enhanced debris phagocytosis was also achieved at comparable levels for the combination of feline AIM and mouse KIM-1 or mouse AIM and feline KIM-1 (Fig. 1f). Thus, the collaborative function of feline AIM and KIM-1 in the enhancement of debris engulfment was essentially normal.

Impaired recovery from AKI in AIM felinised mice. Then, to assess directly the impact of feline AIM in the poor prognosis of kidney disease in cats, we attempted to felinise AIM in mice and induce AKI. For this purpose, we generated transgenic mice expressing feline AIM under the regulation of the mouse AIM promoter on a mouse AIM⁻ background. Since the function of 3-SRCR and 4-SRCR feline AIM appeared to be comparable in debris clearance (Fig. 1g), we here employed 3-SRCR feline AIM for the transgene. AIM is produced by tissue macrophages under transcriptional regulation by nuclear receptor liver X receptor (LXR)/retinoid X receptor heterodimers^{9,10}. As the LXR response element is located at position -5404 of the AIM promoter¹⁰, we used a genomic DNA fragment of approximately 7 kb upstream of the *AIM* gene to construct the transgene (Fig. 2a). We confirmed that the expression distribution of transgenic feline AIM was comparable to that of endogenous mouse AIM, which is dominant in the resident macrophage-rich liver and spleen⁷, by QPCR and immunohistochemistry (Supplementary Fig. 2a,b). Transgenic feline AIM protein associated with mouse IgM pentamers (Fig. 2b). Although AIM associates predominantly with IgM, it also binds to IgA¹². In serum of AIM felinised mice, AIM

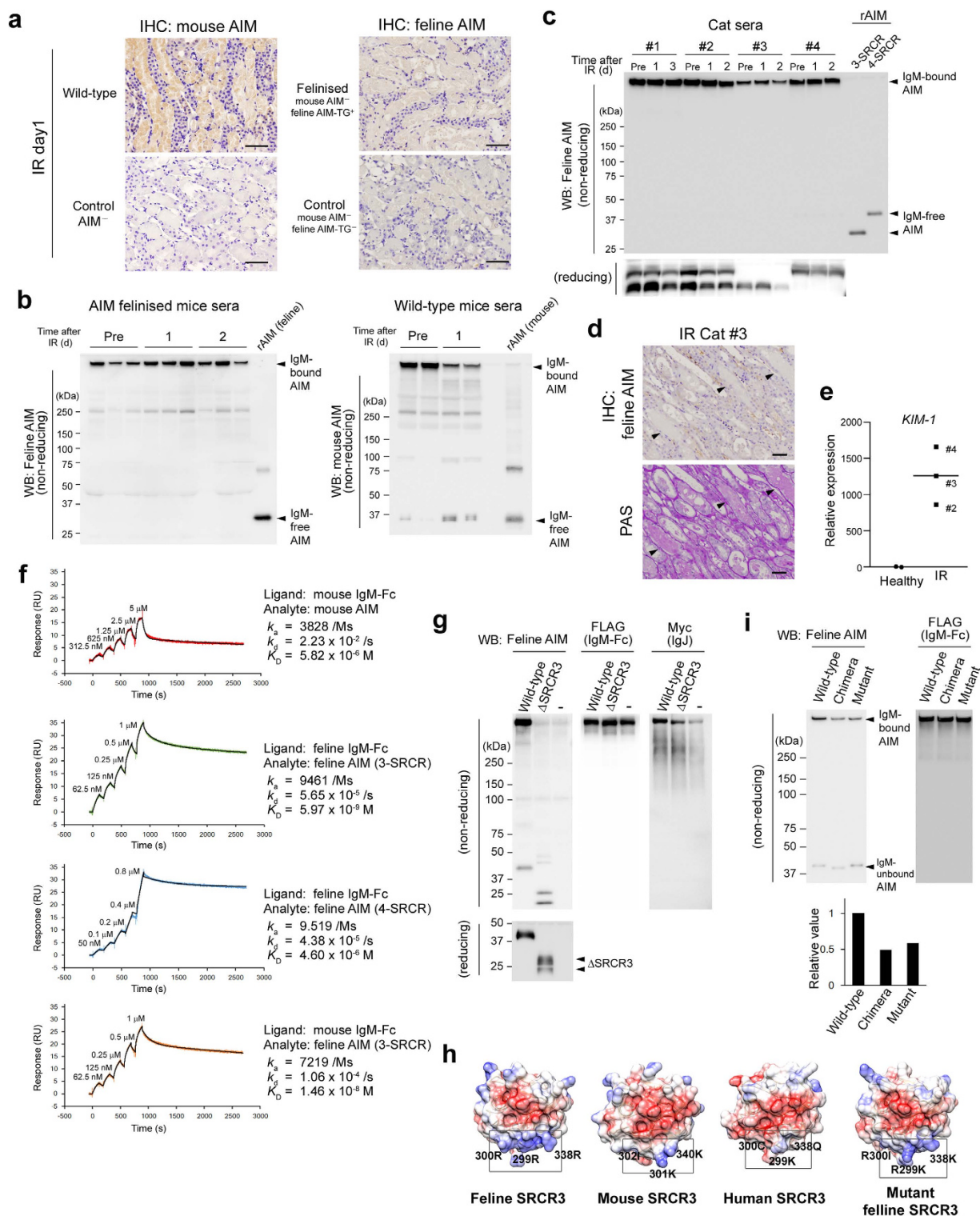


Figure 3. No dissociation of feline AIM from IgM pentamer. (a) Immunohistostaining of the kidney at day 1 after IR. Kidney specimens of AIM⁻ mouse were also immunostained for mouse or feline AIM. Signals were visualised by HRP/DAB. Five mice each were analysed. (b) Immunoblotting in non-reducing conditions for feline AIM (AIM felinised mice; n = 3), and mouse AIM (wild-type mice; n = 2). Pre: before IR. 3-SRCR feline rAIM (15 ng) and mouse rAIM (20 ng) were loaded as controls. (c) Immunoblotting of 4 IR cats in a non-reducing (upper panel) or reducing (lower panel) condition for AIM. Genotype: #1 and 2, 3/4-SRCR; #3, 3/3-SRCR, #4, 4/4-SRCR. rAIM (15 ng each) were loaded as controls. A longer-exposed blot is in Supplementary (b). (d) Immunostaining for feline AIM at the corticomedullary junction in the kidney of an IR cat (#3) at day 2. Scale bars, 50 μm . Black arrows: intraluminal debris. Results from other IR cats are in Supplementary (d). (e) Relative *KIM-1* mRNA levels in the kidney of 3 cats after IR. Kidneys from healthy cats (26 months of age, male, n = 2) were analysed as controls. (f) SPR analysis for AIM (as analyte) and IgM pentamers (as ligand).

Three sets of experiments were performed and similar results were obtained. **(g)** Immunoblotting in a non-reducing (upper panel) or reducing (lower panel) condition for feline AIM. Two different sizes of Δ SRCR-3 are indicated by arrows. Wild-type: wild-type feline AIM (3-SRCR). **(h)** Three-dimensional mapping of the charge distribution of amino acids for the SRCR3 domain. Positively charged: blue; negatively charged: red; neutrally charged: white. The feline AIM-specific cluster of positively charged amino acids is gated. Feline-specific arginine residues, and the corresponding amino acids in mouse, human and mutant feline AIM, are demonstrated. **(i)** Binding of wild-type, chimeric, and mutant feline AIM with feline IgM Fc pentamer assessed by immunoblotting (non-reducing conditions). Signal intensities of the IgM-Fc bound AIM were quantified using NIH ImageJ software, and the relative values to that in wild-type feline AIM are presented in a graph.

was also detected in association with IgA (Fig. 2b). The serum levels of feline AIM in AIM felinised mice were comparable to those of mouse AIM in wild-type mice (approximately 2 μ g/mL) when assessed by immunoblotting (Supplementary Fig. 2c).

Wild-type and AIM felinised mice were subjected to a standard ischemia/reperfusion (IR) protocol to induce AKI that is usually reversible in wild-type mice²². Serum CRE levels peaked at day 1 and recovered thereafter in wild-type mice (Fig. 2c) and 75% of mice survived at day 5 (Fig. 2d). In striking contrast, in AIM felinised mice, serum CRE increased to similar levels as in wild-type mice at day 1 after IR, but continued to increase thereafter (Fig. 2c), resulting in the death of 100% mice by day 3 (Fig. 2d).

One of the important pathological characteristics often observed in AKI is renal tubular obstruction by debris from dead tubular epithelial cells^{27–29}. In histology, on day 3 after IR, there was significant less periodic acid-Schiff (PAS)-stained intraluminal debris at proximal tubules located at the corticomedullary junction in wild-type mice than in AIM felinised mice (Fig. 2e). Similarly, recovery of the brush border in the proximal tubules was apparent in wild-type mice but not in AIM felinised mice (Fig. 2e). The overall acute tubular necrosis (ATN) score³¹ at day 3 was significantly worse in AIM felinised mice than in wild-type mice (Fig. 2f). Thus, AIM felinised mice showed impaired recovery from AKI.

Feline AIM does not dissociate from IgM during AKI. During AKI, serum AIM dissociates from IgM pentamer and the IgM-free AIM translocates to urine. The urinary AIM accumulates at intraluminal debris, thereby promoting its clearance²². To our surprise, immunohistochemical analysis of the kidney from IR-AIM felinised mice revealed that the intraluminal debris was not stained for AIM, in sharp contrast to the AIM-positive debris in wild-type mice, suggesting that AIM did not appear in urine after IR in AIM felinised mice (Fig. 3a). In support, immunoblotting of sera showed that IgM-free AIM was not detected after IR in AIM felinised mice, as observed in cats, whereas it was increased in wild-type mice (Fig. 3b). Thus, it appeared that feline AIM did not dissociate from IgM pentamer upon AKI induction.

We therefore assessed IgM-free AIM in sera of 4 cats with AKI artificially induced by IR³² at different injury levels. The information of the AKI cats is presented in Supplementary Table 2. All IR cats showed poor recovery from AKI, as determined by the serum levels of creatinine (Cre) and blood urea nitrogen (BUN) (Supplementary Fig. 3a). Immunoblotting of sera from mice with AKI in non-reducing conditions shows substantially increased amounts of IgM-free AIM (<40 kDa) with decreased IgM-bound AIM (>600 kDa) compared to those of non-AKI mice (Fig. 3b right, also shown in ref. 22). In contrast to this observation, sera from cats with AKI had no IgM-free AIM. Note that a small amount of IgM-free 4-SRCR AIM was detected even before IR (in #1, 2 and #4 cats), but its level did not increase upon IR (Fig. 3c, a longer-exposed blot is presented in Supplementary Fig. 3b). In addition, we examined urine from IR AKI-cats for the presence of AIM using an enzyme-linked immunosorbent assay (ELISA). In humans and mice, AIM is not detectable in urine samples from healthy individuals, and its levels increase under AKI²². Consistent with the serum results above, AIM was not detectable in urine of IR AKI-cats, indicating no urinary AIM excretion under AKI (Supplementary Fig. 3c). Furthermore, histological analysis demonstrated no AIM accumulation in the intraluminal debris of cats after IR (Fig. 3d, Supplementary Fig. 3d), in contrast to humans and mice (Fig. 3a left) with AKI where AIM accumulation is observed in debris²². Moreover, we analysed sera from several hospitalized cats with spontaneous AKI caused by various events such as renal stones. Again, no IgM-free AIM was observed in serum from these cats (Supplementary Fig. 3e). Thus, in cats, it is likely that feline AIM did not dissociate from IgM pentamers adequately during AKI and was not excreted in urine, causing less sufficient recovery from AKI due to inefficient intraluminal debris clearance. It is noteworthy that *KIM-1* mRNA expression was highly increased in the kidney in IR-cats, as assessed by quantitative PCR (QPCR) (Fig. 3e). Importantly, *KIM-1* mRNA levels continued to increase in all IR cats (Fig. 3e), in accordance with the observations that the serum levels of Cre and BUN did not improve after IR (Supplementary Fig. 3a). This is similar to the situation in AIM felinised mice (and also AIM[−] mice²²) after IR, in contrast to wild-type mice, where these elements improve promptly.

Extremely high binding affinity of feline AIM with IgM. To obtain a mechanistic view for the *undisociativeness* of feline AIM from IgM, we employed surface plasmon resonance (SPR) analysis to quantitatively compare the binding affinity constant of AIM/IgM-Fc pentamers in feline and mouse AIM. The resulting association rate constant (k_a) was 9461/molar second (Ms), the dissociation rate constant (k_d) was 5.65×10^{-5} /s, and the dissociation-association rate (K_D) was 5.97×10^{-9} M in feline AIM (3-SRCR)/IgM-Fc, whereas in mouse AIM/IgM-Fc they were 3828/Ms, 2.23×10^{-2} /s, and 5.82×10^{-6} M (Fig. 3f). Thus, in cats, AIM associates with IgM approximately 1000-fold more strongly than in mice. Interestingly, the 4-SRCR feline AIM and IgM-Fc, the k_a (that shows the binding efficient) was markedly lower (9.519/Ms) compared to that for 3-SRCR feline AIM and IgM-Fc (9461/Ms), whereas the k_d (that shows the dissociation efficiency) was high (4.38×10^{-5} /s), comparable

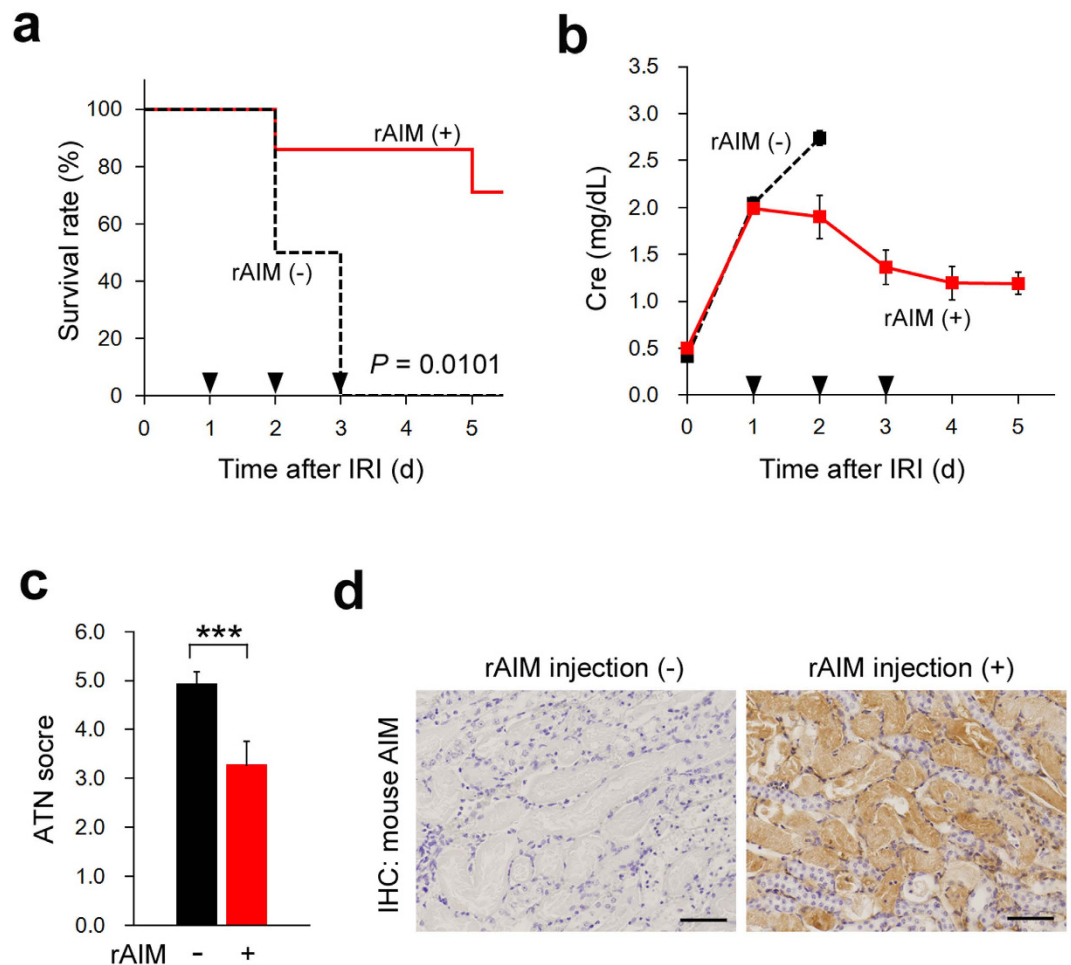


Figure 4. Amelioration of AKI in AIM feline mouse by rAIM. (a) Survival of AIM feline (n = 4, black dotted line) and mouse rAIM-injected AIM feline mice (n = 7, red solid line) after IR (Kaplan-Meier method). ▼; rAIM (mouse; 1 mg) injected. P: statistical significance between each type of mouse was calculated by using the log-rank test. (b) The Cre levels on the indicated days after IR. Black squares, AIM feline mice (n = 4); red squares, rAIM (mouse; 1 mg)-injected AIM feline mice (n = 7). Male mice at 8–10 weeks of age were used. (c) The ATN score in the kidney of IR-treated AIM feline mice (n = 3) and rAIM-injected AIM feline mice (n = 3) on day 3 after IR. (d) Accumulation of AIM on intraluminal debris in rAIM-injected AIM feline mice at day 1 after IR. Signals were visualised by HRP/DAB. Scale bars, 50 μ m.

to that for 3-SRCR feline AIM/IgM-Fc (5.65×10^{-5} /s) (Fig. 3f). This result suggests that 4-SRCR feline AIM is less efficient in binding with IgM-Fc than 3-SRCR feline AIM, but its dissociation from IgM may be as inefficient as that of 3-SRCR feline AIM. Therefore, it is likely that 4-SRCR feline AIM may not be released efficiently upon AKI. This result is consistent with the finding that a level of IgM-free 4-SRCR AIM was detected in non-AKI cat serum, but it did not increase upon AKI induction (Fig. 3c, Supplementary Fig. 3b). Since the binding affinity constant between feline AIM and mouse IgM-Fc pentamers was also high (Fig. 3f), it is likely that the strong association was brought about by feline AIM, and not by feline IgM. This is consistent with the observation that feline AIM was not released from mouse IgM pentamer in AIM feline mice with AKI (Fig. 3b). It is also possible that the strong binding of feline AIM with IgM might contribute partly to the increased serum AIM levels in cats.

In addition, we found that the third SRCR domain (SRCR3) appeared to be responsible for IgM binding, since its deletion abrogated the association with IgM (Fig. 3g; similar results for mouse and human AIM will be published elsewhere). In particular, three-dimensional mapping of the charge distribution of amino acids revealed that SRCR3 domain of feline AIM possessed a cluster of positively charged amino acids (indicated by blue) on the surface of the molecule, which was not observed in mouse and human AIM (Fig. 3h, gated area). Thus, it is possible that the feline AIM-specific cluster of positively charged amino acids might be involved in the increase its binding affinity with IgM. This is reminiscent of the fact that complement C1q also binds to the Fc region of IgM, and the positively charged amino acids of C1q are involved in its association with IgM³³. To test this hypothesis, we created a chimeric feline AIM in which SRCR3 was substituted to that of mouse AIM, and, furthermore, a mutant feline AIM in which feline-specific arginine residues consisting the positively charged amino acid cluster (at 299, 300, and 338 in Fig. 1a, Supplementary Fig. 3f) were substituted to the corresponding amino acids of mouse sequence, and verified their binding affinity with feline IgM-Fc. Intriguingly, although we first performed

SPR analysis, both chimeric and mutant feline AIM showed a high level of non-specific binding to the surface of the SPR sensor chip, which was not observed for wild-type feline or mouse AIM. This artificial response interfered with the SPR quantification of the AIM/Fc binding affinity (Supplementary Fig. 3g). Therefore, instead, we evaluated their binding biochemically, by immunoblotting the IgM Fc-bound AIM in the supernatant after a co-culture of feline Fc-expressing HEK293T cells, and cells expressing either wild-type, chimeric, or mutant feline AIM. Both chimeric and mutant feline AIM exhibited a decreased binding to the Fc pentamers, suggesting that the positively charged amino acid-cluster in feline SRCR3 domain is responsible to the increase in the binding affinity of feline AIM and IgM-Fc (Fig. 3i).

Administration of rAIM facilitates recovery from AKI in AIM felinised mice. Lastly, we tested whether the administration of rAIM facilitates AKI recovery in AIM felinised mice. We injected mouse rAIM intravenously into AIM felinised mice on days 1, 2 and 3 after IR. The survival of rAIM-injected mice was improved markedly to 80% (Fig. 4a). The serum Cre level in rAIM-injected mice decreased by day 5, whereas in non-treated AIM felinised mice, it continued to increase until the mice died (Fig. 4b). By histology, the ATN score of rAIM-injected AIM felinised mice was improved at day 5 (Fig. 4c). We also confirmed by immunohistochemistry that injected rAIM accumulated at intraluminal debris (Fig. 4d). Thus, rAIM administration ameliorated AKI in AIM felinised mice, implying its therapeutic potential in cats.

Discussion

Of the various important findings obtained in the current study, the evidence suggesting the linkage between the unique characteristics of feline AIM and the impaired recovery from AKI in cats may be especially noteworthy (Summarized in Supplementary Fig. 4). This finding strongly supports that the effective dissociation of AIM during AKI is necessary for recovery from AKI. Feline AIM was not released from IgM pentamers during AKI, and therefore, did not appear in urine. One might argue that the mechanism responsible for AIM dissociation from IgM, which remains unknown, might be defective in cats. This is not likely, however, because feline AIM did not dissociate from IgM in mice with AKI as shown in AIM felinised mice, where the releasing mechanism must be active. Certainly, however, we cannot exclude a possibility that feline AIM might exhibit not only increased binding affinity to IgM, but also unresponsiveness to the releasing mechanism. If so, it will be interesting to determine whether the both characteristics of feline AIM are brought about by feline SRCR3 domain, particularly by the unique cluster of positively charged amino acids. Future clarification of the releasing mechanism will answer this question.

Although the transition from AKI to CKD is still controversial^{34–37}, AKI certainly predisposes patients to CKD development^{38–43}. Thus, inadequate recovery of AKI due to feline AIM must be an important risk factor for the high susceptibility of cats to CKD. In support of this hypothesis, we observed in a previous study that after mild IR, AIM[−] mice (whose kidney phenotype during AKI is essentially the same with that in AIM felinised mice) showed chronically persisting inflammation and progressive fibrosis, as well as higher *KIM-1* mRNA levels in the kidney compared to wild-type mice²². Of note, many primary diseases that cause cat CKD have the properties of AKI-causing incidents such as renal stones, infectious nephritis and peritonitis, and exposure to renal toxins including some antibiotics and anti-inflammatory reagents^{44–46}. In this aspect, AIM administration during such acute events would be a potential therapy to prevent CKD in cats. Certainly, however, a proportion of cat CKD may occur without a preceding history of AKI. For instance, obesity and obesity-associated diseases including diabetes are also risk factors for CKD, and in fact, the prevalence of obesity is increasing rapidly in cats^{47–49}. AIM also possesses an anti-obesity effect: IgM-free AIM is incorporated into adipocytes and induces lipolysis by reducing the enzymatic activity of fatty acid synthase within the cells^{14,17}. Indeed, in AIM[−] mice fed a high-fat diet, hypertrophy of adipocytes occurs more abundantly and the mass of visceral adipose tissues is greater, leading to more advanced obesity than in wild-type mice¹⁴. Thus, it is possible that feline AIM, which is characterised by its less efficient release from IgM, may also predispose cats to obesity, thereby contributing to CKD development. In this regard, AIM administration might also be effective for CKD prevention indirectly by suppressing the progression of obesity. Certainly, it is worthwhile to assess whether AIM administration would be directly effective for the amelioration of AKI-independent CKD. In conclusion, in addition to our previous study in AKI mice, our current findings may further corroborate the impact of AIM and its efficient release from IgM on protection from kidney disease, and thus, support its future therapeutic availability in humans. Moreover, our study provides a basis for the development of a next-generation therapy against renal failure in cats that might support their health and longevity.

Methods

General experimental approaches. No samples, mice, or data points were excluded from the reported analyses. The samples were not randomised to experimental groups. Analyses were not performed in a blinded fashion, except the evaluation of intraluminal debris and ATN score based on histologic results.

Mice. AIM[−] mice⁷ had been backcrossed to C57BL/6 (B6) for 15 generations before they were used for experiments. All animal experiments were carried out in strict accordance with the recommendations in the Guide for the Care and Use of Laboratory Animals of the National Institutes of Health. The protocol was approved by the Committee on the Ethics of Animal Experiments of the University of Tokyo (Permit Number: P10-143). All surgeries were performed under sodium pentobarbital anaesthesia, and all efforts were made to minimise suffering.

Cat samples and experiments. Serum and plasma samples were collected from healthy cats after obtaining informed consent in writing from their owners. The cat IR experiments were performed under anaesthesia,

and all efforts were made to minimise suffering. The study protocol was approved by the Ethics Committee of the Nippon Veterinary and Life Science University (Permission Numbers: #27-1 & #s27-1), and of Kitasato University (Permission Number: #15-056).

Antibodies and reagents. The antibodies and reagents used for histology and biochemical experiments are as follows. Primary antibodies specific for mouse AIM (rab2 rabbit polyclonal) and feline AIM (PAC11 rabbit polyclonal, clone #33 and #56 mouse monoclonal) were established in our laboratory. Antibodies for cats were generated by immunizing mice (monoclonal) or rabbits (polyclonal) with feline AIM recombinant proteins tagged with an HA peptide. PAC-11 was used for immunoblotting in reducing conditions, whereas #33 and #56 were used for immunoblotting in non-reducing conditions, immunohistochemistry, and ELISA. We also used antibodies against mouse IgM (polyclonal; Santa Cruz Biotechnology), cat IgM (goat polyclonal; Bethyl Laboratories), HA (clone: 3F10; Roche), FLAG (clone: M2; SIGMA), and c-myc (goat polyclonal; QED Bioscience Inc). In addition, goat anti-rabbit IgG (H+L) secondary antibody, HRP conjugate (Thermo Fisher Scientific), were used. Cat serum IgM was quantified using a Cat IgM ELISA Quantification Set (Bethyl Laboratories, Inc., Montgomery, USA).

Cloning of feline AIM cDNA. The 5'-cDNA was synthesised from a cat spleen RNA pool using a SMARTer™ RACE cDNA Amplification Kit (Clontech Lab., Inc.). The 5'-cDNA of feline AIM was amplified using the Universal Primer A Mix supplied with the kit and a reverse primer (fSRCR1_rev) designed for SRCR1 of the predicted feline *cd5l* mRNA sequence (XM_011291200.1; GI:755808687). After obtaining the correct sequence of the 5'-end of feline AIM mRNA, we then amplified full-length feline AIM cDNA from a cat tissue cDNA pool (purchased from Zyagen Laboratories, Inc., CA) by PCR. By this method, 3-SRCR feline AIM cDNA was obtained. It is suspected that the cat cDNA pool used was from a 3-SRCR AIM homozygote cat (or cats). The 4-SRCR feline AIM cDNA was isolated by RT-PCR using splenic RNA isolated from a cat that possessed 4-SRCR AIM in serum. PCR primers were designed for SRCR1 (including ATG) and SRCR3 (including TAG) according to the sequence of 3-SRCR feline AIM (fAIM_5_f_EcoRI and fAIM_3_r_XhoI). Sequences of the oligonucleotides used are presented in Supplementary Table 4.

SPR analysis. The interaction of AIM proteins with immobilised IgM-Fc pentamers were examined at 25 °C using a Biacore T100 SPR instrument (GE Healthcare, Little Chalfont, UK). IgM-Fc pentamers were covalently immobilised on a CM5 sensor chip (GE Healthcare) at pH 4.5 to 1000 resonance units using the amine coupling procedure at a flow rate of 10 µL/min. A reference flow cell, on which BSA (albumin, bovine, fraction V; Sigma) was immobilised, was used to record the background response, which was subtracted from each sample. Measurements were performed according to single-cycle kinetic analysis. Recombinant AIMs were diluted in a running buffer (Dulbecco phosphate-buffered saline [D-PBS (–)] pH 7.4; Nacalai Tesque) at a concentration ranging from 50 nM to 5 µM, and injected into flow cells at a flow rate of 30 µL/min for 120 sec repeating five times at 60-sec intervals. After injection, AIMs were allowed to dissociate in the running buffer for 30 min. The results were analysed using Biacore T100 evaluation software. The differences in binding responses on the IgM-Fc-immobilised flow cell and the control BSA flow cell were fit to the heterogeneous ligand model to determine kinetic and affinity constants.

Homology modeling. Homology models of AIM SRCR3 domains were generated using the Swiss-Model server (<http://swissmodel.expasy.org/SWISS-MODEL>)^{50,51}. The SWISS-MODEL template library (SMTL version 2016-04-06, PDB release 2016-04-01) was searched with Blast⁵² and HHBlits⁵³ for evolutionary related structures matching the target sequences. The template with the highest quality (5a2e.1.A; PDB ID: 5A2E) was selected for model building. Models were built based on target-template alignment using Promod-II⁵⁴. Coordinates which were conserved between the target and template were copied from the template to the model. Insertions and deletions were remodelled using a fragment library. Side chains were then rebuilt. Finally, the geometry of the resulting model was regularised by using a force field. When loop modelling with ProMod-II did not give a satisfactory result, an alternative model was built with MODELLER⁵⁵. The global and per-residue model quality was assessed using the QMEAN scoring function⁵⁶. The obtained QMEAN4 scores were –1.95, –2.36, –2.34, and –2.36 for feline, mouse, human, and mutant feline models, respectively. Molecular graphics images were produced using the UCSF Chimera package⁵⁷ from the Computer Graphics Laboratory, University of California, San Francisco (supported by NIH P41 RR-01081).

Induction of AKI in mice. IR induction was performed as described previously^{22,58}. Briefly, mice (male, 8–12 weeks old) were anaesthetised by intraperitoneal Avertin injection (250 µg/kg bodyweight), and both kidneys were exposed through a small flank incision. Both renal arteries and veins were occluded with clamps at 37 °C. After the ischaemic period, the clamps were released to induce blood reperfusion. The number of mice tested was chosen to be sufficient to ensure statistical analysis of intra- and inter-group comparisons.

Induction of AKI in cats. Cat IR was performed based on an established protocol for mouse IR after approval from the Ethics Committee of the Kitasato University, Japan. A recent report by Schmiedt *et al.*³² was also referred to. Briefly, the cats were anaesthetised by subcutaneous injection of atropine (0.05 mg/kg bodyweight) and intravenous injection of propofol (10 mg/kg bodyweight; Mylan Pharmaceutical Co., Ltd., Tokyo) followed by inhalation of isoflurane during the operation. The cats were boarded in the supine position, and both kidneys were exposed through an incision at the xiphoid to umbilicus midline. Both renal arteries and veins were separated from the surrounding tissue and occluded with Satin ski forceps for 1 h. After the ischemic period, the clamps were released to induce blood reperfusion. Ampicillin (20 mg/kg bodyweight) was administered

intravenously before and after the operation. Occasionally, intravenous administration of buprenorphine (20 µg/kg bodyweight) was used for pain management. After the experiments, the cats were euthanised by intravenous injection of pentobarbital sodium (75 mg/kg bodyweight) and a saturated amount of potassium chloride.

Evaluation of renal function in mice and cats. Serum mouse Cre concentrations were measured by using a Lab-Assay™ Creatinine Kit (Wako Pure Chemical Industries, Ltd., Osaka, Japan). BUN and Cre levels were analysed using the SIMENS Dimension RXL Max System (SIEMENS Co., Ltd.) or JCA-BM2250 (JEOL Ltd., Tokyo, Japan). The mice were selected for blood drawing in a randomised manner.

Histology. *PAS staining:* cat kidneys were fixed in 10% neutral-buffered formalin for 48 h, and mouse kidneys were fixed in 4% paraformaldehyde (PFA) in PBS for 24 h, followed by embedding in paraffin. PAS staining was performed on 4-µm sections of paraffin-embedded kidney blocks. *AIM detection on intraluminal debris:* 4-µm-sections of PFA-fixed paraffin-embedded kidney were immunostained with a rabbit anti-AIM polyclonal antibody (rab2) for mice or an anti-feline AIM monoclonal antibody (#33) for cats, followed by incubation with Histofine Simple Stain Mouse MAX-PO (R) (NICHIREI, Japan) for 30 min (for mouse specimens) or with Histofine Simple Stain MAX-PO (MULTI) (NICHIREI, Japan) for 30 min (for cat specimens). A Histofine Mouse Stain Kit (NICHIREI, Japan) was also used to detect cat AIM in felinised mouse tissues. After staining with DAB, the sections were counter-stained with hematoxylin. The specimens were subjected to analysis using an inverted microscope (FSX-100; Olympus).

Evaluation of renal damage. The kidney specimens were stained with PAS, and PAS-stained debris (at corticomedullary junction) or brush border (corticomedullary junction and cortex region) were quantified by using NIH ImageJ software, and their percentage areas per whole section in a slide are presented. Five different sections of the kidney from at least 3 different mice of each group were examined. The ATN score was graded by proximal tubule dilation, brush border damage, proteinaceous casts, interstitial widening, and necrosis (0, none; 1, <11%; 2, 11–25%; 3, 26–45%; 4, 46–75%; 5, >75%)²². Evaluation of the specimens was performed in a blinded manner to mouse strain.

Quantitative PCR assay. Quantitative evaluation of mRNA was performed by the $\Delta\Delta C_T$ method using a QuantStudio 3 Real-Time PCR system (Invitrogen; CA, USA). Sequences of the oligonucleotides used are presented in Supplementary Table 4.

Production of recombinant proteins. For mouse AIM (mAIM), HEK293T cells were transfected with the pCAGGS-mAIM plasmid and cultured in DMEM + GlutaMax medium (Gibco, CA) supplemented with 5% foetal bovine serum (FBS) for 3 days. rAIM was purified from culture supernatant using a rat anti-mouse AIM monoclonal antibody (clone #36, made in house) -conjugated HiTrap NHS-activated HP column (GE Healthcare Life Sciences, PA). Bound protein was eluted with 0.1 M glycine-HCl pH 2.3 and neutralised with 1 M Tris-HCl pH 8.5. Protein was concentrated using Amicon Ultra filter concentrators (Millipore, MA), and stored at –80 °C in PBS. Endotoxin levels were measured using Limulus Color KY Test Wako (Wako Pure Chemical Industries, Ltd., Osaka). Protein concentration was determined by a bicinchoninic acid (BCA) assay according to the manufacturer's protocol (Pierce, Rockford, IL). Recombinant feline AIM and AIM-HA were produced essentially by the same protocol except as follows. The pCAGGS-feline AIM and pCAGGS-feline AIM-HA plasmids were used; a mouse anti-feline AIM monoclonal antibody (clone #33) was used for purification of feline AIM, and bound protein was eluted with 0.1 M glycine-HCl pH 2.0; anti-HA matrix (Roche) was used for feline AIM-HA purification.

Feline and mouse Fc pentamers were produced as follows. HEK293T cells were co-transfected with pCAGGS-FLAG-IgM-Fc and pCAGGS-IgJ-Myc plasmids, then cultured in DMEM + GlutaMax medium (Gibco, CA) with 5% FBS for 3 days. Pentameric FLAG-Fc protein associated with Myc-IgJ was purified from culture supernatant using an ANTI-FLAG® M2 Affinity Gel (Sigma-Aldrich), and then eluted with glycine-HCl pH 3.0, followed by neutralisation with 1 M Tris-HCl pH 8.5. Protein was concentrated using Amicon Ultra filter concentrators (Millipore, MA), and stored at 4 °C in PBS. Protein concentration was determined by a BCA assay according to the manufacturer's protocol (Pierce, Rockford, IL).

Immunoprecipitation and deglycosylation. Five microlitres of cat plasma were incubated with 10 µL anti-feline AIM monoclonal antibody (clone #33) conjugated with protein G Sepharose (GE Healthcare Life Sciences, PA) at 4 °C overnight. The precipitates were washed 5 times with a wash buffer (1% NP-40 in PBS containing protease inhibitors) and eluted in 40 µL glycine-HCl pH 2.0 followed by neutralisation with 1 M Tris-HCl pH 8.5. Deglycosylation was performed using an Enzymatic Protein Deglycosylation Kit (Sigma-Aldrich) following the manufacturer's protocols. The samples were heated at 95 °C for 5 min with 2 × SDS sample buffer containing 2-mercaptoethanol, and loaded on an SDS-PAGE for immunoblotting.

Analysis of AIM and IgM levels. Feline AIM in cat plasma and AIM felinised mouse serum were measured by immunoblotting using an anti-feline AIM rabbit polyclonal serum (PAC-11). The lower limit of quantification assessed using feline rAIM as a standard was 2.5 ng/mL for cat AIM. All ELISA assays were performed in a duplicated manner. Cat urinary AIM was measured by ELISA using mouse anti-cat AIM monoclonal antibodies (mouse IgG, clones #33 and #56; generated in our laboratory). The lower limit of quantification assessed by using recombinant AIM protein as a standard was 1.9531 ng/mL for cat AIM. For feline IgM, a Cat IgM ELISA Quantification Set (Bethyl Laboratories, Inc., Montgomery, USA) was used. To analyse feline AIM by ELISA, the urine samples were diluted at 1/10. To analyse plasma IgM, the samples were diluted at 1/3000.

In vitro phagocytosis assay. An *in vitro* phagocytosis assay was performed as described previously²². Briefly, to prepare dead cell debris, mProx24 cells, a murine renal proximal tubular epithelial cell line derived from C57BL/6J adult mouse kidney⁵⁹ and kindly provided by Dr Takeshi Sugaya (CMICS Co., Tokyo), were heat-killed by incubation at 65 °C for 20 min in PBS. The dead cells were then labelled with Fixable Viability Dye (FVD, eBioscience) eFluor[®] 780 followed by being crushed by pipetting vigorously. The fluorescence-labelled dead cell debris were coated with AIM proteins by incubating in serum-free DMEM/F12 (1:1) containing none (control), mouse AIM, or feline AIM (3-SRCR or 4-SRCR) at a concentration of 100 µg/mL at 37 °C for 1 h. To prepare KIM-1-expressing epithelial cells, mProx24 cells were transfected with a mouse or feline KIM-1-IRES-EGFP or EGFP expression vector as a control. At 24 h after transfection, the cells were mixed with the fluorescence-labelled AIM protein-coated dead cell debris as described above in serum-free DMEM/F12 (1:1) supplemented with 5 µg/mL insulin, 5 µg/mL transferrin, and 5 ng/mL selenous acid for 30 min at 37 °C. After incubation, the cells were harvested, washed 3 times with ice-cold PBS, resuspended in PBS containing DAPI, and analysed by flow cytometry (LSRII; BD). Live mProx24 cells were identified as DAPI-negative and the cells overexpressing mouse or feline KIM-1 were determined by EGFP expression. The proportion of engulfment of eFluor[®] 780 (incorporated dead cell signal)-positive dead cell debris in DAPI-negative EGFP-positive mProx24 cells is presented.

Generation of AIM felinised mice. A 7-kb mouse genomic DNA fragment of the AIM (*cd5l*) gene promoter region was isolated from C57BL/6 mouse tail DNA by PCR. The promoter fragment and a feline AIM cDNA fragment were inserted into a rabbit β-globin non-coding exon/intron cassette. After removing the vector sequence, the resulting DNA fragment was injected into the pronuclei of C57BL/6 mouse fertilised eggs to produce transgenic mice. The feline AIM transgenic mice were crossbred onto AIM⁻ mice to generate AIM felinised mice.

Statistical analysis. A power calculation was not performed to predetermine sample size. We did not use randomisation to determine samples or mice to be allocated to experiments. Assessment of ATN scores was performed in a blind fashion. *In vitro* experiments were repeated at least three times. Data were analysed using BellCurve for Excel (Social Survey Research Information Co., Ltd.) and are presented as mean values ± s.e.m. unless otherwise specified. Paired results were assessed using parametric tests such as Student's t-test. Comparisons between multiple groups were analysed using two-way analysis of variance followed by Bonferroni's post-hoc test. For the Kaplan-Meier curves, *P* values were determined with the log-rank test. Unless otherwise specified, the significance code was added to each figure legend.

References

1. The American Veterinary Medical Foundation. Total Pet Ownership and Pet Population. *U.S. Pet Ownership & Demographics Sourcebook*. Section 1 (2012).
2. The European Pet Food Industry. *Facts and Figures 2014*. http://www.fedial.org/fileadmin/user_upload/Secretariat/facts_and_figures_2014.pdf. (2014).
3. White, J. D., Norris, J. M., Baral, R. M. & Malik, R. Naturally-occurring chronic renal disease in Australian cats: a prospective study of 184 cases. *Aust. Vet. J.* **84**, 188–194 (2006).
4. White J. D., Malik, R. & Norris, J. M. Feline chronic kidney disease: can we move from treatment to prevention? *Vet. J.* **190**, 317–322 (2011).
5. Lulich, J. P., O'Brien, T. D., Osborne, C. A. & Polzin, D. J. Feline renal failure: questions, answers, questions. *Compendium on Continuing Education for the Practicing Veterinarian*. **14**, 127–152 (1992).
6. Brown, S. A. Linking treatment to staging in chronic kidney disease. *Consultations in Feline Internal Medicine* (edit. By August, J.R.). **6**, 475–482 (2010).
7. Miyazaki, T., Hirokami, Y., Matsushashi, N., Takatsuka, H. & Naito, M. Increased susceptibility of thymocytes to apoptosis in mice lacking AIM, a novel murine macrophage-derived soluble factor belonging to the scavenger receptor cysteine-rich domain superfamily. *J. Exp. Med.* **189**, 413–422 (1999).
8. Yamazaki, T. *et al.* Circulating AIM as an indicator of liver damage and hepatocellular carcinoma in humans. *PLoS One* **9**, e109123 (2014).
9. Arai, S. *et al.* A role of the apoptosis inhibitory factor AIM/Spα/Ap16 in atherosclerosis development. *Cell Metab.* **1**, 201–213 (2005).
10. Joseph, S. B. *et al.* LXR-dependent gene expression is important for macrophage survival and the innate immune response. *Cell* **119**, 299–309 (2004).
11. Valledor, A. F. *et al.* Activation of liver X receptors and retinoid X receptors prevents bacterial-induced macrophage apoptosis. *Proc. Natl. Acad. Sci. USA* **101**, 17813–17818 (2004).
12. Arai, S. *et al.* Obesity-associated autoantibody production requires AIM to retain IgM immune complex on follicular dendritic cells. *Cell Rep.* **3**, 1187–1198 (2013).
13. Sarrias, M. R. *et al.* A role for human Spα as a pattern recognition receptor. *J. Biol. Chem.* **280**, 35391–35398 (2005).
14. Kurokawa, J. *et al.* AIM is endocytosed into adipocytes and decreases lipid droplets via inhibition of fatty acid synthase activity. *Cell Metab.* **11**, 479–492 (2010).
15. Tissot, J. D. *et al.* IgM are associated to Spα (CD5 antigen-like). *Electrophoresis*. **23**, 1203–1206 (2002).
16. Maehara, N. *et al.* Circulating AIM prevents hepatocellular carcinoma through complement activation. *Cell Rep.* **9**, 61–74 (2014).
17. Iwamura, Y. *et al.* Apoptosis inhibitor of macrophage (AIM) diminishes lipid droplet-coating proteins leading to lipolysis in adipocytes. *Biochem. Biophys. Res. Commun.* **422**, 476–481 (2012).
18. Arai, S. & Miyazaki, T. Impacts of the apoptosis inhibitor of macrophage (AIM) on obesity-associated inflammatory diseases. *Semin. Immunopathol.* **36**, 3–12 (2014).
19. Kurokawa, J. *et al.* Apoptosis inhibitor of macrophage (AIM) is required for obesity-associated recruitment of inflammatory macrophages into adipose tissue. *Proc. Natl. Acad. Sci. USA* **108**, 12072–12077 (2011).
20. Hamada, M. *et al.* MafB promotes atherosclerosis by inhibiting foam-cell apoptosis. *Nat. Commun.* **5**, 3147 (2014).
21. Mera, K. *et al.* Serum levels of apoptosis inhibitor of macrophage are associated with hepatic fibrosis in patients with chronic hepatitis C. *BMC Gastroenterol.* **13**, 14:27 (2014).
22. Arai, S. *et al.* Apoptosis inhibitor of macrophage protein enhances intraluminal debris clearance and ameliorates acute kidney injury in mice. *Nat. Med.* **22**, 183–193 (2016).
23. Charlton, J. R., Portilla, D. & Okusa, M. D. A basic science view of acute kidney injury biomarkers. *Nephrol. Dial. Transplant.* **29**, 1301–1311 (2014).

24. Amin, R. P. *et al.* Identification of putative gene based markers of renal toxicity. *Environ Health Perspect.* **112**, 465–479 (2004).
25. Ichimura, T. *et al.* Kidney injury molecule-1 (KIM-1), a putative epithelial cell adhesion molecule containing a novel immunoglobulin domain, is up-regulated in renal cells after injury. *J. Biol. Chem.* **273**, 4135–4142 (1998).
26. Kai, T., Yamazaki, T., Arai, S. & Miyazaki, T. Stabilization and augmentation of circulating AIM in mice by synthesized IgM-Fc. *PLoS One* **9**, e97037 (2014).
27. Bonventre, J. V. & Yang, L. Cellular pathophysiology of ischemic acute kidney injury. *J. Clin. Invest.* **121**, 4210–4221 (2011).
28. Humphreys, B. D. *et al.* Intrinsic epithelial cells repair the kidney after injury. *Cell Stem Cell* **2**, 284–291 (2008).
29. Kusaba, T., Lalli, M., Kramann, R., Kobayashi, A. & Humphreys, B. D. Differentiated kidney epithelial cells repair injured proximal tubule. *Proc. Natl. Acad. Sci. USA* **111**, 1527–1532 (2014).
30. Ichimura, T. *et al.* Kidney injury molecule-1 is a phosphatidylserine receptor that confers a phagocytic phenotype on epithelial cells. *J. Clin. Invest.* **118**, 1657–1668 (2008).
31. Ismail, O. Z. *et al.* Kidney Injury Molecule-1 Protects against Gα12 Activation and Tissue Damage in Renal Ischemia-Reperfusion Injury. *Am. J. Pathol.* **185**, 1207–1215 (2015).
32. Schmiedt, C. W., Brainard, B. M., Hinson, W., Brown, S. A. & Brown, C. A. Unilateral Renal Ischemia as a Model of Acute Kidney Injury and Renal Fibrosis in Cats. *Vet. Pathol.* **53**, 87–101 (2016).
33. Gadjeva, M. G. *et al.* Interaction of human C1q with IgG and IgM: revisited. *Biochemistry*. **47**, 13093–13102 (2008).
34. Bedford, M., Farmer, C., Levin, A., Ali, T. & Stevens, P. Acute kidney injury and CKD: chicken or egg? *Am. J. Kidney Dis.* **59**, 485–491 (2012).
35. Rifkin, D. E., Coca, S. G. & Kalantar-Zadeh, K. Does AKI truly lead to CKD? *J. Am. Soc. Nephrol.* **23**, 979–984 (2012).
36. Chawla, L. S. & Kimmel, P. L. Acute kidney injury and chronic kidney disease: an integrated clinical syndrome. *Kidney Int.* **82**, 516–524 (2012).
37. Chawla, L. S., Eggers, P. W., Star, R. A. & Kimmel, P. L. Acute kidney injury and chronic kidney disease as interconnected syndromes. *N. Engl. J. Med.* **371**, 58–66 (2014).
38. Ishani, A. *et al.* Acute kidney injury increases risk of ESRD among elderly. *J. Am. Soc. Nephrol.* **20**, 223–228 (2009).
39. Hsu, C. Y. Linking the population epidemiology of acute renal failure, chronic kidney disease and end-stage renal disease. *Curr. Opin. Nephrol. Hypertens.* **6**, 221–226 (2007).
40. Thakar, C. V., Christianson, A., Himmelfarb, J. & Leonard, A. C. Acute kidney injury episodes and chronic kidney disease risk in diabetes mellitus. *Clin. J. Am. Soc. Nephrol.* **6**, 2567–2572 (2011).
41. Chawla, L. S., Amdur, R. L. & Amodeo, S. The severity of acute kidney injury predicts progression to chronic kidney disease. *Kidney Int.* **79**, 1361–1369 (2011).
42. Murugan, R. & Kellum, J. A. Acute kidney injury: what's the prognosis? *Nat. Rev. Nephrol.* **7**, 209–217 (2011).
43. Amdur, R. L., Chawla, L. S., Amodeo, S., Kimmel, P. L. & Palant, C. E. Outcomes following diagnosis of acute renal failure in U.S. veterans: focus on acute tubular necrosis. *Kidney Int.* **76**, 1089–1097 (2009).
44. Segev, G., Nivy, R., Kass, P. H. & Cowgill, L. D. A retrospective study of acute kidney injury in cats and development of a novel clinical scoring system for predicting outcome for cats managed by hemodialysis. *J. Vet. Intern. Med.* **27**, 830–839 (2013).
45. Lee, Y. J., Chan, J. P., Hsu, W. L., Lin, K. W. & Chang, C. C. Prognostic factors and a prognostic index for cats with acute kidney injury. *J. Vet. Intern. Med.* **26**, 500–505 (2012).
46. Worwag, S. & Langston, C. E. Acute intrinsic renal failure in cats: 32 cases (1997–2004). *J. Am. Vet. Med. Assoc.* **232**, 728–732 (2008).
47. Scarlett, J. M. & Donoghue, S. Associations between body condition and disease in cats. *J. Am. Vet. Med. Assoc.* **212**, 1725–1731 (1998).
48. Courcier, E. A., O'Higgins, R., Mellor, D. J. & Yam, P. S. Prevalence and risk factors for feline obesity in a first opinion practice in Glasgow, Scotland. *J. Feline. Med. Surg.* **12**, 746–753 (2010).
49. Lund, E. M., Armstrong, P. J., Kirk, C. A. & Klausner, J. S. Prevalence and risk factors for obesity in adult cats from private US veterinary practices. *Int. J. Appl. Res. Vet. Med.* **3**, 88–96 (2005).
50. Schwede, T., Kopp, J., Guex, N. & Peitsch, M. C. SWISS-MODEL: An automated protein homology-modeling server. *Nucleic Acids Res.* **31**, 3381–3385 (2003).
51. Arnold, K., Bordoli, L., Kopp, J. T. & Schwede, T. The SWISS-MODEL workspace: A web-based environment for protein structure homology modelling. *Bioinformatics* **22**, 195–201 (2006).
52. Altschul, S. F. *et al.* BLAST and PSI-BLAST: A new generation of protein database search programs. *Nucleic Acids Res.* **25**, 3389–3402 (1997).
53. Remmert, M., Biegert, A., Hauser, A. & Soding, J. HHblits: lightning-fast iterative protein sequence searching by HMM-HMM alignment. *Nat. Methods* **9**, 173–175 (2011).
54. Guex, N. & Peitsch, M. C. SWISS-MODEL and the Swiss-Pdb Viewer: An environment for comparative protein modelling. *Electrophoresis* **18**, 2714–2723 (1997).
55. Šali, A. & Blundell, T. L. Comparative protein modelling by satisfaction of spatial restraints. *J. Mol. Biol.* **234**, 779–815 (1993).
56. Benkert, P., Biasini, M. & Schwede, T. Toward the estimation of the absolute quality of individual protein structure models. *Bioinformatics* **27**, 343–350 (2011).
57. Pettersen, E. F. *et al.* UCSF Chimera - A Visualization System for Exploratory Research and Analysis. *J. Comput. Chem.* **25**, 1605–1612 (2004).
58. Park, K. M. *et al.* Inducible nitric-oxide synthase is an important contributor to prolonged protective effects of ischemic preconditioning in the mouse kidney. *J. Biol. Chem.* **278**, 27256–27266 (2003).
59. Kitamura, S. *et al.* Transforming growth factor-β 1 induces vascular endothelial growth factor expression in murine proximal tubular epithelial cells. *Nephron Experimental Nephrology*. **95**, e79–e86 (2003).

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Author Contributions

R.S. and E.H. carried out the major experiments and contributed to this work equally; S.M. did the SPR analysis and structure modelling of AIM; T.Y. contributed to the biochemical experiments; R.T. helped with histology; Y.I., N.M., Y.O. and T.A. contributed to the cat experiments; N.T. and K.Y. contributed to the production of the transgenic mice; S.A. performed the *in vitro* experiments including the phagocytosis assay; T.M. cloned cat AIM cDNA and constructed most plasmids; S.A. and T.M. designed the experiments; and T.M. supervised the whole study and wrote the paper.

Additional Information

Accession codes: The feline AIM (cd5l) mRNA sequences have been deposited in the DDBJ/EMBL/GenBank under accession numbers LC149874 (3-SRCR variant 1), LC149875 (3-SRCR variant 2), LC149876 (4-SRCR variant 1) and LC149877 (4-SRCR variant 2).

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Apoptosis Inhibitor of Macrophages in Cats: A Potential Link Between an Exon 3 Variant Allele and Progression of Naturally Occurring Chronic Kidney Disease

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Abstract

ABSTRACT

Background

The protein apoptosis inhibitor of macrophages (AIM) is involved in kidney repair. An AIM (fAIM) genetic variant in cats resulting in a domain duplication might abrogate fAIM's protective effect on kidney function.

Objectives

To confirm that the domain duplication previously described in fAIM results from an exon duplication at the genomic level and to determine if cats with chronic kidney disease (CKD) harboring the variant fAIM allele are at higher risk for a decline in renal function relative to fAIM wild-type cats.

Animals

Medical records ($n = 172$) and genomic DNA samples ($n = 100$) from cats presented to Washington State University and having a diagnosis of CKD were analyzed.

Methods

Sequencing and PCR were used to determine fAIM genotype. Based on serum creatinine (SCr) concentrations, cats were phenotyped according to IRIS CKD staging. The phenotype–genotype association was tested using Fisher's exact test.

Results

The 4-domain variant of fAIM was confirmed to be a result of exon 3 duplication and is present in 62% of the DNA samples from cats. Medical records of cats ($n = 50$) with CKD met the inclusion criteria. Cats homozygous for the exon 3 fAIM variant allele are more likely to have worse IRIS stage ($p = 0.01$) and more likely to have experienced renal function deterioration (increasing SCr concentration) than fAIM wild-type cats ($p = 0.03$).

Conclusions and Clinical Importance

The fAIM homozygous variant genotype is associated with declining kidney function in cats with CKD, presumably from deficient fAIM-mediated renal tubular repair.

Keywords: biomarker, cats, fAIM, genetics, renal disease

Abbreviations

apoptosis inhibitor of macrophages

blood urea nitrogen

chronic kidney disease

copy number variation

droplet digital PCR

2 copies of exon 3 of fAIM

3 copies of Exon 3

4 copies of Exon 3

feline apoptosis inhibitor of macrophages

genomic DNA

kidney injury molecule 1

serum creatinine concentration

symmetric dimethylarginine

systemic blood pressure

scavenger receptor cysteine-rich

serum T4 concentration

urine protein to creatinine ratio

urine specific gravity

1.

Introduction

Chronic kidney disease (CKD) in cats is characterized by progressive loss of multiple functions within the kidney [¹, ², ³]. Chronic kidney disease is caused by multiple insulting factors and initiated in different renal structures such as the renal tubules. Renal repair mechanisms are initiated almost immediately within the kidney after a renal insult. Tubular damage results in the accumulation of intratubular cellular debris [⁴, ⁵]. This debris is subsequently and rapidly cleared by phagocytic tubular epithelial cells as part of the normal renal repair response to ensure the continued flow of filtrate [⁵]. Failure to remove debris leads to the death of the associated nephron and irreversible, insidious interstitial fibrosis (maladaptive repair) with associated progressive decline in kidney function [⁵]. Renal fibrosis is inversely correlated with glomerular filtration rate (GFR); as interstitial fibrosis increases, GFR decreases [⁶]. One of the hallmarks of kidney disease is a decrease in GFR. Serum creatinine concentration (SCr) is routinely used as a proxy for GFR for assessing renal function [⁷].

The clinical course and long-term prognosis of CKD in cats are highly variable among individual cats [⁸, ⁹]. After a renal insult and initial decrease in GFR, filtration function might improve in some cats or at least remain stable. In

contrast, it might continue to deteriorate in other cats over time, resulting in progressive CKD [^{8, 9}]. The reason why renal filtration function improves in some animals with CKD but not in others remains unclear, but it is increasingly apparent that genetics influences the outcome of several kidney lesion repair processes and, thus, the pathogenesis of CKD [^{10, 11}]. However, studies evaluating the intersection between genotype and CKD in cats remain largely unexplored.

Apoptosis inhibitor of macrophages (AIM) facilitates kidney repair in humans and mice [^{5, 12, 13}]. Mice that are AIM-deficient (AIM⁻) fail to repair normal renal parenchyma and exhibit progressive renal dysfunction [¹³]. Cats can synthesize a wild-type or variant feline AIM (fAIM) protein or both. The wild-type fAIM protein contains three scavenger receptor cysteine-rich (SRCR) protein domains, whereas the variant contains 4-SRCR domains, resulting from duplication of the SRCR1 domain [^{12, 13}].

Genes are organized in exons and introns (Figure 1). The DNA sequence is transcribed into messenger RNA (mRNA), a process that requires splicing out the introns. The resulting mRNA contains a sequence of exons carrying the nucleotide sequence that codes for amino acids and proteins. In humans and mice, the fAIM mRNA sequence consists of six exons that code for a 37 kDa three-domain AIMa protein. Cats uniquely express the three-domain 37 kDa and a 45 kDa four-domain fAIM [^{5, 12}]. The SRCR1 protein domain is coded by exon 3. Based on the mRNA structure in cats [⁵], the duplicated SRCR1 can result from duplication of exon 3 at the gene level, but this possibility remains to be confirmed.

FIGURE 1

Schematic representation depicting gene exons and introns. In the process of transcription, introns are spliced out resulting in messenger RNA (mRNA), comprised of exons. mRNA exons are translated into protein domains.

Whether the additional SRCR1 domain alters any or all of the core functions of AIM (e.g., attachment to debris, engulfment by phagocytes and association with or dissociation from IgM) remains to be determined, but it is reasonable to postulate that the predicted structural difference between wild-type AIM and the variant would cause functional differences. Thus, considering the role of wild-type fAIM in normal kidney repair and the marked alteration in the variant fAIM protein structure, we hypothesized that cats variant for the exon 3 allele of fAIM might experience impaired kidney lesion repair, which could be reflected in a progressive decline of kidney function. Our objectives were to confirm that the domain duplication previously described in fAIM results from exon duplication at the genomic level and to determine if cats with CKD harboring the variant fAIM allele are at higher risk for decline in renal function relative to fAIM wild-type cats.

2.

Materials and Methods

2.1.

Study Design

The study was carried out in four stages. First, DNA samples from cats collected from a study completed at Washington State University (WSU; protocol #4915) were screened to identify those with each fAIM exon 3 variant: homozygous wild-type, homozygous variant, or heterozygous (results not shown). Then, the fAIM gene in samples

from cats identified as carrying 2, 3, and 4 copies of exon 3 were sequenced as described below. Second, medical records from client-owned cats admitted to the Veterinary Teaching Hospital at WSU were classified phenotypically following the inclusion and exclusion criteria described below (Figure ²). Third, archived (PrIMe DNA Bank-IACUC protocol #6493) genomic DNA from those cats meeting the inclusion criteria were classified as homozygous for the wild-type (2 copies of exon 3) and variant (3 and 4 copies of exon 3, respectively) alleles of the fAIM gene using droplet digital PCR (ddPCR). One of the researchers (NFV) determined the final list of CKD cases included in the study and remained blinded to the genotype. Lastly, a subset of archived genomic DNA from the PrIMe DNA Bank feline samples (IACUC protocol #6493) was genotyped using ddPCR to obtain preliminary information about the frequency of the variant in the cat population.

FIGURE 2

Study design: Medical records of cats admitted to the VTH at WSU were selected based on the inclusion and exclusion criteria. Cats diagnosed with chronic kidney disease were then phenotyped based on two approaches. Once the phenotypic classification was completed, cats were genotyped based on the number of copies of exon 3 in the feline apoptosis inhibitor of macrophages gene.

2.2.

Genomic DNA Isolation

Genomic DNA from feline samples was isolated from cheek swabs using *Quick*-DNA miniprep kit (Zymo Research, Irvine, WA, USA) following the manufacturer's protocol. After extraction, the eluted DNA concentration was determined using Qubit fluorometric quantification following the manufacturers' instructions (Invitrogen, Carlsbad, CA, USA). The DNA samples were stored at -80°C until used.

2.3.

Sequencing of Feline Apoptosis Inhibitor of Macrophage Gene (fAIM) Variants

Genomic DNA from archived feline samples (IACUC protocol #4915) was used for sequence studies. The complete DNA sequence of fAIM was amplified for PacBio sequencing, employing the following PCR parameters: a single cycle at 94°C for 30 s, followed by 30 cycles at 94°C for 30 s and 65°C for 750 s, with a final extension at 65°C for 10 min. In each PCR reaction, approximately 50 ng of genomic DNA template, $1\times$ LongAmp Taq reaction buffer, 0.4 μM of each primer (Fa2 5' CTGCAGACAAGACCTCTGTC 3' and Ra1 5' CAAGGCGTAGGCAGAGTTCAG 3'), 0.3 mM dNTPs, 5% dimethylsulfoxide (DMSO; Sigma–Aldrich, St. Louis, MO, USA), 2.5 U of LongAmp Taq DNA polymerase (New England Biosciences, Ipswich, MA, USA), and 10 μL of nuclease-free water were added for a final reaction volume of 25 μL .

Post-PCR amplification, the product was visualized on a 0.8% agarose/Tris acetate-EDTA (TAE) gel containing 0.5 $\mu\text{g/mL}$ of ethidium bromide. Upon observing a PCR band of the expected size (approximately 11 kb), the PCR product was purified using 0.8 $\times$ Ampure XP magnetic beads (Agencourt Biosciences, Beverly, MA, USA); product quality was assessed on the Agilent 5200 Fragment Analyzer (Agilent, Santa Clara, CA, USA) and subjected to PacBio sequencing at the WSU Laboratory for Biotechnology and Bioanalysis.

For the preparation of the PacBio SMRTbell library, the PCR products of 10–15 kb were employed by utilizing the Pacific Biosciences SMRTbell template preparation kit 1.0 (Pacific Biosciences, CA, USA) according to the

manufacturer's instructions. A separate library was created for each amplicon, and purification was done using 0.8× Ampure XP magnetic beads (Agencourt Biosciences, Beverly, MA, USA). These purified libraries subsequently were sequenced using the P6C4 sequencing chemistry on the RS II platform, with a 240-min movie time and one SMRT cell per library.

The sequence analysis was carried out using the Pacific Biosciences SMRT Analysis software suite for long amplicon analysis and hierarchical genome assembly process (HGAP), resulting in polished amplicon sequences. Further sequence analysis was performed using Lasergene v. 14 (DNASTar, WI, USA).

2.4.

Simulation of the Molecular Structure of 3-SRCR and 4-SRCR Domain Variants of fAIM and Estimation of Their Physicochemical Parameters

The nucleotide sequence for both the 3-SRCR and 4-SRCR domain variants of fAIM was converted to amino acid sequence using EditSeq in Lasergene v. 14 (DNASTar Madison, WI, USA). The corresponding amino acid sequences can be found in Supplemental information (Table S¹). Initially, the amino acid sequence's signal was determined using deep neural networks as implemented through SignalP 5.0 (<http://www.cbs.dtu.dk/services/SignalP/index.php>) [¹⁴]. Utilizing this signal sequence information, the mature protein sequences and the predicted tertiary structures for both fAIM proteins were modeled using deep learning algorithms on the RaptorX server (<http://raptorx.uchicago.edu/>) [¹⁵, ¹⁶]. The predicted tertiary structure images were regenerated using UCSF Chimera v1.14 (<https://www.cgl.ucsf.edu/chimera/download.html>) [¹⁷, ¹⁸].

2.5.

Case Selection

Medical records of cats presented to the Veterinary Teaching Hospital at WSU and diagnosed with CKD between 2011 and 2022 were screened for inclusion in the study (Figure 2). Records that indicated both a diagnosis of CKD and archived DNA were pre-selected. The following data were extracted from medical records: age at diagnosis of CKD, sex, reproductive status, breed, body weight, SCr, blood urea nitrogen concentration (BUN), urine specific gravity (USG), and serum thyroxine concentration (T4).

Cases were eligible for inclusion if they met the following criteria: (1) SCr $\geq$ 1.6 mg/dL together with a low USG (< 1.035) [¹⁹], or SCr $\geq$ 1.6 mg/dL on two occasions a minimum of 3 months apart [¹⁹], and (2) follow-up data for more than 6 months. The follow-up period was defined as the interval between the date of CKD diagnosis (designated baseline) and the date of the last available SCr (designated endpoint).

Cases were excluded if follow-up data was available for less than 6 months. Cats with other diagnosed renal conditions (e.g., ureteral obstruction, pyelonephritis, renal lymphoma, polycystic kidney disease, and subcutaneous ureteral bypass placement) were excluded. Cats with diagnosed hyperthyroidism were ineligible if T4 concentration was outside of the reference concentration range (1–4 µg/dL) during the follow-up period. Cats under treatment for hyperthyroidism with stable T4 concentrations within the reference concentration range (1–4 µg/dL) were not excluded. Follow-up data was not included if dehydration was reported. Serum creatinine concentration data from cats receiving fluid therapy or drugs suspected to alter SCr (e.g., doxorubicin, carboplatin, nonsteroidal anti-inflammatory

drugs, gentamicin), or with a history compatible with an episode of acute disease in a patient with CKD, were excluded from the analysis.

2.6.

Classification of Cases Based on Renal Function

Assessment of renal function was based on SCr reported in the medical records during the follow-up period. We used two approaches for phenotypic classification and analyzed them independently (Figure ²).

2.6.1.

Approach A: International Renal Interest Society (IRIS) Staging

Cats were categorized based on the 1–4 International Renal Interest Society (IRIS) staging criteria [¹⁹]. Cats were assigned an IRIS stage at the beginning and end of the follow-up period. Kidney function was considered worse if there was an increase in stage severity at the end of the follow-up period compared with the baseline. Otherwise, a CKD cat was considered non-progressive.

2.6.2.

Approach B: Percentage Change in SCr From Baseline

The percentage SCr change was estimated based on a ratio of endpoint and baseline SCr. If cats had an increase in SCr > 20% at the endpoint relative to the SCr at baseline, cats were classified as having declining kidney function (worsening). Otherwise, cats were classified as non-progressive.

2.7.

Determination of fAIM Exon 3 Copy Number

After assessment of kidney function, fAIM exon 3 copy number variation was determined using archived (PrIME DNA Bank (IACUC protocol #6493)) genomic DNA samples from CKD cats that met the inclusion criteria.

Exon 3 copy number variation (CNV) was determined using ddPCR. Approximately 100 ng of DNA was combined with a reaction mixture composed of 2× Bio-Rad ddPCR mix and 20× primer and TaqMan probe mixes for fAIM exon 3 as the region of interest (HEX) and an internal (exon 5 of fAIM; exon 5) and external (exon 3 of feline telomerase reverse transcriptase; TERT) reference gene (FAM). Then, droplets were generated in a droplet generator (QX200, Bio-Rad, USA) by loading the reaction mixture containing the genomic DNA (gDNA) sample into DG8 reaction cartridges (Bio-Rad, USA) with 70 µL droplet generation oil (Bio-Rad, USA). After droplet generation, droplets were transferred to a 96-well PCR plate (Bio-Rad, USA). The PCR amplification was performed in a T100 Touch thermal deep-well thermocycler (Bio-Rad, USA) with the following parameters: 96°C for 10 min followed by 45 cycles of 96°C for 15 s and 60°C for 2 min, and finally 96°C for 10 min and a 4°C infinite hold, all with a 2°C/s ramp rate. The PCR plate was cooled to room temperature before loading the plate in the QX200 digital droplet reader (Bio-Rad, USA).

Fluorescence data were acquired and analyzed using QuantaSoft Analysis Pro Software version 1.05.596 (Bio-Rad, USA) following manufacturer recommendations. Copy number variation determination involves quantification of target and reference loci (TERT or exon 5 for our study). The ddPCR software counts positive and negative droplets

for each of the targets (exon 3 in fAIM gene and each of the reference genes), estimating the average CNV of exon 3 per droplet. The software was set to calculate the copy number automatically based on input data. Each study sample was run in combination with both of the reference loci once.

2.8.

Frequency of fAIM Exon 3 Genotypes From Archival DNA

The frequency of cats with 2, 3, and 4 copies of exon 3 of fAIM was determined using 100 archived DNA samples from the PrIME DNA Bank. A subset of samples was selected from > 1500 banked samples by simple randomization. Genotyping for fAIM exon 3 CNV was determined using ddPCR as described above.

2.9.

Statistical Analysis

Age at diagnosis, duration of follow-up period, SCr at baseline and endpoint, percentage change in SCr from baseline, BUN, and body weight for the cats were evaluated descriptively and compared statistically. Shapiro–Wilk and Brown–Forsythe tests were used to verify normality and the homogeneity of variances, respectively. An unpaired *t*-test was used to compare variables at the baseline or endpoint for cats within progressive and non-progressive groups. Mann–Whitney's test was performed if the data was not normally distributed. A paired *t*-test was used to compare body weight and SCr at baseline and endpoint within each group. Spearman's correlation coefficient was used to investigate the correlations between body weight and SCr. The strength of the association between fAIM exon 3 CNV and phenotype (IRIS stage progression and SCr worsening) was determined by calculating the odds ratio (OR) and tested statistically using Fisher's exact test. The OR and 95% confidence interval (CI) were calculated as described previously [20]. Statistical analyses were performed using Microsoft Excel and SigmaPlot version 14.0 (Systat Software Inc., California). For all tests, $p \leq 0.05$ was considered significant.

3.

Results

3.1.

Sequencing of Feline Apoptosis Inhibitor of Macrophage (fAIM) Gene

To ascertain the genomic status and provide a comprehensive understanding of exon 3 structure, three feline gDNA samples were subjected to long-read sequencing. These samples, previously identified with 2, 3, and 4 copies of exon 3 by qPCR screening (data not shown), underwent precise examination. A summary of the fragment analyzer run is included in the Supporting Information (Figure S¹). Fragment analysis of the fAIM PCR product indicates three amplicon sizes between approximately 10 and 12.5 kb pairs (Figure S²). In the sample with approximately 10.4 kb pairs, each allele contained an exon 3 of 321 base pairs, indicating a homozygous wild-type gene (Figure S², Panel A). For the sample with approximately 12.5 kb pairs amplicon, both alleles contained 2 copies of exon 3 of 321 base pairs, reflecting a homozygous state for the mutant allele (Figure S², Panel B). The sample with both size amplicons contains an allele with an exon 3 and the other allele with 2 copies of exon 3, indicative of a heterozygous state for the mutant allele (Figure S², Panel C). Exon 3 copies exhibited a homology range of 96.2%–100%. The altered fAIM gene

consists of a large novel intragenic tandem exon 3 and flanking intron duplication of the fAIM gene (Figure ^{S2}, Panel A, the intron size is approximately 2.5 kb pairs). The wild-type fAIM gene is composed of six exons (exons 1–6; Figure ^{S2}, Panel A). Detailed maps of fAIM introns and exons for both wild-type and mutant alleles are provided in Figure ^{S2}, Panel B.

3.2.

Simulation of the Molecular Structure of 3-SRCR and 4-SRCR Domain Variants of fAIM and Estimation of Their Physicochemical Parameters

The copy number variation of feline exon 3 results in an additional SRCR domain 1 in the protein structure (Figure ^{3A,B}). Thus, affected cats express a 4-SRCR domain variant of fAIM. To explore the potential impact of the extra protein domain on the protein's physicochemical features, we modeled in silico the molecular structure of 3-SRCR and 4-SRCR domain variants of fAIM and estimated their physicochemical parameters. A protein simulation model for both protein variants is shown in Figure ³, Panel A and B. Amino acid sequences for 3-SRCR and 4-SRCR domain fAIM variants are presented in Table ^{S1}. The 4-SRCR domain protein variant results in a duplication of the first domain, resulting in a larger, more negatively charged, and less water-soluble protein than the wild-type 3-SRCR domain fAIM protein (Table ^{S2}).

FIGURE 3

Panel A: Simulation model of both the fAIM protein depicting the wild-type (A) and variant (B) scavenger receptor cysteine-rich domains and the 4-SRCR (B), which is the result of a genomic level duplication of exon 3. Panel B shows the extra protein domain; Domain 1': cornflower blue; Domain 1' (duplicated): light sea green; Domain 2: medium blue; Domain 3: purple. Initially, the signal sequence for the amino acid sequence was determined utilizing SignalP 5.0 (<http://www.cbs.dtu.dk/services/SignalP/index.php>). The protein sequence of the predicted tertiary structure for both feline apoptosis inhibitor of macrophages proteins was modeled using deep learning algorithms through the RaptorX server (<http://raptorx.uchicago.edu/>). The predicted tertiary structure images were regenerated using UCSF Chimera v1.14 (<https://www.cgl.ucsf.edu/chimera/download.html>). Panel B: Schematic representation of PacBio sequencing results of feline apoptosis of macrophages (fAIM) gene from a heterozygous cat showing a large intragenic tandem exon 3 and flanking intron duplication of the fAIM gene. Duplication of exon 3 can occur in one or both alleles of fAIM, resulting in three genotypes: Those homozygous for the wild-type fAIM exon 3 allele, those homozygous for the variant fAIM exon 3 allele, and heterozygotes. The wild-type fAIM gene codes for a protein consisting of three cysteine-rich domains. The duplication of exon 3 leads to the duplication of the scavenger receptor cysteine-rich domains (SRCR) protein subunit 1. The figure depicts the base pairs (bp) for each exon and intron component based on DNA from a sample of 1 cat.

3.3.

Phenotypic Case Selection

Initially, 172 medical records of cats diagnosed with CKD during the designated time period were identified. Ninety-one of these also had archived DNA, and thus, only those medical records were reviewed. Fifty cases met the inclusion criteria (Table ^{S3}). Twenty cases had SCr ≥ 1.6 mg/dL and USG < 1.035 at diagnosis, and 30 had SCr ≥ 1.6 mg/dL at least twice ≥ 3 months apart. Forty-one cases were excluded because follow-up data duration was < 6 months

($n = 24$), other renal conditions were present ($n = 6$), T4 concentrations were abnormal ($n = 6$) or clinical information was incomplete ($n = 5$, Table ^{S4}).

The most common breeds represented were domestic short hair (29/50; 58%), domestic long hair (7/50; 14%), and Siamese or Siamese-mix (6/50; 12%). Other breeds were Bengal (2/50; 4%), Ragdoll (2/50; 4%), Bombay (1/50; 2%), domestic medium hair (1/50; 2%), Egyptian Mau (1/50; 2%), and Persian (1/50; 2%; Table ^{S3}). Of 50 cats, 28 (56%) were neutered males, 21 (42%) were spayed females, and 1 (2%) was an intact male (Table ¹). Age of cats at CKD diagnosis ranged from 8 to 15 years for all cats, and no significant difference was found in age at diagnosis between cats with progressively increased SCr versus those that remained non-progressive ($p = 0.90$, Table ¹).

TABLE 1

Clinical data median (range) for CKD cats included in the study and classified based on the progression of serum creatinine concentration ($n = 50$).

	Cats classified as worsen CKD	Cats classified as non-progressive CKD	<i>p</i>
<i>n</i>	24	26	
Sex	7 FS 16 MC 1 M	14 FS 12 MC	$p = 0.08$
Age on diagnosis (years)	13 (8.0–15)	12 (9.0–15)	$p = 0.90$
Follow-up period (months)	39 (18–70)	28 (16–46)	$p = 0.35$
Weight on baseline (kg)	4.6 (4.0–5.3)	5.0 (3.9–6)	$p = 0.28$
Weight on endpoint (kg)	3.9 (3.5–4.8)	4.1 (3.4–5.3)	$p = 0.63$
sCr on baseline (mg/dL)	1.9 (1.7–2.1)	2.1 (1.9–2.7)	$p = 0.06$
sCr on endpoint (mg/dL)	2.8 (2.6–3.7)	1.8 (1.6–2.5)	$p < 0.001^*$
Change of sCr (%)	58 (29–91)	–12 (–23–1.5)	$p < 0.001^*$
BUN on baseline (mg/dL)	31 (26–44)	28 (24–37)	$p = 0.26$
BUN on endpoint (mg/dL)	52 (45–81)	37 (27–42)	$p < 0.001^*$

Abbreviations: FS, females spayed; M, male intact; MC, males castrated.

*

Significantly different ($p < 0.05$).

3.4.

Classification of Patients According to Renal Function

3.4.1.

Approach A: IRIS Staging

Cats were classified as IRIS stage 2 (*n* = 43) and stage 3 (*n* = 7) at baseline. Most of the cats (*n* = 29) remained in the same IRIS stage at the end of the follow-up period. Twelve (24%) cats progressed from stage 2 to 3 and two cats from stage 2 to 4. Four cats changed from IRIS stage 2 to 1 and two cats from IRIS stage 3 to 2 (Figure ^{4A} and Table ^{S3}).

FIGURE 4

Panel A: Schematic depicting the evolution of IRIS staging from baseline to the end of the follow-up period for cats with 2 copies (homozygous), 3 copies (heterozygous) and 4 copies (homozygous) of exon 3 of fAIM. Each row represents a single cat that met the inclusion criteria. The blue boxes represent the IRIS stage at diagnosis, the red box shows a worsening of the IRIS staging and green box improvement of the IRIS staging. Arrows indicate the direction of the change from baseline. The odds of worsening the IRIS stage were 17 (95% confidence interval: 1.55 to 196, *p* = 0.01 Table ²) times higher in cats homozygous for the fAIM variant allele than in wild type cats. Panel B: Percentage of change of serum creatinine concentration at the end of the follow-up period relative to concentration at diagnosis for cats with 2 copies (homozygous), 3 copies (heterozygous), and 4 copies (homozygous) of exon 3 of fAIM. The odds of worsening the serum creatinine concentration were 9 (95% confidence interval: 1.38 to 67, *p* = 0.03, Table ²) times higher in cats homozygous for the fAIM variant allele than in wild type cats.

3.4.2.

Approach B: Percentage Change in SCr From Baseline

Twenty-four cats had progressively worsened in the SCr, whereas the remaining cats remained non-progressive (*n* = 26; Figure ^{4B} and Table ^{S3}). No correlation was found between body weight and SCr (*p* > 0.05). Results for SCr and BUN changes are presented in Tables ¹ and ^{S3}.

3.5.

Number of Copies of fAIM Exon 3

Fifteen of 50 (30%) cats were homozygous for the wild-type allele of fAIM, and 35 of 50 (70%) cats were variant-type (> 2 copies of exon 3, Table ^{S3}). Within the fAIM variant group, 26 of 50 (52%) cats were heterozygous for the variant allele of fAIM (3 copies of exon 3), whereas 9 (18%) cats were homozygous for the variant allele of fAIM (4 copies of exon 3, Table ^{S3}).

3.6.

Association Between Copies of fAIM Exon 3 and Change in SCr

A significant association was found between fAIM exon 3 copy number variation and IRIS staging change (Figure ⁴, Panels A and B, Table ^{S2}). Cats with 4 copies of exon 3 of fAIM had higher odds of worsening (*p* = 0.01) IRIS stage (approach A) and SCr than cats with wild-type fAIM exon 3 (approach B, *p* = 0.03, Table ²).

TABLE 2

Phenotype–genotype statistical association: The strength of the association between phenotype and fAIM genotype was determined by calculating the OR (95% CI) and tested statistically using Fisher's exact test.

Genotype	Phenotype	
Worsen IRIS stage vs. non-progressive IRIS stage	Worsen Serum creatinine concentration vs. non-progressive SCr	
Homozygous + heterozygous variant vs. wild-type	7.3 (0.85 to 62) $p = 0.07$	3.66 (0.97 to 13) $p = 0.05$
Homozygous variant vs. wild-type	17 (1.55 to 196) $p = 0.01$	9.6 (1.38 to 67) $p = 0.03$

3.7.

Frequency of fAIM Exon 3 Genotypes in a Subset of DNA Samples From WSU DNA Bank

Of the selected 100 banked feline DNA samples, 38% were homozygous for the wild-type fAIM exon 3 allele, 20% of the cats were homozygous for the variant fAIM exon 3 allele, and the remaining 42% were heterozygous.

4.

Discussion

We determined that some cats harbor an altered fAIM gene consisting of a large intragenic tandem exon 3 and flanking intron duplication of the fAIM gene (Figures ³, Panel B and ^{S2}). Duplication of exon 3 can occur in one or both alleles of fAIM, resulting in three genotypes: those homozygous for the wild-type fAIM exon 3 allele, those homozygous for the variant fAIM exon 3 allele, and heterozygotes. The wild-type fAIM gene codes for a protein consisting of three cysteine-rich domains referred to as scavenger receptor cysteine-rich domains (SRCR1, 2, 3, Figure ³, Panel B) [⁵]. On the other hand, the duplication of exon 3 leads to the duplication of the SRCR protein subunit 1 (Figure ³, Panels A and B).

The most clinically relevant result of our study is that the fAIM genotype is associated with deteriorating renal function (Table ²). Most of the cats with CKD (83%, 20/24, Table ^{S3}) that showed a progressive decline in kidney function were cats in the population with the variant fAIM genotype (Table ² and Figures ³ and ^{S2}). The altered structure of the fAIM variant protein might affect its function in vivo. In a previous study [⁵], one of many functions of fAIM was investigated [²¹] and it was shown that both the wild-type and mutant fAIM variants support comparable macrophage phagocytic capability in vitro. However, the authors did not investigate the differences in pro-repair renal function of the wild-type and variant fAIM proteins in vivo.

Based on computer simulations, we identified potential physicochemical differences between the fAIM protein variants (Table ^{S2}). The 4-SRCR domain variant of the fAIM protein (variant protein resulting from an exon 3 duplication) is not only larger than the wild-type 3-SRCR domain protein, but the extra domain also increases the

overall negative charge. Because negative charge hinders the renal filtration of molecules [22], a larger and more negatively charged variant fAIM might not be filtered into the renal tubules (its site of action) as efficiently as wild-type fAIM, thereby compromising its pro-repair function in fAIM variant cats. Comparing fAIM urine concentrations of genotyped cats would provide valuable insight into potential glomerular filtration impairment. Alternatively, the larger fAIM variant might reach its sites of action in the tubular lumen, but the additional SRCR domain might sterically hinder its interaction with ligands (e.g., kidney injury molecule-1 or cluster differentiation 36) in vivo [5, 23, 24, 25]. Although these possibilities have yet to be investigated, they offer plausible mechanisms consistent with our findings.

In our study, 50% of the heterozygous cats showed a $\geq 20\%$ increase in SCr at the end of the follow-up period with respect to baseline. This finding could be a result of differential gene expression leading to the synthesis of the wild-type fAIM protein only or higher proportional quantities of wild-type fAIM than variant fAIM protein. Considering the complexity of renal disease, other explanations not involving fAIM could impact our results, but the specific explanations have yet to be identified. Notably, findings from a recent study provide a possible explanation for the worsening of IRIS stage in wild-type cats [26]. The study showed that a high salt diet prevents AIM activation, precluding its renal function [26]. Our study was not designed to evaluate the impact of diet on the effect of wild-type and variant fAIM on CKD cats, but this aspect deserves attention in future research.

Our research suggests a role for the fAIM genetic variant with respect to renal function (as estimated by SCr) in cats with CKD. The potential clinical utility of a gene variant as a marker to identify cats at a higher risk of progressive decline in kidney function would be limited if it were harbored only by a few cat breeds and present in a small fraction of the population. However, the fAIM variant reported here is present in several breeds (Table S3) and was identified in 60% of the cats screened. This finding emphasizes the potential clinical impact of this genetic marker and should prompt additional research efforts.

Our study has some limitations, some of which are inherent to its retrospective design and the complexity of kidney disease. Most of the cases had comorbidities such as hypertension and were receiving various treatments (Table S3) that might have confounded the SCr and study results. Some of these factors were controlled, in part, by the inclusion and exclusion criteria applied, but inclusion and exclusion criteria also might have biased the study results. The other important limitation regards constraints of SCr as a maker of kidney disease. We used two different criteria for phenotyping cats to improve the rigor of the results, but both approaches rely on SCr. Unfortunately, there were no other makers routinely used in clinical practice that we could have applied consistently in our study, but future research would benefit from information provided by other biomarkers.

A larger genetic population study also would help us understand which other breeds carry this genetic change and whether any specific breeds are more likely to harbor the fAIM exon 3 variant genotypes. Also, our study simulated wild-type and variant fAIM protein (Figure 3, Panel B) in silico based on the DNA sequence but did not study the protein structure of fAIM protein isolated from genotyped cats. Therefore, interpretations of the results assume that the altered fAIM gene would produce an altered fAIM protein. All of these limitations emphasize the need for more studies to confirm our findings and clarify the limitations of fAIM genetic testing as a risk factor for progressive CKD or perhaps even CKD itself.

Our study confirms that the previously described 4-domain variant of fAIM resulted from an exon duplication at the genomic level. This finding supports noninvasive sampling methods for identifying cats harboring this fAIM genetic alteration. The exon 3 duplication in the fAIM gene is present in multiple breeds. Our study also found a significant association between exon 3 duplication in the fAIM gene and a decline in kidney function in a group of client-owned cats from the Pacific Northwest. This discovery can potentially change strategies for diagnosing, treating, and preventing kidney disease, creating new opportunities to improve the standard of care for cats. Although results of our study suggest that fAIM variants are associated with CKD progression, supporting experimental evidence would be necessary to determine if fAIM variants are biologically detrimental or functionally inferior to the wild-type fAIM. A large multicenter prospective study is needed to confirm our findings, evaluate the predictive value of fAIM exon 3 variants on progressive CKD, and to understand the global impact of this genetic alteration on CKD cats using a genetic epidemiology study.

Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Approved by the Institutional Animal Care and Use Committee of Washington State University (protocols #6493 and 4915).

Conflicts of Interest

Dr. Villarino might establish a service center at Washington State University that offers genetic testing. The other authors declare no conflicts of interest.

Supporting information

Figure S1: Fragment analysis of the fAIM PCR product reveals 3 amplicon sizes between ~10 and 12.5 k base pairs. Sample A1 and A4 (~12.5 k base pairs amplicon) both alleles contained 2 copies of exon 3 of 321 base pairs corresponding to a homozygous for the mutant allele ($E3 \times 4$). Sample A2 corresponds to a heterozygous. Sample A3 ~10.4 k base pairs each allele contained an exon 3 of 321 base pairs corresponding to a homozygous wild-type gene ($E3 \times 2$). The two bands in A3 are due to the intron sequence variation among wild-types. Specifically, the upper band contains a repetitive segment of Exon 2 along with intron, whereas the lower band lacks these repetitive sequences. As a result, the wildtype produced two bands, including one of smaller size.

Figure S2: fAIM introns and exons maps for homozygous wild-type gene from the sample containing ~10 k base pairs amplicon (panel A) depicting 1 copy of exon 3. Panel B; homozygous mutant allele (2 copies of exon 3 of 321 base pairs) from the sample containing a ~12.5 k base pairs amplicon and a qPCR with 4 copies of Exon 3. Panel C; heterozygous for the mutant allele (one copy of Exon 3 in one of the alleles and 2 copies of exon 3 in the other one).

Table S1: Amino acid sequence for feline apoptosis inhibitor of macrophage. The wild-type fAIM protein contains 3 scavenger receptor cysteine-rich (SRCR) protein domains, while the variant contains 4-SRCR domains.

Table S2: The physicochemical properties of both wild-type 3-SRCR domain and variant 4-SRCR domain fAIM calculated using the ExPASy ProtParam tool following conversion of nucleotide sequence into amino acid sequence.

Table S3: Clinical data of cats diagnosed with chronic kidney disease that meet the inclusion criteria.

Table S4: List of cats that were excluded from the analysis because did not meet the inclusion criteria.

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References

1. Brown C. A., Elliott J., Schmiedt C. W., and Brown S. A., “Chronic Kidney Disease in Aged Cats: Clinical Features, Morphology, and Proposed Pathogeneses,” *Veterinary Pathology* 53, no. 2 (2016): 309–326, 10.1177/0300985815622975.
2. DiBartola S. P., Rutgers H. C., Zack P. M., et al., “Clinicopathologic Findings Associated With Chronic Renal Disease in Cats: 74 Cases (1973-1984),” *Journal of the American Veterinary Medical Association* 190, no. 9 (1987): 1196–1202, 10.2460/javma.1987.190.9.1196.
3. Lawson J. S., Liu H. H., Syme H. M., Purcell R., Wheeler-Jones C. P. D., and Elliott J., “The Cat as a Naturally Occurring Model of Renal Interstitial Fibrosis: Characterisation of Primary Feline Proximal Tubular Epithelial Cells and Comparative Pro-Fibrotic Effects of TGF- β 1,” *PLoS One* 13, no. 8 (2018): e0202577.
4. Sugisawa R., Komatsu G., Hiramoto E., et al., “Independent Modes of Disease Repair by AIM Protein Distinguished in AIM-Felinized Mice,” *Scientific Reports* 8, no. 1 (2018): 13157.
5. Sugisawa R., Hiramoto E., Matsuoka S., et al., “Impact of Feline AIM on the Susceptibility of Cats to Renal Disease,” *Scientific Reports* 6 (2016): 35251.
6. Morais G. B., Viana D. A., Silva F. M. O., et al., “Polarization Microscopy as a Tool for Quantitative Evaluation of Collagen Using Picrosirius Red in Different Stages of CKD in Cats,” *Microscopy Research and Technique* 80, no. 5 (2017): 543–550.
7. Finch N., “Measurement of Glomerular Filtration Rate in Cats: Methods and Advantages Over Routine Markers of Renal Function,” *Journal of Feline Medicine and Surgery* 16, no. 9 (2014): 736–748.
8. Cowgill L. D., Polzin D. J., Elliott J., et al., “Is Progressive Chronic Kidney Disease a Slow Acute Kidney Injury?,” *Veterinary Clinics of North America. Small Animal Practice* 46, no. 6 (2016): 995–1013.
9. Chakrabarti S., Syme H. M., and Elliott J., “Clinicopathological Variables Predicting Progression of Azotemia in Cats With Chronic Kidney Disease,” *Journal of Veterinary Internal Medicine* 26, no. 2 (2012): 275–281.
10. Guo C., Dong G., Liang X., and Dong Z., “Epigenetic Regulation in AKI and Kidney Repair:

Mechanisms and Therapeutic Implications,” *Nature Reviews. Nephrology* 15, no. 4 (2019): 220–239. **11.** Tampe B. and Zeisberg M., “Contribution of Genetics and Epigenetics to Progression of Kidney Fibrosis,” *Nephrology, Dialysis, Transplantation* 29, no. Suppl 4 (2014): iv72–iv79. **12.** Oshima M., Iwata Y., Furuichi K., et al., “Association of Apoptosis Inhibitor of Macrophage (AIM) Expression With Urinary Protein and Kidney Dysfunction,” *Clinical and Experimental Nephrology* 21, no. 1 (2017): 35–42. **13.** Arai S., Kitada K., Yamazaki T., et al., “Apoptosis Inhibitor of Macrophage Protein Enhances Intraluminal Debris Clearance and Ameliorates Acute Kidney Injury in Mice,” *Nature Medicine* 22, no. 2 (2016): 183–193. **14.** Nielsen H., Tsirigos K. D., Brunak S., and von Heijne G., “A Brief History of Protein Sorting Prediction,” *Protein Journal* 38, no. 3 (2019): 200–216. **15.** Xu J., Mcpartlon M., and Li J., “Improved Protein Structure Prediction by Deep Learning Irrespective of Co-Evolution Information,” *Nature Machine Intelligence* 3 (2021): 601–609. **16.** Xu J. and Wang S., “Analysis of Distance-Based Protein Structure Prediction by Deep Learning in CASP13,” *Proteins* 87, no. 12 (2019): 1069–1081. **17.** Pettersen E. F., Goddard T. D., Huang C. C., et al., “UCSF Chimera – A Visualization System for Exploratory Research and Analysis,” *Journal of Computational Chemistry* 25, no. 13 (2004): 1605–1612, 10.1002/jcc.20084. **18.** Gasteiger E., Hoogland C., Gattiker A., et al., “Protein Identification and Analysis Tools on the ExPASy Server,” in *The Proteomics Protocols Handbook*. Springer Protocols Handbooks, ed. Walker J. M. (Humana Press, 2005), 571–607. **19.** International Renal Interest Society, “IRIS Staging of CKD,” (2023), http://www.iris-kidney.com/pdf/2_IRIS_Staging_of_CKD_2023.pdf. **20.** Altman D. G., *Practical Statistics for Medical Research* (Chapman and Hall, 1991). **21.** Sanjurjo L., Aran G., Roher N., Villedor A. F., and Sarrias M. R., “AIM/CD5L: A Key Protein in the Control of Immune Homeostasis and Inflammatory Disease,” *Journal of Leukocyte Biology* 98, no. 2 (2015): 173–184. **22.** Miner J. H., “Glomerular Filtration: The Charge Debate Charges Ahead,” *Kidney International* 74, no. 3 (2008): 259–261. **23.** Yang X., Okamura D. M., Lu X., et al., “CD36 in Chronic Kidney Disease: Novel Insights and Therapeutic Opportunities,” *Nature Reviews. Nephrology* 13, no. 12 (2017): 769–781. **24.** Ichimura T., Asseldonk E. J., Humphreys B. D., Gunaratnam L., Duffield J. S., and Bonventre J. V., “Kidney Injury Molecule-1 Is a Phosphatidylserine Receptor That Confers a Phagocytic Phenotype on Epithelial Cells,” *Journal of Clinical Investigation* 118, no. 5 (2008): 1657–1668, 10.1172/JCI34487. **25.** Matsuura K., Maehara N., Hirota A., et al., “Two Independent Modes of Kidney Stone Suppression Achieved by AIM/CD5L and KIM-1,” *Communications Biology* 5, no. 1 (2022): 783. **26.** Wang C. T., Tezuka T., Takeda N., Araki K., Arai S., and Miyazaki T., “High Salt Exacerbates Acute Kidney Injury by Disturbing the Activation of CD5L/Apoptosis Inhibitor of Macrophage (AIM) Protein,” *PLoS One* 16, no. 11 (2021): e0260449.

Associated Data

Supplementary Materials

Figure S1: Fragment analysis of the fAIM PCR product reveals 3 amplicon sizes between ~10 and 12.5 k base pairs. Sample A1 and A4 (~12.5 k base pairs amplicon) both alleles contained 2 copies of exon 3 of 321 base pairs corresponding to a homozygous for the mutant allele ($E3 \times 4$). Sample A2 corresponds to a heterozygous. Sample A3 ~10.4 k base pairs each allele contained an exon 3 of 321 base pairs corresponding to a homozygous wild-type gene ($E3 \times 2$). The two bands in A3 are due to the intron sequence variation among wild-types. Specifically, the upper band contains a repetitive segment of Exon 2 along with intron, whereas the lower band lacks these repetitive sequences. As a result, the wildtype produced two bands, including one of smaller size.

Figure S2: fAIM introns and exons maps for homozygous wild-type gene from the sample containing ~10 k base pairs amplicon (panel A) depicting 1 copy of exon 3. Panel B; homozygous mutant allele (2 copies of exon 3 of 321 base pairs) from the sample containing a ~12.5 k base pairs amplicon and a qPCR with 4 copies of Exon 3. Panel C; heterozygous for the mutant allele (one copy of Exon 3 in one of the alleles and 2 copies of exon 3 in the other one).

Table S1: Amino acid sequence for feline apoptosis inhibitor of macrophage. The wild-type fAIM protein contains 3 scavenger receptor cysteine-rich (SRCR) protein domains, while the variant contains 4-SRCR domains.

Table S2: The physicochemical properties of both wild-type 3-SRCR domain and variant 4-SRCR domain fAIM calculated using the ExPASy ProtParam tool following conversion of nucleotide sequence into amino acid sequence.

Table S3: Clinical data of cats diagnosed with chronic kidney disease that meet the inclusion criteria.

Table S4: List of cats that were excluded from the analysis because did not meet the inclusion criteria.

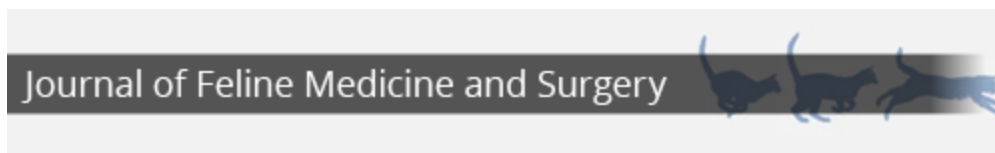
References

1. Brown C. A., Elliott J., Schmiedt C. W., and Brown S. A., “Chronic Kidney Disease in Aged Cats: Clinical Features, Morphology, and Proposed Pathogeneses,” *Veterinary Pathology* 53, no. 2 (2016): 309–326, 10.1177/0300985815622975.
2. DiBartola S. P., Rutgers H. C., Zack P. M., et al., “Clinicopathologic Findings Associated With Chronic Renal Disease in Cats: 74 Cases (1973-1984),” *Journal of the American Veterinary Medical Association* 190, no. 9 (1987): 1196–1202, 10.2460/javma.1987.190.09.1196.
3. Lawson J. S., Liu H. H., Syme H. M., Purcell R., Wheeler-Jones C. P. D., and Elliott J., “The Cat as a Naturally Occurring Model of Renal Interstitial Fibrosis: Characterisation of Primary Feline Proximal Tubular Epithelial Cells and Comparative Pro-Fibrotic Effects of TGF- β 1,” *PLoS One* 13, no. 8 (2018): e0202577.
4. Sugisawa R., Komatsu G., Hiramoto E., et al., “Independent Modes of Disease Repair by AIM Protein Distinguished in AIM-Felized Mice,” *Scientific Reports* 8, no. 1 (2018): 13157.
5. Sugisawa R., Hiramoto E., Matsuoka S., et al., “Impact of Feline AIM on the Susceptibility of Cats to Renal Disease,” *Scientific Reports* 6 (2016): 35251.
6. Morais G. B., Viana D. A., Silva F. M. O., et al., “Polarization Microscopy as a Tool for Quantitative Evaluation of Collagen Using Picrosirius Red in Different Stages of CKD in Cats,” *Microscopy Research and Technique* 80, no. 5 (2017): 543–550.
7. Finch N., “Measurement of Glomerular Filtration Rate in Cats: Methods and Advantages Over Routine Markers of Renal Function,” *Journal of Feline Medicine and Surgery* 16, no. 9 (2014): 736–748.
8. Cowgill L. D., Polzin D. J., Elliott J., et al., “Is Progressive Chronic Kidney Disease a Slow Acute Kidney Injury?,” *Veterinary Clinics of North America. Small Animal Practice* 46, no. 6 (2016): 995–1013.
9. Chakrabarti S., Syme H. M., and Elliott J., “Clinicopathological Variables Predicting Progression of Azotemia in Cats With Chronic Kidney Disease,” *Journal of Veterinary Internal Medicine* 26, no. 2 (2012): 275–281.
10. Guo C., Dong G., Liang X., and Dong Z., “Epigenetic Regulation in AKI and Kidney Repair: Mechanisms and Therapeutic Implications,” *Nature Reviews. Nephrology* 15, no. 4 (2019): 220–239.
11. Tampe B. and Zeisberg M., “Contribution of Genetics and Epigenetics to Progression of Kidney Fibrosis,” *Nephrology, Dialysis, Transplantation* 29, no. Suppl 4 (2014): iv72–iv79.
12. Oshima M., Iwata Y., Furuichi K., et al., “Association of Apoptosis Inhibitor of Macrophage (AIM) Expression With Urinary Protein and Kidney Dysfunction,” *Clinical and Experimental Nephrology* 21, no. 1 (2017): 35–42.

- 13. Arai S., Kitada K., Yamazaki T., et al., “Apoptosis Inhibitor of Macrophage Protein Enhances Intraluminal Debris Clearance and Ameliorates Acute Kidney Injury in Mice,” *Nature Medicine* 22, no. 2 (2016): 183–193.
- 14. Nielsen H., Tsirigos K. D., Brunak S., and von Heijne G., “A Brief History of Protein Sorting Prediction,” *Protein Journal* 38, no. 3 (2019): 200–216.
- 15. Xu J., Mcpartlon M., and Li J., “Improved Protein Structure Prediction by Deep Learning Irrespective of Co-Evolution Information,” *Nature Machine Intelligence* 3 (2021): 601–609.
- 16. Xu J. and Wang S., “Analysis of Distance-Based Protein Structure Prediction by Deep Learning in CASP13,” *Proteins* 87, no. 12 (2019): 1069–1081.
- 17. Pettersen E. F., Goddard T. D., Huang C. C., et al., “UCSF Chimera – A Visualization System for Exploratory Research and Analysis,” *Journal of Computational Chemistry* 25, no. 13 (2004): 1605–1612, 10.1002/jcc.20084.
- 18. Gasteiger E., Hoogland C., Gattiker A., et al., “Protein Identification and Analysis Tools on the ExPASy Server,” in *The Proteomics Protocols Handbook*. Springer Protocols Handbooks, ed. Walker J. M. (Humana Press, 2005), 571–607.
- 19. International Renal Interest Society , “IRIS Staging of CKD,” (2023), http://www.iris-kidney.com/pdf/2_IRIS_Staging_of_CKD_2023.pdf.
- 20. Altman D. G., *Practical Statistics for Medical Research* (Chapman and Hall, 1991).
- 21. Sanjurjo L., Aran G., Roher N., Valledor A. F., and Sarrias M. R., “AIM/CD5L: A Key Protein in the Control of Immune Homeostasis and Inflammatory Disease,” *Journal of Leukocyte Biology* 98, no. 2 (2015): 173–184.
- 22. Miner J. H., “Glomerular Filtration: The Charge Debate Charges Ahead,” *Kidney International* 74, no. 3 (2008): 259–261.
- 23. Yang X., Okamura D. M., Lu X., et al., “CD36 in Chronic Kidney Disease: Novel Insights and Therapeutic Opportunities,” *Nature Reviews. Nephrology* 13, no. 12 (2017): 769–781.
- 24. Ichimura T., Asseldonk E. J., Humphreys B. D., Gunaratnam L., Duffield J. S., and Bonventre J. V., “Kidney Injury Molecule-1 Is a Phosphatidylserine Receptor That Confers a Phagocytic Phenotype on Epithelial Cells,” *Journal of Clinical Investigation* 118, no. 5 (2008): 1657–1668, 10.1172/JCI34487.
- 25. Matsuura K., Maehara N., Hirota A., et al., “Two Independent Modes of Kidney Stone Suppression Achieved by AIM/CD5L and KIM-1,” *Communications Biology* 5, no. 1 (2022): 783.
- 26. Wang C. T., Tezuka T., Takeda N., Araki K., Arai S., and Miyazaki T., “High Salt Exacerbates Acute Kidney Injury by Disturbing the Activation of CD5L/Apoptosis Inhibitor of Macrophage (AIM) Protein,” *PLoS One* 16, no. 11 (2021): e0260449.

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Prevalence of the *AIM* exon 3 duplication variant, a putative biomarker associated with progression of kidney disease, in 1000 cats

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Abstract

Objectives

A protein called apoptosis inhibitor of macrophages (AIM) helps clear waste from the kidney tubules. Some domestic cats harbor a duplication of exon 3 in the *AIM* gene, producing a four-domain AIM protein that includes an additional scavenger receptor cysteine-rich subunit. This variant has been associated with increased risk of chronic kidney disease. Pilot studies suggest that homozygosity for this genetic alteration is present in approximately 20% of the cat population. The objective of the current study was to determine the prevalence of the wild-type, heterozygous and homozygous variant *AIM* exon 3 genotypes in a large population of domestic cats.

Methods

Genomic DNA from 1000 client-owned cats archived in the Program in Individualized Medicine DNA Bank at Washington State University was analyzed by droplet digital PCR to determine *AIM* exon 3 copy number variation. Samples represented cats of any age, sex, neuter status or clinical condition. Genotype frequencies were summarized with 95% Wilson confidence intervals (CIs) and genotype frequencies were tested for deviation from Hardy–Weinberg equilibrium using χ^2 analysis.

Results

Of the 1000 cats, 340 (34.0%, 95% CI 31.1–36.9) were homozygous wild-type (CNV2), 467 (46.7%, 95% CI 43.6–50.0) were heterozygous (CNV3) and 193 (19.3%, 95% CI 17.0–21.9) were homozygous for the variant (CNV4). Genotype frequencies deviate from Hardy–Weinberg equilibrium ($\chi^2 = 4.49$, degrees of freedom = 1; $P = 0.034$).

Conclusions and relevance

Approximately 1/5 cats were homozygous for the *AIM* exon 3 duplication, and nearly half harbored one copy of the variant allele. This genetic alteration is not restricted to a single breed or type. Given the potential role of this variant in feline kidney disease, the results of this study justify further investigations using genetically confirmed and geographically diverse populations to better define global allele distributions and potential breed predispositions.

Keywords: Feline apoptosis inhibitor of macrophages, AIM, biomarker, copy number variation, kidney

Introduction

Apoptosis inhibitor of macrophages (AIM), also known as CD5-like antigen, is a protein synthesized primarily by tissue macrophages that circulates in plasma bound to immunoglobulins.^{1,2}

The wild-type feline AIM protein comprises three scavenger receptor cysteine-rich (SRCR) domains (SRCR1–3).² Some domestic cats exhibit a structural variant arising from duplication of exon 3 in one or both alleles of the feline *AIM* gene. This duplication produces a four-domain AIM protein with an additional SRCR1 subunit and yields three possible genotypes: homozygous wild-type, heterozygous and homozygous variants.³

The exact phenotype associated with the variant has yet to be elucidated. However, a recent study reported that cats homozygous for the exon 3 variant allele had 17× greater odds of advancing to a higher International Renal Interest Society chronic kidney disease (CKD) stage compared with wild-type cats.³ These results imply that *AIM* genotypes likely carry significant clinical relevance with respect to CKD.³

Although the variant appears common and clinically relevant, the practical utility of *AIM* genotyping depends on its allele frequency in the general feline population and among breeds. Preliminary data suggested that a high proportion of cats carry the four-domain variant of *AIM*: 37.5% were heterozygous and 22% homozygous in one pilot study.² Consistent results were observed in another cohort of 100 cats, where 40% were heterozygous and 20% homozygous for the variant.³ However, these earlier studies were limited by small sample sizes. Therefore, the objective of the present study was to determine the prevalence of wild-type, heterozygous and homozygous variant *AIM* exon 3 genotypes in a larger population of 1000 cats.

Materials and methods

Study design

Copy-number frequencies for exon 3 of *AIM* (corresponding to two, three or four copies) were determined from 1000 archived DNA samples obtained from the Program in Individualized Medicine DNA Bank (IACUC protocol #6493). These samples originated from client-owned cats admitted to the Washington State University (WSU) Veterinary Teaching Hospital. Cats of any age, sex, and neuter or clinical status were eligible for inclusion. Samples were randomly selected from a repository of more than 1500 banked feline DNA specimens based on storage availability. Genotyping for *AIM* exon 3 copy number variation (CNV) was performed using the droplet digital PCR protocol described below.³

Genomic DNA isolation

Genomic DNA was isolated from feline cheek-swab samples using the Quick-DNA Miniprep Kit (Zymo Research) according to the manufacturer's protocol. DNA concentrations were quantified fluorometrically using a Qubit double-stranded DNA assay (Invitrogen) following the manufacturer's instructions. Extracted DNA was stored at -80°C until analysis.

Owner-reported breed or type was recorded when available (n = 996/1000); no pedigree verification was performed. Cats reported as domestic shorthair, domestic mediumhair or domestic longhair were classified as random-bred. All other labels (eg, Bengal, Birman, Ragdoll) were grouped as purebred for descriptive purposes.

Determination of feline *AIM* exon 3 copy number

The analytical methodology followed a previously validated approach.³

Statistical analysis

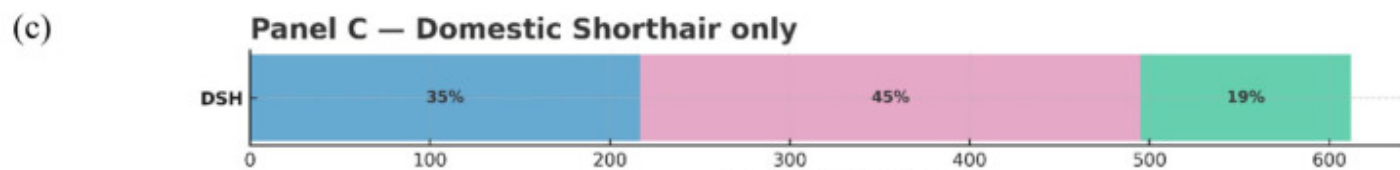
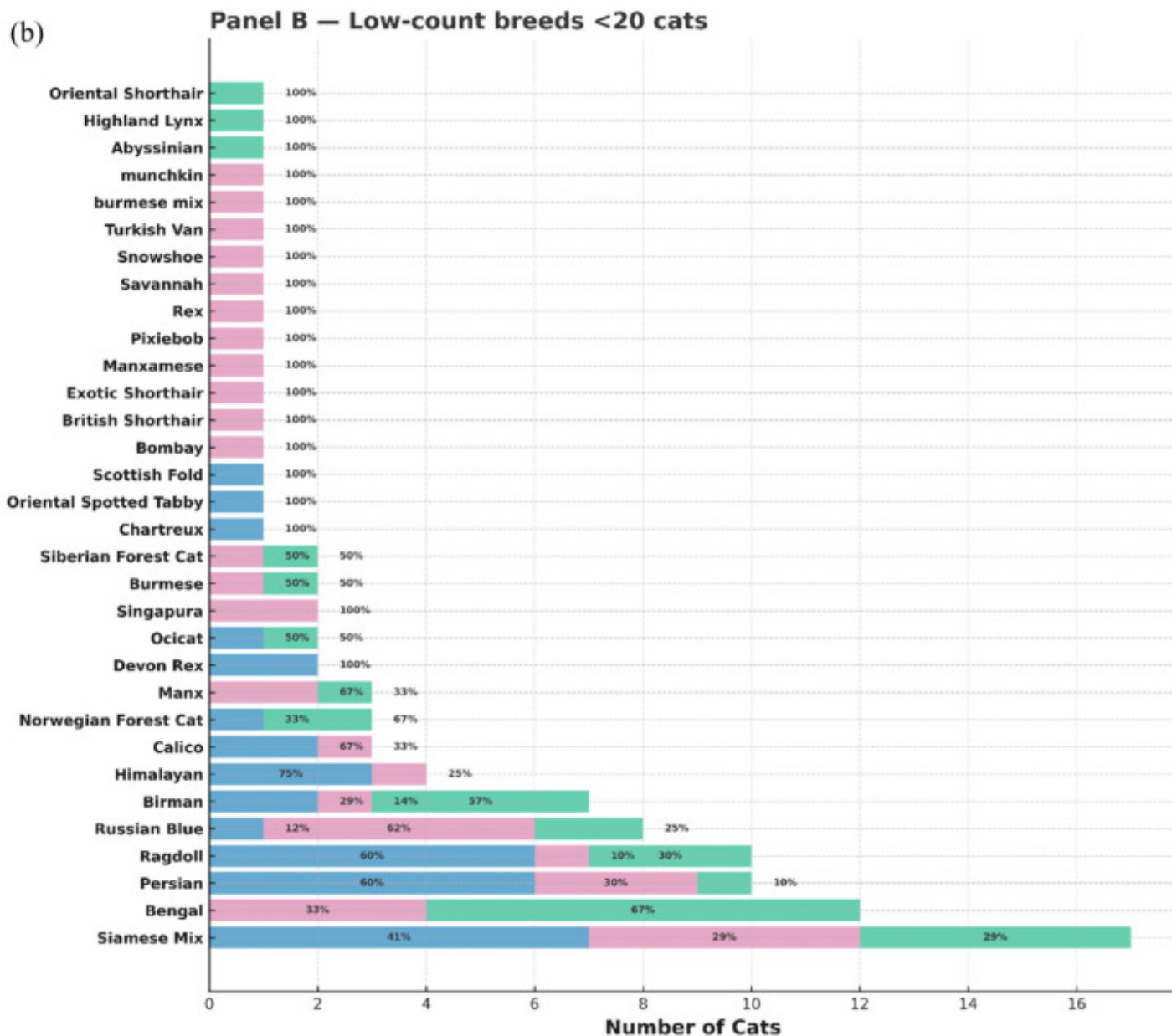
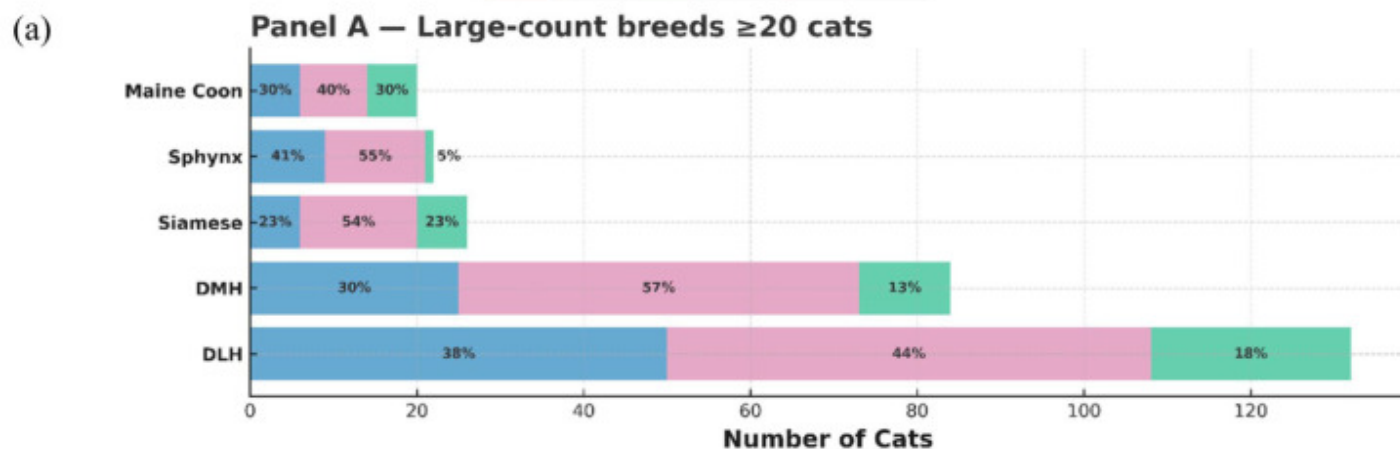
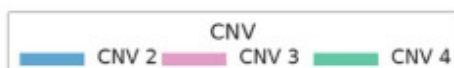
Descriptive statistics were used to summarize genotype frequencies. Genotype proportions were expressed with 95% confidence intervals (CIs) calculated using the Wilson score method,^{4–6} which provides accurate coverage for binomial proportions, particularly for large or moderate sample sizes. Allele frequencies were calculated from observed genotype counts and expected genotype frequencies under Hardy–Weinberg equilibrium (HWE) using p^2 , $2pq$ and q^2 .^{7,8} Departure from HWE was assessed using a χ^2 goodness-of-fit test with one degree of freedom.^{7,8} The significance level was set at $P < 0.05$. All analyses were performed in R (v4.3) and Excel (Microsoft). Because mixing genetically distinct subpopulations can artificially produce heterozygote deficits (Wahlund effect), we assessed the influence of breed structure by identifying purebred groups that contributed disproportionately to the overall χ^2 statistic. High-contributing purebred populations were defined a priori as breeds in which five or more individuals were available and genotype distributions showed an excess of homozygous genotypes (CNV2 or CNV4) relative to random-bred cats. HWE was then re-evaluated in the remaining cohort to determine whether population substructure explained the initial deviation.

Results

Of the 1000 cats analyzed, 340 (34.0%, 95% CI 31.1–36.9) were homozygous wild-type (CNV2), 467 (46.7%, 95% CI 43.6–50.0) were heterozygous (CNV3) and 193 (19.3%, 95% CI 17.0–21.9) were homozygous for the duplicated exon (CNV4). The duplicated allele had an overall frequency of 0.43. The variant was detected across 34 owner-reported breeds or types ([Figure 1](#)). Genotype frequencies in the full cohort deviated from Hardy–Weinberg equilibrium ($\chi^2 = 4.49$; $P = 0.034$), reflecting an excess of homozygous genotypes. Inspection of breed-stratified data showed that this deviation was driven by several purebred groups with skewed genotype distributions. Some breeds had high duplication allele frequency ([Table S1 in the supplementary material](#) [\[7\]](#)). No individual

breed demonstrated significant deviation from Hardy–Weinberg equilibrium when analyzed separately. After removing these high-contributing purebreds (n = 54 cats), the remaining cats (n = 946) no longer deviated from equilibrium ($\chi^2 = 1.43$; $P = 0.23$). In this filtered cohort (n = 946), genotype proportions were 34.6% CNV2 (95% CI 31.6–37.7), 46.8% CNV3 (95% CI 43.7–50.0) and 18.6% CNV4 (95% CI 16.3–21.2).

Figure 1.



Distribution of feline apoptosis inhibitor of macrophages exon 3 genotypes (CNV2, CNV3, CNV4) across owner-reported breeds/type (n = 996). No breed/type was available for 4/1000 cat samples, and therefore these are not represented in this figure. (a) Breeds with ≥ 20 cats. (b) Breeds with < 20 cats. (c) Domestic shorthair (DSH) only. Percentages represent within-breed/type genotype proportions. CNV: copy number variation; DMH = domestic mediumhair; DLH = domestic longhair

Discussion

This large-scale population study revealed that approximately 1/5 domestic cats were homozygous for the *AIM* exon 3 duplication, a variant predicted to introduce an additional SRCR domain through duplication of the SRCR1 subunit.³ Nearly half of all cats carried a single copy of the duplication, confirming that this allele is common in the general population and consistent with earlier pilot observations.³ The duplication was identified across 34 owner-reported breeds/types. Although breed information was based on client reports rather than verified pedigrees, and the study was not designed to estimate breed-specific prevalence, the widespread distribution suggests that breeding practices may influence allele frequencies within certain lineages. Genotype frequencies deviated from Hardy–Weinberg equilibrium. Duplication allele frequencies varied substantially among breeds, with higher frequencies observed in Bengal and Birman cats and lower frequencies in Persian and Sphynx cats, while random-bred cats exhibited intermediate values ([Table S1 in the supplementary material](#) [↗](#)). Pooling these genetically structured subpopulations resulted in an apparent deficit of heterozygotes consistent with a Wahlund effect.^{7–9}

It is important to acknowledge that although the overall data set was large, samples came from a single teaching hospital DNA bank, the number of genetically confirmed purebred cats was limited and not all breeds were represented. Therefore, the absence of HWE deviation among the breeds should not be interpreted as evidence that those populations are in true equilibrium or that selection acting on the *AIM* duplication is absent. Differences in allele frequency – or even selection for or against the duplicated exon – may exist in other purebred lineages or geographic regions not captured here. Larger, breed-specific cohorts will be required to evaluate these possibilities fully.

The *AIM* exon 3 homozygous variant has been associated with a markedly increased risk of progressive CKD (adjusted odds ratio 9.6, 95% CI 1.38–67).³ The high prevalence observed here underscores the potential clinical and translational relevance of this variant. No phenotype data (CKD status, creatinine, symmetric dimethylarginine or age) were available for this cohort; therefore, the present study reports only allele and genotype frequencies, not CKD risk within this population. The results from this study, in combination with prior CKD risk estimates, highlight the need for further work to determine how *AIM* genotyping might inform breeding strategies and clinical monitoring. Recent evidence supports the potential of *AIM*-based therapies for advanced CKD in cats.¹⁰ The effect of these therapies on disease progression in cats harboring the *AIM* variant remains unknown and warrants further investigation, as it may represent a significant advance in precision therapeutics for feline CKD.

This study has several important limitations. First, the cohort represents a hospital-based, convenience sample from a single veterinary teaching institution and therefore may not reflect allele frequencies in the general domestic cat population. In addition, no clinical phenotype data (CKD status, renal biomarkers, age, comorbidities) were available for this cohort, and therefore no genotype–phenotype associations can be inferred from these results; all disease associations are based on previously published work.

Conclusions

This study establishes the *AIM* exon 3 duplication as a common genetic variant in domestic cats. Future investigations involving genetically confirmed purebred populations and broader geographic sampling will be essential to characterize global allele distributions and potential breed-specific predispositions.

Supplemental Material

Table S1

[sj-docx-1-jfm-10.1177_1098612X261445138.docx](#)^(16KB, docx)

Complete, transparent list of duplication allele frequency (q) and sample size (N) for every breed or owner reported type included in the dataset.

Acknowledgments

Figure 1 was created with ChatGPT.

Footnotes

Accepted: 7 April 2026

Supplementary material: The following file is available as supplementary material:

Table S1: Complete, transparent list of duplication allele frequency (q) and sample size (N) for every breed or owner reported type included in the dataset.

Dr Villarino may establish a service center at WSU that offers genetic testing.

Funding: Dr Villarino's kidney research program is supported by the Washington State University College of Veterinary Medicine Kay Yarborough Nelson Distinguished Professor in Feline Health Endowment, a generous gift from Mary Kay Fowler, Washington State University intramural grants program and Washington Research Foundation.

Ethical approval: The work described in this manuscript involved the use of non-experimental (owned or unowned) animals and procedures that differed from established internationally recognized high standards ('best practice') of veterinary clinical care for the individual patient. The study therefore had prior ethical approval from an established (or ad hoc) committee as stated in the manuscript.

Informed consent: Informed consent (verbal or written) was obtained from the owner or legal custodian of all animal(s) described in this work (experimental or non-experimental animals, including cadavers, tissues and samples) for all procedure(s) undertaken (prospective or retrospective studies). No animals or people are identifiable within this publication, and therefore additional informed consent for publication was not required.

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References

1. Arai S, Kitada K, Yamazaki T, et al. Apoptosis inhibitor of macrophage protein enhances intraluminal debris clearance and ameliorates acute kidney injury in mice. *Nat Med* 2016; 22: 183–193. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
2. Sugisawa R, Hiramoto E, Matsuoka S, et al. Impact of feline AIM on the susceptibility of cats to renal disease. *Sci Rep* 2016; 6. DOI: 10.1038/srep35251. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
3. Evangelista GCL, Hwang JK, Broughton-Neiswanger LE, et al. Apoptosis inhibitor of macrophages in cats: a potential link between an exon 3 variant allele and progression of naturally occurring chronic kidney disease. *J Vet Intern Med* 2025; 39. DOI: 10.1111/jvim.70136. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
4. Wilson EB. Probable inference, the law of succession, and statistical inference. *J Am Stat Assoc* 1927; 22: 209–212. [[Google Scholar](#)]
5. Newcombe RG. Two-sided confidence intervals for the single proportion: comparison of seven methods. *Stat Med* 1998; 17: 857–872. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
6. Brown LD, Cai TT, DasGupta A. Interval estimation for a binomial proportion. *Stat Sci* 2001; 16: 101–133. [[Google Scholar](#)]
7. Wigginton JE, Cutler DJ, Abecasis GR. A note on exact tests of Hardy–Weinberg equilibrium. *Am J Hum Genet* 2005; 76: 887–893. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
8. Guo SW, Thompson EA. Performing the exact test of Hardy–Weinberg proportion for multiple alleles. *Biometrics* 1992; 48: 361–372. [[PubMed](#)] [[Google Scholar](#)]
9. Hardy GH. Mendelian proportions in a mixed population. *Science* 1908; 28: 49–50. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
10. Tezuka T, Arakawa H, Kudo K, et al. A clinical impact of apoptosis inhibitor of macrophage on feline chronic kidney disease. *Vet J* 2026; 315. DOI: 10.1016/j.tvjl.2026.106545. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

Table S1

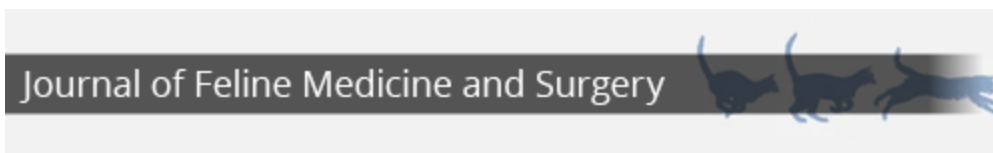
[sj-docx-1-jfm-10.1177_1098612X261445138.docx](#) ^(16KB, docx)

Complete, transparent list of duplication allele frequency (q) and sample size (N) for every breed or owner reported type included in the dataset.

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Feline CKD

Pathophysiology and risk factors — what do we know?

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Abstract

Practical relevance:

Chronic kidney disease (CKD) is one of the most frequently encountered disorders in cats, having increased in prevalence in recent decades. Although the underlying cause is rarely identified, the common final outcome of feline CKD is tubulointerstitial fibrosis. Knowledge of CKD pathophysiology is necessary for optimal individualised patient management, especially with regard to diagnosis and treatment of extrarenal complications.

Patient group:

CKD is most common in senior and geriatric cats, but should be considered in any feline patient with ureterolithiasis, hyperthyroidism, retrovirus infection, systemic hypertension, cardiovascular disease or urinary tract infection.

Evidence base:

Most of our knowledge of the pathogenesis of CKD is extrapolated from human nephrology and experimental animal studies. There is, therefore, a need for further studies in cats. The prevalence of clinical signs in feline CKD is well documented. Several concurrent diseases associated with CKD have also been reported in cats, especially in the geriatric population, but there is no or only limited published evidence demonstrating a cause-and-effect relationship between most of these conditions and CKD. Studies performed over the past 15 years have nevertheless allowed identification of major risk factors (proteinuria, plasma phosphate and plasma creatinine) influencing the progression of feline CKD.

Clinical challenges:

Clinical signs occur in the late stages of renal disease, so populations at higher risk of CKD should be screened routinely. CKD-associated complications (systemic hypertension, secondary renal hyperparathyroidism, hypokalaemia, anaemia, metabolic acidosis) must not be overlooked as they may affect the progression of disease. Disease progression is itself unpredictable and renal function may remain stable for extended periods. Most cats with early CKD do not progress to end-stage CKD before they die.

Audience:

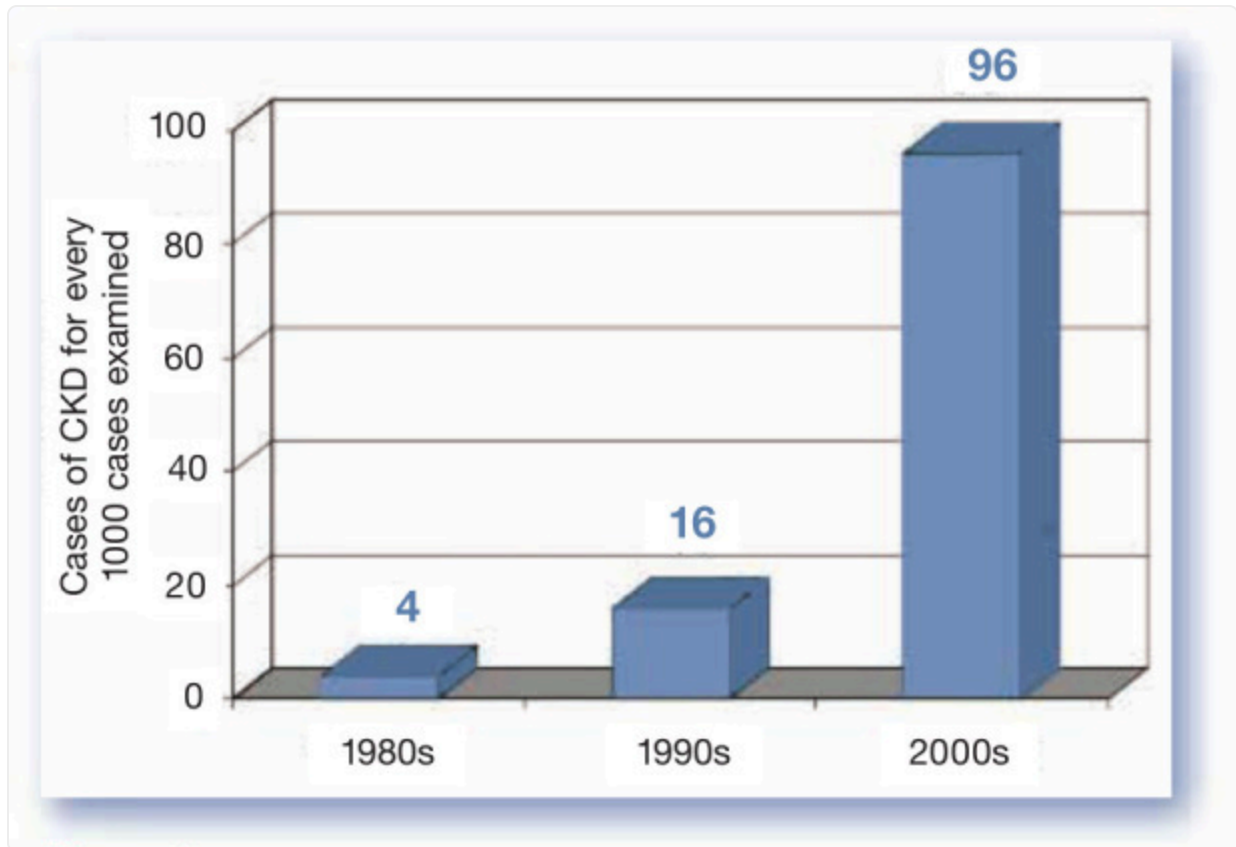
General practitioners play a major role in screening feline patients at risk of development or progression of CKD.

Introduction

Chronic kidney disease (CKD) is the most common renal disease in cats and is defined as structural and/or functional impairment of one or both kidneys that has been present for more than 3 months.¹ CKD may result from any condition that causes progressive and irreversible damage to the kidneys and indeed is intimately associated with an entire spectrum of diseases that occur following initiation of kidney damage. The severity of CKD is denoted by the International Renal Interest Society (IRIS) staging system according to the degree of azotaemia, with stage 1 representing non-azotaemic disease, through to stage 4 representing severe end-stage renal azotaemia.

The prevalence of feline CKD is high and seems to be continuing to rise rapidly ([Figure 1](#)).

Figure 1.



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Prevalence of CKD in recent decades among cats of all ages recorded within the Purdue Veterinary Medical Database^{2,3}

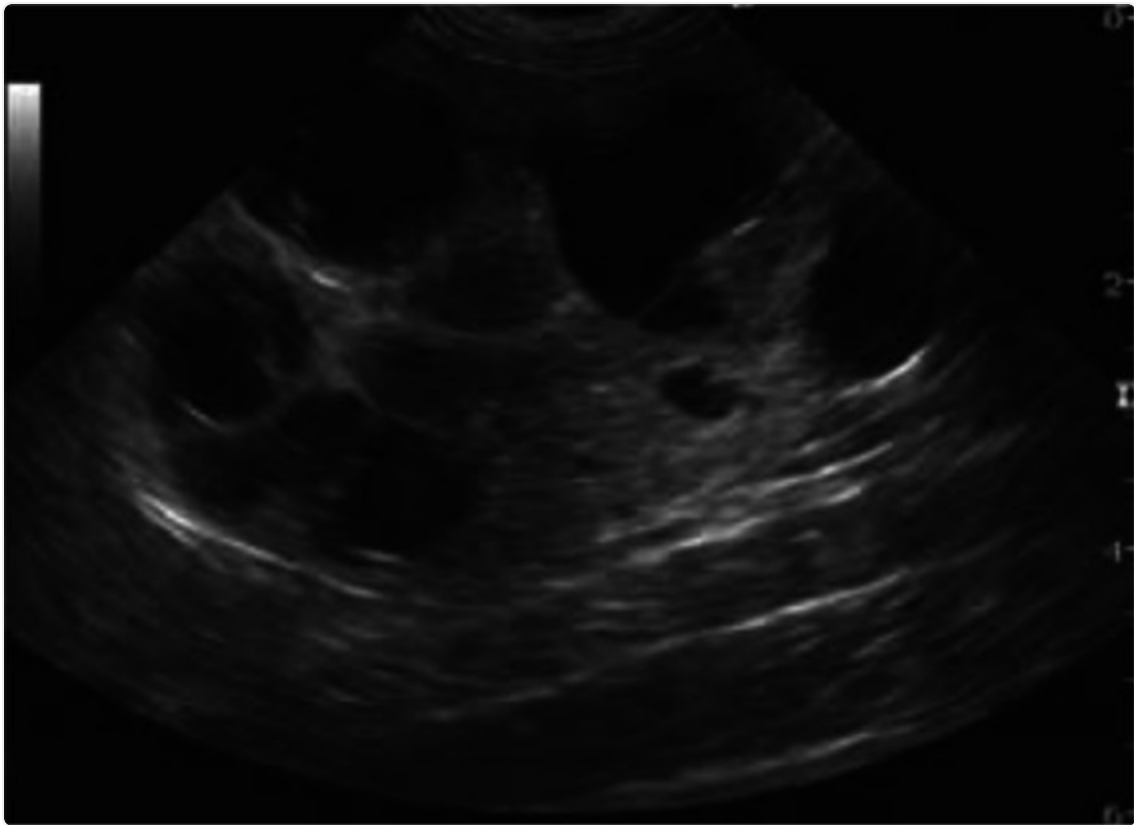
Aetiology of feline CKD

CKD is a non-specific term that does not indicate the cause of kidney damage and/or impaired kidney function. The causes are heterogeneous and most often not identified. They can be divided into two groups – congenital and acquired.

Congenital diseases

Congenital diseases are present at birth and include inherited disorders. The most common genetic disease in cats is autosomal dominant polycystic kidney disease ([Figure 2](#)), which in the past has affected up to 38% of Persian and Persian-cross cats (recent data suggest the prevalence is lower as a result of selective breeding efforts).⁴ Other congenital feline kidney diseases are renal amyloidosis in Abyssinian, Siamese and Oriental cats,⁴ glomerular disease in Abyssinian cats⁵ and juvenile renal dysplasia.⁶

Figure 2.



[Open in a new tab](#)

Ultrasonographic appearance of a polycystic kidney in a cat

Acquired diseases

Acquired diseases have been identified, or are suspected, in many cases of CKD in cats.

- **Upper urinary tract uroliths** were diagnosed in two studies in 15%⁷ and 29%² of cats with CKD. Azotaemia and hyperphosphataemia are frequent findings in cats with ureteral calculi. Interestingly, 76% of cats with unilateral ureteral calculi are azotaemic and 56% show a small contralateral kidney ([Figure 3](#)),⁸ suggesting that pre-existing renal disease is common in cats with ureterolithiasis. After treatment, serum creatinine remains elevated in approximately half of the cats.⁹ It is, however, difficult to disentangle whether ureterolithiasis is the cause or the consequence of CKD ([Figure 4](#)).
- **Renal lymphoma** ([Figure 5](#)), the most common primary renal neoplasm, may cause CKD. It has been reported in 16% of cats with CKD,^{3,10} and 60% of cats with renal lymphoma are azotaemic.¹¹
- **Hyperthyroidism** is the most common endocrine disease of aged cats and may be observed concurrently with CKD. Based on several studies, the reported prevalence of pre-existing CKD in hyperthyroid cats ranges from 14–40%.¹² The ratio of urinary retinol binding protein (RBP) to creatinine concentrations is increased in cats with hyperthyroidism.¹³ As RBP is freely filtered and normally totally reabsorbed, this increase reflects tubular lesions and dysfunction, and may therefore indicate that hyperthyroidism could induce kidney damage. Weight loss, vomiting, polyuria, polydipsia, proteinuria and low urine specific gravity are frequently encountered in both diseases.
Hyperthyroidism can mask and even exacerbate co-existing CKD, with renal disease becoming apparent in 15.3–39% of cats after treatment of hyperthyroidism.^{12,14,15} Proteinuria is an indicator of all-cause mortality in hyperthyroid cats after treatment but not a mediator of progression of CKD in these patients.¹⁵
- **Infections** might well play a causal role in some cases of CKD. There is some evidence of a link between the development of CKD and feline immunodeficiency virus (FIV)¹⁶ and feline leukaemia virus (FeLV) infections.¹⁷ Experimental FIV infection induces renal lesions¹⁶ and clinical CKD has been reported in 9.3% of FIV-infected cats.¹⁸ Mild to severe proteinuria, but not azotaemia, is also associated with FIV infection and has been reported in 25% of cases.^{16,19} More recently, a previously undescribed paramyxovirus, feline morbillivirus, was also shown to be associated with tubulointerstitial nephritis.²⁰
- **Vaccines** containing viruses grown on Crandell Rees feline kidney cells may induce transient renal inflammation.²¹ They contain some feline kidney cell antigens and stimulate production of autoantibodies in vaccinated cats.²² No evidence, however, exists to demonstrate that these vaccine-induced responses may initiate CKD in cats.
- **Nephrotoxic drugs** (eg, aminoglycosides, non-steroidal anti-inflammatory drugs, antineoplastic agents) or **poisoning** (eg, lily toxicity) can cause acute kidney injury (AKI). Incomplete renal recovery after AKI could lead to CKD and accelerate the progression to end-stage renal disease.

- Exclusive feeding of an inappropriate **diet** (high in protein and acid, depleted in potassium) for several years was shown to induce CKD in healthy cats.^{[23](#)} Ad libitum feeding and increased ash intake could be associated with an increased risk of CKD.^{[24](#)} It has been hypothesised that high salt diets might adversely affect renal function, principally by inducing systemic hypertension. However, a recent study of aged cats fed a high sodium diet for 2 years showed no effect on renal function.^{[25](#)}

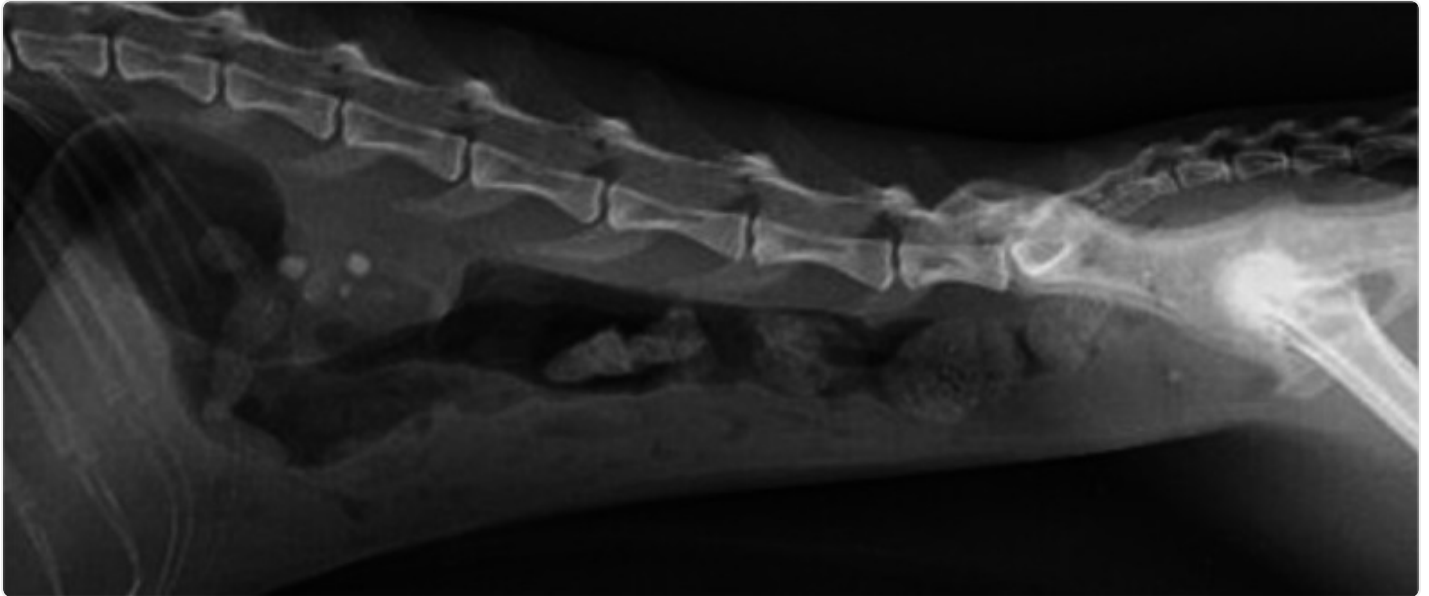
Figure 3.



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'Big kidney-little kidney' syndrome in an azotaemic cat. Abdominal radiographic findings such as these are indicative of ureteral obstruction

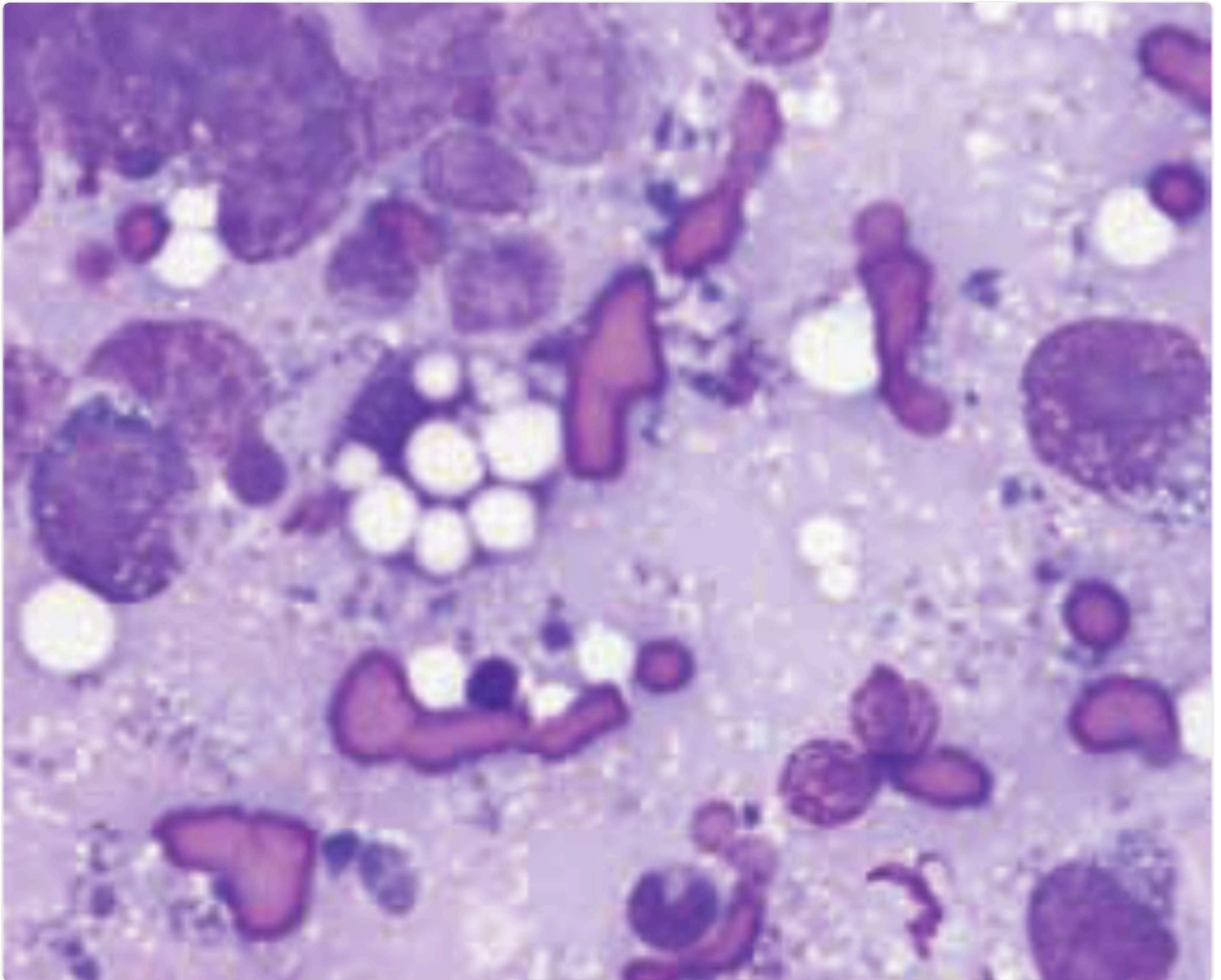
Figure 4.



[Open in a new tab](#)

Abdominal radiograph showing multiple nephroliths in a cat with advanced CKD

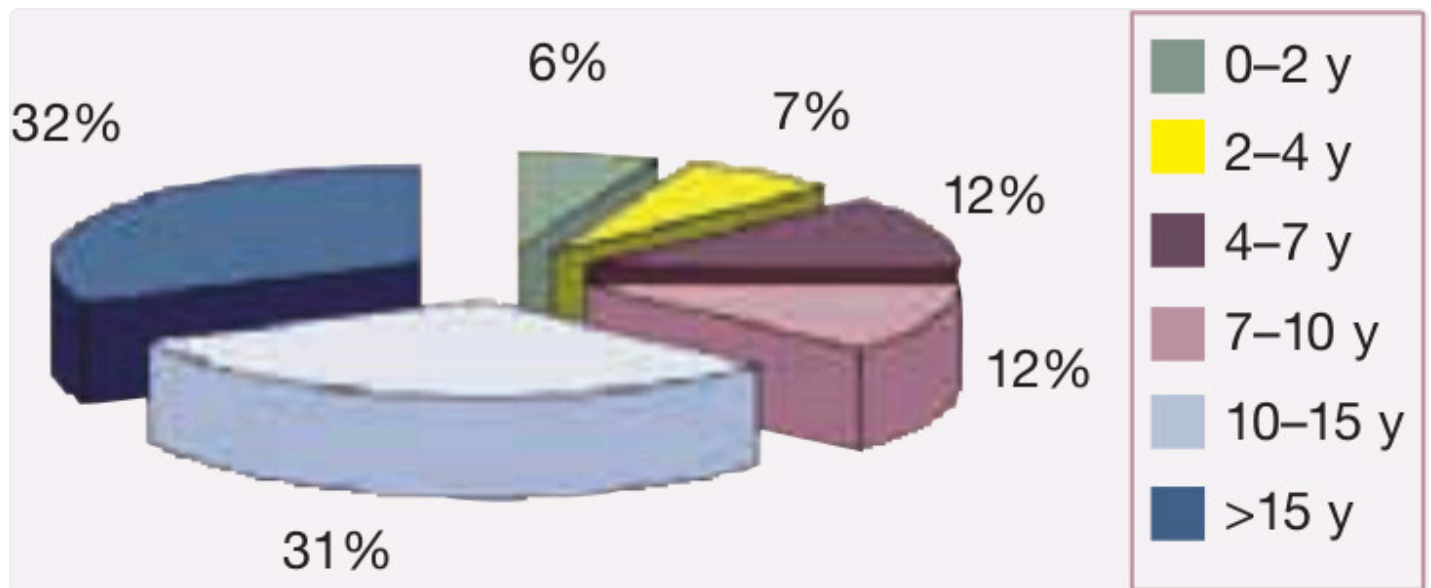
Figure 5.



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Cytology of a fine-needle aspirate from the kidney of a cat with renal lymphoma. Large blastic lymphoid cells typically have a high nucleocytoplasmic ratio, microvacuolated basophilic cytoplasm and a round nucleus with reticulated chromatin and prominent basophilic nucleoli

Figure 6.



[Open in a new tab](#)

Distribution of 'renal failure' in cats of different ages based on 2228 case records from patients diagnosed at 23 US veterinary colleges³

Ageing

Ageing is considered a risk factor for the development of CKD in cats, with CKD indeed being more frequently diagnosed in senior and geriatric cats (Figure 6).^{3,10,26–28} The mortality rate due to CKD is also greater in cats over 9 years old, and cats of any age that die from non-renal causes frequently have renal lesions identified.²⁶ The lesions are mainly tubulointerstitial,^{28,29} and their severity increases with age.²⁹ In one study, 30.5% of aged, apparently healthy cats became azotaemic within 12 months; only plasma creatinine concentration and proteinuria at presentation were predictive of cats at risk of developing azotaemia within this time period.³⁰ However, cats that developed azotaemia were not older than those that remained non-azotaemic, suggesting that ageing per se is not the primary cause of development of azotaemia. The role of systemic hypertension and proteinuria in the development of azotaemia in ageing cats remains unclear. Systolic blood pressure (SBP) has been reported to be minimally affected by ageing in cats.²⁶ The urine protein:creatinine ratio (UPC) is associated with the development of azotaemia,³⁰ but ageing in cats with CKD is not a risk factor for increasing proteinuria.³¹ One study found that, among cats that died from non-renal causes, those that had tubulointerstitial lesions had a longer life span than those with no renal changes.²⁸ Consequently, it was suggested that ageing-associated renal changes might even be protective.²⁸ This finding requires further investigation, however, as the study in question had several potential limitations.³²

Breed

Breed appears to be a potential risk factor for the development of CKD, with Siamese, Abyssinian, Persian, Maine Coon and Burmese cats reportedly more likely to be affected.^{3,33} The strength of evidence for a breed-associated risk is low, however.

Gender

Gender is not a risk factor for the development of CKD.^{3,10,33,34} However, male cats with CKD are younger and overrepresented between the ages of 9 and 11 years.³⁴

Systemic hypertension

Systemic hypertension (as well as cardiovascular disease and urinary tract infections [UTIs] – see below) is a frequent concurrent disease in aged cats with CKD. These various conditions could also contribute to the development of renal disease, although conclusive

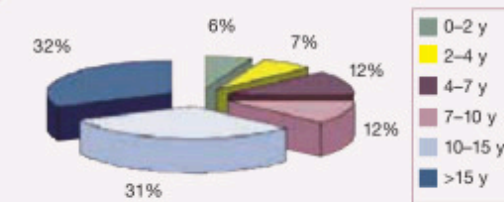


Figure 6 Distribution of 'renal failure' in cats of different ages based on 2228 case records from patients diagnosed at 23 US veterinary colleges³

cause-and-effect relationships have yet to be demonstrated. Systemic hypertension is diagnosed in about 20–65% of cats with CKD.³⁵ Azotaemia is also observed in 61% of hypertensive cats.³⁶ In feline CKD, systemic hypertension is associated with proteinuria but is not predictive of survival.^{31,37} Moreover, it is not associated with interstitial fibrosis or inflammation, suggesting that systemic hypertension is not a primary factor in the progression of

CKD, at least in cats receiving amlodipine therapy.³⁸ However, if cats remain untreated, hypertension might lead to more severe lesions.³⁸

Cardiovascular disease

Cardiovascular diseases resulting from, or independent of, systemic hypertension have been reported in cats with CKD.^{39,40} The prevalence of left ventricular hypertrophy has been estimated to be 46.6% in cats with CKD.³⁹ About 59% of cats with hypertrophic cardiomyopathy (HCM) are azotaemic and 12.7% have CKD. Systolic arterial pressure is also higher in azotaemic than in non-azotaemic cats with HCM.⁴¹ Investigations are needed to document the pathophysiology of concomitant CKD and cardiac disease.

Primary hyperaldosteronism

Primary hyperaldosteronism might also mediate the development of CKD. Mild azotaemia, hypokalaemia and systemic hypertension with associated ocular signs are very common in cats with primary hyperaldosteronism.^{42,43} Moreover, a gradual increase in azotaemia and the development of CKD has been observed in such patients and CKD was reported as a cause of euthanasia.^{42,43} Histopathological renal changes were similar to those observed in humans with hyperaldosteronism.⁴²

Urinary tract infections

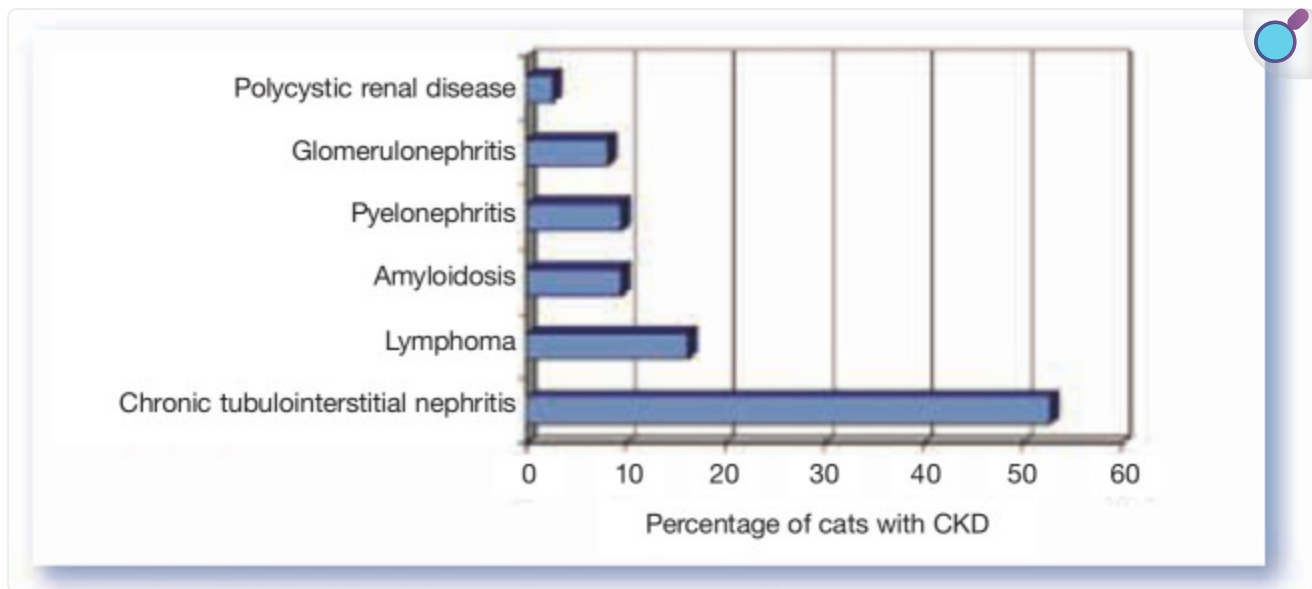
UTIs are considered a risk factor for the development of CKD. Experimental UTI induces chronic pyelonephritis, nephritis and renal fibrosis in cats.⁴⁴ Conversely, in cats with CKD, decreased local and systemic defence mechanisms, urodynamic alterations and changes in the composition of urine might lead to UTI. In feline CKD, the prevalence of UTI is reported to be 17–33%^{3,7,10,45} and that of chronic pyelonephritis has been estimated at 9.5–42%.^{10,46} It has also been reported that two-thirds of cats with UTIs older than 10 years were affected by CKD.⁴⁷ However, in cats with CKD, no association is observed between UTI and markedly elevated serum creatinine concentration or other urine variables.^{7,45}

Pathogenesis of feline CKD

Tubulointerstitial fibrosis appears to be the most common final outcome of feline CKD (Figures 7 and 8).^{2,3,10,48} Primary glomerular diseases are rare – glomerular disease develops secondarily to systemic disease in most cases, specifically neoplastic (eg, leukaemia, lymphoma), infectious and non-infectious inflammatory disorders (eg, pancreatitis, immune-mediated diseases).⁴⁹ The presence of interstitial fibrosis is the strongest histomorphometric predictor of plasma creatinine concentration.³⁸ Tubular damage occurs early in the course of disease, before elevations in serum

urea and creatinine concentrations.⁵⁰ Although cats are able to maintain urine concentrating ability after subtotal nephrectomy,⁵¹ urine specific gravity decreases as plasma creatinine increases in cats with CKD,^{52,53} suggesting a close relationship between glomerular and tubular function during the progression of feline CKD.

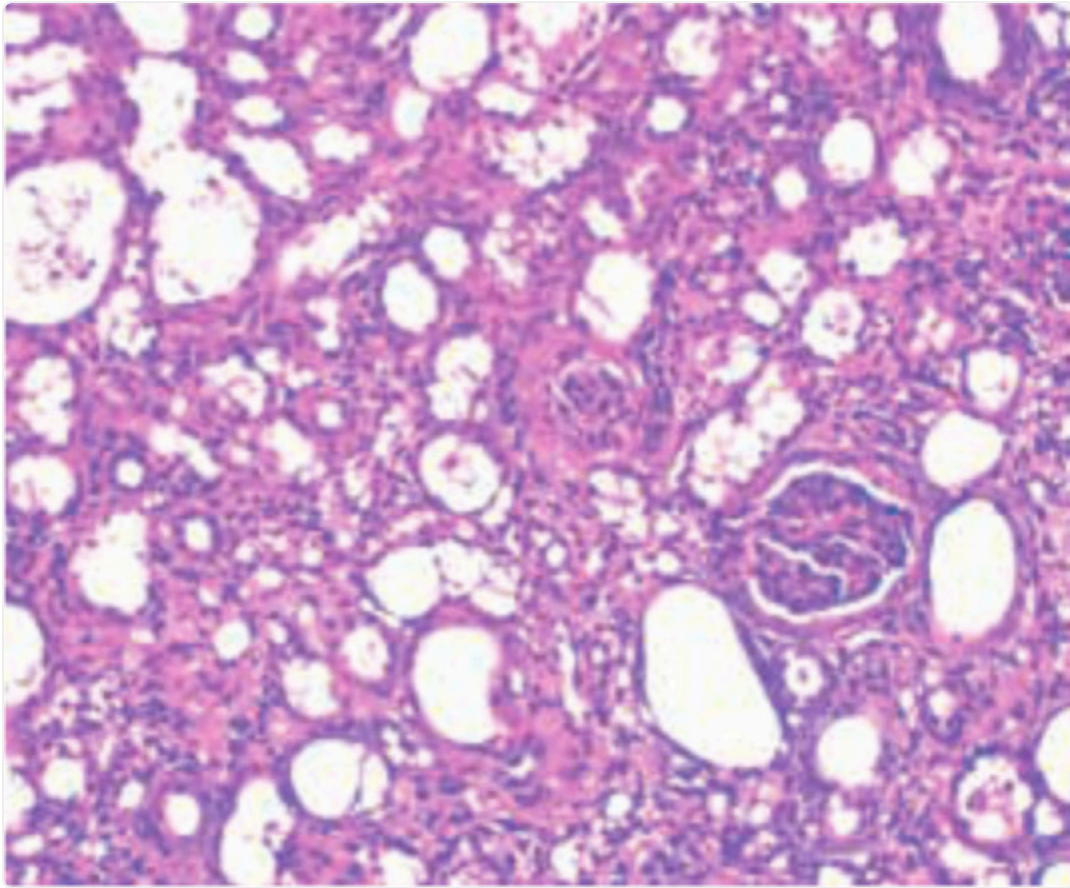
Figure 7.



[Open in a new tab](#)

Distribution of renal lesions in 64 cats with CKD¹⁰

Figure 8.



[Open in a new tab](#)

Renal histology in a cat with end-stage CKD. Note the interstitial inflammation, fibrosis and tubule dilation. H&E, original magnification x200. Courtesy of Dr A Poujade and Dr C Dally, Laboratoire d'Anatomie Pathologique Vétérinaire du Sud-Ouest, Toulouse, France

Although most of the factors that initiate kidney damage remain unknown in cats, it is hypothesised that the process is similar to that described in other species (see box below).⁵⁴

Sustained and chronic kidney injury induces irreversible inflammation with infiltration of inflammatory cells, which in turn serves as a primer that initiates renal fibrogenesis. Inflammatory cells produce profibrotic cytokines (eg, transforming growth factor [TGF- β 1]) which promote fibrogenesis by activation of matrix-producing cells. Urine excretion of TGF- β 1 is increased in feline CKD.⁵⁵

Stages of tubulointerstitial fibrosis

Sustained and chronic kidney injury



Infiltration of inflammatory cells



Fibroblast activation and expansion from various sources



Production and deposition of extracellular matrix



Fibrosis

Duration of injury



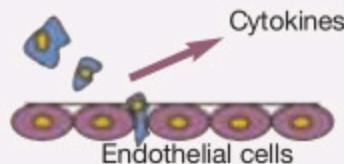
Reversibility of fibrosis



Injury



Tubular epithelial cells



Endothelial cells

Interstitial fibroblasts

Myofibroblasts

Pericytes, fibrocytes, epithelial-mesenchymal transformation

Collagen
Fibronectin

Activated fibroblasts, also referred to as myofibroblasts, produce large amounts of extracellular matrix components (collagen, fibronectin). Other sources of myofibroblasts are pericytes, fibrocytes and tubular epithelial cells that undergo epithelial-mesenchymal transformation. Myofibroblasts express α -smooth muscle actin and vimentin. In cats with CKD, interstitial expression of these markers correlates with the extent of tubulointerstitial lesions, interstitial fibrosis, interstitial fibronectin deposition and renal dysfunction.^{56,57} Chronic hypoxia is often also present and is associated with increased oxidative stress,⁵⁴ which is likely to be important in the pathogenesis of feline CKD.^{58,59}

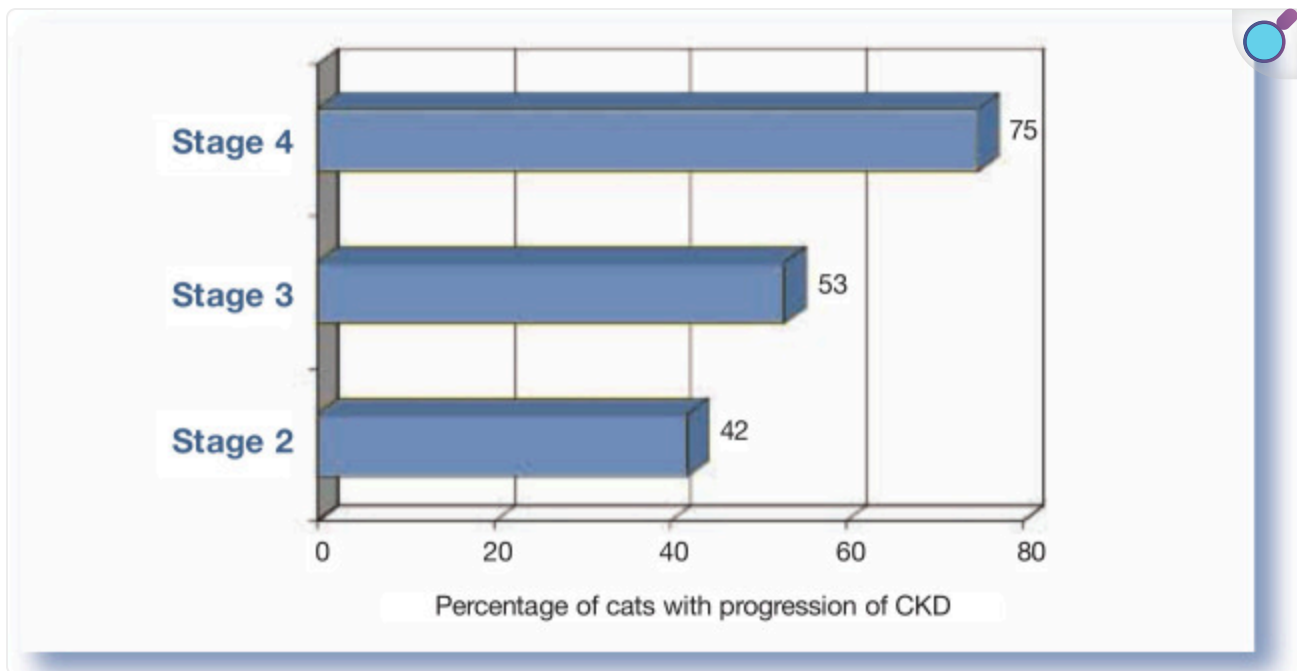
The renin–angiotensin–aldosterone system (RAAS) is additionally activated in cats with naturally occurring or experimental CKD.^{39,60–63} Angiotensin II induces vasoconstriction of the efferent arteriole (and consequently glomerular hypertension) and accumulation of extracellular matrix mediated by TGF- β 1.⁶⁴ In cats with CKD, angiotensin-converting enzyme (ACE) inhibitors decrease glomerular capillary pressure, the ratio of efferent to afferent arteriolar vascular resistance and proteinuria, but have not been shown to improve histopathological changes in the kidneys or survival times.^{46,65} Insufficient statistical power might, however, explain the lack of effect on survival. Aldosterone could also contribute to the development of CKD via several pathogenic mechanisms (see box on the right).⁶⁶

Progression of feline CKD

The progression of CKD is postulated to result from a self-perpetuating vicious cycle. It is attributed to maladaptive functional and structural changes in surviving (or remnant) nephrons after renal injury; in cats with experimental CKD, these are principally increases in single nephron glomerular filtration rate (GFR), glomerular pressure and glomerular volume.⁶⁷

Because of the large reserve capacity of the kidneys, at least 75% of renal function needs to be impaired before azotaemia is detected.⁶⁸ In cats, GFR remains stable for up to 1 year after induction of experimental CKD.^{69–71} Cats with spontaneous CKD show an unpredictable time course of the disease, with extended periods of stable renal function before decompensation (Figures 9 and 10).^{27,40,72,73} In one study, for example, 81% and 37% of cats with stages 2 and 3 CKD, respectively, did not progress to stage 4 before they died.⁷³ The median survival time for advanced stage 2 (serum creatinine 205–250 μ mol/l, 2.3–2.8 mg/dl), 3 (serum creatinine 251–440 μ mol/l, 2.9–5.0 mg/dl) and 4 (serum creatinine >440 μ mol/l, >5.0 mg/dl) was estimated at 1151, 778 and 103 days, respectively, but large individual variation was observed.³³

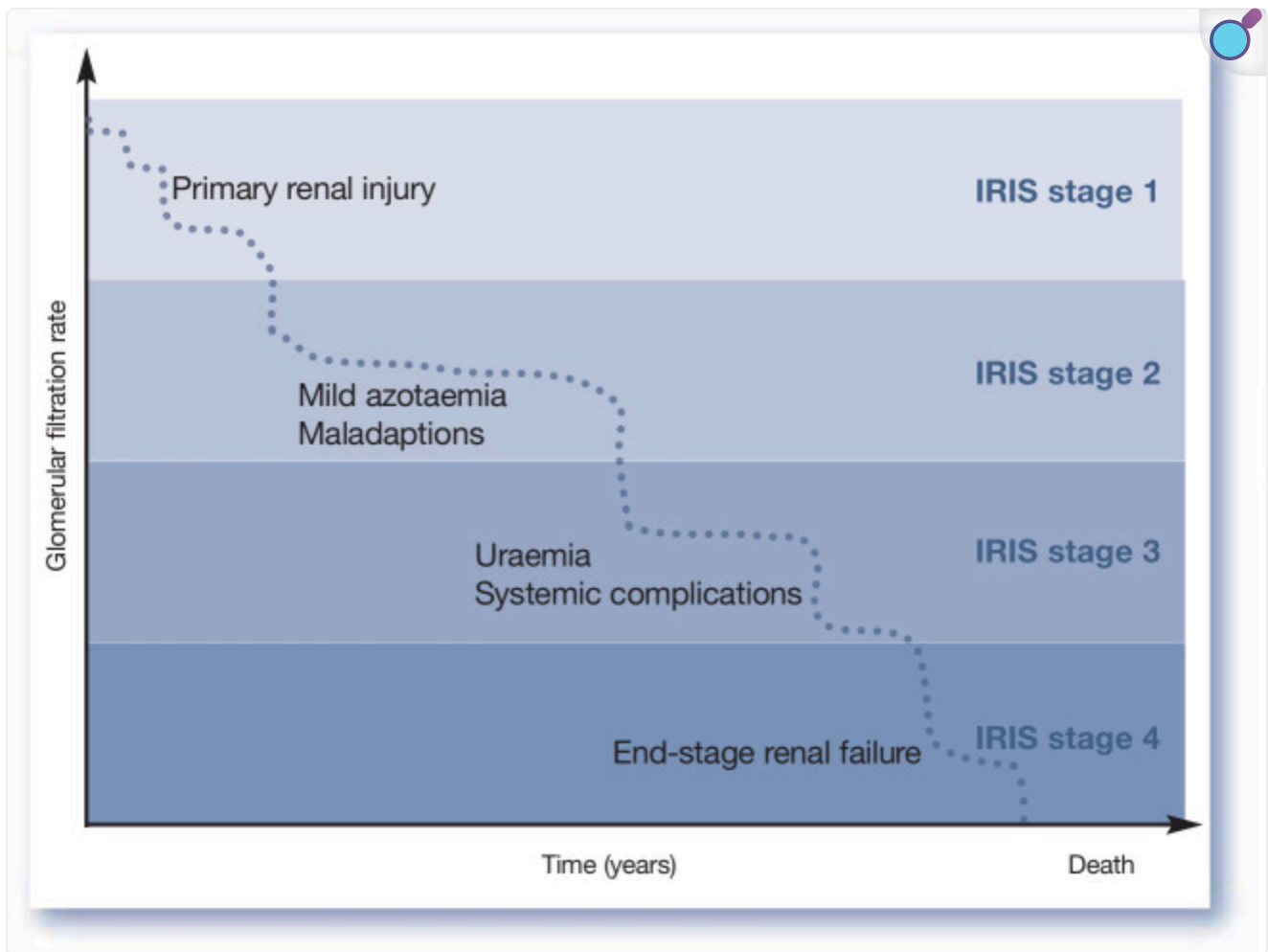
Figure 9.



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Percentage of cats with progression of CKD (ie, $\geq 25\%$ increase in plasma creatinine concentration relative to the concentration at diagnosis) within 1 year of diagnosis, according to IRIS stage of renal disease [73](#)

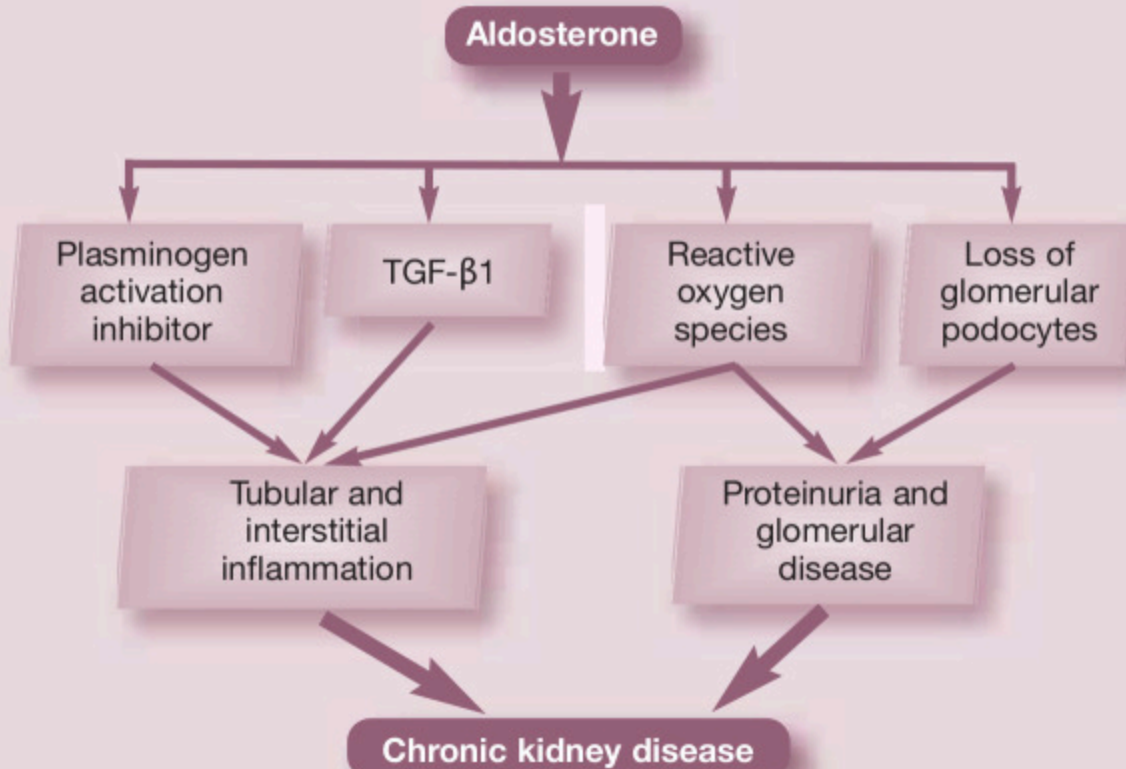
Figure 10.



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Decline in glomerular filtration rate (dotted line) during the progression of CKD from IRIS stage 1 to stage 4. The rate of progression varies according to the nature of the kidney damage and among individual cats. Cats in stages 2 and 3 can have stable renal function for sustained periods until decompensation

Adverse renal effects of aldosterone in CKD



Risk factors for progression

Factors affecting patient survival time and disease progression in cats with CKD have been evaluated in different studies (see accompanying boxes).^{26,31,33,53,73,74} Increased plasma concentrations of creatinine,^{31,53,73} phosphate^{33,53,73} and urea,^{53,73} increased UPC,^{31,53,73} decreased blood haemoglobin concentration and haematocrit,^{31,53,73} and increased blood leukocyte counts⁵³ are associated with a shorter survival time in cats with CKD. For example, an increase in UPC of 0.1 in cats with stage 2 CKD was associated with a 23% increase in the risk of progression within 1 year of diagnosis.⁷³ In one study, laboratory data obtained from cats with CKD at initial evaluation and later within 1 month before death were retrospectively compared.⁷⁴ An increase in plasma creatinine and UPC and a decrease in packed cell volume (PCV) were observed.

However, no single variable has been identified that reliably predicts the survival time in a given cat with CKD.^{53,73} Survival time in relation to various clinical criteria (Table 1) is also very variable.³³

Factors shown not to affect CKD progression

- ❖ Breed⁵³
- ❖ Plasma calcium³³
- ❖ Plasma bicarbonate³³
- ❖ Plasma potassium^{33,53}
- ❖ Presence of UTI⁷³
- ❖ Presence of systemic hypertension^{31,73}

Factors with conflicting results regarding influence on CKD progression

- ❖ Age^{31,33}
- ❖ Body weight^{53,73}
- ❖ Plasma albumin^{33,73}

Factors found to be associated with decreased survival time in CKD patients

- ❖ High UPC^{31,53,73}
- ❖ High plasma creatinine^{31,53,73}
- ❖ High plasma phosphate^{33,53,73}
- ❖ High plasma urea^{53,73}
- ❖ High blood leukocyte counts⁵³
- ❖ Low blood haemoglobin and haematocrit^{31,53,73}

No single variable has been identified that reliably predicts the survival time in a given cat with CKD.



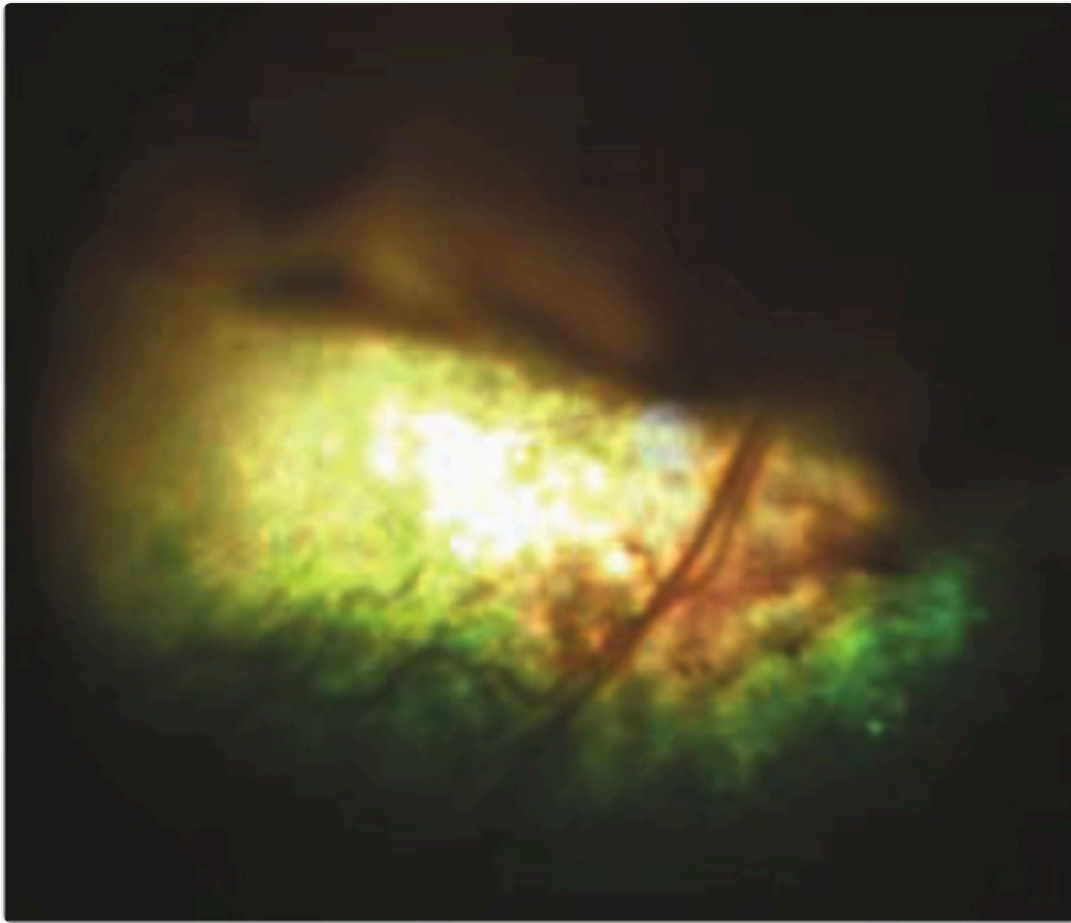
Pathophysiology of secondary changes occurring in feline CKD

The clinical signs and complications of CKD progress with declining GFR due to decreased elimination of, and hence overexposure to, endogenous substances (so-called uraemic toxins). These complications are also considered in themselves to be major contributors to the progression of renal disease, and increase the risk for morbidity and mortality in patients with CKD.

Systemic hypertension

Systemic hypertension is commonly associated with CKD. The pathogenesis of hypertension in cats with CKD is multifactorial and includes impaired excretion of sodium, RAAS activation, stimulation of the sympathetic nervous system, arterial structural changes, endothelial dysfunction and oxidative stress.³⁵ Systemic hypertension may adversely affect renal function by inducing glomerular hypertension and proteinuria; it also increases the risk of other end-organ damage (ie, ocular, neurological and cardiac) ([Figure 11](#)).⁷⁵

Figure 11.



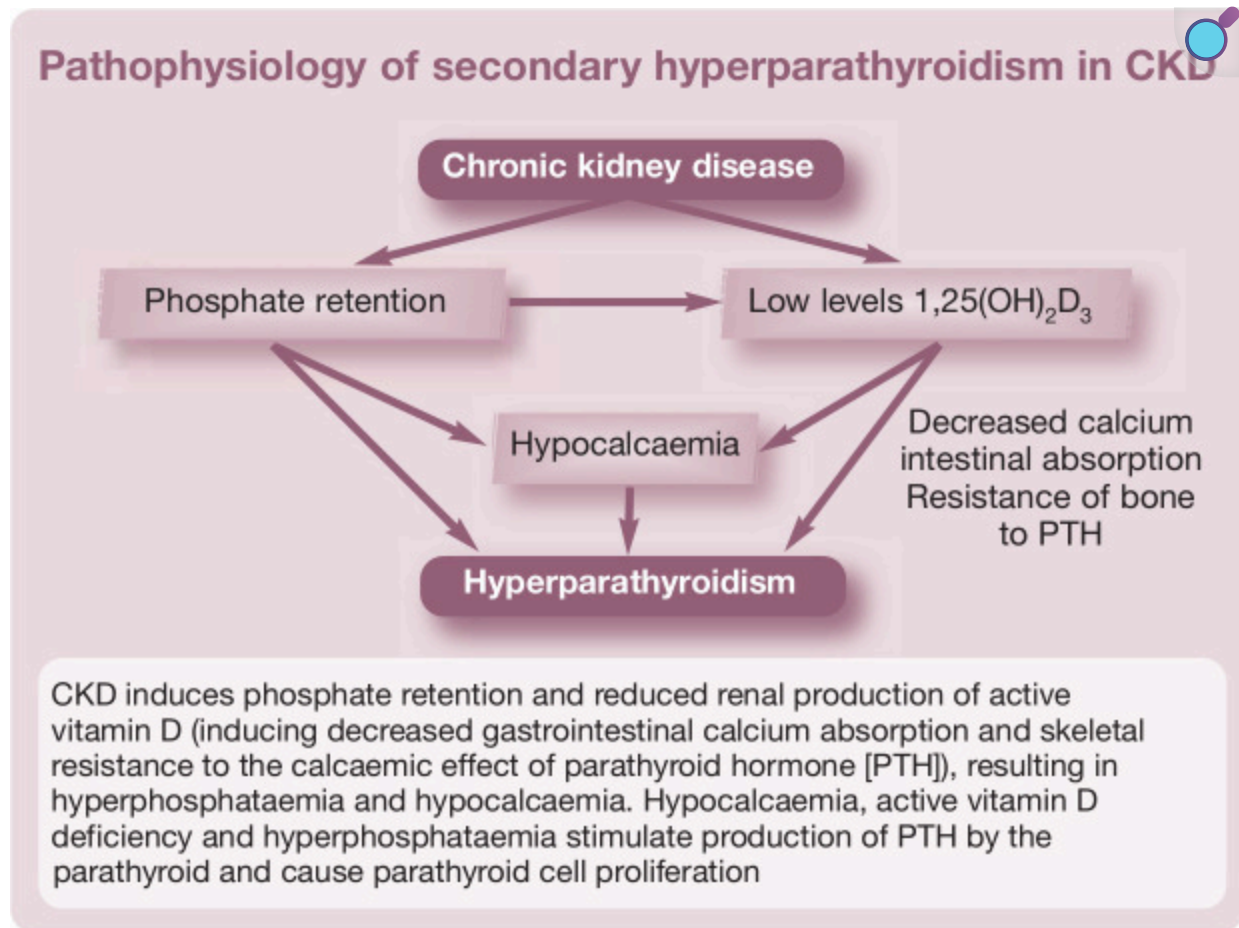
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Dorsal retinal detachment in a cat with systemic arterial hypertension

Renal secondary hyperparathyroidism

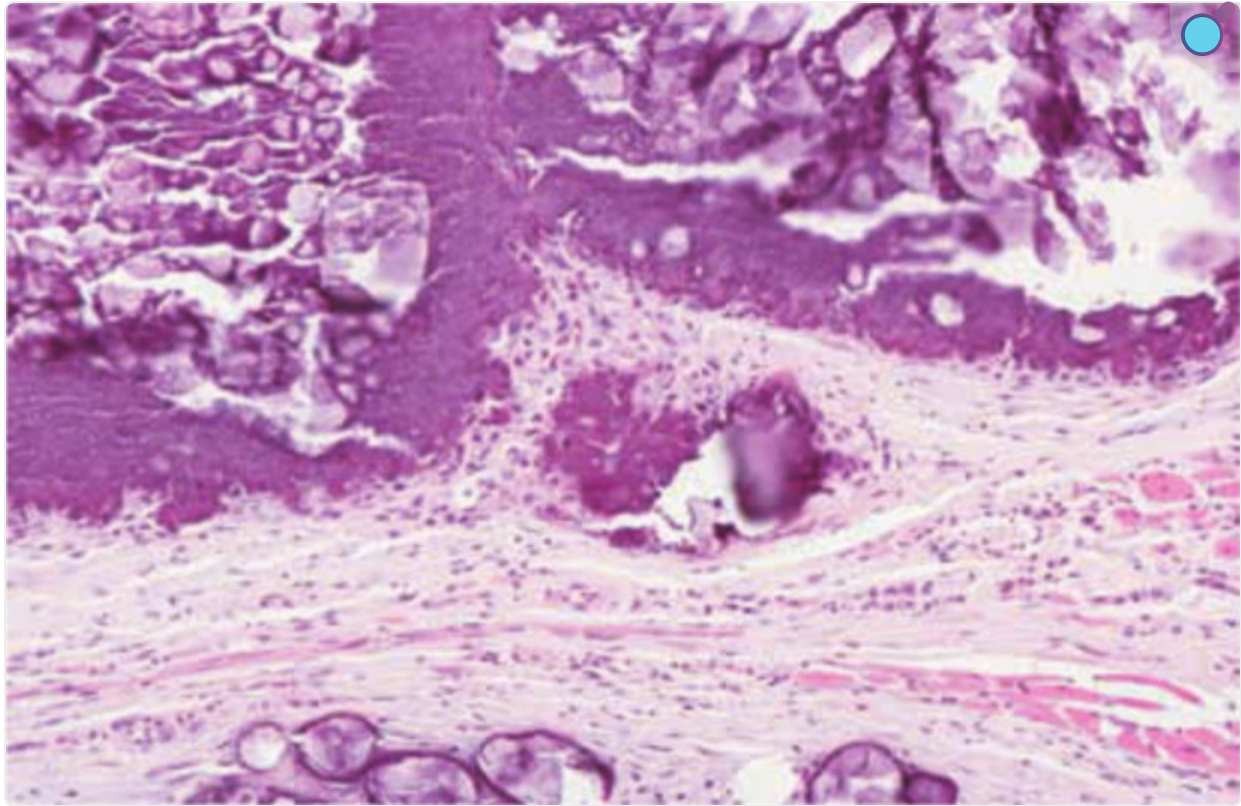
Renal secondary hyperparathyroidism (RHPH) is an adaptive and ultimately maladaptive process that develops in response to declining kidney function, phosphate retention and failure to bioactivate vitamin D (see box above).⁷⁶ It is well recognised that phosphate retention is a major contributor to the progression of CKD. Hyperphosphataemia and increased plasma parathyroid hormone (PTH) concentrations are observed in 13–100% and 47–100%, respectively, of cats with CKD.^{3,53,77} RHPH is particularly common in stages 3 and 4 CKD, as the severity of RHPH increases with the degree of azotaemia.^{3,53,77} Plasma fibroblast growth factor 23 (FGF-23) is a

phosphaturic hormone involved in the pathogenesis of RHPTH in humans. In cats with CKD, plasma FGF-23 is strongly correlated with plasma creatinine concentration and PTH, whereas there is an inverse relationship between FGF-23 and GFR.^{78,79}



Clinical signs induced by RHPTH may result from calcium phosphate precipitation into the tissues (Figure 12) but are relatively uncommon in cats with CKD. Metastatic paw calcification⁸⁰ and osteoporosis¹⁰ have been reported.

Figure 12.



[Open in a new tab](#)

Soft tissue mineralisation in the tongue of a cat with end-stage CKD. H&E, original magnification x200

Hypokalaemia

Hypokalaemia is a common finding in cats with CKD (prevalence 18–30%), especially those in IRIS stages 2 and 3.^{3,10,26,27,52,81} CKD is also the most common cause of hypokalaemia in cats. Cats with severe hypokalaemia are 3.5 times more likely to have CKD.⁸² Hypokalaemia in feline CKD may result from multiple causes (see box below). It may also develop in renal-impaired cats fed a high protein diet.⁸³ Potassium depletion may induce metabolic acidosis, kidney dysfunction (hypokalaemic nephropathy), muscle weakness and cardiac arrhythmia.^{81,84} Therefore, CKD could lead to potassium depletion, which would in turn contribute to progression of the renal disease and worsening of the clinical condition.

By contrast, hyperkalaemia is uncommon in cats with CKD (prevalence 5.8%),^{46,53} except in end-stage oliguric CKD.⁸¹

Anaemia of renal disease

A normochromic, normocytic, hypoproliferative anaemia is frequently diagnosed in cats with CKD, especially in end stages of the disease;^{85,86} about 30–65% of cats with CKD develop anaemia during progression of their renal disease.⁸⁶ The PCV decreases as the severity of CKD increases.^{52,53} CKD-induced anaemia results mainly from inadequate production of erythropoietin by the peritubular fibroblast type-1 interstitial cells. Other contributing factors to anaemia include malnutrition (eg, iron deficiency), shortened red blood cell life span, myelofibrosis, metabolic alterations affecting erythropoiesis, uraemic toxins, hyperparathyroidism and gastrointestinal blood loss.⁸⁶ Anaemia adversely affects the quality of life of cats with CKD.⁸⁶

Potential causes of hypokalaemia in cats with CKD

Increased loss of potassium

- ✦ Decreased renal potassium reabsorption
- ✦ Diet-induced hypokalaemic nephropathy
- ✦ Renal tubular acidosis
- ✦ Hyperaldosteronism

Translocation of potassium from extracellular to intracellular fluid

- ✦ Hypothermia

Decreased intake of potassium

- ✦ Anorexia

Adapted from DiBartola and Autran De Morais⁸¹

Metabolic acidosis

Metabolic acidosis is also frequent, with a prevalence of 22–88% dependent on the severity of CKD.^{3,10,52} Metabolic acidosis generally occurs late in the time course of CKD.⁷² An increase in anion gap due to low plasma bicarbonate and chloride concentrations is observed,⁵² with development of hypochloraemia preceding acidosis.⁷² Metabolic acidosis may result from potassium depletion,⁸⁴ but hypokalaemic cats are generally able to maintain normal venous blood pH and bicarbonate concentration.⁵² Renal ammoniogenesis, the main renal process for urinary hydrogen ion excretion, could also be impaired in CKD.⁵² Hypochloraemia may result from the loss of chloride through the gastrointestinal tract due to gastric hyperacidity or vomiting.⁷²

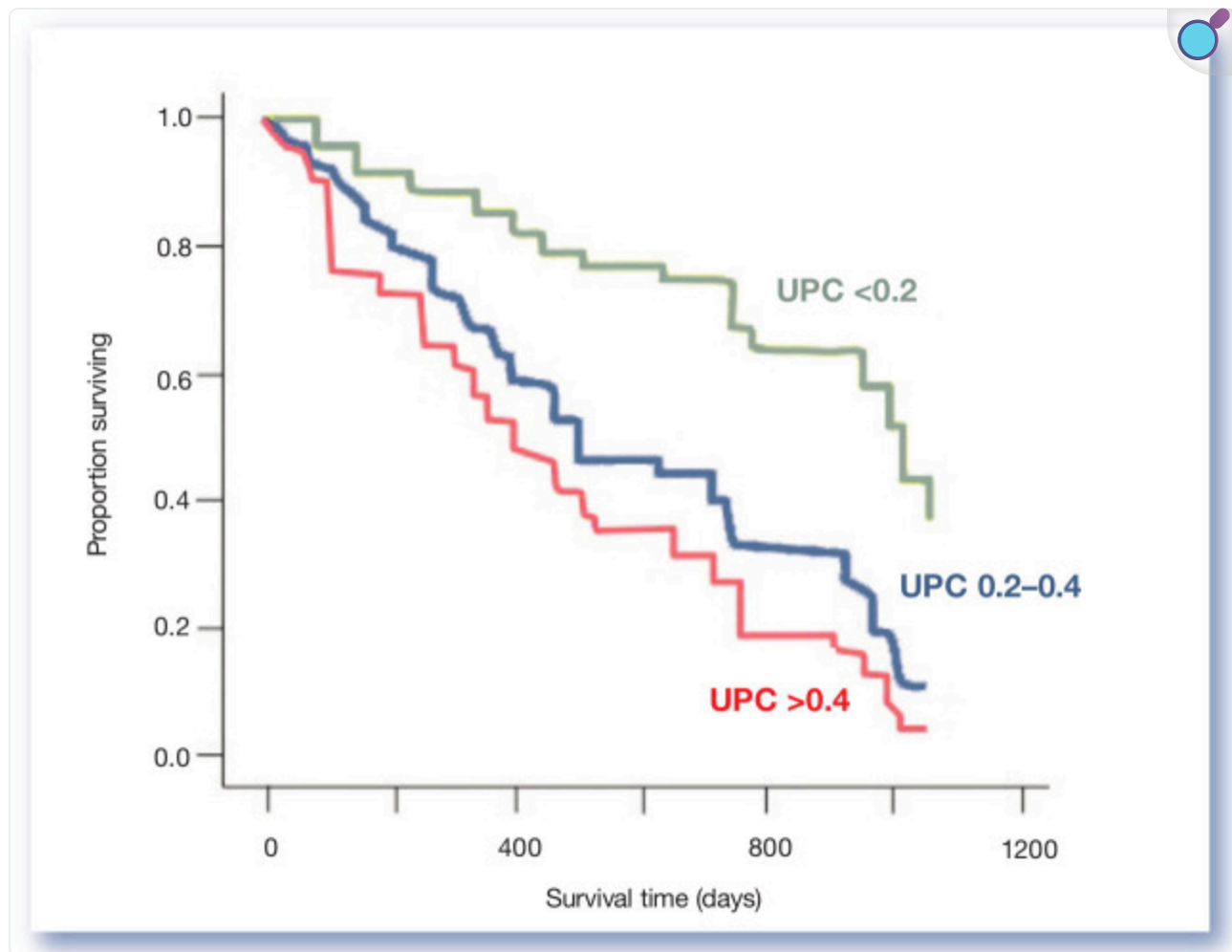
Metabolic acidosis promotes many adverse clinical effects in the renal patient, including anorexia, vomiting, lethargy, weakness and malnutrition.²⁷

Role of proteinuria in disease progression

Proteinuria is a sign of kidney damage, but also a strong indicator for progression of CKD.^{87,88} Small proteins, such as albumin, are filtered by the glomeruli and reabsorbed by the proximal tubule cells by receptor-mediated endocytosis. Dysfunction of these two processes results in proteinuria. Proteinuria accelerates progression of CKD by induction of tubular chemokine expression and complement activation; this, in turn, leads to inflammation and fibrogenesis. Proteinuria is associated with interstitial fibrosis and glomerular hypertrophy,³⁸ and might, therefore, be a marker of tubular dysfunction associated with tubulointerstitial fibrosis.

Proteinuria in cats with naturally occurring CKD is generally mild compared with their human counterparts,^{31,34,46} with 90% and 49% of cats with CKD having a UPC of <1.0 and <0.25, respectively.³¹ The severity of proteinuria, however, has prognostic significance in terms of survival time (Figure 13).^{31,46} Despite proteinuria being generally mild, the ACVIM consensus statement on the treatment of proteinuria recommends therapeutic intervention when UPC $\geq$ 0.4 in cats with CKD causing azotaemia.⁸⁷ Plasma creatinine concentration and SBP are positively correlated with increasing UPC.³¹

Figure 13.



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Effect of proteinuria on survival in cats with CKD. ³¹ The hazard ratio for death or euthanasia was 2.9 and 4.0 for a UPC of 0.2–0.4 and >0.4, respectively, compared with the baseline group (UPC <0.2)

Whether proteinuria is contributing to CKD progression or whether proteinuric CKD is intrinsically more rapidly progressive remains unknown.

Major clinical signs of feline CKD

The prevalence of clinical signs in cats with CKD is presented in [Tables 2](#) and [3](#).

Table 2.

Prevalence of clinical signs detected by owners in cats with CKD

Prevalence [*]	Clinical sign(s)
Very high (>75%)	Lethargy Anorexia Polydipsia Weight loss
High (40–75%)	Depression Vomiting Weakness Polyuria
Moderate (10–39%)	Dysphagia, oral discomfort and buccal cavity lesions Halitosis Constipation
Low (<10%)	Haematuria Dysuria Diarrhoea Neurological signs

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^{*}Maximum percentage of cases showing the clinical sign.

The prevalence can vary between studies^{3,10,26,45,46} and according to the stage of renal disease. Most of the animals included in these studies were cats with advanced-stage CKD

Table 3.

Prevalence of physical examination findings in cats with CKD

Prevalence [*]	Clinical sign(s)
High (40–75%)	Dehydration Cachexia and weight loss
Moderate (10–39%)	Periodontal disease, gingivitis and oral ulceration Hair coat in poor condition Cardiac murmur Large kidneys Small irregular kidneys Palpable goitre Tachycardia Hypothermia Tachypnoea Pallor of the mucous membranes
Low (<10%)	Hyperthermia Retinal lesions

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*Maximum percentage of cases showing the clinical sign.

The prevalence can vary between studies^{[3,10,26,45,46](#)} and according to the stage of renal disease. Most of the animals included in these studies were cats with advanced-stage CKD

Table 1.

Survival time in relation to various clinical criteria

Criteria	Median survival (days)
Diagnosis of CKD	771
Onset of weight loss	401
Start of subcutaneous fluids	142
Onset of anaemia (PCV <25%)	100
>25% weight loss	83
Clinical decompensation	40
Anaemia intervention	25

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PCV = packed cell volume

Adapted from Boyd et al [33](#)

Clinical signs may be non-specific and manifest late in the disease process, once major renal tissue damage has occurred. Many cats remain almost symptom-free until they reach advanced stages of CKD. The frequency of clinical signs is similar in IRIS stages 2 and 3, but stage 4 is associated with increased frequency and severity of signs. [46](#)

Polyuria and polydipsia represent the earliest and most common clinical manifestations of CKD. [27](#) Cats with CKD are more likely to have had polydipsia and polyuria in the year before they are diagnosed. [89](#) Polydipsia, which is a compensatory response to polyuria, is more easily recognised by owners. In advanced stages, polyuria results in dehydration as fluid loss from the body exceeds fluid intake.

Gastrointestinal signs are the most prominent indicators of uraemia and probably represent the main reason for the owner to consult a veterinarian. Hypergastrinaemia, reported in cats with CKD, [90](#) may induce gastric hyperacidity, uraemic gastritis, gastrointestinal bleeding, anorexia and

vomiting. While diarrhoea is uncommon, constipation is more frequent in cats with CKD than in dogs. It is attributable to dehydration but also to the use of phosphate binding agents.⁹¹

Other reported uraemic clinical complications are lethargy, weakness, depression, tremor, myoclonus, seizures, myopathies, uraemic pericarditis and pneumonitis, hypothermia, anaemia and renal osteodystrophy. These disorders are all multifactorial and their pathogenesis has not generally been investigated in cats.

Footnotes

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Key Points

- Whenever CKD is diagnosed in a cat, particular effort should be made to identify underlying conditions that may have a specific treatment.
- Most often, the cause of CKD remains unidentified and only signs of a non-specific tubulointerstitial fibrosis can be detected at the time of diagnosis.
- Age, breed, systemic hypertension, cardiac disease and urinary tract infection are all considered to be risk factors for CKD, although definitive evidence is lacking.
- Whatever its origin, sustained renal injury leads to self-perpetuated maladaptive changes that drive CKD progression.
- Tubulointerstitial fibrosis is the most common histopathological finding in cats with CKD.
- The time course of CKD is typically unpredictable in cats. However, selected clinicopathological variables and clinical criteria may be predictive of a shorter survival time. Notably, the magnitude of proteinuria has negative prognostic significance.
- Cats with CKD should be individually screened for systemic arterial hypertension, renal secondary hyperparathyroidism, hypokalaemia and metabolic acidosis. Appropriate treatments should be implemented on a case-by-case basis.
- Clinical signs of CKD appear late in the course of the disease and are mostly non-specific.

References

1. Bartges JW. Chronic kidney disease in dogs and cats. *Vet Clin North Am Small Anim Pract* 2012; 42: 669–692. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
2. Ross SJ, Osborne CA, Kirk CA, Lowry SR, Koehler LA, Polzin DJ. Clinical evaluation of dietary modification for treatment of spontaneous chronic kidney disease in cats. *J Am Vet Med Assoc* 2006; 229: 949–957. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
3. Lulich JP, Osborne CA, O'Brien RD, Polzin DJ. Feline renal failure: questions, answers, questions. *Compend Contin Educ Pract Vet* 1992; 14: 127–152. [[Google Scholar](#)]
4. Lees GE. Congenital kidney disease. In: Bartges J, Polzin DJ. (eds). *Nephrology and urology of small animals*. Ames: Blackwell Publishing, 2011, pp 568–576. [[Google Scholar](#)]
5. White JD, Norris JM, Bosward KL, Fleay R, Lauer C, Malik R. Persistent haematuria and proteinuria due to glomerular disease in related Abyssinian cats. *J Feline Med Surg* 2008; 10: 219–229. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
6. Aresu L, Zanatta R, Pregel P, Caliari D, Tursi M, Valenza F, et al. Bilateral juvenile renal dysplasia in a Norwegian Forest cat. *J Feline Med Surg* 2009; 11: 326–329. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
7. Bailiff NL, Westropp JL, Nelson RW, Sykes JE, Owens SD, Kass PH. Evaluation of urine specific gravity and urine sediment as risk factors for urinary tract infections in cats. *Vet Clin Pathol* 2008; 37: 317–322. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
8. Kyles AE, Hardie EM, Wooden BG, Adin CA, Stone EA, Gregory CR, et al. Clinical, clinicopathologic, radiographic, and ultrasonographic abnormalities in cats with ureteral calculi: 163 cases (1984–2002). *J Am Vet Med Assoc* 2005; 226: 932–936. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
9. Kyles AE, Hardie EM, Wooden BG, Adin CA, Stone EA, Gregory CR, et al. Management and outcome of cats with ureteral calculi: 153 cases (1984–2002). *J Am Vet Med Assoc* 2005; 226: 937–944. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
10. DiBartola SP, Rutgers HC, Zack PM, Tarr MJ. Clinicopathologic findings associated with chronic renal disease in cats: 74 cases (1973–1984). *J Am Vet Med Assoc* 1987; 190: 1196–1202. [[PubMed](#)] [[Google Scholar](#)]
11. Gabor LJ, Canfield PJ, Malik R. Haematological and biochemical findings in cats in Australia with lymphosarcoma. *Aust Vet J* 2000; 78: 456–461. [[DOI](#)] [[PubMed](#)] [[Google](#)]

12. van Hoek I, Daminet S. Interactions between thyroid and kidney function in pathological conditions of these organ systems: a review. *Gen Comp Endocrinol* 2009; 160: 205–215. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#) 
13. van Hoek I, Daminet S, Notebaert S, Janssens I, Meyer E. Immunoassay of urinary retinol binding protein as a putative renal marker in cats. *J Immunol Methods* 2008; 329: 208–213. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#) 
14. van Hoek I, Lefebvre HP, Peremans K, Meyer E, Croubels S, Vandermeulen E, et al. Short- and long-term follow-up of glomerular and tubular renal markers of kidney function in hyperthyroid cats after treatment with radioiodine. *Domest Anim Endocrinol* 2009; 36: 45–56. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#) 
15. Williams TL, Peak KJ, Brodbelt D, Elliott J, Syme HM. Survival and the development of azotemia after treatment of hyperthyroid cats. *J Vet Intern Med* 2010; 24: 863–869. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#) 
16. Poli A, Tozon N, Guidi G, Pistello M. Renal alterations in feline immunodeficiency virus (FIV)-infected cats: a natural model of lentivirus-induced renal disease changes. *Viruses* 2012; 4: 1372–1389. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#) 
17. Glick AD, Horn RG, Holscher M. Characterization of feline glomerulonephritis associated with viral-induced hematopoietic neoplasms. *Am J Pathol* 1978; 92: 321–332. [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#) 
18. Ishida T, Washizu T, Toriyabe K, Motoyoshi S, Tomoda I, Pedersen NC. Feline immunodeficiency virus infection in cats in Japan. *J Am Vet Med Assoc* 1989; 194: 221–225. [\[PubMed\]](#) [\[Google Scholar\]](#) 
19. Baxter KJ, Levy JK, Edinboro CH, Vaden SL, Tompkins MB. Renal disease in cats infected with feline immunodeficiency virus. *J Vet Intern Med* 2012; 26: 238–243. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#) 
20. Woo PC, Lau SK, Wong BH, Fan RY, Wong AY, Zhang AJ, et al. Feline morbillivirus, a previously undescribed paramyxovirus associated with tubulointerstitial nephritis in domestic cats. *Proc Natl Acad Sci USA* 2012; 109: 5435–5440. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#) 

21. Lappin MR, Basaraba RJ, Jensen WA. Interstitial nephritis in cats inoculated with Crandell Rees feline kidney cell lysates. *J Feline Med Surg* 2006; 8: 353–356. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
22. Whittemore JC, Hawley JR, Jensen WA, Lappin MR. Antibodies against Crandell Rees feline kidney (CRFK) cell line antigens, alpha-enolase, and annexin A2 in vaccinated and CRFK hyperinoculated cats. *J Vet Intern Med* 2010; 24: 306–313. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
23. DiBartola SP, Buffington CA, Chew DJ, McLoughlin MA, Sparks RA. Development of chronic renal disease in cats fed a commercial diet. *J Am Vet Med Assoc* 1993; 202: 744–751. [[PubMed](#)] [[Google Scholar](#)]
24. Hughes KL, Slater MR, Geller S, Burkholder WJ, Fitzgerald C. Diet and lifestyle variables as risk factors for chronic renal failure in pet cats. *Prev Vet Med* 2002; 55: 1–15. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
25. Reynolds BS, Chetboul V, Nguyen P, Testault I, Concordet D, Carlos Sampedrano C, et al. Effects of dietary salt intake on renal function: a 2 year study in healthy aged cats. *J Vet Intern Med* 2013; 27: 507–515. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
26. Elliott J, Barber PJ. Feline chronic renal failure: clinical findings in 80 cases diagnosed between 1992 and 1995. *J Small Anim Pract* 1998; 39: 78–85. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
27. Polzin DJ. Chronic kidney disease. In: Bartges J, Polzin DJ. (eds). *Nephrology and urology of small animals*. Ames: Wiley-Blackwell, 2011, pp 433–471. [[Google Scholar](#)]
28. Lawler DF, Evans RH, Chase K, Ellersieck M, Li Q, Larson BT, et al. The aging feline kidney: a model mortality antagonist? *J Feline Med Surg* 2006; 8: 363–371. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
29. Weiss DJ, Gagne JM, Armstrong PJ. Relationship between inflammatory hepatic disease and inflammatory bowel disease, pancreatitis, and nephritis in cats. *J Am Vet Med Assoc* 1996; 209: 1114–1116. [[PubMed](#)] [[Google Scholar](#)]
30. Jepson RE, Brodbelt D, Vallance C, Syme HM, Elliott J. Evaluation of predictors of the development of azotemia in cats. *J Vet Intern Med* 2009; 23: 806–813. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

31. Syme HM, Markwell PJ, Pfeiffer D, Elliott J. Survival of cats with naturally occurring chronic renal failure is related to severity of proteinuria. *J Vet Intern Med* 2006; 20: 528–535. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
32. Buffington CA. Invited commentary on ‘The aging feline kidney: a model mortality antagonist?’ *J Feline Med Surg* 2006; 8: 361–362. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
33. Boyd LM, Langston C, Thompson K, Zivin K, Imanishi M. Survival in cats with naturally occurring chronic kidney disease (2000–2002). *J Vet Intern Med* 2008; 22: 1111–1117. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
34. White JD, Norris JM, Baral RM, Malik R. Naturally-occurring chronic renal disease in Australian cats: a prospective study of 184 cases. *Aust Vet J* 2006; 84: 188–194. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
35. Jepson RE. Feline systemic hypertension: classification and pathogenesis. *J Feline Med Surg* 2011; 13: 25–34. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
36. Chetboul V, Lefebvre HP, Pinhas C, Clerc B, Boussouf M, Pouchelon JL. Spontaneous feline hypertension: clinical and echocardiographic abnormalities, and survival rate. *J Vet Intern Med* 2003; 17: 89–95. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
37. Jepson RE, Elliott J, Brodbelt D, Syme HM. Effect of control of systolic blood pressure on survival in cats with systemic hypertension. *J Vet Intern Med* 2007; 21: 402–409. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
38. Chakrabarti S, Syme HM, Brown CA, Elliott J. Histomorphometry of feline chronic kidney disease and correlation with markers of renal dysfunction. *Vet Pathol* 2013; 50: 147–155. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
39. Taugner E, Baatz G, Nobiling R. The renin-angiotensin system in cats with chronic renal failure. *J Comp Pathol* 1996; 115: 239–252. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
40. Elliott J, Rawlings JM, Markwell PJ, Barber PJ. Survival of cats with naturally occurring chronic renal failure: effect of dietary management. *J Small Anim Pract* 2000; 41: 235–242. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
41. Gouni V, Chetboul V, Pouchelon JL, Sampedrano CC, Maurey C, Lefebvre HP. Azotemia in cats with feline hypertrophic cardiomyopathy: prevalence and relationships with

echocardiographic variables. J Vet Cardiol 2008; 10: 117–123. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

42. Javadi S, Djajadiningrat-Laanen SC, Kooistra HS, van Dongen AM, Voorhout G, van Sluijs FJ, et al. Primary hyperaldosteronism, a mediator of progressive renal disease in cats. Domest Anim Endocrinol 2005; 28: 85–104. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

43. Andrew R, Harvey AM, Tasker S. Primary hyperaldosteronism in the cat: a series of 13 cases. J Feline Med Surg 2005; 7: 173–182. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

44. Kelly DF, Lucke VM, McCullagh KG. Experimental pyelonephritis in the cat. 1 Gross and histologic findings. J Comp Pathol 1979; 89: 125–139. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

45. Mayer-Roenne B, Goldstein RE, Erb HN. Urinary tract infections in cats with hyperthyroidism, diabetes mellitus and chronic kidney disease. J Feline Med Surg 2007; 9: 124–132. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

46. King JN, Gunn-Moore DA, Tasker S, Gleadhill A, Strehlau G; Benazepril in Renal insufficiency in Cats Study Group. Tolerability and efficacy of benazepril in cats with chronic kidney disease. J Vet Intern Med 2006; 20: 1054–1064. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

47. Bartges JW, Barsanti JA. Bacterial urinary tract infections in cats. In: Kirk RW. (ed). Current veterinary therapy. 13th ed. Philadelphia: WB Saunders, 2000, pp 880–882. [[Google Scholar](#)]

48. Ichii O, Yabuki A, Sasaki N, Otsuka S, Ohta H, Yamasaki M, et al. Pathological correlations between podocyte injuries and renal functions in canine and feline chronic kidney diseases. Histol Histopathol 2011; 26: 1243–1255. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

49. Vaden SL. Glomerular disease. Top Companion Anim Med 2011; 26: 128–134. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

50. Sato R, Soeta S, Syuto B, Yamagishi N, Sato J, Naito Y. Urinary excretion of N-acetyl-beta-D-glucosaminidase and its isoenzymes in cats with urinary disease. J Vet Med Sci 2002; 64: 367–371. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

51. Ross LA, Finco DR. Relationship of selected clinical renal function tests to glomerular filtration rate and renal blood flow in cats. Am J Vet Res 1981; 42: 1704–1710. [[PubMed](#)] [[Google Scholar](#)]

52. Elliott J, Syme HM, Reubens E, Markwell PJ. Assessment of acid-base status of cats with naturally occurring chronic renal failure. *J Small Anim Pract* 2003; 44: 65–70. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
53. King JN, Tasker S, Gunn-Moore DA, Strehlau G; BENRiC (Benazepril in Renal insufficiency in Cats) Study Group. Prognostic factors in cats with chronic kidney disease. *J Vet Intern Med* 2007; 21: 906–916. [[PubMed](#)] [[Google Scholar](#)]
54. Liu Y. Cellular and molecular mechanisms of renal fibrosis. *Nat Rev Nephrol* 2011; 7: 684–696. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
55. Arata S, Ohmi A, Mizukoshi F, Baba K, Ohno K, Setoguchi A, Tsujimoto H. Urinary transforming growth factor- β 1 in feline chronic renal failure. *J Vet Med Sci* 2005; 67: 1253–1255. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
56. Sawashima K, Mizuno S, Mizuno-Horikawa Y, Shimada A, Kudo T, Kurosawa T. Expression of α -smooth muscle actin and fibronectin in tubulointerstitial lesions of cats with chronic renal failure. *Am J Vet Res* 2000; 61: 1080–1086. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
57. Yabuki A, Mitani S, Fujiki M, Misumi K, Endo Y, Miyoshi N, et al. Comparative study of chronic kidney disease in dogs and cats: induction of myofibroblasts. *Res Vet Sci* 2010; 88: 294–299. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
58. Brown SA. Oxidative stress and chronic kidney disease. *Vet Clin North Am Small Anim Pract* 2008; 38: 157–166. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
59. Keegan RF, Webb CB. Oxidative stress and neutrophil function in cats with chronic renal failure. *J Vet Intern Med* 2010; 24: 514–519. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
60. Jensen J, Henik RA, Brownfield M, Armstrong J. Plasma renin activity and angiotensin I and aldosterone concentrations in cats with hypertension associated with chronic renal disease. *Am J Vet Res* 1997; 58: 535–540. [[PubMed](#)] [[Google Scholar](#)]
61. Mishina M, Watanabe T, Fujii K, Maeda H, Wakao Y, Takahashi M. Non-invasive blood pressure measurements in cats: clinical significance of hypertension associated with chronic renal failure. *J Vet Med Sci* 1998; 60: 805–808. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
62. Pedersen KM, Pedersen HD, Häggström J, Koch J, Ersbøll AK. Increased mean arterial pressure and aldosterone-to-renin ratio in Persian cats with polycystic kidney disease. *J Vet Intern Med* 2003; 17: 21–27. [[PubMed](#)] [[Google Scholar](#)]

63. Watanabe T, Mishina M. Effects of benazepril hydrochloride in cats with experimentally induced or spontaneously occurring chronic renal failure. *J Vet Med Sci* 2007; 69: 1015–1023. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
64. Lefebvre HP, Toutain PL. Angiotensin-converting enzyme inhibitors in the therapy of renal diseases. *J Vet Pharmacol Ther* 2004; 27: 265–281. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
65. Brown SA, Brown CA, Jacobs G, Stiles J, Hendi RS, Wilson S. Effects of the angiotensin converting enzyme inhibitor benazepril in cats with induced renal insufficiency. *Am J Vet Res* 2001; 62: 375–383. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
66. Schrier RW, Masoumi A, Elhassan E. Aldosterone: role in edematous disorders, hypertension, chronic renal failure, and metabolic syndrome. *Clin J Am Soc Nephrol* 2010; 5: 1132–1140. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
67. Brown SA, Brown CA. Single-nephron adaptations to partial renal ablation in cats. *Am J Physiol* 1995; 269: R1002–R1008. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
68. Brown SA, Crowell WA, Brown CA, Barsanti JA, Finco DR. Pathophysiology and management of progressive renal disease. *Vet J* 1997; 154: 93–109. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
69. Ross LA, Finco DR, Crowell WA. Effect of dietary phosphorus restriction on the kidneys of cats with reduced renal mass. *Am J Vet Res* 1982; 43: 1023–1026. [[PubMed](#)] [[Google Scholar](#)]
70. Adams LG, Polzin DJ, Osborne CA, O'Brien TD, Hostetter TH. Influence of dietary protein/calorie intake on renal morphology and function in cats with 5/6 nephrectomy. *Lab Invest* 1994; 70: 347–357. [[PubMed](#)] [[Google Scholar](#)]
71. Finco DR, Brown SA, Brown CA, Crowell WA, Sunvold G, Cooper TL. Protein and calorie effects on progression of induced chronic renal failure in cats. *Am J Vet Res* 1998; 59: 575–582. [[PubMed](#)] [[Google Scholar](#)]
72. Elliott J, Syme HM, Markwell PJ. Acid-base balance of cats with chronic renal failure: effect of deterioration in renal function. *J Small Anim Pract* 2003; 44: 261–268. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
73. Chakrabarti S, Syme HM, Elliott J. Clinicopathological variables predicting progression of azotemia in cats with chronic kidney disease. *J Vet Intern Med* 2012; 26: 275–281.

[\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

74. Kuwahara Y, Ohba Y, Kitoh K, Kuwahara N, Kitagawa H. Association of laboratory data and death within one month in cats with chronic renal failure. *J Small Anim Pract* 2006; 47: 446–450. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

75. Brown S, Atkins C, Bagley R, Carr A, Cowgill L, Davidson M, et al. Guidelines for the identification, evaluation, and management of systemic hypertension in dogs and cats. *J Vet Intern Med* 2007; 21: 542–558. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

76. Kidder A, Chew D. Treatment options for hyperphosphataemia in feline CKD. *J Feline Med Surg* 2009; 11: 913–924. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

77. Barber PJ, Elliott J. Feline chronic renal failure: calcium homeostasis in 80 cases diagnosed between 1992 and 1995. *J Small Anim Pract* 1998; 39: 108–116. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

78. Finch NC, Geddes RF, Syme HM, Elliott J. Fibroblast growth factor 23 (FGF-23) concentrations in cats with early nonazotemic chronic kidney disease (CKD) and in healthy geriatric cats. *J Vet Intern Med* 2013; 27: 227–233. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

79. Geddes RF, Finch NC, Elliott J, Syme HM. Fibroblast growth factor 23 in feline chronic kidney disease. *J Vet Intern Med* 2013; 27: 234–241. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

80. Bertazzolo W, Toscani L, Calcaterra S, Crippa L, Caniatti M, Bonfanti U. Clinicopathological findings in five cats with paw calcification. *J Feline Med Surg* 2003; 5: 11–17. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

81. DiBartola SP, Autran De Morais H. Disorders of potassium: hypokalemia and hyperkalemia. In: DiBartola SP. (ed). *Fluid, electrolyte, and acid-base disorders in small animal practice*. 4th ed. St Louis: Elsevier Saunders, 2012, pp 92–119. [\[Google Scholar\]](#)

82. Dow SW, Fettman MJ, Curtis CR, LeCouteur RA. Hypokalemia in cats: 186 cases (1984–1987). *J Am Vet Med Assoc* 1989; 194: 1604–1608. [\[PubMed\]](#) [\[Google Scholar\]](#)

83. Adams LG, Polzin DJ, Osborne CA, O'Brien TD. Effects of dietary protein and calorie restriction in clinically normal cats and in cats with surgically induced chronic renal failure. *Am J Vet Res* 1993; 54: 1653–1662. [\[PubMed\]](#) [\[Google Scholar\]](#)

84. Dow SW, Fettman MJ, Smith KR, Hamar DW, Nagode LA, Refsal KR, et al. Effects of dietary acidification and potassium depletion on acid-base balance, mineral metabolism

and renal function in adult cats. J Nutr 1990; 120: 569–578. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

85. Cowgill LD. Pathophysiology and management of anemia in chronic progressive renal failure. Semin Vet Med Surg (Small Anim) 1992; 7: 175–182. [[PubMed](#)] [[Google Scholar](#)]

86. Chalhoub S, Langston C, Eatroff A. Anemia of renal disease: what it is, what to do and what's new. J Feline Med Surg 2011; 13: 629–640. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

87. Lees GE, Brown SA, Elliott J, Grauer GE, Vaden SL. Assessment and management of proteinuria in dogs and cats: 2004 ACVIM Forum Consensus Statement (small animal). J Vet Intern Med 2005; 19: 377–385. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

88. Syme HM. Proteinuria in cats. Prognostic marker or mediator? J Feline Med Surg 2009; 11: 211–218. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

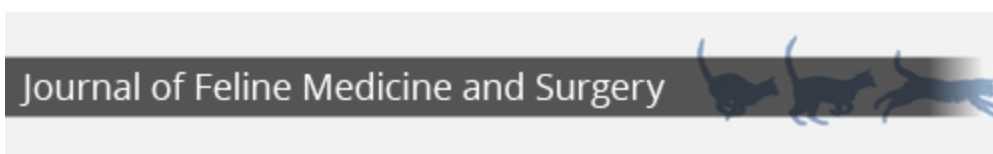
89. Bartlett PC, Van Buren JW, Bartlett AD, Zhou C. Case-control study of risk factors associated with feline and canine chronic kidney disease. Vet Med Int 2010; Article ID: 957570. DOI: 10.4061/2010/957570. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

90. Goldstein RE, Marks SL, Kass PH, Cowgill LD. Gastrin concentrations in plasma of cats with chronic renal failure. J Am Vet Med Assoc 1998; 213: 826–828. [[PubMed](#)] [[Google Scholar](#)]

91. Schulman RL, Krawiec DR. Gastrointestinal complications of uremia. In: Bonagura JD. (ed). Kirk's current veterinary therapy. XIII Small animal practice. Philadelphia: WB Saunders, 2000, pp 864–866. [[Google Scholar](#)]

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Prevalence and classification of chronic kidney disease in cats randomly selected from four age groups and in cats recruited for degenerative joint disease studies

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Abstract

Chronic kidney disease (CKD) and degenerative joint disease are both considered common in older cats. Information on the co-prevalence of these two diseases is lacking. This retrospective study was designed to determine the prevalence of CKD in two cohorts of cats: cats randomly selected from four evenly distributed age groups (RS group) and cats recruited for degenerative joint disease studies (DJD group), and to evaluate the concurrence of CKD and DJD in these cohorts. The RS group was randomly selected from four age groups from 6 months to 20 years, and the DJD group comprised cats recruited to four previous DJD studies, with the DJD group excluding cats with a blood urea nitrogen and/or serum creatinine concentration >20% (the upper end of normal) for two studies and cats with CKD stages 3 and 4 for the other two studies. The prevalence of CKD in the RS and DJD groups was higher than expected at 50% and 68.8%, respectively. CKD was common in cats between 1 and 15 years of age, with a similar prevalence of CKD stages 1 and 2 across age groups in both the RS and DJD cats, respectively. We found significant concurrence between CKD and DJD in cats of all ages, indicating the need for increased screening for CKD when

selecting DJD treatments. Additionally, this study offers the idea of a relationship and causal commonality between CKD and DJD owing to the striking concurrence across age groups and life stages.

Introduction

Chronic kidney disease (CKD) in cats is a prolonged process marked by irreversible loss of kidney function, usually without an identifiable cause, and has been increasingly diagnosed over the past three decades, particularly in older cats.¹⁻⁴ Because glomerular filtration rate (GFR) is not commonly measured in the clinical setting, serum creatinine concentration is most often used as a marker of GFR.⁵ The International Renal Interest Society (IRIS) has developed a staging system in which the stages are based on the serum creatinine concentration in animals with stable CKD. The staging system is used as a guide for therapeutic intervention, as well as prognostication.^{5,6} The first stage is often overlooked because the serum creatinine concentrations fall within the normal reference interval for most laboratories; therefore, CKD stage 1 can be identified by finding proteinuria, normoglycemic glucosuria, isosthenuria (1.008–1.012), poorly concentrated urine (1.013–1.035) or evidence of CKD via renal imaging.^{1,5,6} CKD stages 2, 3 and 4 represent progressive increases in serum creatinine concentrations and clinical signs.

Degenerative joint disease (DJD) is the progressive destruction of one or more components of synovial or cartilaginous joints.^{7,8} Radiographic DJD is very common in cats; however, these findings do not correlate well with results of orthopedic examination,⁸⁻¹⁰ making the diagnosis of DJD-associated pain and impairment challenging. There are many proposed treatments for feline DJD.^{11,12} Although non-steroidal anti-inflammatory drugs (NSAIDs) have been shown to be efficacious in controlling pain associated with DJD in cats, there is concern regarding the use of NSAIDs in cats with DJD and concurrent CKD¹³ owing to the recognized role prostanoids play in renal function.¹⁴

Previously published reports on the prevalence of CKD in cats that predate the IRIS staging system used a serum creatinine concentration >1.8 or 2 mg/dl (159 or 177 μ mol/l) as the indicator for CKD.^{2-4,15} The objectives of this study were to use multiple indicators of renal disease to describe the prevalence of CKD in a cohort of cats randomly selected from four age groups and a group of cats recruited for multiple DJD studies, the concurrence of CKD with DJD in the latter group and assign IRIS stage classification to both cohorts of cats if CKD was present.

Materials and methods

This is a retrospective study combining data from five studies at our institution. The Institutional Animal Care and Use Committee at North Carolina State University approved all studies used to collect the data reported herein. The studies were explained to all pet owners, who then signed an informed consent.

Part 1: Randomly selected group

The randomly selected (RS) group consisted of a population of randomly selected cats from a single practice, as described previously.^{9,16} This database of cats was divided into four age groups (6 months to 5 years; 5–10 years; 10–15 years; and 15–20 years) and each cat was assigned a number within the given age group. The numbers were randomly ranked using a computer program and the first 25 cats in each age group were selected to come in for a health screen. This yielded a cohort of 100 cats with 25 cats that had been randomly selected in each age group.⁹ Although selected at random, no cat was acutely ill at the time of data collection. Age, gender, body weight and body condition score (BCS) using a five-grade body index system¹⁷ were recorded for all cats. All cats underwent physical, orthopedic and neurological examinations. Orthogonal radiographs were made of all appendicular joints and spinal segments while cats were under sedation. Samples were subsequently collected for complete blood count (CBC), serum chemistry and urine analysis (by cystocentesis).⁹

Part 2: DJD group

The DJD group consisted of 128 cats recruited to one of four different studies of reduced activity or impaired mobility conducted at our institution: a trial of a feline therapeutic diet^{18,19} (study A); a study evaluating a nutritional supplement (study B; unpublished data, subjects described previously¹⁹); a study assessing the reliability of a novel feline musculoskeletal pain index²⁰ (study C), and an ongoing study investigating a treatment for DJD-associated chronic pain (study D; unpublished data). As described previously, patient history, physical examination, CBC, serum chemistry and urine analysis were used to screen cats for other systemic disorders and a detailed client questionnaire was used to determine whether there was mobility impairment.^{9,18,19} Age, gender, body weight and BCS using a nine-grade body index system²¹ were recorded for all cats.

For all studies, cats were included if (i) they lived only indoors, (ii) no anti-inflammatory medications had been given 4 weeks prior to the study, (iii) no clinically significant changes in CBC or serum chemistry results had occurred within the previous 4 weeks, and (iv) a client-specific outcome measures form indicated impairment for activities, as described previously.¹¹ Cats were enrolled in studies A, B and C if pain was elicited in at least one joint during orthopedic

examination and there was radiographic evidence of DJD.^{9,16} Cats with at least two painful joints and overlapping radiographic evidence of DJD in these joints were enrolled into study D. Cats in studies A and B only had radiographs made when joint pain was elicited, whereas those in studies C and D had radiographs made of all appendicular joints and spinal segments.

Cats were excluded from studies A and B if the blood urea nitrogen and/or serum creatinine was >20% above the upper limit of the reference interval.¹⁸ Cats were excluded from studies C and D if the serum creatinine was >2.8 mg/dl (249 μ mol/l).²⁰ In summary, cats recruited to studies A, B, C and D were all cats that exhibited mobility impairment due to DJD-associated pain, excluding cats with the most severe CKD.

CKD evaluation

As part of the present study, cats in the RS and DJD groups were placed in one of three categories based on the results of CBC, serum chemistry, urine analysis, and radiographic determination of degenerative kidney changes and kidney size (evidence of CKD [CKD_{pos}], small kidney, or no evidence of CKD [CKD_{neg}]) (Supplementary material). Because urine specific gravity (USG) is not a reliable indicator of feline kidney function in isolation, and other factors, such as water intake and other diseases, can affect this, cats with a USG of 1.013–1.034 without other indicators of kidney disease were categorized as CKD_{neg}. Two authors (CLM and SLV) went over the above results for each cat to determine category placement. If the features of parenchymal mineralization, irregular margins and/or misshapen kidneys were seen radiographically, then cats were considered to have degenerative kidney changes. A board-certified radiologist reviewed all radiographs for kidney abnormalities. The left and right kidneys were visualized on the ventrodorsal view of spinal radiographs and the lengths were measured to the nearest millimeter (Efilm Workstation 3.3.0; Soundelkin). The kidney lengths were compared to the length of a single lumbar vertebra (L2) measured from endplate to endplate by a single investigator (CLM).^{22–24} A ratio was calculated for each kidney; a range of 2–3 times the length of L2 was considered normal.^{3,22,23,25–27} Cats with CKD were staged according to the IRIS guidelines⁵; without urine protein to creatinine ratios and blood pressure measurements, IRIS substaging was not possible in this study.

Data analysis

Descriptive statistics were used to characterize demographic data, including gender, body weight, BCS and breed. The prevalence of CKD was determined as the ratio of CKD_{pos} cats to the total number of cats and the total in each age group. For the CKD_{pos} cats, CKD stage was determined and the prevalence of each stage was reported as a ratio to the total number of CKD_{pos} cats and the

total number of cats in both the RS and DJD groups (population totals). For the purpose of this study, DJD group cats were placed in one of four life stages (immature, mature, senior or geriatric), with the immature life stage including kitten, junior and prime, according to the American Association of Feline Practitioners guidelines.^{28,29} Henceforth, the RS cohort divisions are referred to as age groups and the DJD cohort divisions are referred to as life stages.

Results

Part 1: RS group

Fourteen cats were excluded from the study because either a urine sample was not collected for urine analysis or the kidneys were not visible on radiographs, precluding full evaluation of CKD. Of the 86 remaining, 24 cats were aged 0–4.9 years, 22 cats were aged 5.0–9.9 years, 19 cats were aged 10.0–14.9 years and 21 cats were aged 15–20 years ([Table 1](#)). Demographic data for these 86 cats are presented in [Table 2](#). Sixty cats were domestic shorthair, five domestic longhair, two mixed and 19 purebred (seven Siamese, six Maine Coon Cats, three Abyssinians, one each of Persian, Tonkinese and British Shorthair). Of the RS cats, 50% (43/86) were CKD_{pos}, 11.6 % (10/86) were in the small kidney group and 38.4 % (33/86) were CKD_{neg} ([Figure 1](#)). Of the 33 CKD_{neg} cats, six had a USG <1.035 with no other evidence of CKD. For the 43 CKD_{pos} cats, 27.9% (12/43) were CKD stage 1, 65.1% (28/43) stage 2 and 7.0% (3/43) stage 3; none of the cats was CKD stage 4 ([Table 3](#)). All small kidney cats were CKD stage 1 and were not included in the CKD_{pos} cat group. The total prevalence (CKD_{pos}/RS population total) for CKD stages 1, 2 and 3 were 13.9% (12/86), 32.6% (28/86) and 3.5% (3/86), respectively ([Table 3](#)). Five Siamese (71.4%) and three Maine Coon Cats (50%) were CKD_{pos}, and another Maine Coon Cat was in the small kidney group. All Abyssinians in the RS group were CKD_{neg}.

Table 1.

Prevalence of cats with chronic kidney disease (CKD_{pos}) and International Renal Interest Society (IRIS) CKD stage in each age group and life stage for the randomly selected (RS) and degenerative joint disease (DJD) groups. Percentages presented in parentheses

Age (y)	RS group				DJD group			
	0.0-4.9	5.0-9.9	10.0-14.9	15.0-20.0	0-6	7-10	11-14	>15
Life stage					Immature [‡]	Mature	Senior	Geriatric
	n = 24	n = 22	n = 19	n = 21	n = 8	n = 32	n = 59	n = 29
CKD _{pos} *	9/24 (37.5)	9/22 (40.9)	8/19 (42.1)	17/21 (80.9)	4/8 (50.0)	21/32 (65.6)	38/59 (64.4)	25/29 (86.2)
CKD stage 1 [†]	3/9 (33.3)	3/9 (33.3)	3/8 (37.5)	3/17 (17.6)	1/4 (25.0)	9/21 (42.9)	16/38 (42.1)	3/25 (12.0)
CKD stage 2 [†]	6/9 (66.7)	6/9 (66.7)	4/8 (50.0)	12/17 (70.6)	3/4 (75.0)	12/21 (57.1)	22/38 (57.9)	21/25 (84.0)
CKD stage 3 [†]			1/8 (12.5)	2/17 (11.8)				1/25 (4.0)

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*Percentage represents the number of CKD_{pos} cats out of the total number of cats in each age group and life stage

[†]Percentage represents the number of cats in each IRIS CKD stage out of the total number of CKD_{pos} cats in each age group and life stage

[‡]Combines kitten, junior and prime American Association of Feline Practitioners' life stages

Table 2.

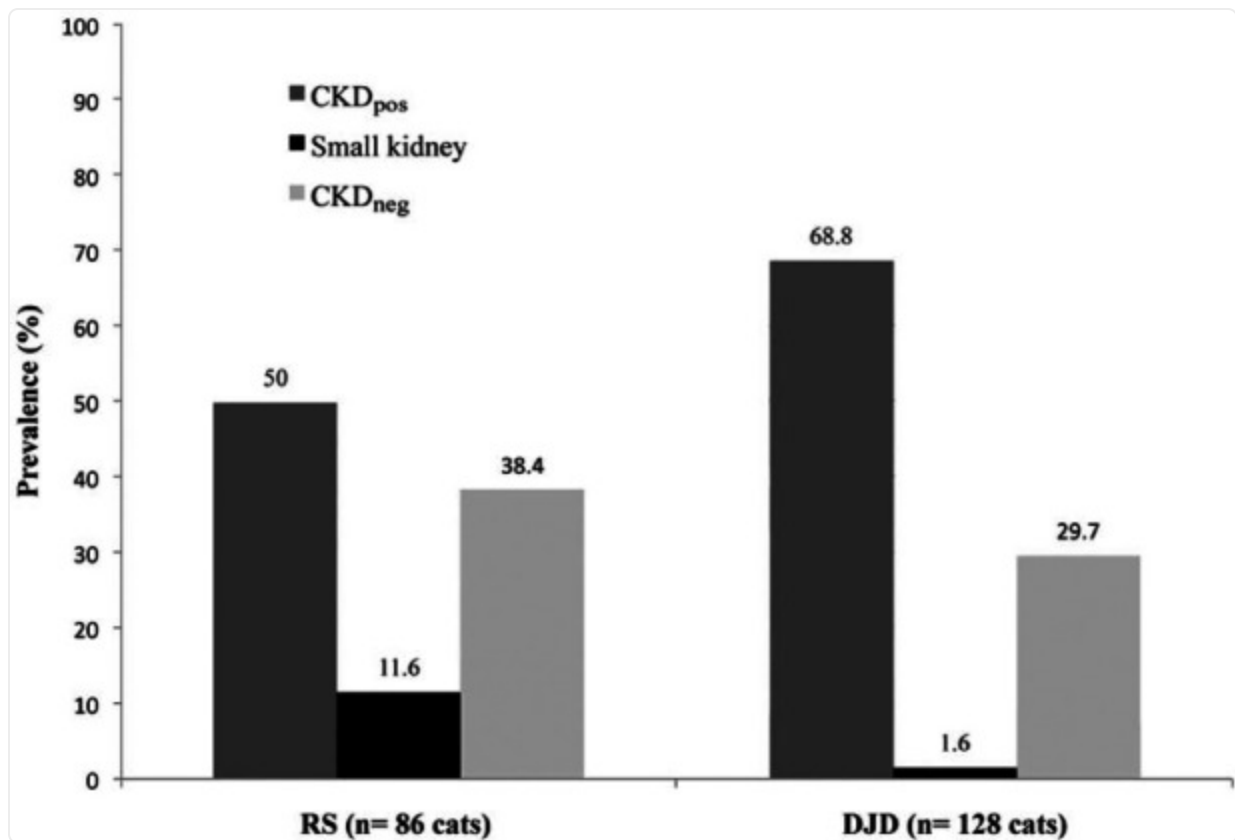
Demographic data from the 86 cats in the randomly selected (RS) group summarized for the whole group, as well as cats with chronic kidney disease (CKD_{pos}) (including International Renal Interest Society CKD stages), the small kidney group and cats without CKD (CKD_{neg})

		All cats	CKD _{pos}			Small kidney	CKD _{neg}
			CKD stage 1	CKD stage 2	CKD stage 3		
			n = 86	n = 12	n = 28		
Age (y)	Mean ± SD	9.1 ± 5.1	9.4 ± 5.2	10.9 ± 5.4	15.7 ± 2.5	7.4 ± 4.1	7.3 ± 4.4
	Range	1.0–20.0	1.6–11.0	1.0–20.0	12.8–17.3	1.4–15.0	1.0–16.2
Weight (kg)	Mean ± SD	5.0 ± 1.6	5.1 ± 1.0	4.8 ± 1.9	4.7 ± 0.4	3.9 ± 1.1	5.4 ± 4.4
	Range	2.1–10.2	3.0–6.3	2.1–9.8	4.4–5.2	2.4–6.1	3.4–10.2
Sex	MC	30	1	10	1	1	17
	FS	56	11	18	2	9	16
BCS (1–5)	Median (min, max)	3 (1, 5)	4 (2, 5)	3 (1, 5)	3 (1, 4)	2 (1, 4)	3 (2, 5)

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BCS = body condition score; MC = male castrated; FS = female spayed

Figure 1.



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Prevalence of cats with chronic kidney disease (CKD_{pos}), small kidneys, and without CKD (CKD_{neg}) in the randomly selected (RS) and degenerative joint disease (DJD) groups. Total prevalence (percentage) is shown above each bar. The prevalence was calculated as the number of cats in each category out of the total number of cats in the RS and DJD groups

Table 3.

Prevalence of each International Renal Interest Society chronic kidney disease (CKD) stage in cats with CKD in the randomly selected (RS) and degenerative joint disease (DJD) groups. 'CKD_{pos}/CKD_{pos} total' represents the number of cats in each CKD stage per the total number of CKD_{pos} cats in the RS and DJD groups. 'CKD_{pos}/pop total' is the number of cats in each CKD stage over the total number of cats in the RS and DJD groups

		CKD stage 1	CKD stage 2	CKD stage 3
RS group	CKD _{pos} /CKD _{pos} total (%)	12/43 (27.9)	28/43 (65.1)	3/43 (7.0)
	CKD _{pos} /RS pop total (%)	12/86 (13.9)	28/86 (32.6)	3/86 (3.5)
DJD group	CKD _{pos} /CKD _{pos} total (%)	29/88 (33.0)	58/88 (65.9)	1/88 (1.1)
	CKD _{pos} /DJD pop total (%)	29/128 (22.7)	58/128 (45.3)	1/128 (0.8)

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Part 2: DJD group

One hundred and twenty-eight cats were included from studies A (24 cats), B (37 cats), C (24 cats) and D (43 cats). Eight were immature, 32 mature, 59 senior and 29 geriatric. Demographic data for this cohort are presented in [Table 4](#). One hundred cats were domestic shorthair, nine were domestic longhair, one was mixed and 18 were purebred (seven Maine Coon, five Siamese, two Scottish Folds, and one each of Abyssinian, British Shorthair, Manx and Himalayan). Of the 128 cats, 68.8% (88/128) were CKD_{pos}, 1.6% (2/128) were in the small kidney group and 29.7% (38/128) were CKD_{neg} ([Figure 1](#)). Of the 38 CKD_{neg} cats, 13 had a USG <1.035 with no other evidence of CKD. For the 88 CKD_{pos} cats, 33.0% (29/88) were CKD stage 1, 65.9% (58/88) stage 2, 1.1% (1/88) stage 3 and none were stage 4 ([Table 3](#)). The total prevalence (CKD_{pos}/DJD population total) for CKD stages 1, 2 and 3 were 22.7% (29/128), 45.3% (58/128) and 0.8% (1/128) respectively ([Table 3](#)). Four Siamese (80%), six Maine Coon Cats (85.6%) and the Abyssinian were CKD_{pos}.

Table 4.

Demographic data from the 128 cats in the degenerative joint disease group summarized for the whole group, as well as cats with chronic kidney disease (CKD_{pos}) (including International Renal Interest Society CKD stages), the small kidney group and cats without CKD (CKD_{neg})

		All cats n = 128	CKD _{pos}			Small kidney n = 2	CKD _{neg} n = 38
			CKD stage 1 n = 29	CKD stage 2 n = 58	CKD stage 3 n = 1		
Age (y)	Mean ± SD	12.5 ± 3.6	11.7 ± 2.6	13.4 ± 3.8	18.3	11.8 ± 0.8	11.7 ± 3.5
	Range	2.5–21.2	6.0–16.9	3.8–21.1	–	10.9–12.8	2.6–21.2
Weight (kg)	Mean ± SD	5.3 ± 2.8	5.0 ± 1.0	5.3 ± 1.4	5.8	4.6 ± 0.8	5.8 ± 1.7
	Range	2.8–11.0	2.9–7.3	2.8–8.7	–	4.0–5.1	3.3–11.0
Sex	MC	52	10	26		1	15
	FS	76	19	32	1	1	23
BCS (1–9)	Median (min, max)		5 (3, 8)	6 (3, 9)	5	4.5 (4, 5)	7 (2, 9)

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BCS = body condition score; MC = male castrated; FS = female spayed

Discussion

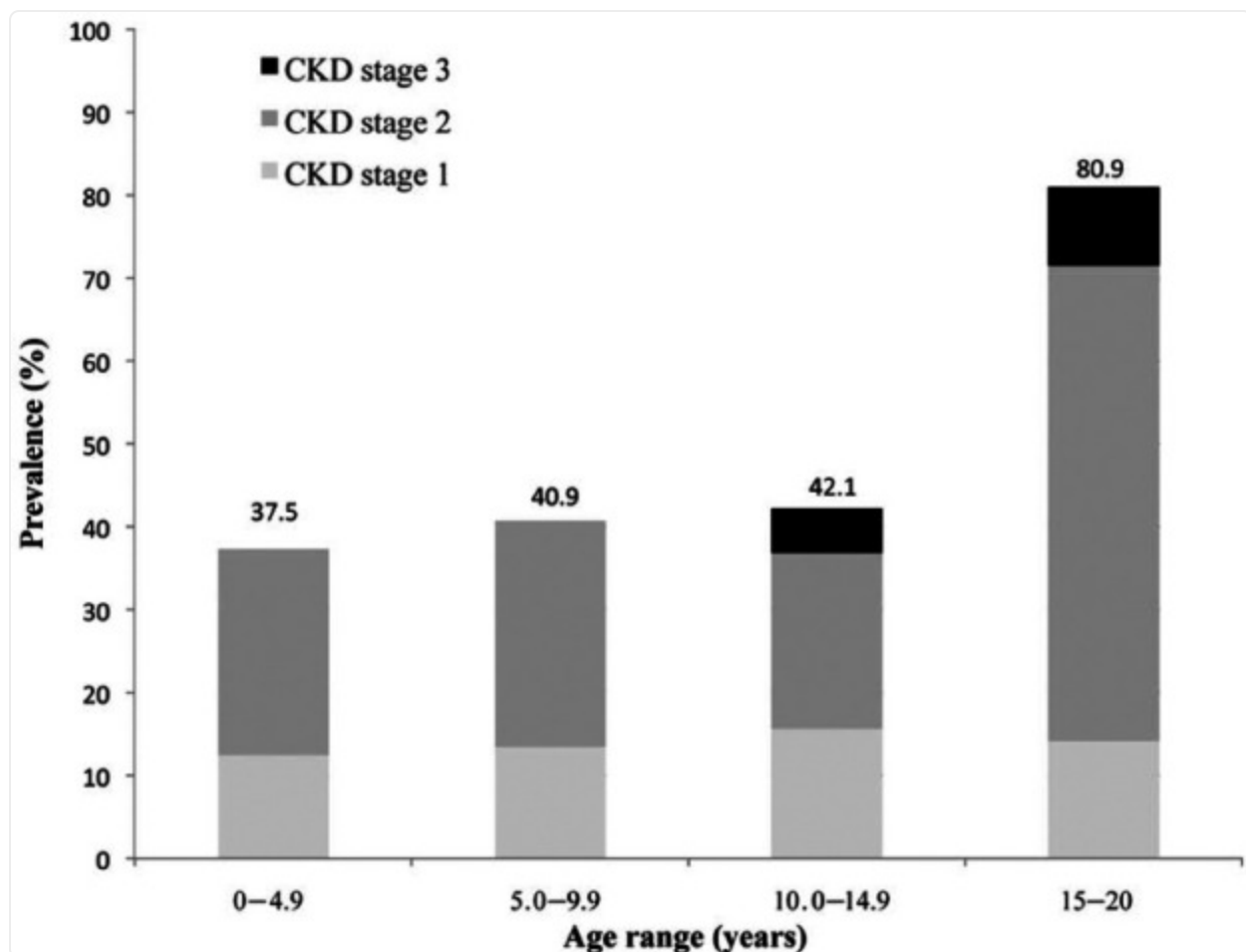
The objectives of this study were to determine the prevalence of CKD in two different cohorts of cats (the RS group and the DJD group), report the concurrence of CKD and DJD in the DJD group, and assign CKD stage using IRIS guidelines. The 50% prevalence of CKD in the RS cats was much higher than previously reported, with the majority of cats being CKD stage 2 (32.6%). The prevalence of CKD in the DJD cats was even higher, with 68.8% of cats in this group having CKD and 45.3% of these cats classified as CKD stage 2. In both the RS and DJD groups, the prevalence of CKD was similar in the younger age groups and life stages, with an apparent increase in prevalence in cats aged 15 years and older. However, direct comparisons of the data from the two groups must be made very cautiously as the inclusion criteria were very different for the two groups.

Demographic data for cats with CKD was similar to previous reports.²⁻⁴ More females than males were CKD_{pos} in the RS and DJD groups. Previously reported gender ratios are variable.^{2,3,16,30,31} The variability in this study is likely a result of the small sample size. Consistent with previous studies both groups comprised mainly domestic short- and longhair cats.^{3,4,16} Siamese, Maine Coon cats, Abyssinian, Russian Blue and Burmese cats have been previously reported to have twice the likelihood of being affected with CKD than domestic shorthair cats.³ Findings presented here support the previously reported increased risk for CKD in purebred cats; however, the small number of purebred cats in this study precluded the ability to identify breed predispositions or to determine if the breed distribution in CKD_{pos} cats differed between the RS and DJD cats.

The prevalence of CKD in the RS cats is strikingly higher than previously reported. The prevalence of kidney disease in cats seen at academic institutions is reported to be 1.18–3.05%, but only 1.9% in cats evaluated in a private practice setting.^{3,4} However, these studies used azotemia to make the diagnosis of kidney disease and likely missed non-azotemic cats with CKD, particularly those with CKD stages 1 or 2. Nonetheless, the prevalence of azotemic cats in the present study (CKD stage 2 and 3) was higher than previously reported (36.2%). Kidney size does not always correlate with kidney disease;²³ however, small kidneys may be a risk factor for subsequent kidney disease.³ In the current study, cats in the small kidney group had a lower BCS and weight than CKD_{pos} and CKD_{neg} cats. This may account for the smaller kidney size without the other appreciated changes (Tables 2 and 4). The increase in CKD found in this study is likely multi-factorial. Veterinarians and pet owners have a heightened awareness of feline CKD.²⁸ Fewer pet owners take their cat to the veterinarian for preventive care, even though 75% of owners admit that routine care is important for their cat.³² Thus, the early stages of CKD are probably missed in many cats. Additionally, other factors may represent important biases. The RS group of cats was recruited from a single feline-only practice in the USA. Feeding and maintenance of these cats may differ from other geographical locations, and may affect the prevalence of CKD.

Unexpectedly in the RS cats, the prevalence of CKD in the three age groups <15 years was very similar (37.5%, 40.9%, 42.1%) ([Figure 2](#)). One study found the prevalence of CKD to be 13% in cats <4 years, 24% in cats aged 4–10 years, 31% in cats aged 10–15 years and 32% in cats ≥15 years.³ The CKD stages in the RS group of cats did indicate more advanced CKD (stage 3) in cats >10 years of age; however, this only encompassed 3.5% of the RS cats. Of interest was the similar prevalence of CKD stages 1 and 2 in the three age groups <15 years of age. The reason for this even distribution of CKD stage among the younger cats is not known; however, there are no historical data to compare our data with. Previous studies have not evaluated the severity of azotemia, nor have they included cats with a creatinine concentration <1.6 mg/dl (140 μmol/l). Though difficult to diagnose, recognizing CKD stage 1 and 2 cats is likely critical to finding an inciting cause and implementing appropriate therapy.⁶ Histopathology and long-term follow up may elucidate why so many younger cats have CKD. Our data indicate that cats of all ages should be screened for CKD to identify at-risk cats and adjust treatment accordingly.

Figure 2.



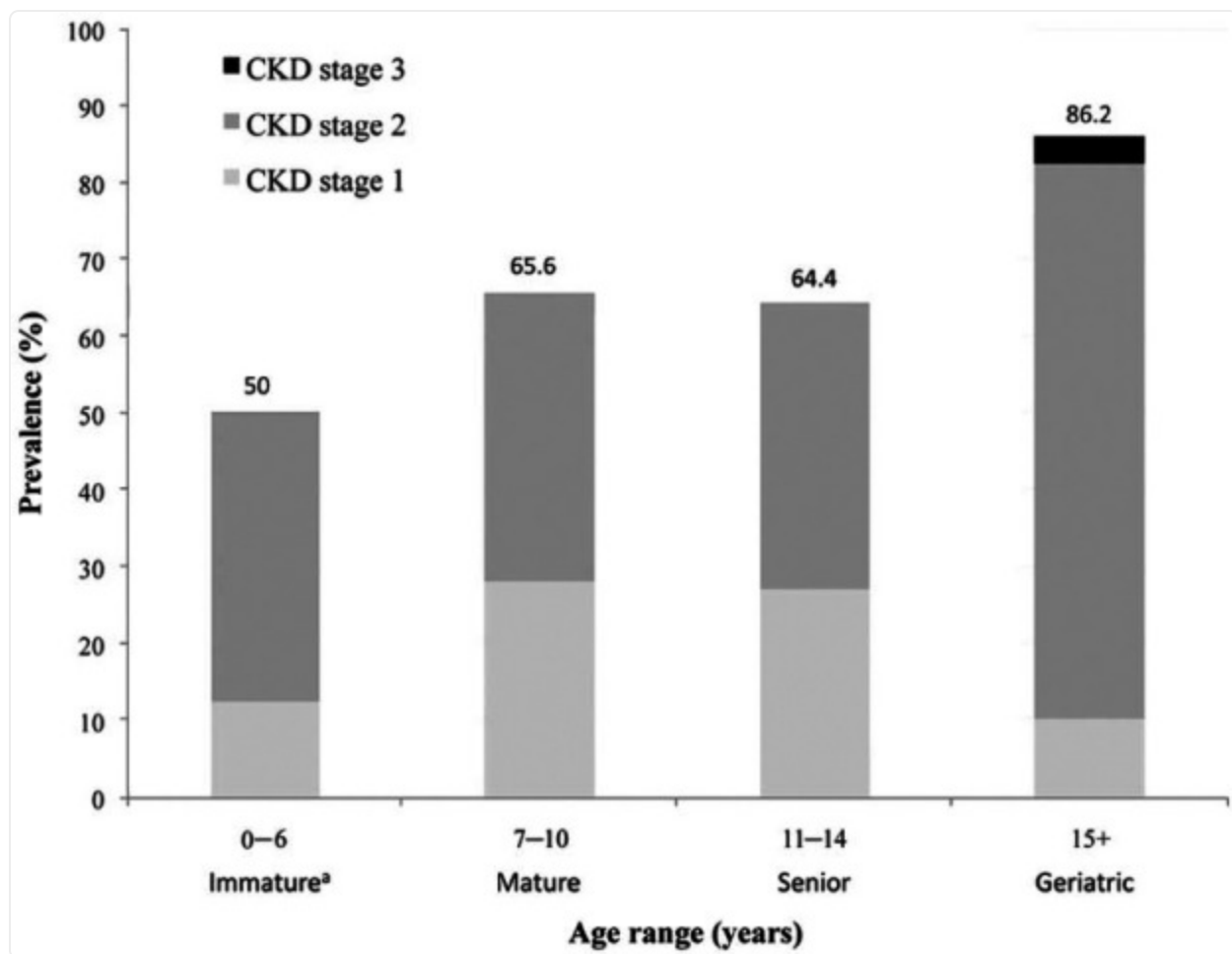
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Prevalence of cats with chronic kidney disease (CKD_{pos}) in each age group of the randomly selected group cats with International Renal Interest Society CKD stage contribution designated by different shades within each bar. The total prevalence (percentage) of CKD_{pos} cats out of the total number of cats in the respective age group is shown above each bar. The number of CKD_{pos} cats in each age group is presented in [Table 2](#)

An even higher prevalence of CKD was found in the DJD cats than RS cats (68.8%); however, because of the difference in the way cats were selected for the RS and DJD groups, a direct comparison cannot be made between the two groups nor between age groups and life stages. Increasing age is highly correlated with the presence and severity of radiographic DJD, as well as an increased chance of finding crepitus, effusion or joint thickening on orthopedic

examination;^{10,17,33} age-related progression of this DJD is likely to be the cause of mobility impairment, and explains why the majority of cats in the DJD group were older. Despite the low number of cats in the immature life stage (n = 8), 50% of these cats with DJD were CKD_{pos}, 75% of which were CKD stage 2 (3/4) ([Figure 3](#)). Similarly, the majority of CKD_{pos} cats in other life stages were CKD stage 2. A single CKD stage 3 cat was enrolled in the DJD group erroneously owing to differing laboratory ranges. Although this cat was included in our descriptive statistics, and its data presented in this report, inclusion does not affect the data presented on CKD stages 1 and 2 cats. It is known that pathological kidney changes may already be present even when the serum creatinine concentration is <2 mg/dl (177 µmol/l), which would allow cats at the lower end of CKD stage 2 to be missed if simply evaluating serum creatinine concentration.³⁴ Together, our findings indicate a strong concurrence between these two disease processes and the need to screen for CKD, particularly when considering NSAID treatment for DJD, in any aged cat.

Figure 3.



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Prevalence of cats with chronic kidney disease (CKD_{pos}) in each life stage of the degenerative joint disease group with the International Renal Interest Society CKD stage contribution designated by different shades within each bar. The total prevalence (percentage) of CKD_{pos} cats out of the total number of cats in the respective life stage is shown above each bar. CKD stage contribution within each life stage is represented by different shading. The number of CKD_{pos} cats in each life stage is presented in [Table 2](#).

^aCombines kitten, junior and prime American Association of Feline Practitioners' life stages

The high prevalence of CKD in cats with DJD has not been documented previously. This has important implications for treatment.³⁵ Avoiding NSAIDs in cats with both DJD and CKD would mean few cats with DJD would be treated with what is currently assumed to be the most effective

class of analgesics for the treatment of chronic musculoskeletal pain in the cat. Two recent studies have suggested that cats with CKD can be safely treated with low doses of NSAIDs; however, in these studies the authors report the decision to treat cats with CKD with NSAIDs was biased, with the veterinarian choosing cats that 'looked' able to tolerate NSAIDs.^{36,37} Additionally, the doses used in these studies (0.02 mg/kg/day, on average) are not known to be effective in alleviating DJD-associated pain. Further understanding of the effect of NSAIDs on the compromised feline kidney is needed, and an understanding of what alternative therapies are efficacious for managing feline DJD-associated pain is required.

The high co-prevalence of CKD and DJD is interesting, and could be explained in several ways. One explanation for the concurrency could be behavioral. In the DJD studies, owners presented cats for inclusion because of perceived changes in activity level and interaction with their cats. It is possible that CKD may have been a reason for these behavioral changes in the cats (in any life stage); when screened for the study, these cats were found to have DJD. Another possible explanation for the concurrency could be shared etiology of CKD and DJD. The cause of primary DJD in cats is not currently known, although inflammatory and immune-mediated etiologies have been suggested,³⁸ and immune dysfunction has been seen in association with DJD in a genomic/proteomic study.³⁹ A previous study of DJD in cats revealed an increase in the prevalence of azotemia, but this significant association was no longer statistically significant when controlled for age.¹⁰ Histologically, end-stage CKD is associated with tubulointerstitial fibrosis.³⁰ Chronic inflammation is often implicated in the development of fibrosis; however, the inciting cause is rarely identified because of the chronicity of the disease.^{30,40} It would be interesting to know whether this inflammation results from targeting of a specific antigen within the kidney or if the inflammation is due to immune system dysfunction. It is possible that inflammation or immune dysfunction may be a common mechanism behind both DJD and CKD, and this requires further study.

Limitations of this study included lack of ultrasonographic evaluation of the kidneys to document chronic degenerative changes. Radiographs are often the best tool for measuring kidney size, but subtle changes to the renal parenchyma that can be missed with radiographs might be detected via ultrasonic evaluation. Ultrasonography might have provided additional information about the small kidney cats and whether they should have been included in the CKD_{pos} group. Cats with a USG of 1.013–1.035 were not included in the CKD_{pos} group because additional diagnostics (eg, abdominal ultrasound) were needed to confidently diagnose CKD in the absence of other indicators; therefore, the true prevalence may actually have been underestimated by the exclusion of these cats. Some cats had albuminuria. Urine protein to creatinine ratios and blood pressure measurements were not routinely measured in the original studies. Future studies should include

these two variables to allow for IRIS CKD substaging, which may potentially discern subtle differences between age groups and life stages. The RS cats were selected at random, 25 within each of four age groups, and thus the prevalence may not represent the prevalence in the general population. Additionally, the RS cats were selected from a single feline-only practice, and the DJD cats were required to be predominantly indoor cats, and this may have biased the results.

Conclusions

The 50% prevalence of CKD in a randomly selected population of cats from a single feline-only practice was significantly higher than previously reported, including cats with CKD stage 1, 2 and 3. The prevalence of CKD was similar in cats up to 15 years of age, but there was an increased prevalence among cats older than 15 years. The prevalence of CKD in cats recruited to DJD studies was 68.8%, with the majority of these cats being CKD stage 2. Future studies are needed to determine if CKD and DJD are related, and additional studies should aim to define treatment options for cats with DJD that are safe in cats with concurrent CKD.

Supplemental Material

Click here for Supplementary Appendix

[sj-pdf-1-jfm-10.1177_1098612X13511446.pdf](#)^(48.2KB, pdf)

Algorithm to determine cats with chronic kidney disease (CKDpos), small kidney and CKDneg groups

Acknowledgments

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Footnotes

Supplementary material: Algorithm to determine cats with chronic kidney disease (CKD_{pos}), small kidney and CKD_{neg} groups.

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References

1. Brown SA. Evaluation of chronic renal disease: a staged approach. *Compend Educ Pract Vet* 1999; 21: 752–763. [[Google Scholar](#)]
2. Elliott J, Barber PJ. Feline chronic renal failure: clinical findings in 80 cases diagnosed between 1992 and 1995. *J Small Anim Pract* 1998; 39: 78–85. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
3. Lulich JP, O'Brien TD, Osborne CA, et al. Feline renal failure: questions, answers, questions. *Compend Educ Pract Vet* 1992; 14: 127–152. [[Google Scholar](#)]
4. Lund EM, Armstrong PJ, Kirk CA, et al. Health status and population characteristics of dogs and cats examined at private veterinary practices in the United States. *J Am Vet Med Assoc* 1999; 214: 1336–1341. [[PubMed](#)] [[Google Scholar](#)]
5. Polzin DJ. Evidence-based step-wise approach to managing chronic kidney disease in dogs and cats. *J Vet Em and Crit Care* 2013; 23: 205–215. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
6. Novartis Animal Health Inc. IRIS staging of CKD. http://www.iris-kidney.com/guidelines/en/staging_ckd.shtml (2007, accessed July 7, 2013).

7. Lascelles BD, Robertson SA. DJD-associated pain in cats: what can we do to promote patient comfort? *J Feline Med Surg* 2010; 12: 200–212. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
8. Lascelles BD. Feline degenerative joint disease. *Vet Surg* 2010; 39: 2–13. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
9. Lascelles BD, Henry JB, III, Brown J, et al. Cross-sectional study of the prevalence of radiographic degenerative joint disease in domesticated cats. *Vet Surg* 2010; 39: 535–544. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
10. Hardie EM, Roe SC, Martin FR. Radiographic evidence of degenerative joint disease in geriatric cats: 100 cases (1994–1997). *J Am Vet Med Assoc* 2002; 220: 628–632. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
11. Lascelles BD, Hansen BD, Roe S, et al. Evaluation of client-specific outcome measures and activity monitoring to measure pain relief in cats with osteoarthritis. *J Vet Intern Med* 2007; 21: 410–416. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
12. Guillot M, Moreau M, Heit M, et al. Characterization of osteoarthritis in cats and Meloxicam efficacy using objective chronic pain evaluation tools. *Vet J* 2013; 196: 360–367. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
13. FDA Center for Veterinary Medicine. FDA announces addition of boxed warning to METACAM® (Meloxicam) labels.
<http://www.fda.gov/AnimalVeterinary/NewsEvents/CVMUpdates/ucm231254.htm> (2010, accessed March 20, 2013).
14. Brown SA. What we know about NSAIDs and cat kidneys. In: Small animal and exotics proceedings of the North American veterinary conference; 2008 Jan 19–23. Gainesville: NAVC, 2008, pp 733–735. [[Google Scholar](#)]
15. DiBartola SP, Rutgers HC, Zack PM, et al. Clinicopathologic findings associated with chronic renal disease in cats: 74 cases (1973–1984). *J Am Vet Med Assoc* 1987; 190: 1196–1202. [[PubMed](#)] [[Google Scholar](#)]
16. Lascelles BD, Dong YH, Marcellin-Little DJ, et al. Relationship of orthopedic examination, goniometric measurements, and radiographic signs of degenerative joint disease in cats. *BMC Vet Res* 2012; 8: 10. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

17. German A, Martin L. Feline obesity. In: Pibot P, Biourge V, Elliot DA. (eds). Encyclopedia of feline clinical nutrition. Aimargues: Royal Canin, 2008, p 16. [[Google Scholar](#)]
18. Lascelles BD, DePuy V, Thomson A, et al. Evaluation of a therapeutic diet for feline degenerative joint disease. J Vet Intern Med 2010; 24: 487–495. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
19. Benito J, Gruen ME, Thomson A, et al. Owner-assessed indices of quality of life in cats and the relationship to the presence of degenerative joint disease. J Feline Med Surg 2012; 14: 863–870. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
20. Benito J, Hansen B, DePuy V, et al. Feline musculoskeletal pain index (FMPI): responsiveness and criterion validity testing. J Vet Intern Med 2013; 27: 474–482. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
21. Laflamme D. Development and validation of a body condition score system for cats: a clinical tool. Feline Pract 1997; 25: 13–18. [[Google Scholar](#)]
22. Thrall DE, Robertson ID. The abdomen. In: Thrall DE, Robertson IE. (eds). Atlas of normal radiographic anatomy and anatomic variants in the dog and cat. St Louis, MO: Saunders/Elsevier, 2011, pp 181–183. [[Google Scholar](#)]
23. Feeney DA, Johntson GR. The kidneys and ureters. In: Thrall DE. (ed). Textbook of veterinary diagnostic radiology. 5th ed. St Louis, MO: Saunders/Elsevier, 2007, pp 693–707. [[Google Scholar](#)]
24. Lee R, Leowijuk C. Normal parameters in abdominal radiology of the dog and cat. J Small Anim Pract 1982; 23: 251–269. [[Google Scholar](#)]
25. Burk RL, Feeney DA. The abdomen. In: Burk RL, Feeney DA. (eds). Small animal radiology and ultrasonography: a text and atlas. St Louis, MO: Saunders/Elsevier, 2003, pp 249–476. [[Google Scholar](#)]
26. Dibartola SP, Rutgers HC. Diseases of the kidney. In: Sherding RG. (ed). The cat diseases and clinical management. 2nd ed. New York: Churchill Livingstone, 1989, pp 1353–1395. [[Google Scholar](#)]
27. Shiroma JT, Gabriel JK, Carter RL, et al. Effect of reproductive status on feline renal size. Vet Radiol Ultrasound 1999; 40: 242–245. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

28. Vogt AH, Rodan I, Brown M, et al. Feline life stage guidelines. *J Feline Med Surg* 2010; 12: 43–54. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
29. American Animal Hospital Association. AAFP/AAHA feline life stage guidelines. <https://www.aahanet.org/Library/FelineLife.aspx> (2010, accessed July 7, 2013). [[DOI](#)] [[PubMed](#)]
30. Chakrabarti S, Syme HM, Elliott J. Clinicopathological variables predicting progression of azotemia in cats with chronic kidney disease. *J Vet Intern Med* 2012; 26: 275–281. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
31. White JD, Norris JM, Baral RM, et al. Naturally-occurring chronic renal disease in Australian cats: a prospective study of 184 cases. *Aust Vet J* 2006; 84: 188–194. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
32. Buns K. AVMA report details pet ownership, veterinary care. *J Am Vet Med Assoc* 2013; 242: 280. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
33. Clarke SP, Mellor D, Clements DN, et al. Prevalence of radiographic signs of degenerative joint disease in a hospital population of cats. *Vet Rec* 2005; 157: 793–799. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
34. Garreres A, Laroute V, Farge F, et al. Disposition of plasma creatinine in non-azotaemic and moderately azotaemic cats. *J Feline Med Surg* 2007; 9: 89–96. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
35. Bennett D, Zainal Ariffin SM, Johnston P. Osteoarthritis in the cat: 2. how should it be managed and treated? *J Feline Med Surg* 2012; 14: 76–84. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
36. Gowan RA, Lingard AE, Johnston L, et al. Retrospective case-control study of the effects of long-term dosing with Meloxicam on renal function in aged cats with degenerative joint disease. *J Feline Med Surg* 2011; 13: 752–761. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
37. Gowan RA, Baral RM, Lingard AE, et al. A retrospective analysis of the effects of Meloxicam on the longevity of aged cats with and without overt chronic kidney disease. *J Feline Med Surg* 2012; 14: 876–881. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
38. Allan GS. Radiographic features of feline joint diseases. *Vet Clin North Am Small Anim Pract* 2000; 30: 281–302. [[PubMed](#)] [[Google Scholar](#)]

39. Gao X, Lee J, Malladi S, et al. Feline degenerative joint disease: a genomic and proteomic approach. J Feline Med Surg 2013; 15: 466–477. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
40. Chakrabarti S, Syme HM, Brown CA, et al. Histomorphometry of feline chronic kidney disease and correlation with markers of renal dysfunction. Vet Pathol 2012; 50: 147–155. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

Click here for Supplementary Appendix

[sj-pdf-1-jfm-10.1177_1098612X13511446.pdf](#)^(48.2KB, pdf)

Algorithm to determine cats with chronic kidney disease (CKDpos), small kidney and CKDneg groups

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Epidemiology of Chronic Kidney Disease (CKD) in Cats: An Analysis of the Factors Involved

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Abstract

Chronic Kidney Disease (CKD) is among the most frequent and clinically relevant metabolic disorders in cats, affecting a substantial proportion of the adult and geriatric feline population. The prevalence of CKD increases with age, being estimated at 20–50% in cats over 10 years of age. The etiology of this disorder involves genetic predisposition, dietary factors, as well as exposure to toxins or concomitant diseases that may accelerate the progression of renal lesions. The present study synthesizes data from the scientific literature together with clinical-epidemiological analyses performed on a cohort of 120 cats over a four-year period. The results highlighted an overall prevalence of CKD in the studied cohort, with a significantly higher frequency in cats over nine years of age, particularly affecting purebred animals. A slightly higher prevalence was observed in males compared to females, while reproductive status indicated an increased risk of CKD in neutered patients. Living environment and nutritional status were also influential, with a higher disease prevalence in indoor-only cats and in those fed nutritionally deficient diets. The

multifactorial nature of CKD pathogenesis, combined with the nonspecific clinical manifestations in its early stages, explains the frequent late diagnosis of most cases. Consequently, an in-depth understanding of the epidemiology and associated risk factors constitutes an essential prerequisite for the development of preventive strategies, early screening protocols, and personalized therapeutic management aimed at optimizing quality of life and longevity in feline patients.

Keywords: chronic renal failure, IRIS staging, cats, prevalence, SDMA, creatinine

1. Introduction

Chronic kidney disease (CKD) in cats is a progressive, irreversible condition that impairs renal function and leads to systemic complications. The epidemiology of feline CKD varies depending on the population studied and the diagnostic criteria applied. Among internal disorders encountered in veterinary practice, feline CKD represents one of the most important conditions, exerting a major impact on both longevity and quality of life in companion animals. The disease is characterized by the progressive and irreversible loss of functional nephrons, resulting in a decline in glomerular filtration rate and the accumulation of nitrogenous waste products in the body [1].

The prevalence of feline CKD varies considerably according to population and diagnostic criteria. Epidemiological studies conducted in the United States and Europe have reported an overall prevalence of 1–3% in the general cat population, but this may exceed 30% in geriatric patients [2,3]. Certain breeds, such as Persians and Siamese, also appear to be more susceptible to the disease, suggesting a genetic component in its pathogenesis [4].

Clinical signs associated with CKD often develop late, when renal function is already significantly compromised. The most common manifestations include polyuria, polydipsia, weight loss, anorexia, and lethargy [5,6]. This insidious onset makes early diagnosis difficult, yet crucial for establishing an appropriate therapeutic plan. Consequently, the identification of renal insufficiency at an early stage requires not only careful clinical evaluation, but also the integration of laboratory tests (renal profile, complete blood count, serum biochemistry) and imaging studies such as abdominal ultrasonography.

Recent advances in biomarkers have highlighted the role of symmetric dimethylarginine (SDMA), a sensitive serum marker that reflects a decline in glomerular filtration rate earlier than creatinine [7,8]. Thus, the combined use of SDMA, creatinine, and urea currently represents an essential tool for evaluating renal function in cats, complementing clinical and imaging assessments. Early

diagnosis, through the monitoring of SDMA and creatinine, allows timely therapeutic interventions that may improve prognosis [9].

In addition, the recommendations developed by the International Renal Interest Society (IRIS) [10] provide a standardized framework for staging CKD in cats and dogs, enabling both the monitoring of disease progression and the implementation of personalized therapeutic strategies.

The multifactorial etiology of chronic kidney disease, the long subclinical course, and the variable rate of progression, make CKD a major challenge in feline medicine. To date, several epidemiological and cohort studies have investigated CKD prevalence and risk factors in cats from various geographic regions (e.g., the United Kingdom, North America, and Japan). However, regional data from Central and Eastern Europe remain scarce.

This study provides the first integrated epidemiological and biochemical overview of feline CKD in a Central–Eastern European population, extending current evidence beyond Western cohorts and highlighting region-specific patterns of disease occurrence and management. In this context, the study aimed to describe the prevalence of CKD in a clinical population of 120 cats examined over four years; identify demographic and environmental factors associated with CKD (age, breed, sex, reproductive status, environment, and diet); and correlate these findings with serum biomarkers such as creatinine and SDMA.

The present study integrates both retrospective and prospective data on feline chronic kidney disease (CKD), aiming to provide a comprehensive overview of its epidemiological distribution, clinical presentation, and biochemical patterns in a defined regional population. By aligning local observations with internationally reported trends, the study contributes to the consolidation of current evidence on CKD while emphasizing the clinical and diagnostic implications of disease progression in feline patients. Rather than introducing new biomarkers or experimental diagnostic criteria, this research strengthens the translational relevance of existing knowledge within a real-world clinical context.

In this context, we designed a four-year cohort study to assess the prevalence, risk factors, and biochemical profiles associated with feline CKD in a regional clinical population.

2. Materials and Methods

The study comprised two complementary components: a retrospective literature analysis and a clinical investigation of feline chronic kidney disease (CKD). A structured search of PubMed, Scopus, and Web of Science was performed using the keywords “feline chronic kidney disease,” “cats with CKD,” “epidemiology,” “SDMA,” “IRIS staging,” and “risk factors in CKD”. The search covered publications from 1992 to 2024. Studies were eligible for inclusion if they reported prevalence estimates or risk factors related to feline CKD, while case reports, experimental studies without epidemiological data, and non-peer-reviewed sources were excluded. The review aimed to summarize the most relevant epidemiological findings and to provide a comparative framework for interpreting the clinical results of the present cohort.

The clinical component included 120 cats diagnosed with CKD and examined at the University Veterinary Hospital (Iași, Romania) between 2021 and 2024. Diagnosis was established based on serum biochemical analyses (creatinine, urea, SDMA), urinalysis, and abdominal ultrasonography, in accordance with the IRIS guidelines (2021) [[10](#)].

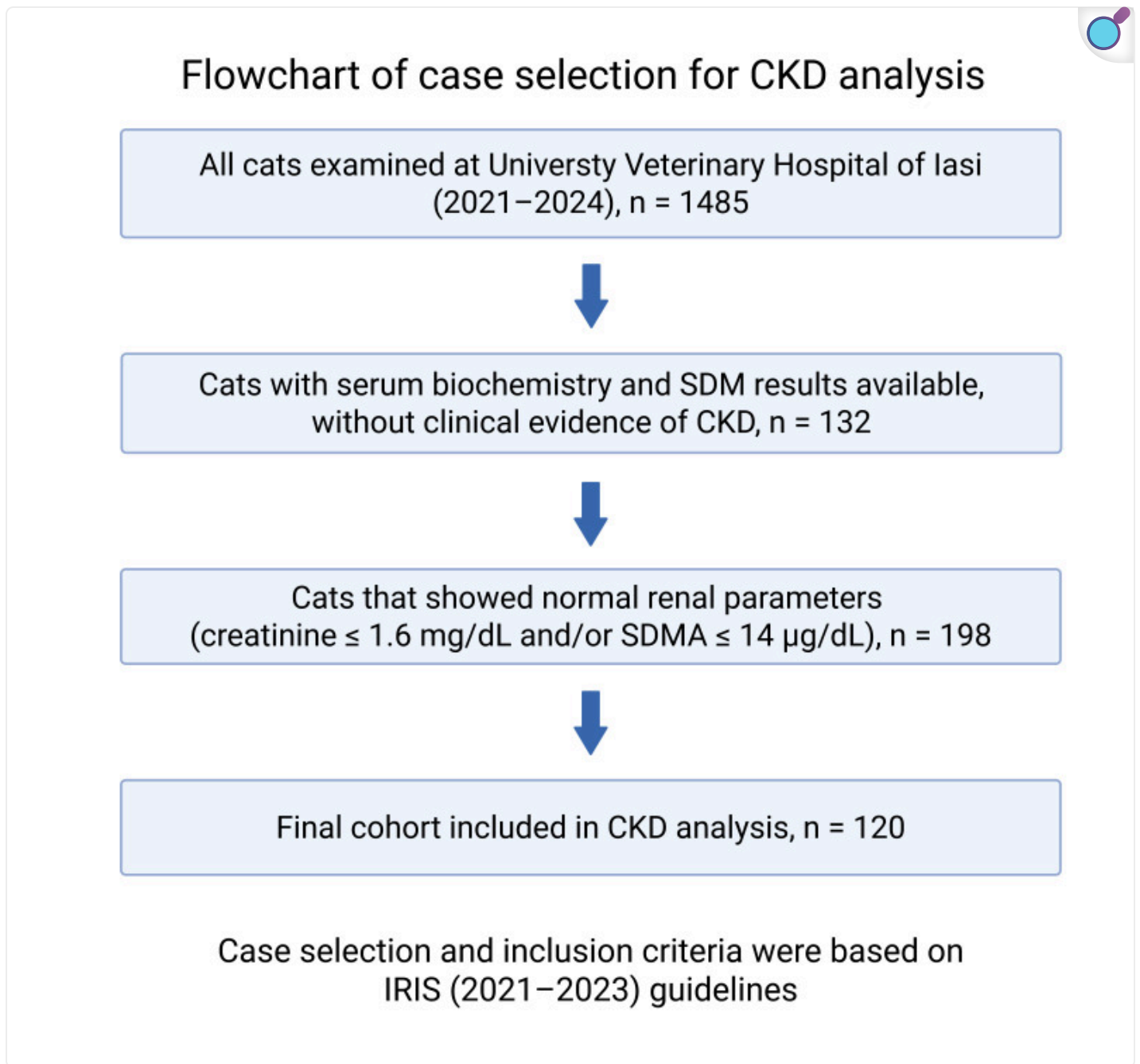
2.1. Study Design

The study was designed as a combined retrospective–prospective observational analysis of feline chronic kidney disease (CKD) cases presented to the University Veterinary Hospital for Companion Animals, Iasi University of Life Sciences (IULS), Romania. The retrospective phase covered cases recorded between January 2021 and December 2022, while the prospective phase extended from January 2023 to December 2024. The cohort included both routine and referred clinical cases and therefore represents a clinical-based, rather than a population-based, sample.

2.2. Source Population and Case Selection

Between January 2021 and December 2024, a total of 1485 cats were examined at the University Veterinary Hospital from Iasi. Among these, 120 cats (8.1%) fulfilled the inclusion criteria for CKD. Case identification followed a two-step process: (1) initial detection of azotemia or abnormal SDMA values in laboratory records, and (2) clinical confirmation according to IRIS guidelines. A detailed flowchart of the selection process is provided in [Figure 1](#). Inclusion and exclusion criteria were based on IRIS (2021–2023) guidelines [[10](#)]. The final cohort was stratified by stage (IRIS I–IV).

Figure 1.



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Flowchart illustrating the selection process of feline CKD cases from the total hospital population (2021–2024).

2.3. Inclusion and Exclusion Criteria

Cats were included if they presented persistent azotemia and/or elevated SDMA on two or more consecutive occasions, separated by at least two weeks, and if clinical evaluation excluded pre-renal or post-renal causes. Diagnosis and staging followed the IRIS (2021) guidelines, with CKD defined as serum creatinine >1.6 mg/dL and/or SDMA >14 μ g/dL, persisting for a minimum of two weeks. This composite criterion allowed the inclusion of cats across all IRIS stages while improving sensitivity for detecting early CKD cases.

A sensitivity analysis using the standard IRIS creatinine-only threshold was performed to confirm diagnostic consistency ([Table 1](#)). Cats with acute kidney injury, severe dehydration, or systemic conditions associated with reversible azotemia were excluded. Systemic diseases such as hyperthyroidism, diabetes mellitus, and hepatic disorders were excluded based on serum total T4 measurement, fasting glucose, and liver enzyme profiling (ALT, ALP). Thyroid testing (T4) was performed in 92% ($n = 110$) of cats, while glucose and hepatic profiles were available for all cases. Cats with serum T4 concentrations exceeding 50 nmol/L or fasting glucose levels above 8.0 mmol/L were excluded, as these findings indicated concurrent endocrine disorders that could influence renal function markers.

Table 1.

Inclusion and exclusion criteria and diagnostic thresholds for feline chronic kidney disease (CKD) according to IRIS guidelines (2021–2023).

Criterion	Parameter	Threshold/Range	Basis	Applied Rule
Inclusion	Serum creatinine	>1.6 mg/dL (measured twice, ≥2 weeks apart)	IRIS 2021 Guidelines	Persistent azotemia without pre-/post-renal cause
Inclusion	SDMA	>14 µg/dL (measured twice, ≥2 weeks apart)	IRIS 2021	Used as early marker; CKD confirmed if SDMA or creatinine elevated
Inclusion	Clinical course	Chronic (>3 months)	Literature/Clinical records	Required for case confirmation
Inclusion	Age	>1 year	IRIS 2023	Juvenile cases excluded
Exclusion	Acute kidney injury	Rapid onset (<2 weeks), reversible after rehydration	IRIS 2023	Excluded
Exclusion	Post-renal obstruction	Urolithiasis, urinary obstruction	Clinical/imaging data	Excluded
Exclusion	Dehydration/infection	Azotemia resolved after therapy	Clinical evaluation	Excluded
Exclusion	Systemic illness	Hyperthyroidism, diabetes, hepatic disease	Laboratory data	Excluded

Criterion	Parameter	Threshold/Range	Basis	Applied Rule
Sensitivity analysis	Creatinine-only threshold	>1.6 mg/dL (IRIS)	IRIS 2021	Alternative CKD definition

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[Table 1](#) summarizes the diagnostic thresholds and selection criteria applied for inclusion and exclusion of feline cases in the CKD cohort, as well as the results of the sensitivity analysis based on the IRIS creatinine-only definition.

2.4. Laboratory Analyses

Blood and urine samples were analyzed in the hospital's Clinical Pathology Laboratory. Biochemical parameters (creatinine, urea, SDMA) were measured using an IDEXX Catalyst Dx Analyzer (IDEXX Laboratories, Westbrook, ME, USA), calibrated daily according to the manufacturer's protocol. Internal quality control was performed before each analytical batch. Urine specific gravity (USG) was measured using a manual refractometer (Atago PAL-09S, Tokyo, Japan), and the urine protein-to-creatinine ratio (UPC) was determined spectrophotometrically (IDEXX method, reference range < 0.4).

Median UPC values and their distribution across CKD stages were calculated. Where available, repeated urine samples were analyzed (n = 28) to verify persistent proteinuria, consistent with IRIS 2023 recommendations.

2.5. Variables and Statistical Analysis

Demographic (age, breed, sex, neuter status, environment, diet) and clinical variables were recorded. Categorical variables were analyzed using chi-square or Fisher's exact tests; continuous data were summarized as median [IQR] and compared using Mann-Whitney or Kruskal-Wallis tests, depending on distribution. Effect sizes were reported with 95% confidence intervals. Logistic regression models were used to assess adjusted associations between CKD and selected risk factors (breed, sex, indoor/outdoor status). All statistical analyses were performed using SPSS v.27.0 (IBM Corp., Armonk, NY, USA). A significance threshold of $p < 0.05$ was adopted.

2.6. Ethical Approval

This study involved the retrospective and prospective analysis of clinical data from privately owned cats. No experimental procedures were performed, and no interventions beyond routine clinical care were conducted. Therefore, formal ethical committee approval was not required under national and institutional regulations. Written informed consent was obtained from all owners prior to inclusion, authorizing the use of anonymized data for scientific purposes. In accordance with Romanian national and institutional regulations (IULS), separate ethical committee approval was not required.

The study protocol followed the principles of good veterinary practice and patient confidentiality, in accordance with the recommendations of the European College of Veterinary Internal Medicine [\[11\]](#).

3. Results

The prevalence of CKD has increased considerably in recent decades, particularly among geriatric cats. This trend has been attributed both to the increase in life expectancy and to the improvement of diagnostic tools available in veterinary practice. Among the variables studied were age, sex, diet, and lifestyle, each of which may directly or indirectly influence the development of renal dysfunction. Distribution by breed and age group revealed significant differences, which are comparable to findings previously reported in the literature by Marino et al. (2014) [\[3\]](#).

3.1. Clinical Examination Findings

Clinical examination revealed variable changes depending on disease stage and degree of renal decompensation. In stage II (early phase), only mild alterations in general condition were observed, while stage III and IV cases showed marked systemic signs such as apathy, weight loss, dehydration, and anorexia. Less frequent manifestations included oral ulcerations, pale mucous membranes, unkempt coat, and irregular kidneys on palpation, hypertension, uremic halitosis, and hypersalivation ([Table 2](#)).

Table 2.

Clinical changes observed at various stages of CKD in cats.

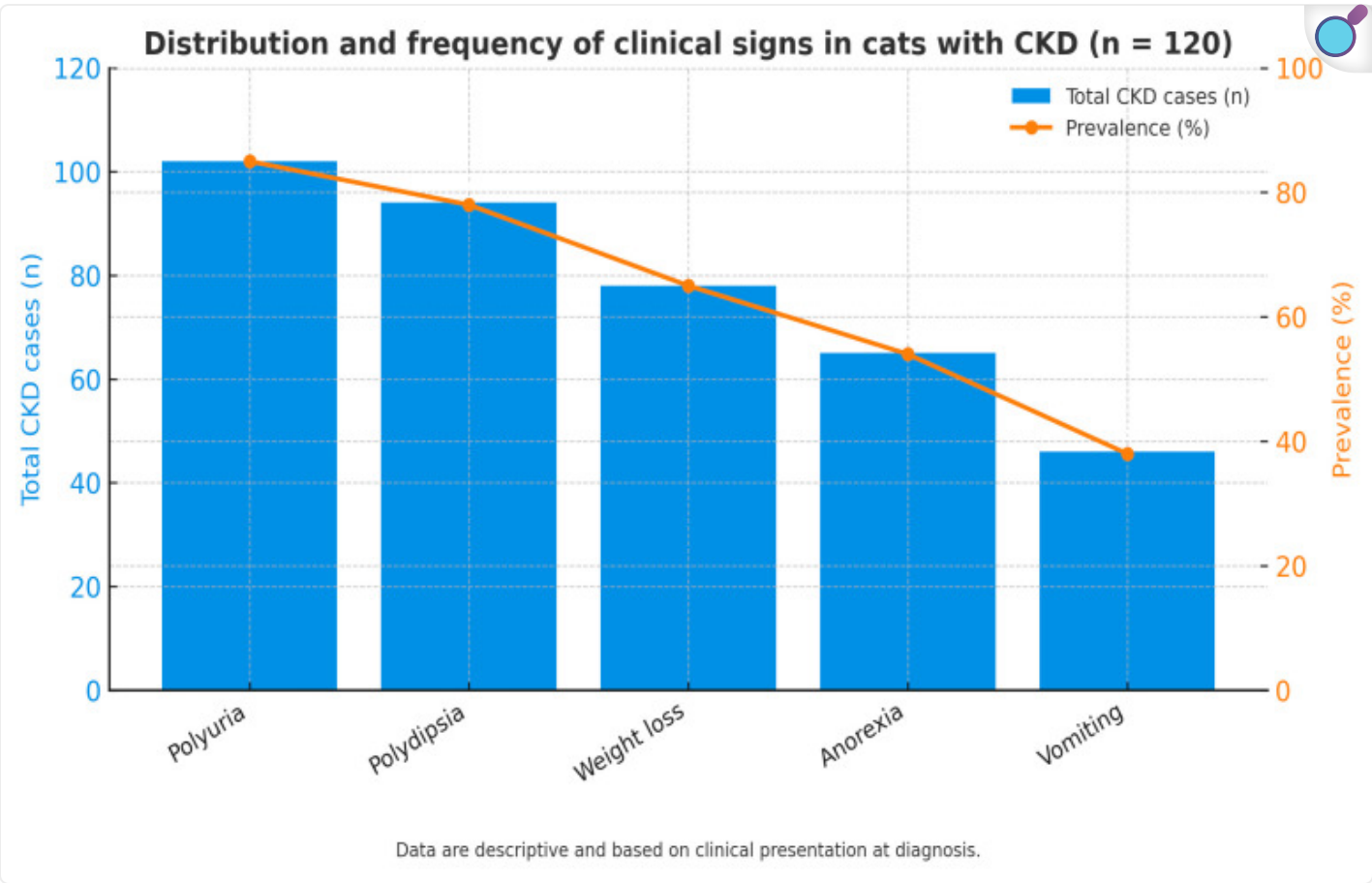
CKD Stage	Main Clinical Signs
Initial (compensated)	Mild polydipsia and polyuria (diluted urine) Mild weight loss Coat less shiny Kidneys normal or enlarged at palpation Variable appetite
Intermediate (moderate decompensation)	Polydipsia and polyuria Progressive weight loss Dehydration Anorexia Uremic halitosis Oral ulcers, hypersalivation Nausea, vomiting Small, irregular kidneys at palpation Anemia (pale mucous membranes) Early signs of hypertension (cardiac murmur, ocular changes)
Advanced/Terminal (severe uremia)	Anorexia Frequent vomiting Marked cachexia Severe lethargy/apathy Dull, unkempt coat Intense halitosis, multiple oral ulcers Oliguria or anuria

CKD Stage	Main Clinical Signs
	Severe hypertension → retinal hemorrhages, detachment, blindness
	Neurological signs: seizures, tremors, ataxia
	Kidneys small, fibrotic, difficult to palpate

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[Figure 2](#) illustrates the distribution of the main clinical signs observed among cats diagnosed with chronic kidney disease (CKD). Blue bars represent the number of affected cats (left axis), while the orange line indicates the percentage within the total CKD cohort (right axis). Polyuria (85%, n = 102) and polydipsia (78%, n = 94) were the most frequent findings, followed by weight loss (65%, n = 78), anorexia (54%, n = 65), and vomiting (38%, n = 46). These signs generally appeared at the time of biochemical decompensation, confirming that diagnosis typically coincided with the onset of clinical manifestations. Data are descriptive and based on clinical presentation at diagnosis.

Figure 2.



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Distribution and frequency of clinical signs in cats with CKD (n = 120).

3.2. Distribution of CKD by Age Group

Age-related distribution is presented in [Table 3](#). The majority of CKD cases occurred in cats aged 10–14 years, corresponding to the highest-risk period for disease manifestation. The incidence increased progressively with age, confirming the degenerative nature of CKD.

Table 3.

Distribution of CKD cases by age group and stage.

Age Group (Years)	Cats Examined (n)	CKD Cases (n)				Prevalence (%)
		Stage I (%)	Stage II (%)	Stage III (%)	Stage IV (%)	
<2	2	2 (1.70%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	1.70%
2–4	4	3 (2.5%)	1 (0.83%)	0 (0.00%)	0 (0.00%)	3.33%
5–7	7	1 (0.83%)	5 (4.17%)	1 (0.83%)	0 (0.00%)	5.83%
8–9	14	1 (0.83%)	9 (7.50%)	3 (2.50%)	1 (0.83%)	11.66%
10–11	37	4 (3.33%)	13 (10.83%)	18 (15.00%)	2 (1.70%)	30.86%
12–14	41	6 (5.00%)	16 (13.33%)	14 (11.67%)	5 (4.17%)	34.17%
≥15	15	2 (1.70%)	5 (4.17%)	7 (5.83%)	1 (0.83%)	12.53%
Total	120	n = 19 (15.83%)	n = 49 (40.83%)	n = 43 (35.83%)	n = 9 (7.5%)	100%

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Approximately 6% of cases (n = 7) were recorded in the 5–7 years age group, suggesting that renal impairment may have a subclinical onset during this period. This emphasizes the importance of periodic monitoring of renal function in middle-aged patients, particularly in the presence of risk factors such as inadequate diets, recurrent urinary tract infections, or genetic predisposition. Only 5% of patients (n = 6) were younger than five years, indicating that the disease is relatively rare in

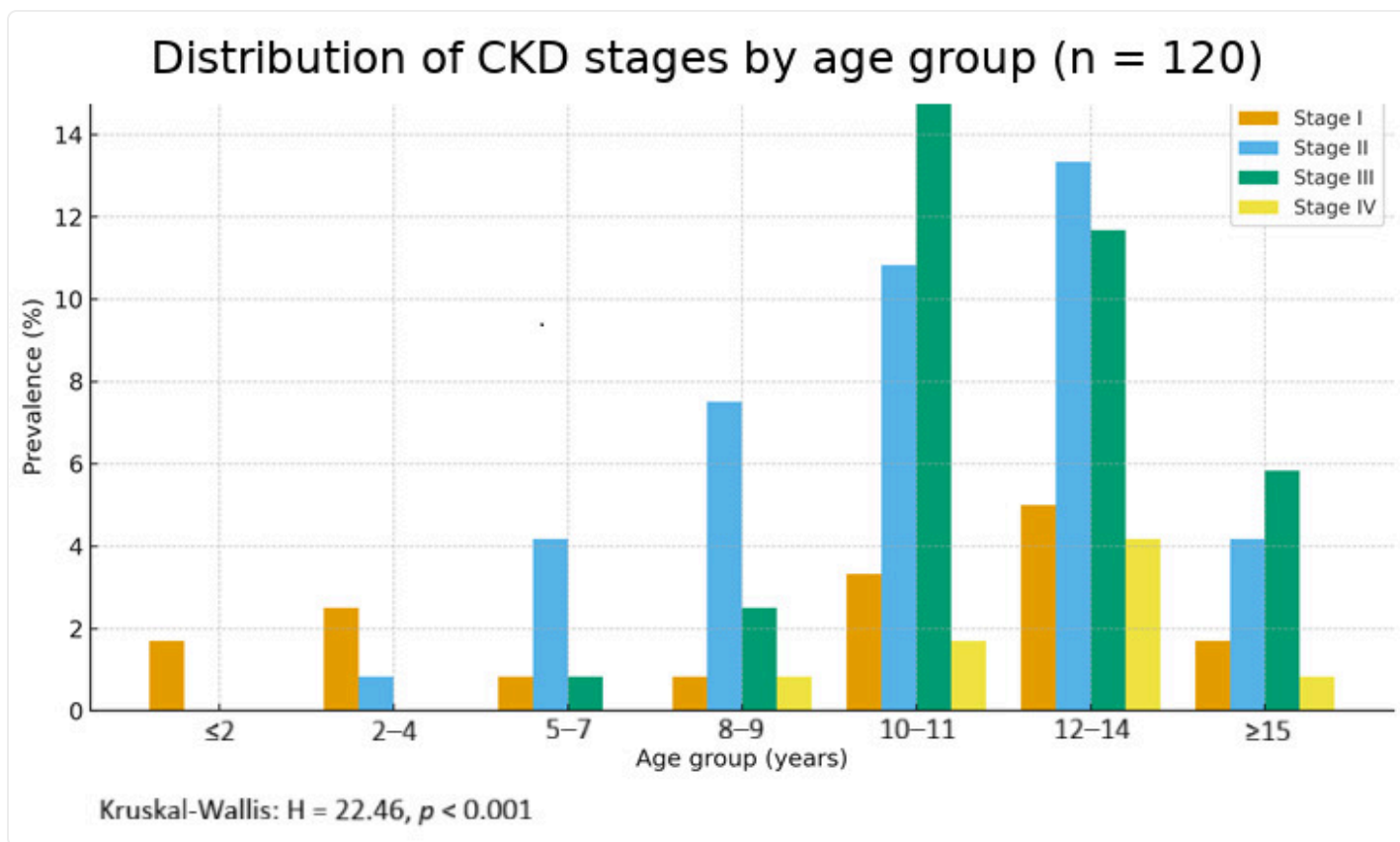
young animals, though not negligible. In such cases, the etiology may differ and include congenital disorders, acute intoxication, or severe infectious processes.

A predominance of stage II (~41%) and stage III (~36%) disease was observed among cats diagnosed with CKD, demonstrating that in these stages, clinical manifestations recognizable by owners—such as polyuria, polydipsia, weight loss, anorexia, and vomiting—were already present, accompanied by clear biochemical alterations. In stage I, which was relatively rarely reported, clinical signs were subtle (reduced appetite and decreased activity), yet biochemical results revealed elevated values of an early diagnostic parameter (SDMA).

In 89.2% of cases (107/120), IRC was diagnosed in cats over seven years of age. Only 5% (n = 6) were under five years of age, suggesting rare but possible early-onset forms due to congenital or toxic causes.

The distribution of disease stages by age group shows a clear increase in prevalence with advancing age ([Figure 3](#)). The clustered bar graph shows the prevalence of IRIS stages I-IV by age group. Percentages are calculated within the full IRC cohort (denominator = 120). A significant trend towards higher stage prevalence with advancing age was observed (Kruskal–Wallis $H = 22.46, p < 0.001$).

Figure 3.



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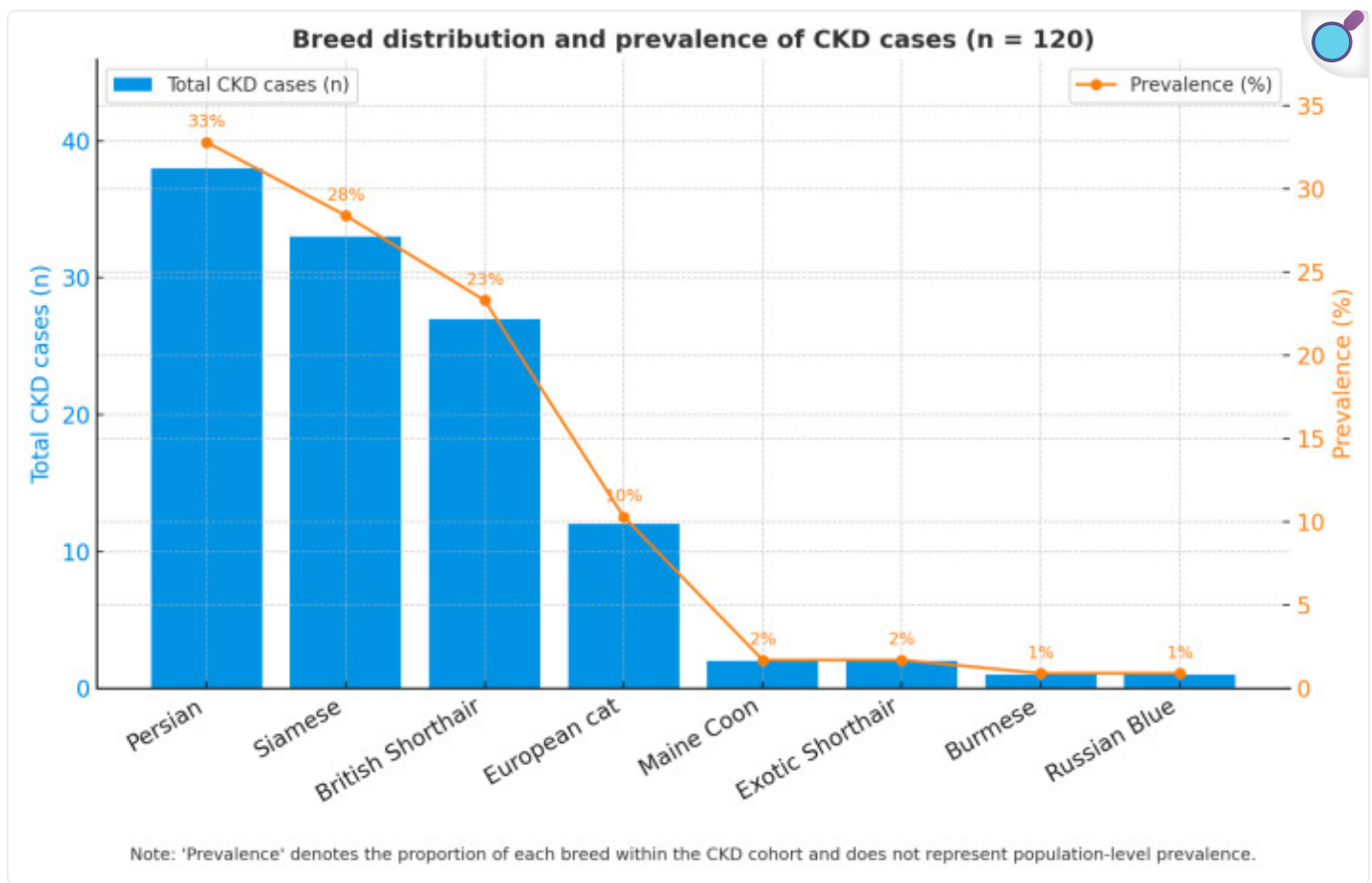
Distribution of CKD stages by age group (n = 120). Note: 89.2% (107/120) of CKD cases were in cats ≥7 years.

These findings are consistent with data from the literature, which support that the prevalence of CKD increases significantly with advancing age and that early detection plays a key role in improving prognosis.

3.3. Breed Distribution and Predisposition

Breed composition among cats with CKD is shown in [Figure 4](#). Within the CKD cohort (n = 120), the most frequently represented breeds were Persian (n = 38), Siamese (n = 33) and British Shorthair (n = 27), followed by European cats (n = 12) and isolated cases of Other breed, Maine Coon, Exotic Shorthair, Burmese and Russian Blue.

Figure 4.



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Distribution of CKD by breed.

These values represent the distribution of CKD cases within the study cohort and likely mirror the case-mix of the clinic's clientele; therefore, they should not be interpreted as breed-specific risk in the absence of hospital-wide denominators (i.e., total cats of each breed seen during the study period).

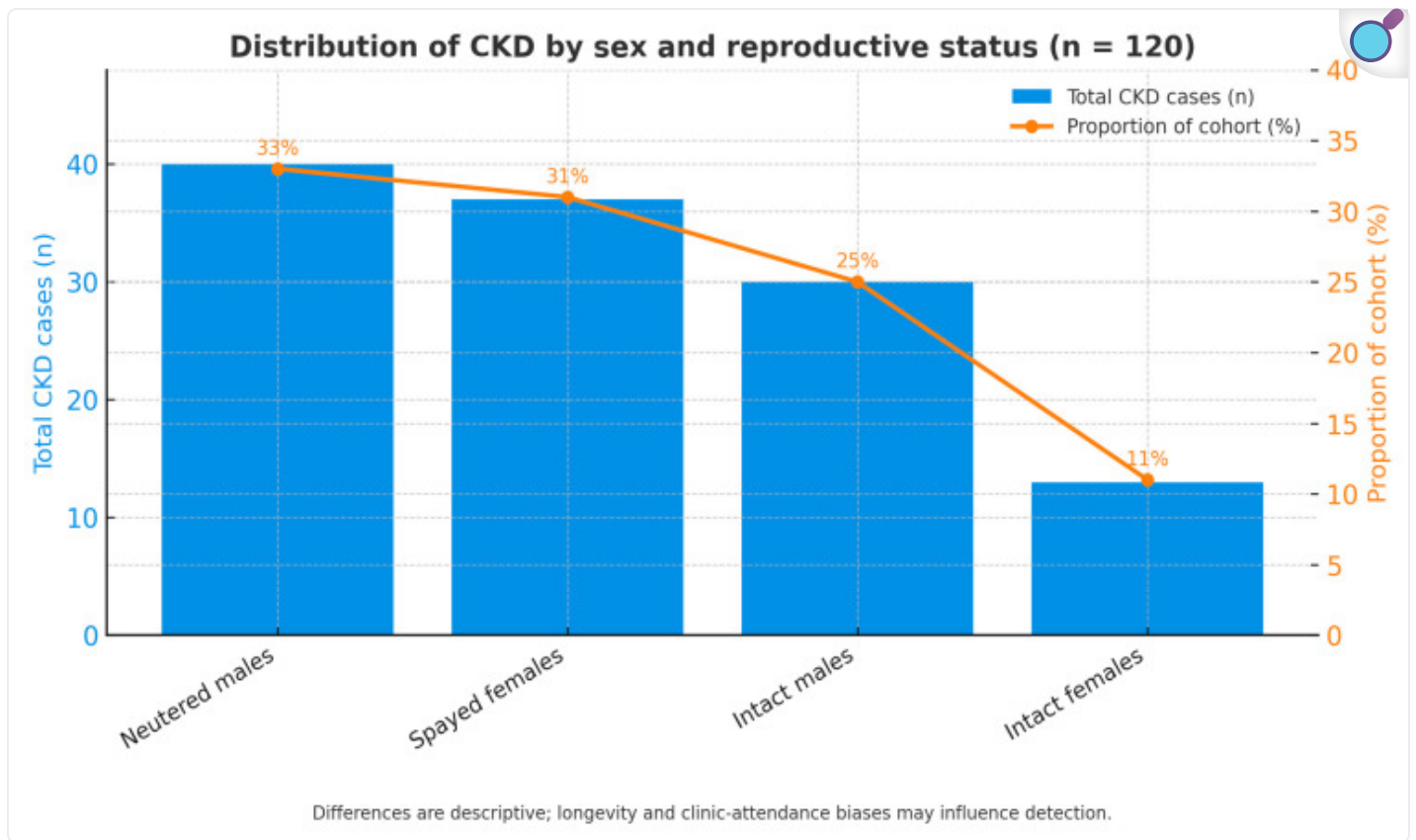
Consistent with previous literature, several pure breeds have been reported to harbor congenital or familial nephropathies (e.g., Persians, Siamese, British Shorthair) [3,4]; however, our data are presented descriptively and no causal inference is made.

Although the observed differences were statistically significant after adjustment, the interpretation remains descriptive, as referral bias and breed representation in the hospital population may influence the apparent prevalence. Nevertheless, the data indicate that purebred status, particularly in Persian and Siamese cats, constitutes an important pre-disposing factor for CKD in clinical populations.

3.4. Distribution of CKD by Sex and Reproductive Status

Information on the distribution of chronic kidney disease (CKD) by sex and reproductive status is illustrated in [Figure 5](#). Among the 120 feline patients diagnosed with CKD, notable variation was observed among the four categories analyzed. The largest proportion of cases was recorded in neutered males (33%, n = 40), followed by spayed females (31%, n = 37) and intact males (25%, n = 30), while intact females showed the lowest frequency (11%, n = 13).

Figure 5.



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Distribution of CKD by sex and reproductive status.

Overall, males represented 58% of the total population ($n = 70$) compared with 42% females ($n = 50$), indicating a slight predominance of male cats regardless of reproductive status. The distribution between neutered and intact individuals within both sexes was relatively even, suggesting that sex hormones are unlikely to act as independent determinants of CKD onset.

These findings are descriptive and should be interpreted cautiously, as they may be influenced by longevity and clinic-attendance biases. Neutered cats generally live longer and undergo more regular clinical monitoring than intact cats, which may increase the probability of CKD detection rather than reflect a true biological predisposition. Thus, the sex-related distribution of CKD was heterogeneous but not significantly skewed toward any single group, showing only a slight trend toward higher prevalence among males. Further studies using adjusted multivariate models are

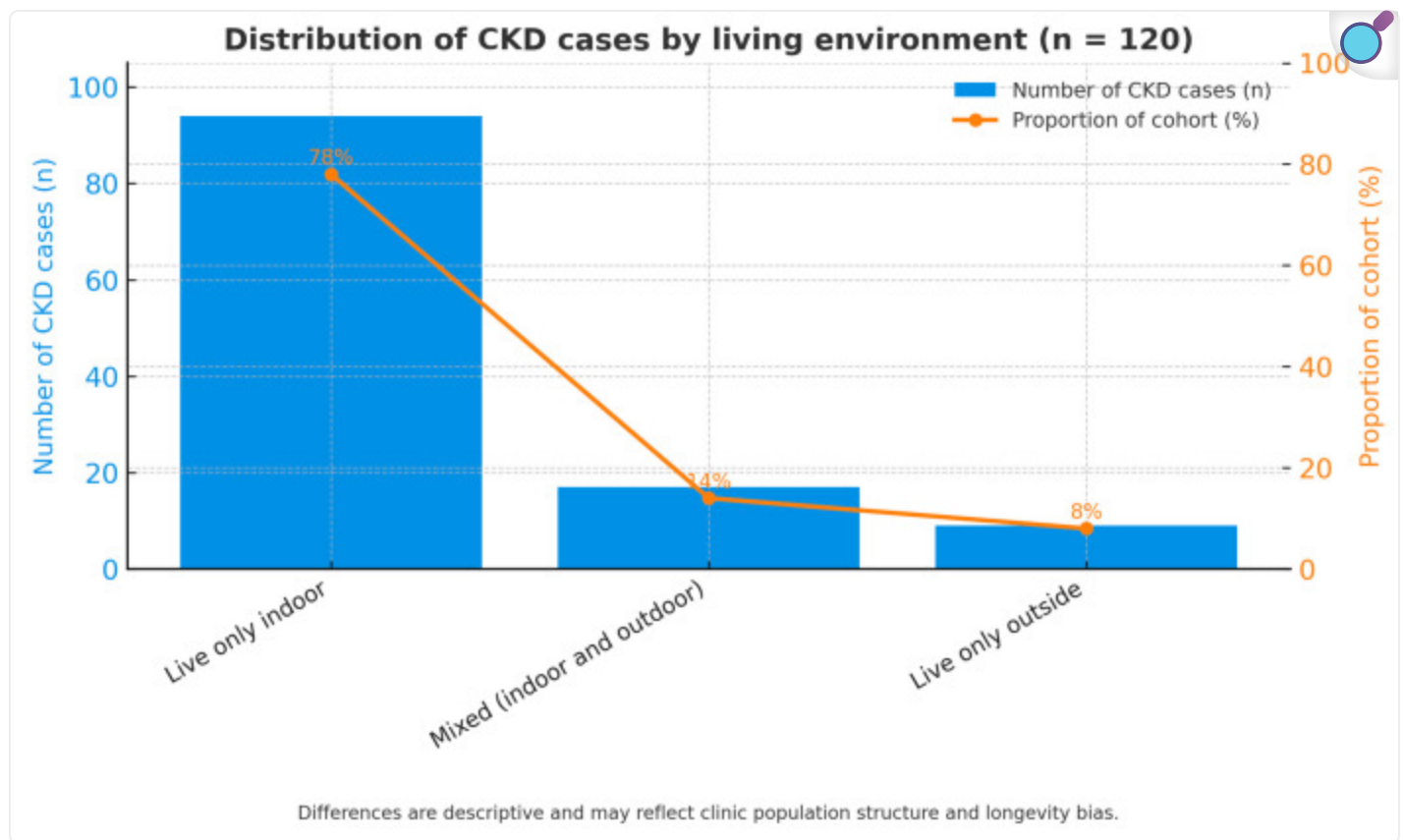
needed to clarify whether sex or hormonal status independently influence CKD risk once age and other confounders are controlled.

Combined bar–line chart showing the total number of cats diagnosed with chronic kidney disease (CKD) across four categories: neutered males, spayed females, intact males, and intact females. Percentages indicate each category’s proportion within the total CKD cohort (n = 120). Differences are descriptive and may reflect longevity and clinic-attendance biases rather than true biological susceptibility.

3.5. Distribution of CKD by Living Environment

The distribution of chronic kidney disease (CKD) according to living environment is presented in [Figure 6](#). The chart illustrates the distribution of the 120 feline patients diagnosed with CKD based on their predominant housing conditions. According to the data, the majority of cases—78% (n = 94)—were cats living exclusively indoors. A smaller proportion, 14% (n = 17), consisted of cats living primarily outdoors, while the remaining 8% (n = 9) belonged to a mixed category alternating between indoor and outdoor environments.

Figure 6.



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Distribution of CKD cases by living environment.

This distribution indicates that most CKD patients were indoor cats, which likely reflects the general composition of the hospital population rather than a true environmental predisposition. Indoor cats are more frequently monitored, have longer life expectancy, and are more often presented for routine health checks, which may lead to earlier CKD detection. Conversely, outdoor cats—although less represented in this sample—may be underrepresented due to higher mortality from acute causes such as trauma or infectious diseases, which limits their inclusion in chronic disease statistics.

The mixed-exposure group (8%) provides an intermediate perspective, but its small sample size prevents firm conclusions regarding the influence of this lifestyle pattern. Therefore, the observed

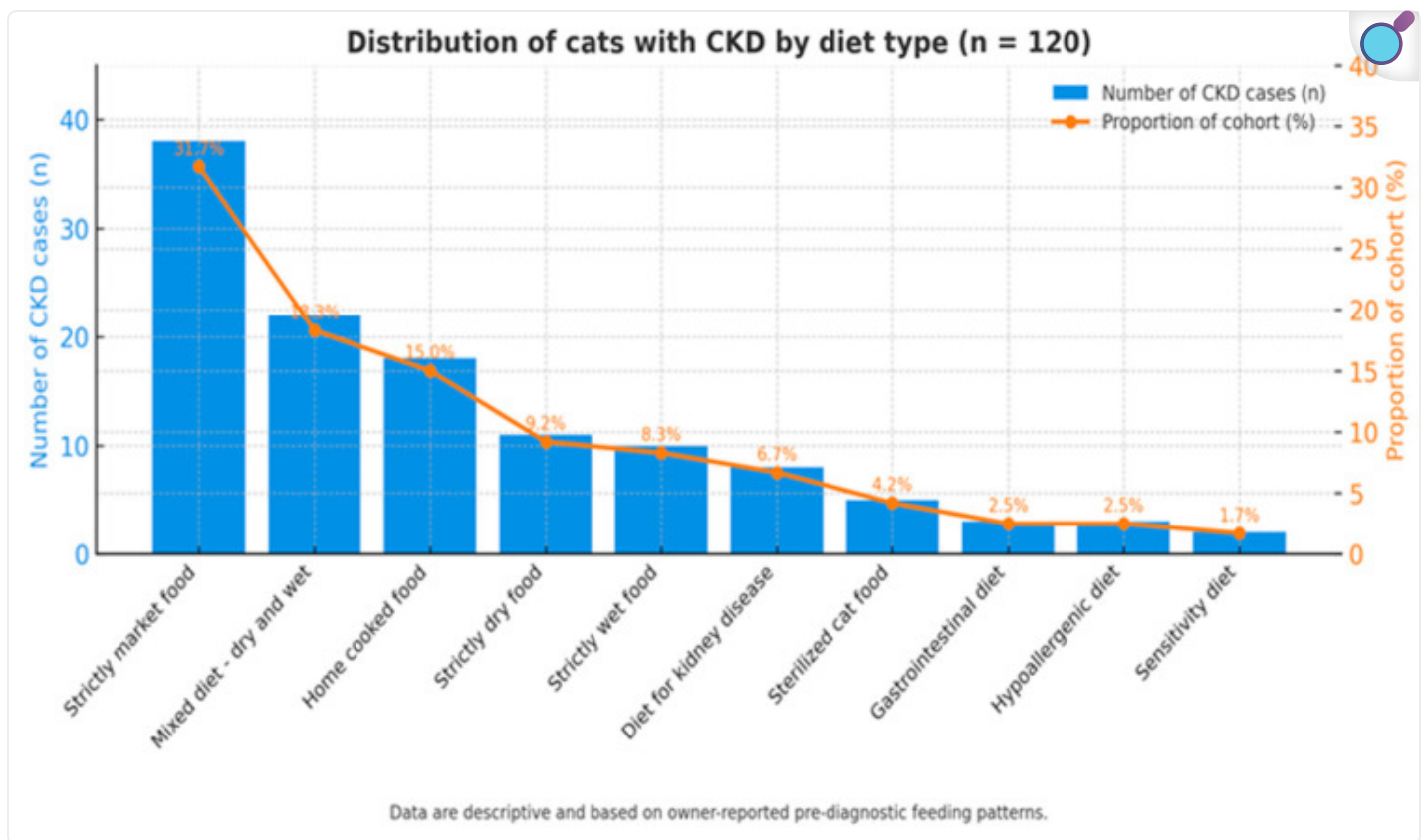
predominance of indoor cats among CKD cases should be interpreted as descriptive and reflective of clinic attendance patterns, not as evidence of environmental risk.

Bar chart illustrating the proportion of cats diagnosed with chronic kidney disease (CKD) by living environment: indoor (78%), outdoor (14%), and mixed (8%). The predominance of indoor cats among CKD cases likely reflects clinic population structure and differences in longevity and medical monitoring rather than true environmental risk.

3.6. Distribution by Diet

The analysis of dietary regimens ([Figure 7](#)) among the 120 feline patients diagnosed with chronic kidney disease (CKD) revealed considerable diversity in feeding practices prior to diagnosis. The most common diets were standard commercial foods from super-markets, as many cats readily accepted this type of feed while refusing other complete formulations.

Figure 7.



[Open in a new tab](#)

Distribution of dietary regimens among cats diagnosed with CKD.

Bar chart showing the number and percentage of cats according to predominant diet type prior to CKD diagnosis. The most frequent categories were supermarket commercial diets (32%) and nutritionally unbalanced home-prepared regimens (18%). Data are descriptive and based on owner-reported feeding habits before diagnosis, not on controlled dietary intervention.

Dietary information was collected retrospectively from owner reports during anamnesis and referred to the period prior to CKD diagnosis. The reported feeding regimens were classified into five categories: (1) commercial supermarket diets, (2) nutritionally unbalanced home-prepared diets, (3) mixed wet and dry feeding, (4) exclusively wet or exclusively dry commercial diets, and (5) therapeutic or veterinary-prescribed diets (renal, hypoallergenic, or gastrointestinal formulations).

The distribution of feeding categories among the 120 CKD cases is summarized in [Table 4](#).

Table 4.

Classification of feeding regimens and distribution of cats diagnosed with CKD prior to dietary intervention.

Diet Type	n	%	Notes
Commercial supermarket diets	38	32%	Most common; non-specialized dry/wet products
Nutritionally unbalanced home-prepared diets	22	18%	Owner-prepared; variable composition
Mixed feeding (wet + dry)	26	22%	Combined commercial foods, no supplementation
Exclusive commercial (wet or dry only)	20	17%	Cats refusing alternative types
Therapeutic/prescribed diets	14	11%	Provided after previous illness
Total	120	100%	—

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The majority of cats (32%) were fed standard supermarket commercial diets, while 18% were maintained on nutritionally unbalanced home-prepared diets. These regimens may result in suboptimal nutrient intake (low omega-3, antioxidants, B-complex vitamins) and higher phosphorus exposure, but no adjusted analysis was performed to confirm such associations.

Therefore, these findings should be interpreted as descriptive, reflecting owner-reported pre-diagnostic feeding habits rather than causal relationships. The high prevalence of standard commercial food categories likely reflects product accessibility and feeding habits within the clinic population.

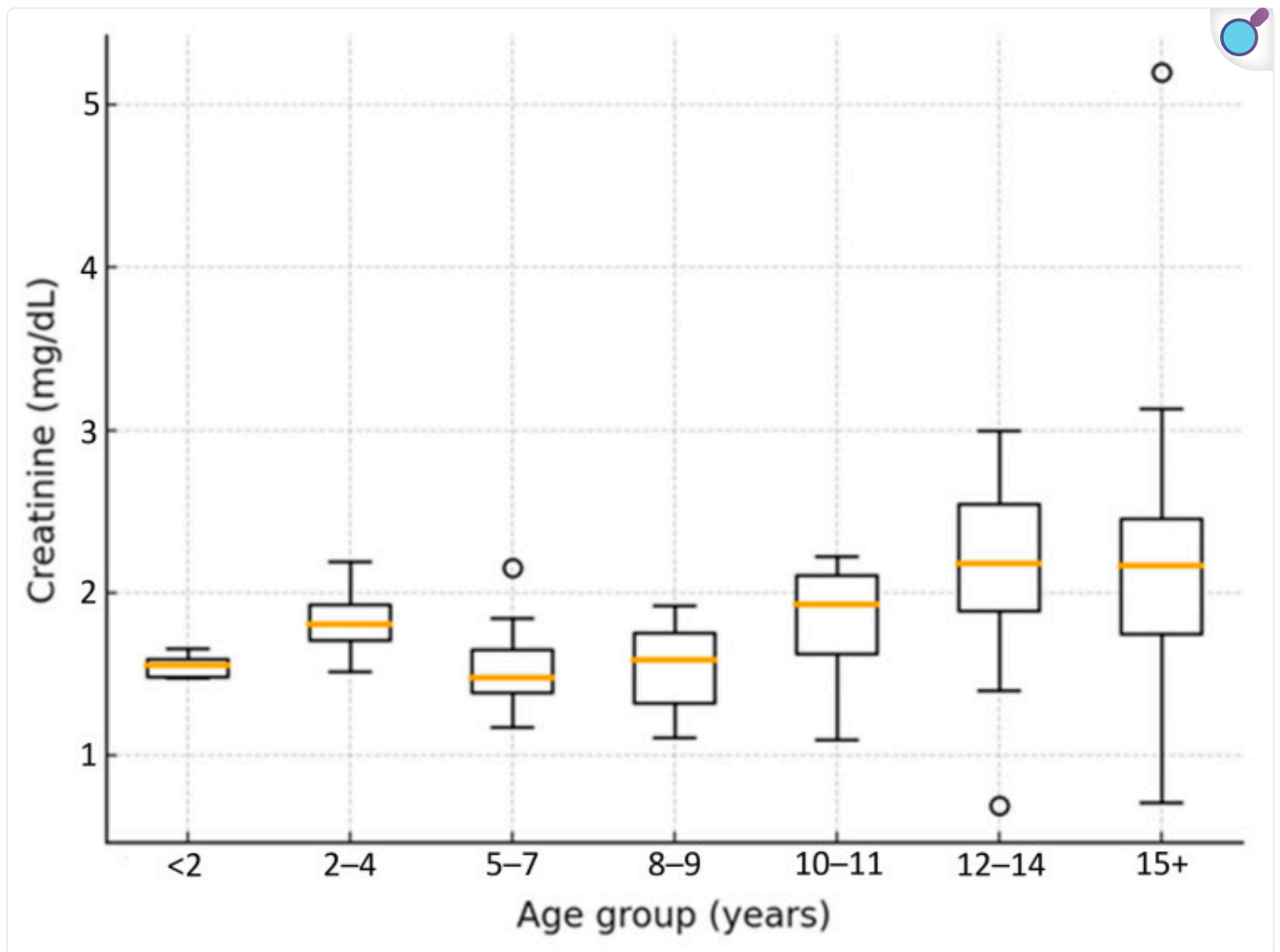
In addition to the descriptive distribution of feeding practices, it is important to note that optimal dietary management for CKD involves more than protein restriction. According to current clinical nutrition evidence, renal-supportive diets should include: controlled but highly digestible protein levels; restricted phosphorus content (<0.6–0.8% DM) to reduce renal mineral load; increased omega-3 fatty acids (EPA/DHA) to alleviate glomerular hypertension and inflammation; reduced sodium to minimize systemic hypertension; balanced potassium levels to correct CKD-associated hypokalemia; enhanced B-complex vitamin supplementation due to increased urinary losses; and antioxidants such as vitamin E and β -carotene to mitigate oxidative stress. Fermentable fibers and prebiotics (e.g., FOS, MOS) are also beneficial, improving nitrogen excretion through the gastrointestinal tract. These nutritional principles form the foundation of therapeutic renal diets and have demonstrated efficacy in slowing CKD progression and improving survival in affected cats.

In conclusion, the distribution of feeding practices prior to CKD diagnosis indicates a predominance of standard commercial diets and limited use of renal-supportive formulations. These results provide insight into real-world feeding patterns among cats later diagnosed with CKD but do not establish causal links between specific diet types and disease occurrence.

3.7. Results of Biochemical Markers

The analysis of serum creatinine concentrations ([Figure 8](#)) across age groups revealed a progressive, non-linear increase with advancing age. Median values rose from 1.8 mg/dL [IQR 1.4–2.2] in cats younger than seven years to 3.8 mg/dL [IQR 2.9–5.1] in cats aged twelve years or older. The median difference between these two age categories was 2.0 mg/dL (95% CI: 1.3–2.6; Mann–Whitney U = 214.5, $p < 0.001$), indicating a strong age-related increase in creatinine concentrations.

Figure 8.



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Distribution of serum creatinine values by age group.

Urea concentrations followed a similar trend, increasing progressively with advancing disease stage. Elevated urea levels (>80 mg/dL) were most frequently observed in IRIS stages III–IV, consistent with the presence of azotemia secondary to reduced glomerular filtration rate, as described in IRIS (2023) guidelines. Although less specific than creatinine or SDMA, urea remains a supportive indicator of impaired renal clearance and provides complementary diagnostic information in combination with other biochemical markers.

[Figure 8](#) illustrates the distribution of creatinine values across age categories. In cats over twelve years of age, some values exceeded 5 mg/dL, while in others they remained below 2 mg/dL. This confirms that the highest prevalence of elevated creatinine values occurred in cats older than ten to twelve years, reflecting the greater prevalence of CKD in this age group. However, considerable individual variability was observed. Increases in median values were already evident in cats aged two to four years (~1.9 mg/dL), whereas a broader spread of creatinine values, ranging up to 3.0 mg/dL, was noted in cats over twelve years, indicating wide inter-individual variability.

These findings demonstrate that serum creatinine and urea increase progressively with age, confirming the well-established correlation between aging and renal function decline. The results are consistent with the hypotheses proposed by Polzin (2011) [[1](#)], Brown et al. (2014) [[12](#)], and Pérez-López et al. (2019) [[13](#)], which describe CKD as a progressive, age-associated degenerative process characterized by nephron loss. The non-parametric analysis indicates that age exerts a significant and clinically relevant effect on biochemical markers of renal function, with a median difference of approximately 2 mg/dL in creatinine between younger and older cats.

Boxplot showing serum creatinine concentrations (mg/dL) across feline age categories. Median values and variability increased progressively with age, with the highest values recorded in cats aged ≥ 12 years. Non-parametric Kruskal–Wallis analysis confirmed significant differences between groups ($H = 22.46, p < 0.001$).

Relationship Between Biochemical Markers (SDMA vs. Creatinine)

A positive and statistically significant correlation was identified between SDMA and creatinine (Spearman's $\rho = 0.84, p < 0.001$). Linear regression yielded $\text{SDMA} = 6.41 + 6.77 \times \text{Creatinine}$ (mg/dL) with $R^2 = 0.88$ (95% CI for slope 6.31–7.22; 95% CI for intercept 5.09–7.74), indicating that 88% of SDMA variability was explained by creatinine. The slope was significantly greater than zero ($p < 0.001$). After adjustment for age, the association remained significant (partial $\rho = 0.73, p < 0.001$), suggesting that the SDMA–creatinine relationship is not driven by age alone. Bland–Altman analysis showed minimal bias ($-0.00 \mu\text{g/dL}$) and narrow limits of agreement (-4.48 to $+4.48 \mu\text{g/dL}$) between measured and regression-predicted SDMA values, supporting good agreement across the measurement range.

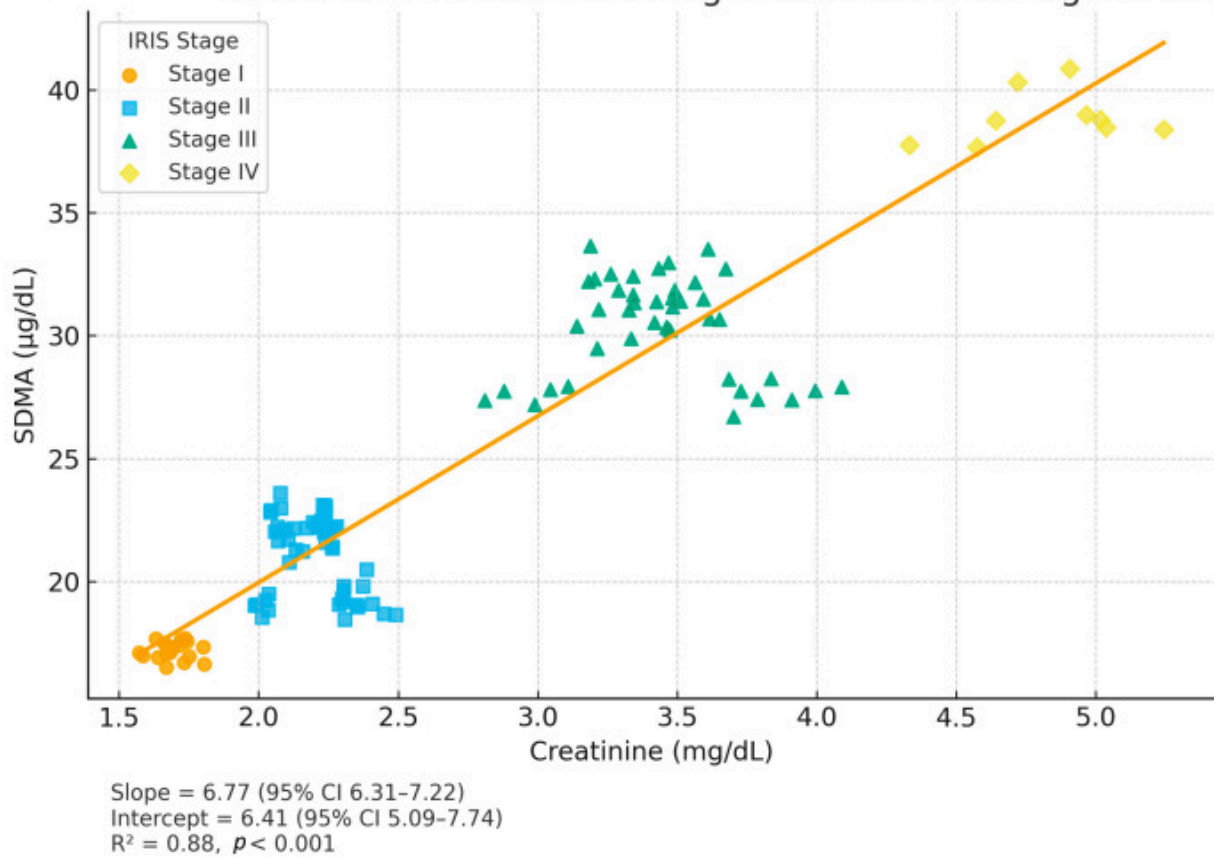
Graphical analysis ([Figure 9a,b](#)) showed greater dispersion of SDMA values at moderate creatinine levels (2–4 mg/dL), consistent with inter-individual differences in glomerular filtration rate and metabolic adaptation. At higher creatinine values (>5 mg/dL), the relationship became more linear and parallel, reflecting advanced nephron loss and reduced renal clearance. Clinically, these

findings emphasize the complementary diagnostic value of the two biomarkers: SDMA serves as an early indicator of reduced glomerular filtration rate, while creatinine remains a confirmatory marker for established renal insufficiency. The combined assessment of both parameters enhances diagnostic sensitivity and staging accuracy in feline CKD.

Figure 9.

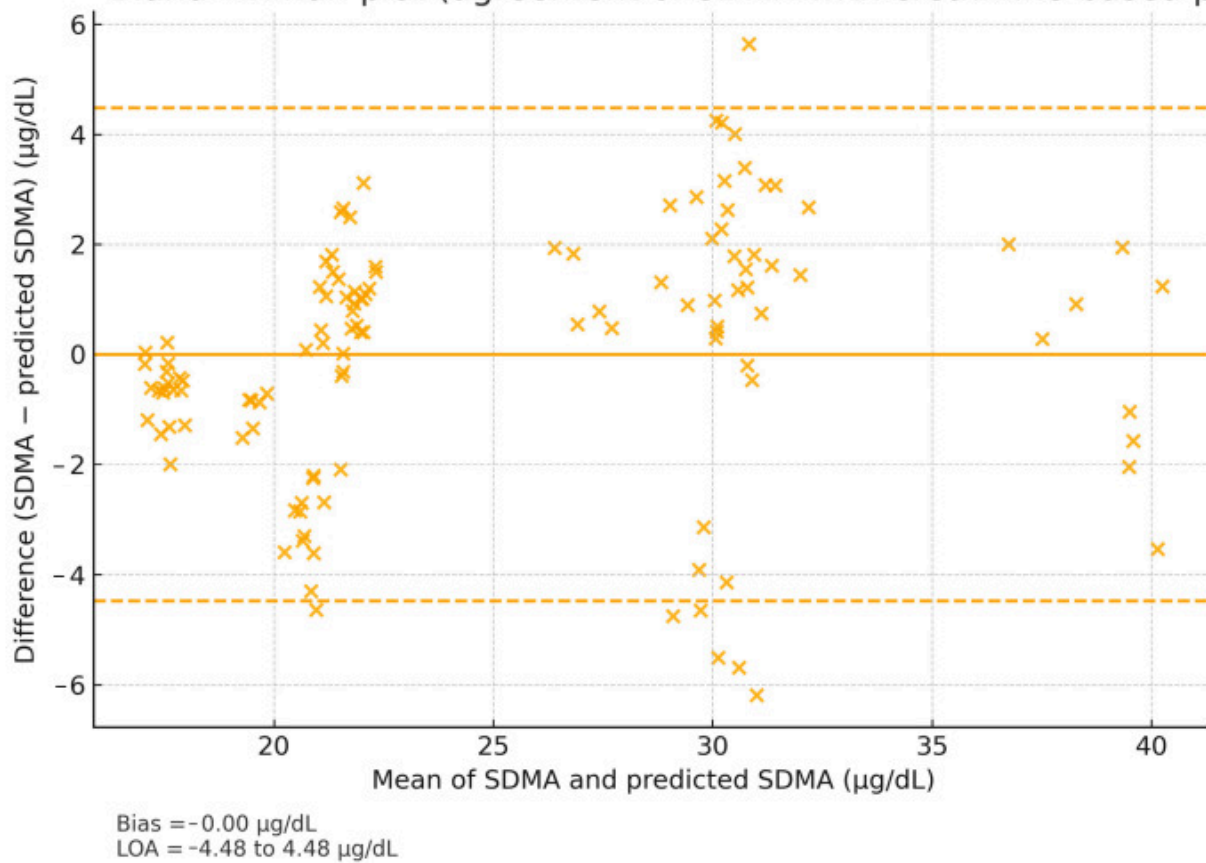


SDMA vs. creatinine with regression line and stage overlays



(a)

Bland-Altman plot (agreement of SDMA with creatinine-based prediction)



(b)

Relationship and agreement between SDMA and creatinine in cats with CKD. **(a)** Scatter plot with IRIS stage overlays and regression line (95% CI). Regression parameters: $\text{SDMA} = 6.41 + 6.77 \times \text{Creatinine (mg/dL)}$; $R^2 = 0.88$, $p < 0.001$; 95% CI for slope 6.31–7.22 and for intercept 5.09–7.74. **(b)** Bland–Altman plot showing bias = $-0.00 \mu\text{g/dL}$ and 95% limits of agreement = -4.48 to $+4.48 \mu\text{g/dL}$, indicating minimal systematic deviation and good agreement between measured and regression-predicted SDMA.

Clinically, these findings emphasize the complementary diagnostic value of the two biomarkers: SDMA serves as an early indicator of reduced glomerular filtration rate, while creatinine remains a confirmatory marker for established renal insufficiency. The combined assessment of both parameters enhances diagnostic sensitivity and staging accuracy in feline CKD. The combined interpretation of SDMA and creatinine thus provides a more comprehensive view of renal function, allowing differentiation between incipient, subclinical dysfunction and overt azotemia. This integrative approach improves diagnostic sensitivity, staging accuracy according to the IRIS guideline, and supports therapeutic decision-making at an earlier stage of the disease. Although the present study did not include a formal analysis of sensitivity and specificity against a gold standard, the observed evolutionary concordance between SDMA and creatinine values confirms their complementary nature and descriptive relevance in monitoring the progression of chronic kidney disease in cats.

3.8. Urinalysis and Blood Pressure Results

Building on the biochemical results that revealed age- and stage-related increases in creatinine and SDMA ([Table 5](#)), a complementary evaluation of urinary and hemodynamic parameters was performed. This approach aimed to integrate functional and structural indicators of renal damage and to improve staging accuracy in feline CKD.

Table 5.

Sensitivity analysis of CKD definition (comparison of creatinine-, SDMA-, and combined-based diagnosis).

Diagnostic Criterion	CKD Cases (n)	% of Cohort	Cohen’s κ (Agreement)
Creatinine > 1.6 mg/dL (IRIS criterion)	115	95.8%	—
SDMA > 14 µg/dL (standalone)	103	85.8%	—
Combined (creatinine and/or SDMA)	120	100%	0.91
Additional cases detected with SDMA	+5	4.2%	—

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Comparative analysis between diagnostic criteria based on creatinine, SDMA, and their combined use. The inclusion of SDMA identified 4.2% of additional early CKD cases compared with creatinine-only criteria. Cohen’s κ indicates excellent agreement between the two definitions.

Building upon the biochemical findings described above, urinalysis and blood pressure assessments were further analyzed to provide complementary insights into renal function and disease progression. These parameters reflect glomerular and tubular alterations that cannot be fully captured through serum biomarkers alone.

Urinalysis and systolic blood pressure (BP) measurements provided additional insight into renal function and disease progression in cats with chronic kidney disease (CKD). Urine specific gravity (USG) values ranged between 1.008 and 1.030, consistent with impaired ability to concentrate. Median USG decreased gradually with advancing IRIS stage, from 1.025 [1.020–1.030] in stage I to 1.010 [1.008–1.014] in stage IV (Table 6). Although low USG values can occasionally occur in non-renal conditions (e.g., fluid therapy, administration of diuretics), results were interpreted in conjunction with serum biochemical findings and clinical context.

Table 6.

Urinalysis summary by IRIS stage (USG, UPC, persistent proteinuria).

IRIS Stage	Median USG [IQR]	Median UPC [IQR]	Persistent Proteinuria (Yes/No)
I	1.025 [1.020–1.030]	0.20 [0.15–0.25]	No (borderline in 2/19)
II	1.018 [1.012–1.022]	0.28 [0.22–0.35]	Yes (7/49)
III	1.014 [1.010–1.018]	0.38 [0.30–0.46]	Yes (18/43)
IV	1.010 [1.008–1.014]	0.45 [0.39–0.55]	Yes (6/9)

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Proteinuria was a frequent finding among CKD cats, reflecting glomerular membrane injury and possible inflammatory processes. The urine protein-to-creatinine ratio (UPC) ranged from 0.20 [0.15–0.25] in stage I to 0.45 [0.39–0.55] in stage IV, indicating progressive protein loss with disease severity. Persistent proteinuria was documented in 28 cats (23%), confirmed by two or more consecutive urinalyses showing UPC > 0.4 obtained at intervals ≥14 days. The presence of proteinuria was associated with advanced stages of chronic kidney disease (CKD) and higher blood pressure values.

Summary of urinalysis parameters according to IRIS stage. USG = urine specific gravity; UPC = urine protein-to-creatinine ratio. Proteinuria became more frequent and pronounced with advancing CKD stage.

Taken together, these urinalysis findings are consistent with the progressive biochemical alterations described in the previous section. The parallel decrease in USG and increase in UPC with advancing IRIS stage support the concept of cumulative nephron loss and glomerular damage. When integrated with the combined interpretation of SDMA and creatinine, these urinary parameters provide a coherent functional profile of CKD progression, reinforcing the complementary diagnostic value of serum and urine markers in feline renal assessment.

Systolic BP showed a progressive increase with IRIS stage, from 135 [125–145] mmHg in stage I to 170 [155–185] mmHg in stage IV ([Table 7](#)). The prevalence of hypertension (≥ 160 mmHg) rose from 10% in stage I to 65% in stage IV. A moderate positive correlation was observed between UPC and BP ($\rho = 0.42, p = 0.003$), suggesting that glomerular hypertension and proteinuria are interrelated features of CKD progression. These findings highlight the complementary diagnostic value of urinalysis and blood pressure monitoring in staging and prognostic evaluation of feline CKD.

Table 7.

Blood pressure summary by IRIS stage (median BP, prevalence of hypertension, correlation with UPC).

IRIS Stage	Median BP [IQR] (mmHg)	Hypertension (≥ 160 mmHg) (%)	Correlation with UPC (ρ , p -Value)
I	135 (125–145)	10%	$\rho = 0.18, p = 0.25$
II	145 (135–155)	15%	$\rho = 0.25, p = 0.10$
III	160 (145–170)	35%	$\rho = 0.39, p = 0.01$
IV	170 (155–185)	65%	$\rho = 0.42, p = 0.003$

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Systolic blood pressure (BP) distribution across CKD stages. Hypertension prevalence increased from 10% in stage I to 65% in stage IV. BP showed a moderate positive correlation with UPC, supporting the association between glomerular hypertension and proteinuria.

The integrated assessment of biochemical, urinary, and hemodynamic parameters aligns with IRIS recommendations and emphasizes the multifactorial pathophysiology of feline CKD ([Table 5](#), [Table 6](#) and [Table 7](#)).

3.9. Ultrasonographic Findings

Abdominal ultrasonography was performed in 45 cats (37.5%) diagnosed with CKD to assess structural renal alterations and to complement clinical and biochemical findings. The ultrasonographic appearance of the kidneys varied according to disease stage and chronicity.

In early CKD (IRIS I–II), renal dimensions were typically within reference limits, although mild cortical echogenicity and subtle corticomedullary blurring were occasionally noted. In contrast, cats in advanced stages (IRIS III–IV) displayed irregular renal outlines, cortical thinning, and increased parenchymal echogenicity, consistent with fibrosis and chronic parenchymal remodeling. Small, rounded kidneys were recorded in 18/45 cats (40%), predominantly in stage IV, whereas diffuse cortical hyperechogenicity occurred in 32/45 (71%), frequently associated with medullary hyperechogenicity (24/45, 53%).

Hydronephrosis or pyelectasia was observed in 7 cats (16%), generally interpreted as secondary to chronic inflammatory or fibrotic obstruction. No cystic or neoplastic lesions were recorded.

Cats exhibiting diffuse cortical hyperechogenicity had higher median serum creatinine (4.1 mg/dL [IQR 3.3–5.3]) and SDMA (31 µg/dL [IQR 26–35]) values compared with cats showing near-normal renal architecture ($p < 0.01$). These observations reinforce the value of ultrasonography as a non-invasive tool for evaluating renal structural remodeling and supporting disease staging in feline CKD.

4. Discussion

Chronic kidney disease (CKD) remains a leading cause of morbidity in geriatric cats and represents a major clinical and welfare concern in companion animal practice. The findings of this study reinforce previous epidemiological data indicating that CKD is strongly age-associated, progressive, and irreversible. In our cohort, over 89% of CKD cases occurred in cats older than seven years, supporting the concept of CKD as a degenerative, senescence-associated disorder rather than an acute-onset pathology, in agreement with Polzin (2011) [1] and Marino et al. (2014) [3].

The clinical profile—dominated by polyuria, polydipsia, and weight loss—is consistent with previous reports [2,5,10,14,15,16], confirming that clinical recognition typically coincides with biochemical decompensation. However, these signs are nonspecific and overlap with other endocrine or metabolic disorders (e.g., diabetes mellitus, hyperthyroidism), emphasizing the diagnostic value of integrated biochemical screening in older cats. Elliott and Barber (1998) [5]

similarly reported that nonspecific clinical signs may guide clinicians toward biochemical investigation in suspected renal cases.

Age was confirmed as the primary determinant of CKD risk, consistent with Pérez-López et al. (2019) [13] and other epidemiological studies [17,18]. The rising prevalence beyond ten years of age highlights the importance of annual or biannual biochemical and urinary monitoring in senior cats, which can facilitate early detection and improve clinical outcomes. The age distribution in our population aligns closely with the prevalence ranges described by Brown et al. (2014) [12], who identified CKD in 30–40% of cats over 10 years.

Breed predisposition was evident, with higher proportions of purebred cats (Persian, Siamese, British Shorthair) affected compared with mixed-breed cats, a finding consistent with Marino et al. (2014) [3]. Although this pattern likely reflects genetic susceptibility, clinic-based population bias cannot be excluded. Sex and reproductive status did not independently influence CKD prevalence; the apparent predominance among neutered animals is more plausibly attributed to their longer life expectancy rather than to hormonal influences, as also suggested by White et al. (2006) [9].

Nutritional history emerged as a potentially modifiable factor influencing renal health. Cats maintained on unbalanced commercial or home-prepared diets showed higher disease prevalence, aligning with previous reports linking excessive dietary phosphorus and protein intake to renal dysfunction [19,20,21]. Conversely, renal diets with controlled phosphorus and moderate protein content have demonstrated measurable benefits in slowing CKD progression [22,23,24,25,26]. Hall et al. (2019) [21] reported that cats on a renal diet gained weight and maintained stable renal function compared to those fed standard commercial diets. These findings reinforce the importance of early dietary management as both a preventive and therapeutic strategy.

Biochemical analysis confirmed the diagnostic relevance of both creatinine and SDMA, the latter providing superior sensitivity for early renal dysfunction. The strong correlation observed between SDMA and creatinine ($\rho = 0.84$, $p < 0.001$; $R^2 = 0.88$) supports the concurrent use of both biomarkers. SDMA increases even with mild GFR decline and thus offers a valuable diagnostic window before overt azotemia develops [7,13,27,28]. Hall et al. (2014) [7] reported that SDMA elevation precedes creatinine increases in most cats with early CKD, and Pérez-López et al. (2019) [13] further confirmed SDMA's role as an early biomarker independent of muscle mass. Routine inclusion of SDMA testing in health screening protocols for cats older than seven years could substantially improve early CKD detection rates.

Ultrasonographic evaluation, although performed in a subset of cats, revealed a consistent relationship between structural remodeling (cortical thinning, hyperechogenicity) and biochemical severity, with higher creatinine and SDMA values observed in cats with marked cortical changes. These findings are in accordance with Elliott and Barber (1998) [5] and Bartlett et al. (2010) [4], who described similar ultrasonographic patterns in advanced CKD. This reinforces ultrasonography as a valuable non-invasive adjunct in the diagnostic and prognostic assessment of feline CKD [29].

Collectively, these findings underscore the multifactorial nature of feline CKD, influenced by aging, breed genetics, nutritional management, and clinical monitoring practices. They also emphasize the importance of integrating biochemical markers, urinalysis, and imaging findings to achieve accurate staging and prognosis.

4.1. Study Limitations

The study's retrospective, single-center design and limited sample size may constrain external validity. As the cohort originated from a referral hospital, selection bias toward more advanced cases cannot be excluded. Longitudinal, multicenter studies are warranted to validate these observations and to explore additional risk modifiers such as genetic polymorphisms, comorbid conditions, and early dietary interventions.

4.2. Clinical Implications

From a practical standpoint, the results highlight the need for routine screening of cats older than seven years using both creatinine and SDMA, coupled with urinalysis and BP measurement. Early detection and nutritional modulation remain the cornerstone of CKD management, capable of delaying progression and improving quality of life.

5. Conclusions

Chronic kidney disease in cats is a common condition with significant prevalence, strongly influenced by age and breed. Its prevalence rises markedly with age, reaching peak levels after 10–12 years.

The use of modern biomarkers, particularly SDMA, provides new opportunities for early diagnosis and effective monitoring of disease progression. The results of this study are consistent with the

international literature and highlight the need for standardized screening and treatment protocols.

A clear understanding of disease prevalence and etiological factors, combined with precise early diagnosis, is essential for the efficient management of feline CKD, ultimately improving both quality and duration of life in affected animals.

Future research should focus on genetic and nutritional risk factors as well as on validating personalized therapeutic strategies.

Author Contributions

Conceptualization, M.G. and V.N.; methodology, M.G., C.H. and R.C.; software, R.C.; validation, C.H., C.R. and O.T.; formal analysis, R.C.; investigation, M.G., C.H. and C.R.; resources, V.N. and O.T.; data curation, R.C.; writing—original draft preparation, M.G. and R.C.; writing—review and editing, R.C., C.H., C.R. and V.N.; visualization, R.C.; supervision, M.G., R.C., V.N. and O.T. All authors have read and agreed to the published version of the manuscript.

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The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

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References

1. Polzin D.J. Chronic Kidney Disease in Small Animals. *Vet. Clin. N. Am. Small Anim. Pract.* 2011;41:15–30. doi: 10.1016/j.cvsm.2010.09.004. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
2. Sparkes A.H., Caney S., Chalhoub S., Elliott J., Finch N., Gajanayake I., Langston C., Lefebvre H.P., White J., Quimby J. ISFM Consensus Guidelines on the Diagnosis and Management of Feline Chronic Kidney Disease. *J. Feline Med. Surg.* 2016;18:219–239. doi: 10.1177/1098612X16631234. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
3. Marino C.L., Lascelles B.D.X., Vaden S.L., Gruen M.E., Marks S.L. Prevalence and Classification of Chronic Kidney Disease in Cats Randomly Selected from Four Age Groups and in Cats Recruited for Degenerative Joint Disease Studies. *J. Feline Med. Surg.* 2014;16:465–472. doi: 10.1177/1098612X13511446. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
4. Bartlett P.C., Van Buren J.W., Bartlett A.D., Zhou C. Case-Control Study of Risk Factors Associated with Feline and Canine Chronic Kidney Disease. *Vet. Med. Int.* 2010;2010:957570. doi: 10.4061/2010/957570. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
5. Elliott J., Barber P.J. Feline Chronic Renal Failure: Clinical Findings in 80 Cases Diagnosed between 1992 and 1995. *J. Small Anim. Pract.* 1998;39:78–85. doi: 10.1111/j.1748-5827.1998.tb03598.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
6. Bartges J.W. Chronic Kidney Disease in Dogs and Cats. *Vet. Clin. N. Am. Small Anim. Pract.* 2012;42:669–692. doi: 10.1016/j.cvsm.2012.04.008. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

7. Hall J.A., Yerramilli M., Obare E., Yerramilli M., Jewell D.E. Comparison of Serum Concentrations of Symmetric Dimethylarginine and Creatinine as Kidney Function Biomarkers in Cats with Chronic Kidney Disease. *J. Vet. Intern. Med.* 2014;28:1676–1683. doi: 10.1111/jvim.12445. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
8. Braff J., Obare E., Yerramilli M., Elliott J., Yerramilli M. Relationship between Serum Symmetric Dimethylarginine Concentration and Glomerular Filtration Rate in Cats. *J. Vet. Intern. Med.* 2014;28:1699–1701. doi: 10.1111/jvim.12446. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
9. White J., Norris J., Baral R., Malik R. Naturally-Occurring Chronic Renal Disease in Australian Cats: A Prospective Study of 184 Cases. *Aust. Vet. J.* 2006;84:188–194. doi: 10.1111/j.1751-0813.2006.tb12796.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
10. IRIS Staging System. [(accessed on 15 July 2025)]. Available online: <https://www.iris-kidney.com/iris-staging-system>.
11. European College of Veterinary Internal Medicine—Companion Animals. [(accessed on 15 July 2025)]. Available online: <https://www.ebvs.eu/colleges/european-college-veterinary-internal-medicine-companion-animals>.
12. Brown C.A., Elliott J., Schmiedt C.W., Brown S.A. Chronic Kidney Disease in Aged Cats: Clinical Features, Morphology, and Proposed Pathogeneses. *Vet. Pathol.* 2016;53:309–326. doi: 10.1177/0300985815622975. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
13. Pérez-López L., Boronat M., Melián C., Saavedra P., Brito-Casillas Y., Wägner A.M. Assessment of the Association between Diabetes Mellitus and Chronic Kidney Disease in Adult Cats. *J. Vet. Intern. Med.* 2019;33:1921–1925. doi: 10.1111/jvim.15559. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
14. Piyarungsri K., Pusoonthornthum R. Risk and Protective Factors for Cats with Naturally Occurring Chronic Kidney Disease. *J. Feline Med. Surg.* 2017;19:358–363. doi: 10.1177/1098612X15625453. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
15. Polzin D.J. Evidence-Based Step-Wise Approach to Managing Chronic Kidney Disease in Dogs and Cats. *J. Vet. Emerg. Crit. Care.* 2013;23:205–215. doi: 10.1111/vec.12034. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
16. Freeman L.M., Lachaud M.-P., Matthews S., Rhodes L., Zollers B. Evaluation of Weight Loss Over Time in Cats with Chronic Kidney Disease. *J. Vet. Intern. Med.* 2016;30:1661–

1666. doi: 10.1111/jvim.14561. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

17. Polzin D.J. Chronic renal failure. In: Ettinger S.J., Feldman E.C., editors. Textbook of Veterinary Internal Medicine. 7th ed. WB Saunders; Philadelphia, PA, USA: 2010. pp. 1990–2021. [[Google Scholar](#)]

18. Lulich J., Osborne C., O'Brien T., Polzin D. Feline Renal Failure: Questions, Answers, Questions. *Compend. Contin. Educ. Pract. Vet.* 1992;14:121–152. [[Google Scholar](#)]

19. Schauf S., Coltherd J.C., Atwal J., Gilham M., Carvell-Miller L.J., Renfrew H., Elliott J., Elliott D., Bijsmans E.S., Biourge V.C., et al. Clinical Progression of Cats with Early-Stage Chronic Kidney Disease Fed Diets with Varying Protein and Phosphorus Contents and Calcium to Phosphorus Ratios. *J. Vet. Intern. Med.* 2021;35:2797–2811. doi: 10.1111/jvim.16263. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

20. Elliott J., Rawlings J.M., Markwell P.J., Barber P.J. Survival of Cats with Naturally Occurring Chronic Renal Failure: Effect of Dietary Management. *J. Small Anim. Pract.* 2000;41:235–242. doi: 10.1111/j.1748-5827.2000.tb03932.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

21. Hall J.A., Fritsch D.A., Jewell D.E., Burris P.A., Gross K.L. Cats with IRIS Stage 1 and 2 Chronic Kidney Disease Maintain Body Weight and Lean Muscle Mass When Fed Food Having Increased Caloric Density, and Enhanced Concentrations of Carnitine and Essential Amino Acids. *Vet. Rec.* 2019;184:190. doi: 10.1136/vr.104865. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

22. Ross S.J., Osborne C.A., Kirk C.A., Lowry S.R., Koehler L.A., Polzin D.J. Clinical Evaluation of Dietary Modification for Treatment of Spontaneous Chronic Kidney Disease in Cats. *J. Am. Vet. Med. Assoc.* 2006;229:949–957. doi: 10.2460/javma.229.6.949. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

23. Jacob F., Polzin D.J., Osborne C.A., Allen T.A., Kirk C.A., Neaton J.D., Lekcharoensuk C., Swanson L.L. Clinical Evaluation of Dietary Modification for Treatment of Spontaneous Chronic Renal Failure in Dogs. *J. Am. Vet. Med. Assoc.* 2002;220:1163–1170. doi: 10.2460/javma.2002.220.1163. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

24. Machado D.P., Ruberti B., Teixeira F.A., Vendramini T.H.A., Pfrimer K., Chacar F.C., Balieiro J.C.C., Pontieri C.F.F., Brunetto M.A. Body Composition of Healthy Cats and Cats with Chronic Kidney Disease Fed on a Dry Diet Low in Phosphorus with Maintenance Protein. *Toxins.* 2022;14:865. doi: 10.3390/toxins14120865. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

25. Dobenecker B., Webel A., Reese S., Kienzle E. Effect of a High Phosphorus Diet on Indicators of Renal Health in Cats. *J. Feline Med. Surg.* 2018;20:339–343. doi: 10.1177/1098612X17710589. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
26. Plantinga E.A., Everts H., Kastelein A.M.C., Beynen A.C. Retrospective Study of the Survival of Cats with Acquired Chronic Renal Insufficiency Offered Different Commercial Diets. *Vet. Rec.* 2005;157:185–187. doi: 10.1136/vr.157.7.185. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
27. Kongtasai T., Paepe D., Meyer E., Mortier F., Marynissen S., Stammeleer L., Defauw P., Daminet S. Renal Biomarkers in Cats: A Review of the Current Status in Chronic Kidney Disease. *J. Vet. Intern. Med.* 2022;36:379–396. doi: 10.1111/jvim.16377. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
28. Michael H., Szlosek D., Clements C., Mack R. Symmetrical Dimethylarginine: Evaluating Chronic Kidney Disease in the Era of Multiple Kidney Biomarkers. *Vet. Clin. N. Am. Small Anim. Pract.* 2022;52:609–629. doi: 10.1016/j.cvsm.2022.01.003. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
29. Yan G.-Y., Chen K.-Y., Wang H.-C., Ma T.-Y., Chen K.-S. Relationship between Ultrasonographically Determined Renal Dimensions and International Renal Interest Society Stages in Cats with Chronic Kidney Disease. *J. Vet. Intern. Med.* 2020;34:1464–1475. doi: 10.1111/jvim.15814. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Data Availability Statement

The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.



Article

A Real-World Data Analysis on Feline Chronic Kidney Disease in Greece: Clinical Profiles, Comorbidities, and Quality of Life

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Simple Summary

Chronic Kidney Disease is a condition affecting cats, usually of older age, that has a severe impact on their health and daily life. In this study, more than 200 cats with CKD were investigated to evaluate their health problems and how the disease affected their wellbeing. The results showed that most cats (67%) had other health issues as well, like dental disease and anemia, and that this causes a significant reduction in the cat's comfort and happiness. The quality of life of the animal significantly declines as the disease advances (according to the IRIS stage). These findings can help veterinarians and pet owners understand the impact of chronic kidney disease on a cat's life, and the importance of early detection of the disease, as well as treatment of other health issues, in order to significantly improve the animal's quality of life.

Abstract

Feline chronic kidney disease (fCKD) is highly prevalent in geriatric cats, affecting up to 80%, and is a leading cause of morbidity in cats with a challenging diagnosis. This study aimed to characterize the clinical profiles, comorbidities, and quality of life of cats with fCKD in real-world clinical settings in Greece. Real-world data from 208 client-owned cats with fCKD were analyzed, including demographics, clinical findings, body and muscle condition scores, and practitioner-assigned stage according to International Renal Interest Society (IRIS) guidelines. Additionally, owners completed the VetMetrica™ Health-Related Quality of Life (HRQL) questionnaire. Associations between comorbidities, IRIS stage, and HRQL domains were assessed using appropriate statistical analysis. At least one comorbidity was present in 66.8% of cats, with oral/dental disease (29.81%) and anemia (15.38%) being the most common. HRQL scores declined significantly with advancing IRIS stage ($p < 0.001$). Notably, anemia showed the strongest association with reduced HRQL across all domains, including an 18.79-point reduction in Emotional Wellbeing ($p < 0.001$). Oral/dental disease was also significantly associated with poorer Comfort (OR 5.16) and Emotional Wellbeing (OR 7.10). Overall, comorbidities independently contributed to impaired quality of life in cats with fCKD. These findings highlight that early detection of fCKD and targeted management of comorbidities alongside renal care may substantially improve patient welfare.



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Keywords: cat; chronic kidney disease; Greece; quality of life; real-world data

1. Introduction

Feline chronic kidney disease (fCKD) is among the most common and clinically significant disorders in cats, particularly in older individuals [1,2]. It represents a major cause of morbidity and mortality, with a prevalence that exceeds that reported in dogs [3–5]. According to the International Renal Interest Society (IRIS) [6], the prevalence of fCKD in the general feline population ranges between 1.0 and 3.0% [2,5,7,8], while in geriatric cats it may reach up to 80% [1,2,7,9].

Chronic kidney disease (CKD) is characterized by structural and/or functional damage of one or both kidneys that persists for at least three months. This irreversible loss of renal structure and/or function may remain stable for variable periods, but progresses over time [9]. Tubulointerstitial nephritis is the most commonly observed histopathologic lesion in fCKD; however, the precise underlying cause remains unclear [1,2,10]. Current evidence supports a multifactorial nature, shaped by genetic predisposition, individual susceptibility, and environmental influences [10].

Several factors increase CKD risk or contribute to its progression. Age is the most consistent risk factor [1,2,9–12], while certain breeds [1,8,13,14] and nutritional statuses [1,8,14] have also been implicated. Exposure to therapeutic or diagnostic agents [1] and the presence of comorbidities—such as hyperthyroidism, cardiovascular disease, urinary tract infections, or viral infections like FIV and FeLV—further contribute to disease progression [1,8,10,14,15].

Diagnosis is based on history, physical examination, clinicopathological data, imaging findings, and, when indicated, histopathology results [16,17]. Serum creatinine (sCr), though widely used, is insensitive in the early disease course and influenced by non-renal factors such as muscle mass, hydration, age, diet, and variation in measurement methods used [16,18,19]. More recently, symmetric dimethylarginine (SDMA) and the urine protein-to-creatinine (UP/C) ratio have been adopted as additional markers in clinical practice [16,20]. However, the diagnosis of fCKD is often delayed, usually established only once clinical signs emerge [1,16]. Low sensitivity of commonly used biomarkers, challenges in urine collection, and owners' tendency to postpone veterinary consultation contribute to the underdiagnosis of early disease [19]. These diagnostic challenges are suggestive of the need for large-scale, real-world data to better understand how fCKD presents and progresses in a typical clinical setting.

Once a diagnosis is achieved, disease staging becomes essential for clinical management. Based on the biomarkers mentioned above, IRIS has established a four-stage grading system for fCKD. Cats are classified into IRIS stages according to sCr and/or SDMA values when they are stable and hydrated, and are further substaged based on proteinuria and blood pressure measurements [20].

Clinical signs of CKD in cats are often non-specific and may not appear until significant renal damage has occurred [1]. Polyuria and polydipsia are usually the earliest and most common manifestations [21]. As the disease advances, other signs such as weight loss, gastrointestinal disturbances, lethargy, and dehydration become increasingly common [1,22], reflecting the systemic impact of reduced renal function and the decline in overall condition.

CKD negatively affects a cat's quality of life (QoL), leading to reduced activity, poor appetite, and difficulties with dietary modifications or medication administration. Therefore, assessing QoL provides an additional dimension for evaluating disease severity and the effectiveness of therapeutic interventions [23]. Insights from human medicine further

reinforce its importance: studies consistently demonstrate that CKD reduces QoL compared to healthy individuals [24,25], with progressive deterioration in advanced stages [26].

The chronic nature of CKD also impacts owners' mental health. Higher levels of stress, depression, and anxiety, as well as reduced QoL, have been reported among caregivers of pets with chronic or terminal conditions [27]. These findings highlight that the wellbeing of cats with CKD and their owners is closely linked, emphasizing the need to address both in clinical management and research.

In recent years, the emergence of real-world data (RWD) and real-world evidence (RWE) has created new opportunities in medical research [28]. The FDA [29] defines RWD as health-related data from sources such as medical records, insurance claims, or disease registries, while RWE is the clinical evidence generated from their analysis. In veterinary medicine, however, only a limited number of studies have incorporated RWE into routine practice [30,31]. Applying these methods to fCKD could provide new perspectives on both the clinical presentation and QoL of cats and their owners, improve disease management, and support more informed decision-making in daily practice.

This study uses a real-world data collection system for cats to comprehensively characterize the clinical profiles, comorbidities, and quality of life of cats with fCKD in Greece, representing the first multicenter real-world analysis to integrate clinician-reported data with owner-reported quality-of-life outcomes in this population. The study seeks to generate practical knowledge that can support veterinarians in diagnosis and disease management.

2. Materials and Methods

2.1. Development of Data Collection Forms

The following forms were developed to systematically collect data on fCKD cases. Their content was based on an extensive literature review in both human and veterinary studies related to kidney disease [32–35].

(a) "Screener Case Report Form" that included the specific study inclusion and exclusion criteria for fCKD cases (Table S1, Supplementary Materials) and the cat's IRIS stage with a description of the basic guidelines for staging (Table S2, Supplementary Materials).

(b) "Veterinarian-Completed Form" to gather feline patient demographics, such as sex, age, breed, weight, Body Condition Scoring (BCS), Muscle Condition Scoring (MCS), and any comorbidities (Table S3, Supplementary Materials).

(c) "Veterinarian/Owner Completed Case Form" to be filled out by the veterinarian and the cat owner to record data about the cat's diet, medical history, and treatment. It also included a checklist for owners to identify existing clinical signs and to report which of these they found most troublesome for themselves and their cat (Tables S4 and S5, Supplementary Materials).

(d) "Owner-Completed Form" to anonymously gather demographic data from the participating cat owners (Table S6, Supplementary Materials) who were required to be over 18 years old.

(e) "VetMetrica health-related quality of life (HRQL) Questionnaire" to evaluate the association of fCKD severity (as defined by IRIS Stages) on the cat QoL using the VetMetrica™ Feline HRQL assessment. Owners completed the questionnaire independently on the day of the consultation. This tool consists of 20 behavior-based questions answered on a 7-point Likert scale (0 = Not at all to 6 = Could not be more) [35–37]. This results in scores in three HRQL domains, including Vitality, Comfort, and Emotional Wellbeing. Summary scores on physical wellbeing were also calculated. For interpretability, raw domain scores (0–6) were normalized to a 0–100 scale.

2.2. Participant Recruitment

Recruitment was conducted by veterinarians in Greece who are part of the Asclepius One Health (AOH) network, an independent non-profit organization operating internationally from Greece [31,38]. Participating veterinarians identified eligible cats (a) by searching their clinic's medical records for already diagnosed fCKD cases and (b) by enrolling new fCKD cases that were admitted to their clinics between October 2023 and July 2024. In both cases, owners who agreed to participate admitted their cats to the veterinary clinics and provided written informed consent during their appointments for clinical examination and blood sampling.

Most cat owners who participated in the study were female (175/208, 84.1%). Both female and male owners exhibited similar mean age (45.7 and 46.4, respectively) and age range (20–78 and 20–74, respectively). Owner demographics, such as sex and age, showed no notable patterns. Detailed demographics are included in the Supplementary Materials (Table S7, Supplementary Materials).

2.3. Enrolment of fCKD Cases

Eligible cats were client-owned, aged one year or older, in a stable health condition, and with a diagnosis of CKD confirmed by the responsible veterinarian at any IRIS stage.

Cats were excluded if they were pregnant, lactating, on an unbalanced diet, or had been hospitalized within seven days prior to enrollment. Further criteria for exclusion were comorbidities that could confound the assessment of fCKD, such as acute kidney injury, endocrine disorders (e.g., uncontrolled hyperparathyroidism), specific congenital or obstructive renal conditions, and suspected or confirmed malignancy. Cats recently treated for parasitic or infectious diseases known to cause kidney damage (e.g., leptospirosis) or concurrently participating in a clinical trial involving the administration of any investigational product were also excluded (Table S1, Supplementary Materials).

Practitioners were encouraged to collect serum and, when feasible, urine samples at two Timepoints [Timepoint 1 (Day 0–7 from first consultation) and Timepoint 2 (Day 30–37 from first consultation)] to assist with cCKD diagnosis and IRIS staging; they were submitted for biochemical and urine analysis, respectively. The previously mentioned data collection forms were completed at both Timepoints.

2.4. Laboratory Analysis

The samples were sent to a commercial veterinary laboratory in Athens, Greece, which is accredited under ISO 17025 [39] and certified with ISO 9001 [40] (Vet in Progress Plus, Attiki, Greece).

Serum biochemical parameters were analyzed using a SIEMENS Dimension EXL with an LM clinical chemistry analyzer (Siemens Healthineers, Erlangen, Germany). Electrolyte concentrations were measured using a JOKOH EX-DS electrolyte analyzer (JOKOH Co., Ltd., Kanagawa, Japan). Serum analysis included blood urea nitrogen (BUN), albumin (ALB), total proteins (TP), phosphorus (P), calcium (Ca and Ca⁺⁺), potassium (K), sodium (Na), and chloride (Cl), along with the Na/K ratio. Blood creatinine (CREA) was measured using a validated enzymatic assay [41,42], for which the laboratory is accredited under ISO 17025. Urinary creatinine, urinary protein, and the urinary protein-to-creatinine ratio (UP/C) were measured in urine samples.

2.5. CKD Diagnosis and IRIS Staging

The final diagnosis of fCKD was based on the results of the biochemical analysis, the cat's medical history, clinicopathological findings, and any additional diagnostic tests performed (e.g., imaging) at the discretion of the responsible veterinarian. The IRIS stage

was determined by the responsible veterinarians according to IRIS guidelines [43]. Serum creatinine concentration was the primary biomarker used for staging in the majority of cases, reflecting routine clinical practice. SDMA values were incorporated into IRIS staging when available, but since they were not consistently available across the entire study population, they were not included in the analysis. Similarly, data required for substaging, such as proteinuria assessment via UP/C, were not uniformly available in the overall study population. The practitioner-assigned IRIS stage was used as a practical indicator of each cat's disease progression in a real-world setting.

The diagnosis of all comorbidities, including anemia and hypertension, was made by the responsible veterinarian based on their clinical assessment and the diagnostic tools available at their individual practice. No standardized diagnostic protocol or specific quantitative thresholds (e.g., a specific value for systolic pressure for hypertension) were provided to the participating veterinarians, thus reflecting the real-world clinical decision-making process.

2.6. Statistical Analysis

Statistical analyses were conducted using IBM SPSS Statistics (Version 29.0.2.0, Armonk, NY, USA) and R (Version 2024.04.1). All statistical tests were two-tailed, and a p -value of <0.05 was considered statistically significant. All analyses utilized only data from Timepoint 1.

Descriptive statistics were employed to summarize demographic and clinical characteristics of the study population. Normality of continuous variables was assessed using the Kolmogorov–Smirnov or Shapiro–Wilk tests. Comorbidities were classified into categories based on the primary organ system, as detailed in Table S8 (Supplementary Materials).

Chi-square tests were applied for categorical variables. Kruskal–Wallis and Mann–Whitney U tests were used for non-parametric comparisons. One-way analysis of variance was followed by post hoc pairwise comparisons using t -tests with a Bonferroni correction, and t -tests were used for parametric comparisons of normally distributed continuous variables. A Benjamini–Hochberg correction was applied to the ANOVA results to control for multiple comparisons. Dunn's test with a Bonferroni correction post hoc test was used for pairwise comparison when statistically significant results were recognized.

Comorbidities were analyzed as categorical (present/absent) variables. A multi-step approach was used to analyze the association of fCKD and comorbidities with HRQL scores.

First, Permutational Multivariate Analysis of Variance (PERMANOVA) was used to assess the overall association of categorical variables (like presence of comorbidity) with multidimensional VetMetrica™ scores.

Next, odds ratios (OR) with 95% confidence intervals (CI) were calculated to evaluate the likelihood of QoL deterioration associated with specific comorbidities. For this analysis, a 'deteriorated' HRQL was defined as a score falling below the median score of the control group (cats without any comorbidities) for that specific domain.

Finally, a multiple linear regression analysis was conducted to determine the independent relationship between the variables (specific comorbidities) and the deterioration in HRQL scores for each domain. Comorbidities in fewer than ten cats in the dataset were merged into a single "All Other" category to ensure adequate sample size and model stability for each predictor.

To evaluate the correlation between the presence of a clinical sign (binary variable) and the severity of fCKD as depicted by the IRIS Stage reported (considered in this analysis as an ordinal variable), the point-biserial correlation coefficient (r_{pb}) was calculated. The equality of proportions of tests was conducted only for clinical signs with at least 20 cases to ensure the validity of the test.

3. Results

3.1. Enrolled fCKD Cases

A total of 42 practitioners provided data on 208 fCKD cases that were admitted to their clinics. The majority (38, 90.4%) were general practitioners, while four of the practitioners were referral clinicians. More than half of the data (119/208, 57.2%) was obtained from 11 major veterinary clinics, while 31 practitioners contributed the remaining 42.7% (89/208) of the fCKD cases. The mean number of cases contributed per clinic was five (range: 1–23).

Most of the fCKD cases (135/208, 64.9%) were classified by the participating veterinarians as IRIS Stages 1 and 2, while 35.1% (73/208) were classified as IRIS Stages 3 and 4 at Timepoint 1. Timepoint 2 measurements were completed for 177 cases following a 30-day interval, during which 24 cats died, and seven were lost during follow-up (owner moved out of town, refused second examination, etc.). The distribution of the recorded cases per IRIS Stage reported by the practitioners for Timepoints 1 and 2 is presented in detail in Table 1. No statistically significant difference was observed between the two timepoints with regard to the frequency of IRIS stages reported (chi-square = 4.36, df = 3, N = 208, $p = 0.226$).

Table 1. Distribution of fCKD cases by IRIS stage as reported by the practitioners (Timepoint 1 and Timepoint 2).

IRIS Stages	Timepoint 1 N (% , 95%CI)	Timepoint 2 N (% , 95%CI)
Stage 1	43 (20.7, 15.2–26.2)	51 (28.8, 22.2–35.4)
Stage 2	92 (44.2, 37.5–51.0)	75 (42.4, 35.1–49.7)
Stage 3	44 (21.2, 15.6–26.7)	34 (19.2, 13.5–24.9)
Stage 4	29 (13.9, 9.2–18.7)	17 (9.6, 5.3–13.9)
Total	208 (100.0)	177 (100.0)

Among the 177 cases with data collected in both Timepoints 1 and 2, the practitioner-assigned IRIS stage remained consistent in 126 cases. Stage migration to a higher IRIS stage at Timepoint 2 was reported in 20 cases (11.3%), and to a lower IRIS stage in 31 cats (17.5%).

3.2. fCKD Cases Demographics

Female (93/208, 44.7%) and male cats (115/208, 55.3%) were almost equally represented in the study population. Among the 208 cats enrolled, most of them were spayed/neutered (198/208, 95.2%). There was no statistically significant association between sex and neuter status (Fisher's exact test $p = 0.157$) or practitioner-assigned IRIS stage (Fisher's exact test $p = 0.245$) (Table 2).

Table 2. Sex distribution of fCKD cases by IRIS Stage.

Category	Stage 1, N (%)	Stage 2, N (%)	Stage 3, N (%)	Stage 4, N (%)	Total, N (%)
Female Intact	0 (0)	1 (1.1)	0 (0)	2 (6.9)	3 (1.4)
Female Spayed	16 (37.2)	41 (44.6)	23 (52.3)	10 (34.5)	90 (43.3)
Male Intact	1 (2.3)	3 (3.3)	3 (6.8)	0 (0)	7 (3.4)
Male Neutered	26 (60.5)	47 (51.1)	18 (40.9)	17 (58.6)	108 (51.9)
Total, N (%)	43 (20.7)	92 (44.2)	44 (21.2)	29 (13.9)	208 (100.0)

Percentages in data cells were calculated using the number of cats within each IRIS stage as the denominator; Percentages in total value cells use the overall study population (N = 208).

Among the eight feline breeds recorded, the Domestic Shorthair (DSH) was the most common (166/208, 79.8%) (Table S9, Supplementary Materials). Breed was not associated with practitioner-assigned IRIS stage (Fischer's exact test $p = 0.192$).

The age distribution was found to be non-normal (K-S normality test = 0.64, $p = 0.037$, Figure S1, Supplementary Materials). The median age of the cats was 10.1 years old (SD = 4.38), with a range of 2 to 20 years. There was no statistically significant difference in the age distribution across the four IRIS stages (Kruskal-Wallis, test statistic = 5.352, $p = 0.148$). The median weight of cats was 4.1 kg (SD = 1.27, range 2–9.1). Upon analysis of feline age and body weight, no statistically significant correlation was observed, although a trend of reducing feline weight as the age increased was observed ($r = -0.131$, $p = 0.059$) (Figure 1).

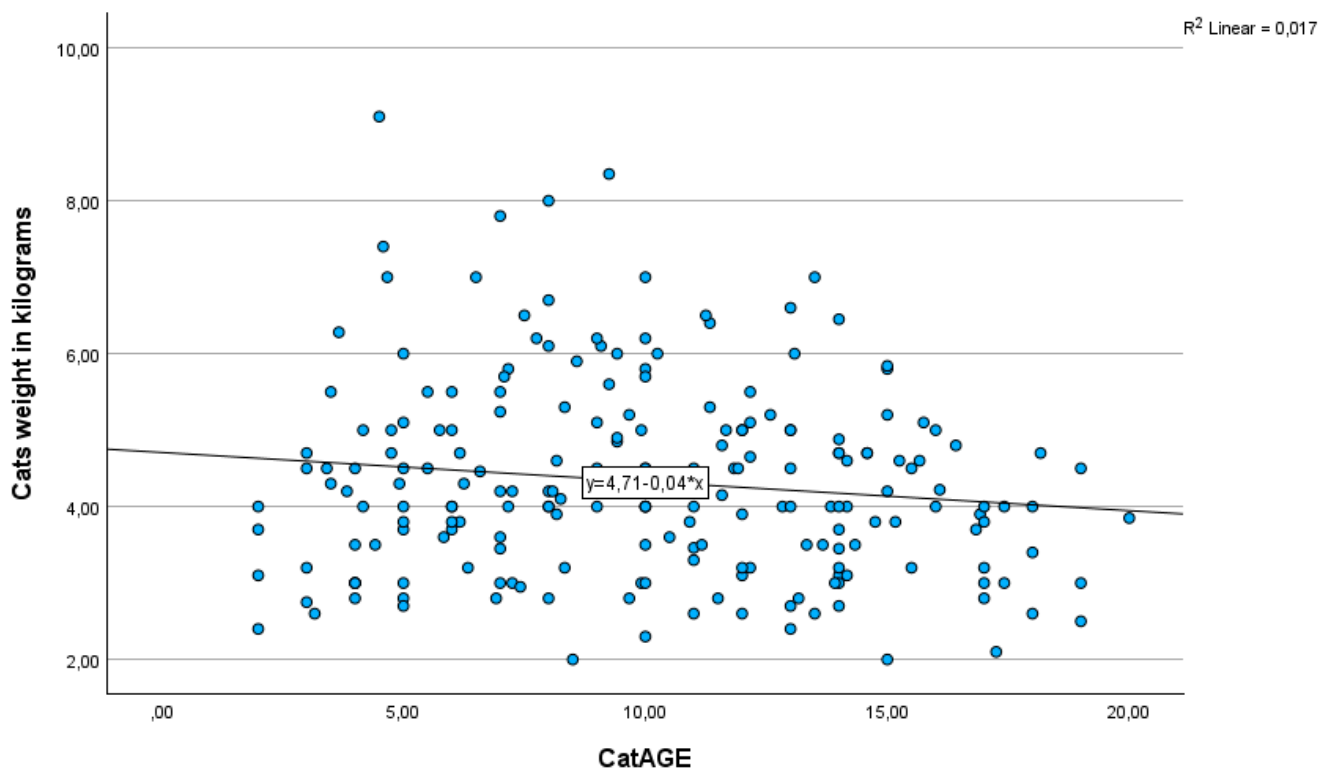


Figure 1. Scatter plot for feline age and weight with Pearson Correlation. The line represents the linear regression fit.

Muscle condition score (A-Normal muscle mass, B-Mild muscle loss, C-Moderate muscle loss, and D-Severe muscle loss) [44,45] was significantly associated with advanced fCKD based on the IRIS staging. More specifically, moderate ($n = 64$) to severe ($n = 6$) muscle loss was more common in late IRIS stages 3 and 4 compared to early IRIS stages 1 and 2 (33/73, 45.2% vs. 37/135 27.4%). Conversely, normal muscle mass ($n = 37$) and/or mild muscle loss ($n = 101$) were more frequently observed in early IRIS stages, with normal muscle mass present in 20.7% (28/135) of early IRIS stages versus 12.3% (9/73) of late IRIS stages, and mild muscle loss in 51.9% (70/135) versus 42.5% (31/73), respectively. The association between muscle condition score and early vs. late IRIS stages was statistically significant (chi-square = 7.20, $df = 2$, $N = 208$, $p = 0.027$).

Similarly, a poorer BCS [44,46] was significantly linked to more advanced IRIS stages. Cats classified as below the ideal BCS (1–3) were more frequent in late IRIS stages (46/73, 63.0%) than in early IRIS stages (64/135, 47.4%), while animals above the ideal BCS (7–9) were more commonly classified at the early IRIS stages (38/135, 28.1% vs. 7/73, 9.6%). For cats with an ideal BCS (scores 4–6), proportions were similar between early and late IRIS stages (33/135, 24.4% vs. 20/73, 27.4%), indicating no difference. The association between BCS and early vs. late IRIS stages was statistically significant (chi-square = 9.89, $df = 2$, $p = 0.007$), with the proportion of animals below the ideal BCS increasing as IRIS stage advanced.

3.3. Clinical Signs

The most prevalent clinical signs reported in cats with fCKD were increased water consumption (105/208, 50.5%), weight loss and muscle wasting (85/208, 40.9%), and polyuria (84/208, 40.4%). These were followed by loss of appetite (38.5%), unkept appearance (30.8%), halitosis (29.3%), and vomiting (26.4%) (Table S10, Supplementary Materials).

A positive correlation, indicating that the frequency increased with advancing IRIS stage, was found for all the clinical signs, except for vomiting, gastroenteritis, mouth ulcers, vision disorders, fragile bones and blood in urine ($p > 0.05$). Significant but weaker correlations were also found for weight loss/muscle wasting ($p = 0.008$), anorexia ($p = 0.002$), halitosis ($p = 0.028$), pale gums ($p = 0.002$), and diarrhea ($p = 0.045$) (Table S11, Supplementary Materials).

Increased water consumption, reduced appetite, unexplained weight loss, increased urination, depressed mood, and weakness/fatigue were most frequently regarded by the owner as troublesome for the cat. In contrast, signs such as halitosis and unkept appearance were considered significantly more disturbing for the owner than for the cat (Table S12, Supplementary Materials).

3.4. Biochemical Examinations and Urinalysis

Biochemical analysis revealed median concentrations for CREA of 2.12 mg/dL (IQR: 1.68–3.48) and for BUN of 43.00 mg/dL (IQR: 31.00–78.50). As expected for non-normally distributed data, a notable difference was observed between the mean and median values (e.g., CREA: 3.04 vs. 2.12 mg/dL), confirming a positive skew due to the presence of individuals with advanced disease. The median P concentration was 5.05 mg/dL (IQR: 4.25–6.40). For ALB, which followed a normal distribution, the mean was 2.96 ± 0.40 g/dL. Other key electrolytes with skewed distributions included K with a median of 4.40 mEq/L (IQR: 4.00–4.90) and Na with a median of 153.00 mEq/L (IQR: 148.00–155.00) (Table S13, Supplementary Materials).

As expected, based on the IRIS staging guidelines, median concentrations of CREA (S1 = 1.34, S2 = 2.00, S3 = 3.49, S4 = 5.18 mg/dL), BUN (S1 = 26, S2 = 39, S3 = 76, S4 = 168 mg/dL), and P (S1 = 4.8, S2 = 4.6, S3 = 5.8, S4 = 10.5 mg/dL) generally increased with advancing disease stage, with statistically significant differences observed between the groups (adjusted $p < 0.001$ for all). In contrast, no significant inter-stage differences were found for ALB, Ca, Ca++, TP, and Na, K, Na/K ratio, and Cl (all adjusted $p > 0.05$) (Table S14, Supplementary Materials).

Urinalysis data were available for a subset of 35 cats. The mean specific gravity (SG) was 1023.77 ± 10.16 . For urine pH, the mean was 6.66 ± 0.73 , and the UP/C ratio had a mean of 0.66 ± 0.69 (Table S15, Supplementary Materials).

3.5. Health-Related Quality of Life Measurement (VetMetrica™) and IRIS Stages

A statistically significant decline in QoL scores across practitioner-assigned IRIS stages was observed for the domains Vitality, Comfort, Emotional Wellbeing, and Physical Wellbeing (Table 3). The Kruskal-Wallis tests revealed highly significant effects of IRIS stage for Vitality ($p < 0.001$), Comfort ($p < 0.001$), Emotional Wellbeing ($p < 0.001$), and Physical Wellbeing ($p < 0.001$). Post hoc multiple comparisons showed that cats in IRIS Stage 4 had significantly lower scores compared with those in Stages 1 and 2 across all domains (adjusted $p < 0.001$ for Vitality, Comfort, Emotional Wellbeing, and Physical Wellbeing). Additional significant differences were observed between Stage 4 and Stage 3 for Vitality (adj. $p = 0.019$), Emotional Wellbeing (adj. $p = 0.011$), and Physical Wellbeing (adj. $p = 0.026$), and between Stage 3 and Stage 2 for Vitality (adj. $p = 0.006$), Comfort (adj. $p = 0.003$), Emotional Wellbeing (adj. $p = 0.003$), and Physical Wellbeing (adj. $p = 0.002$).

Table 3. Comparison of VetMetrica™ health-related quality of life (HRQL) scores across practitioner-assigned IRIS stages.

Domain	H Stat	<i>p</i> -Value	Stage 1 Median	Stage 2 Median	Stage 3 Median	Stage 4 Median	St. Significant Multiple Comparisons
Vitality	44.68	* <0.001	40.42	44.52	35.00	16.09	4-2 (adj. <i>p</i> < 0.001) 4-3 (adj. <i>p</i> = 0.019) 4-1 (adj. <i>p</i> < 0.001) 3-2 (adj. <i>p</i> = 0.006)
Comfort	36.50	* <0.001	38.38	37.08	31.92	25.61	4-1 (adj. <i>p</i> < 0.001) 4-2 (adj. <i>p</i> < 0.001) 3-2 (adj. <i>p</i> = 0.003)
Emotional Wellbeing	45.32	* <0.001	33.09	44.13	27.36	6.88	4-1 (adj. <i>p</i> < 0.001) 4-2 (adj. <i>p</i> < 0.001) 4-3 (adj. <i>p</i> = 0.011) 3-2 (adj. <i>p</i> = 0.003)
Physical Wellbeing	45.69	* <0.001	39.02	40.28	33.03	20.63	4-1 (adj. <i>p</i> < 0.001) 4-2 (adj. <i>p</i> < 0.001) 4-3 (adj. <i>p</i> = 0.026) 3-2 (adj. <i>p</i> = 0.002)

The statistically significant results are depicted with an asterisk.

When comparing cats at early and late IRIS stages, a significant decline in QoL was observed across all domains (Table 4). Mann–Whitney tests revealed significantly lower median scores in cats in a late IRIS stage for Vitality (*p* < 0.001), Comfort (*p* < 0.001), Emotional Wellbeing (*p* < 0.001), and Physical Wellbeing (*p* < 0.001).

Table 4. Comparisons of the various VetMetrica scores for early and late IRIS Stages.

Domain	U Stat	<i>p</i> -Value	Stage 1–2 Median	Stage 3–4 Median
Vitality	2448	* <0.001	43.52	25.13
Comfort	2596.5	* <0.001	36.77	30.00
Emotional Wellbeing	2584.5	* <0.001	41.21	28.16
Physical Wellbeing	2398	* <0.001	39.44	28.16

The statistically significant results are depicted with an asterisk.

3.6. Comorbidities

Most of the cats (139/208, 66.8%) presented with at least one comorbidity. When diagnoses were grouped into broader disease categories, oral/dental diseases were the most prevalent, affecting 29.8% (62/208) of the cats. Other common categories included hematologic abnormalities (15.4%), infectious diseases (10.6%), and both renal/urinary tract and cardiovascular diseases (9.6%) (Table S16, Supplementary Materials). The single most frequently reported individual diagnosis was periodontal disease, which was also present in 46/208 (22.1%) cats. Other common concomitant diseases included stomatitis (16.8%), anemia (15.4%), and mild liver disease (7.2%) (Table S17, Supplementary Materials).

The PERMANOVA analysis revealed a significant effect of comorbidities on the combined QoL measures (*p* = 0.001) (Table S18, Supplementary Materials). Comparisons of VetMetrica QoL scores between cats with and without concomitant diseases revealed that cats with any comorbidity had significantly lower median scores across all four HRQL domains (Mann–Whitney, *p* < 0.001 for all) (Figure 2, Table 5).

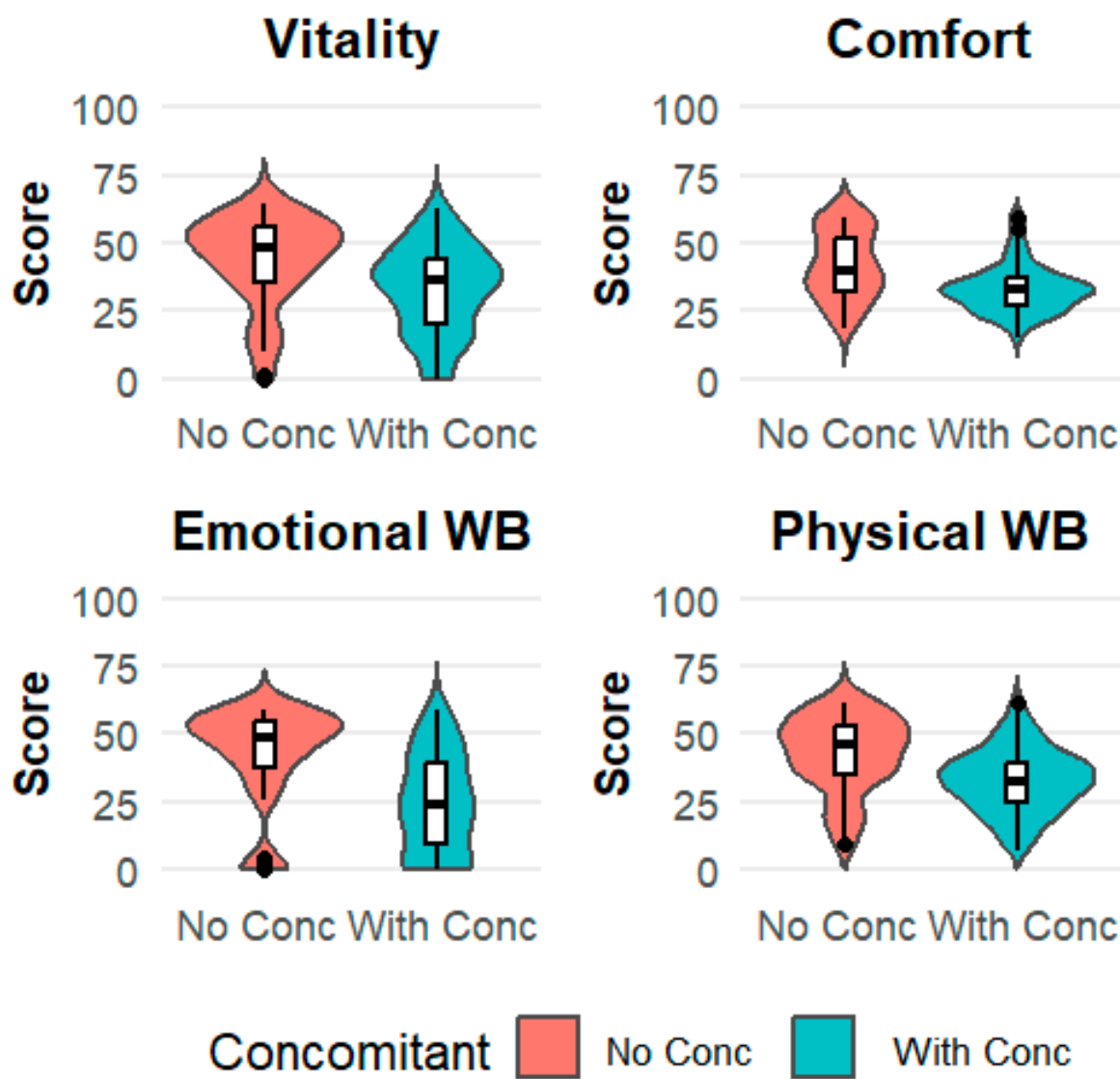


Figure 2. Violin plots with an overlaid boxplot of the three VetMetrica domain scores and the Physical Wellbeing score across the two groups (Cats with fCKD and no concomitant diseases vs. Cats with fCKD and concomitant diseases).

Table 5. Comparisons of the various VetMetrica Scores related to the existence or non-existence of concomitant diseases.

Domain	U Stat	<i>p</i> -Value	No Concomitant Median	With Concomitant Median
Vitality	2863	* <0.001	48.26	35.97
Comfort	2801	* <0.001	39.96	32.56
Emotional Wellbeing	2360	* <0.001	48.83	23.86
Physical Wellbeing	2700	* <0.001	46.25	32.70

The statistically significant results are depicted with an asterisk.

To further explore the specific association of comorbidities with HRQL, an OR analysis was performed. Specifically, cats with anemia were markedly more likely to experience deterioration in HRQL, with an OR of 7.0 (95%CI: 1.62–30.33, *p* = 0.009) for the Vitality domain, 14.85 (95%CI: 1.98–111.51, *p* = 0.009) for Comfort, 12.65 (95%CI: 1.68–95.13, *p* = 0.014) for Emotional Wellbeing, and 13.0 (95%CI: 1.73–97.75, *p* = 0.013) for Physical Wellbeing. Similarly, oral/dental disease as comorbidity were significantly associated with

reduced HRQL scores, with an OR of 2.55 (95%CI: 1.19–5.45, $p = 0.016$) for Vitality, 5.16 (95%CI: 2.08–12.80, $p < 0.001$) for Comfort, 7.10 (95%CI: 2.44–20.71, $p < 0.001$) for Emotional Wellbeing, and 4.43 (95%CI: 1.78–11.02, $p = 0.001$) for Physical Wellbeing (Table 6).

Table 6. Odds Ratios for VetMetrica HRQL domain deterioration by comorbidity category in cats with fCKD.

Category of Comorbidities	Domain Score	OR	95%CI	<i>p</i> -Value
Hematologic	Vitality	7.0	1.6–30.3	* 0.009
Oral/Dental	Vitality	2.6	1.2–5.5	* 0.016
Hematologic (anemia)	Comfort	14.9	2.0–111.5	* 0.009
Oral/Dental	Comfort	5.2	2.1–12.8	* <0.001
Hematologic	Emotional Wellbeing	12.7	1.7–95.1	* 0.014
Oral/Dental	Emotional Wellbeing	7.1	2.4–20.7	* <0.001
Hematologic	Physical Wellbeing	13.0	1.7–97.8	* 0.013
Oral/Dental	Physical Wellbeing	4.4	1.8–11.0	* 0.001

Note: The statistically significant results are depicted with an asterisk.

To control covariates, a multiple linear regression analysis confirmed that anemia, oral, and infectious diseases as comorbidities were significantly and independently associated with deterioration across multiple HRQL domains (Table 7). In the Vitality domain, anemia was associated with an almost 15-point reduction ($p < 0.001$), while oral disease corresponded to a 5.78-point decrease ($p = 0.025$). In the Comfort domain, hematologic, oral, and infectious comorbidities were linked to declines of 7.99 ($p < 0.001$), 5.09 ($p = 0.002$), and 7.77 points ($p = 0.001$), respectively. For Emotional Wellbeing, hematologic, oral, and infectious diseases were linked to reductions of 18.79 ($p < 0.001$), 7.25 ($p = 0.012$), and 9.34 points ($p = 0.026$), respectively. Finally, in the Physical Wellbeing domain, hematologic, oral, and infectious comorbidities were associated with decreases of 11.50 ($p < 0.001$), 5.44 ($p = 0.006$), and 6.11 points ($p = 0.033$), respectively.

Table 7. Results of multiple regression analysis for the association of specific comorbidities with HRQL domain scores.

Domain Score	Comorbidity	Expected Domain Score When No Comorbidity	Deterioration in the Domain Score	SD±	<i>p</i> -Value
Vitality	Hematologic diseases (anemia)	41.64 (±1.68)	−15.00	3.19	* <0.001
	Oral diseases		−5.78	2.55	* 0.025
Comfort	Hematologic diseases (anemia)	40.98 (±1.06)	−7.99	2.01	* <0.001
	Oral diseases		−5.09	1.61	* 0.002
	Infectious diseases		−7.77	2.36	* 0.001
Emotional Wellbeing	Hematologic diseases (anemia)	38.95 (±1.87)	−18.79	3.55	* <0.001
	Oral diseases		−7.25	2.85	* 0.012
	Infectious diseases		−9.34	4.12	* 0.026
Physical Wellbeing	Hematologic diseases (anemia)	41.30 (±1.28)	−11.50	2.43	* <0.001
	Oral diseases		−5.436	1.94	* 0.006
	Infectious diseases		−6.110	2.845	* 0.033

The statistically significant results are depicted with an asterisk.

3.7. Medical History and Diet

Medical history, as recorded in an open text box by the veterinarians, showed that the diagnosis in 39.9% (81/203) of fCKD cases was a result of observable signs of illness by the owner. On the other hand, in 5.4% (11/203) of the cases, CKD was discovered incidentally during routine health screenings or while investigating other medical conditions. In 30% (61/203) of the cases, the medical history was unknown. With regard to diet, commercial dry food and prescribed clinical diets were provided to a large proportion of cats (127/208, 61.1%), while smaller percentages received canned food (81/208, 38.9%), food in pouches (48/208, 23.1%), and/or homemade formulations (25/208, 12.0%).

4. Discussion

This study characterized the clinical presentation, HRQL, and the association of comorbidities in a large, real-world population of cats with fCKD, providing novel real-world evidence from a multicenter cohort that links comorbidities with both clinician-assessed clinical features and owner-reported HRQL. Most of the enrolled fCKD cases were in early IRIS stages, as assigned by the practitioners. The principal finding was the significant and independent negative association of comorbidities with HRQL scores across all domains. While our results confirmed the expected decline in HRQL with advancing IRIS stage, they demonstrated that concurrent diseases are significantly associated with poor QoL, emphasizing the necessity of a holistic approach to the diagnosis and management of fCKD. Our study showed the association of deteriorating physical condition, such as lower body weight and muscle loss, with advanced IRIS stages and identified a discrepancy between the clinical signs considered most troublesome for the cat versus the owner.

A critical finding in our study was the strong association between anemia and poor HRQL, with odds ratios for deterioration exceeding 12.0 for Comfort, Emotional Wellbeing, and Physical Wellbeing. This is clinically relevant, as anemia directly leads to weakness, fatigue, and reduced oxygen-carrying capacity, which would be reflected by the Vitality and Physical Wellbeing domains. Our results, therefore, build on previous observations that anemia adversely affects the quality of life of CKD patients through its overall effects [47] and contributes directly to the clinical expression of uremic syndrome in cats with advanced CKD [48].

These findings are particularly impactful given the complex nature of anemia in the context of CKD. The pathophysiology of CKD-associated anemia is multifactorial, involving reduced erythropoiesis, shortened erythrocyte lifespan, and increased erythrocyte loss [47]. While anemia is known to intensify with advancing IRIS stage due to progressive loss of renal tissue and erythropoietin production [2,49], its direct relationship with survival time is not established, with conflicting evidence across studies [12,50]. By demonstrating a negative association with QoL, our findings elevate the management of anemia in feline CKD to a primary therapeutic target.

Oral and dental diseases emerged as the most common comorbidities, affecting 29.8% of the investigated cats, consistent with previous evidence suggesting a link between oral disease and fCKD. Trevejo et al. (2018) demonstrated that cats with periodontal disease had a significantly higher risk of developing fCKD compared to those without [51]. The greatest risk was observed in cats with IRIS stage 3 or 4, indicating a severity-dependent relationship between oral and renal pathology. Similarly, Greene et al. (2014) identified periodontal disease as a clinical factor associated with increased odds of fCKD diagnosis, alongside thin body condition, weight loss, dehydration, and anesthesia within the preceding year [13]. In contrast, Vetter et al. (2023) investigated renal function in cats with histologically confirmed caudal stomatitis and found no significant difference in the frequency of kidney disease, azotemia, or proteinuria compared with age-matched controls [52]. Interestingly, cats with

caudal stomatitis exhibited biochemical markers of systemic inflammation, such as elevated serum phosphorus, potassium, and globulin concentrations, along with decreased albumin and hematocrit levels, which may represent early subclinical alterations preceding fCKD.

Human medical literature mirrors these findings, demonstrating a consistent association between periodontal disease and CKD, which strengthens with disease severity [53]. Proposed mechanisms include systemic inflammation, oxidative stress, immune dysregulation, and microbial alterations, all of which can contribute to vascular and renal injuries. Although causality has not been established, human studies emphasize the need to determine whether the prevention or treatment of periodontal disease can reduce the incidence or severity of CKD, an issue that holds equal importance in feline medicine [54].

Our study supports these findings as the strong association of oral/dental disease with lower Comfort (OR = 5.16) and Emotional Wellbeing (OR = 7.10) scores indicates that the clinical burden is associated with chronic pain and distress, which directly impairs the cat's QoL.

In our study, the presence of infectious diseases was a significant factor in patient wellbeing, showing a positive correlation with deterioration across all four HRQL domains. Infectious diseases were associated with a 9.34-point decline in Emotional Wellbeing and a 7.77-point decline in Comfort. This positions the negative association of infectious disease as intermediate between the more severe association with anemia (e.g., an 18.79-point drop in Emotional Wellbeing) and the moderate effects of oral comorbidities (e.g., a 7.25-point drop in Emotional Wellbeing).

The specific viral diseases recorded in our study population comprised FIV in 6.73%, FIP in 2.40%, and FeLV in 1.92% of cats. The pathological link is clearest with FIP, which is known to cause granulomatous nephritis [55]. While FeLV is primarily associated with neoplastic processes, a potential link to CKD has also been suggested [1,55]. The role of FIV is more controversial; however, it has been linked to the induction of renal lesions that could plausibly contribute to the decline in HRQL we observed [15,56]. Therefore, managing these infections is critical for quality of life, and this finding aligns with existing guidelines for retroviral diseases, which recommend addressing comorbidities to improve patient outcomes [57].

The systemic burden is visibly reflected in the patient's physical condition. The present study identified a relationship between feline body composition and practitioner-assigned IRIS stage, with BCS and MCS showing signs of decline as the disease progressed. This finding validates these simple, in-clinic assessments as indicators of disease severity and the cat's overall health status. This pattern reflects the multifactorial impact of fCKD on body composition.

Weight loss in affected cats is often gradual and may begin months before clinical diagnosis [1,8,13,22,58]. It is influenced by reduced appetite, nausea, vomiting, and diarrhea—common signs in later stages—as well as by metabolic alterations associated with chronic inflammation and decreased protein or caloric intake [59]. Over time, these mechanisms contribute to both generalized weight loss and selective muscle wasting, manifesting as sarcopenia or cachexia, depending on the underlying cause [9,58,60,61].

Given these interrelated mechanisms, weight alone provides an incomplete picture of nutritional and clinical status in cats with CKD. Incorporating BCS and MCS evaluation allows for a more accurate assessment of disease impact and progression. This combined approach is increasingly emphasized by major veterinary organizations [18,44,58,62], which recommend that each clinical visit include not only weight measurement but also assessment of body and muscle condition. Regular monitoring of these parameters enables earlier detection of nutritional deterioration [18,58] and facilitates timely intervention to preserve lean mass and improve quality of life in cats with CKD. Our study provides quantitative

HRQL evidence to support this combined assessment approach. Disease progression as indicated by the practitioner-assigned IRIS stages was significantly associated with a decline in physical condition (BCS and MCS) as well as lower HRQL scores in the Physical Wellbeing and Vitality domains, indicating the systemic impact of the deterioration of physical condition on the patient's daily life.

This study further documented that several clinical signs showed a positive correlation with disease severity, including loss or decrease in appetite, weakness or fatigue, depressed mood, poor coat appearance, and increased water consumption. Increased urination and weight or muscle loss were also significantly associated with advancing CKD stages, reflecting the progressive systemic impact of renal dysfunction. These findings are consistent with previous reports describing polyuria, polydipsia, anorexia and weight loss as among the earliest and most frequent clinical manifestations of feline CKD [3,8,13,22,58]. Overall, the gradual increase in both the frequency and severity of these signs across CKD stages highlights their diagnostic and prognostic significance. Early recognition of subtle changes in appetite, behavior, or water intake may facilitate timely intervention, dietary modification, and improved quality-of-life outcomes for affected cats.

In addition to the clinical prevalence of specific signs, the present study explored how these manifestations were perceived by owners in terms of their impact on both the cat and themselves. For the cat, the pet owners' answers reflect the key clinical manifestations previously identified as correlating with disease severity. For themselves, the answers reflect their aesthetic or sensory implications in daily interaction. The owners' perceptions are consistent with recent findings that cats with CKD experience a marked reduction in HRQL, particularly in dimensions related to comfort, vitality, and emotional wellbeing. Moreover, these observations align with recent findings emphasizing the relevance of owner perception and emotional burden in the management of chronic feline diseases [23,35,63]. Caregiver stress and frustration may influence adherence and care quality, highlighting the importance of communication strategies that support both the cat's welfare and the owner's wellbeing [23,64]. Our findings show that owners may not recognize the most severe indicators of poor welfare. This suggests the veterinarian's role in educating owners to shift their focus from 'cosmetic' signs to the crucial indicators of wellbeing, such as appetite, interaction, and activity levels.

Our study population was characterized by a predominance of early stage fCKD. At Timepoint 1, 64.9% of the cases were practitioner-assigned as IRIS Stage 1 and 2. This distribution is consistent with population-based or screening studies [7], and suggests that growing awareness and improved diagnostics may result in earlier detection, which is beneficial for disease prognosis and the cats' QoL [16].

Changes in practitioner-assigned IRIS stages classification over time were observed in 28.8% of the 177 cases with data collected at both Timepoints 1 and 2. Overall, 11.3% of the cases moved to a higher stage, while 17.51% shifted to a lower stage. Advancement to a higher stage is expected in a progressive disease, and it can be attributed to a sudden worsening of the clinical condition. A shift to a lower stage, however, may be attributed to the successful management of pre-renal azotemia (e.g., from dehydration or a UTI), the stabilization of a patient after an acute-on-chronic kidney disease event [1] or even the inherent biological variability of creatinine [65].

The median age of fCKD cases in our study was 10 years old. Although fCKD is traditionally considered a disease of geriatric cats [2], our finding that a substantial proportion of cases included middle-aged animals (7–10 years) [66,67] may also reflect this trend of earlier detection. Moreover, it may be suggestive of the multifactorial nature of fCKD beyond age-related renal decline due to progressive nephron loss and tubulointerstitial fibrosis [1]. This contrasts with some studies reporting higher mean ages [7,68] and supports the view

that age alone is not the sole determinant of disease onset or severity. Consistent with the literature, we found no association with sex or breed, suggesting that fCKD is a widespread condition in the cat population [17].

The findings regarding the diet administered to the diseased cats emphasize that a substantial portion of them were not managed strictly with a prescription renal diet. Furthermore, the wide consumption of canned food and pouches indicates that hydration concerns and palatability may influence dietary choices, a critical factor given the risk of food aversion in CKD [18,68]. Despite strong evidence of benefit, long-term adherence to renal diets remains a major clinical challenge, often limited by reduced appetite, food aversion, and difficulties in owner compliance [18,68].

This study has several important limitations inherent to its real-world data design. A primary limitation relates to the integrity of clinical staging. IRIS staging and comorbidity diagnosis were performed by 42 different practitioners, introducing the potential for inter-observer variability in clinical assessment and IRIS staging. This may have contributed to the observed stage migration. The stage shift to lower stages likely reflects not only clinical stabilization but also the variability of practitioner-based staging in a real-world setting. Consequently, the IRIS stages reported should be interpreted as “practitioner-assigned clinical stages” rather than standardized classifications. Future studies employing a centralized review of cases could standardize these variables.

This is further compounded by the lack of SDMA and UP/C data in the overall study population. The lack of uniformly available SDMA data limited the ability to fully assess its contribution to IRIS staging in this real-world study. Seemingly, the lack of uniformly available UP/C data did not allow sub-staging and potentially led to the misclassification of some Stage 1 cats. Financial restrictions that do not allow for assessing a complete panel of renal biomarkers and the challenge of urine collection in cats reflect practical difficulties in a clinical setting. However, future studies should include biomarkers such as SDMA and specialized tools for urine collection in cats.

Another limitation is the lack of standardized diagnostic criteria for the recorded comorbidities. Conditions like anemia and hypertension were reported as present or absent by the practitioners without accompanying quantitative data, such as hematocrit values or specific blood pressure readings. This did not allow the analysis of the association of the severity of these conditions with HRQL. For instance, while our data shows a strong negative association between the presence of anemia and QoL, we cannot differentiate between the impact of mild versus severe anemia. Similarly, diagnoses of infectious diseases such as FIV and FeLV were recorded without accompanying data on the specific testing method or the clinical status of the infection. This limits a more detailed interpretation of how specific pathogens or disease statuses affect HRQL. While the small number of affected cats and their aggregation into a broader ‘Infectious diseases’ category for regression analysis minimizes the statistical impact of this variability, we acknowledge that this lack of detail is a limitation. Future research employing standardized diagnostic protocols would be necessary to clarify these specific relationships.

An additional limitation of this study is the potential for cluster effects related to the 42 different practitioners and clinics contributing data. The study population included cases from both general and referral practitioners. It is expected that different clinics have different protocols or levels of emphasis on diagnostics, such as the frequency of performing dental exams or blood work. This variability could influence the detection rates of certain comorbidities and introduce bias into the observed associations with HRQL. Consequently, the findings should be interpreted with the acknowledgment that some of the observed variation may be attributable to differences between clinics. While we acknowledge that clinic-level cluster effects could influence detection rates, the uneven

distribution of cases across 42 practitioners—many contributing only 1–2 cases—precluded the use of multi-level modeling. Such an approach would have risked model instability and compromised the validity of the reported odds ratios.

Additionally, the generalizability of our findings is limited by the sampling strategy. The study utilized a convenience sample of practitioners, which may not be fully representative of all veterinary clinics in Greece. Veterinarians within the AOH network may represent a more engaged or proactive group, which could contribute to the high proportion of early-stage diagnoses. As such, our findings apply to the specific conditions and practices of this cohort, and caution should be taken when generalizing to other countries or healthcare systems.

Finally, we chose to analyze data from the largest possible sample size and not perform a longitudinal analysis on the 177 cats with data from two timepoints. Thus, the cross-sectional nature of our study allows us to interpret our findings on comorbidities and QoL as associations, not as predictors of progression. Future longitudinal analysis of this cohort is warranted to investigate if the highly associated comorbidities identified here are predictive of more rapid stage progression or a faster decline in HRQL scores over time.

5. Conclusions

In conclusion, this study demonstrates that while fCKD severity is a key determinant of HRQL, the management of comorbidities is equally critical to a cat's welfare. Our real-world data demonstrates that a cat with fCKD is not only a 'kidney patient,' but a complex case whose QoL is led by their comorbidities and their IRIS stage as well, with our data showing these associations are independent of disease stage. These findings have direct clinical implications for routine practice, emphasizing the importance of a holistic management approach that extends beyond renal function alone. Based on these findings, we recommend that clinicians: (1) integrate formal HRQL assessments into routine management to record the patient's QoL, (2) screen for and treat highly associated comorbidities, particularly oral/dental disease and anemia, and (3) educate owners to monitor for the most clinically relevant signs of welfare decline, such as changes in appetite and activity. Such an approach may support more targeted clinical decision-making and improved welfare outcomes in cats with fCKD. Future longitudinal research is warranted to determine if targeted interventions against these factors can improve HRQL scores over time.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/vetsci13020192/s1>, Figure S1: Histogram and kernel density estimation (KDE) for feline age, showing the Free Probability Density Function (PDF) for the normal distribution; Table S1: Eligibility criteria for observational study; Table S2: Description of the International Renal Interest Society (IRIS) staging guidelines in cats; Table S3: Veterinarian-completed form; Table S4: Case sheet of cat with chronic kidney disease: Animal diet, disease history and treatment plan; Table S5: Case sheet of cat with chronic kidney disease: Clinical signs and most troublesome clinical signs for the cat and the cat owner; Table S6: Owner-completed form; Table S7: Owners' demographics; Table S8: Categorization of reported concomitant diseases; Table S9: Feline breeds by IRIS Stage; Table S10: Clinical signs of cats with fCKD, as reported by the practitioners; Table S11: Point-biserial correlation between presence of clinical signs and severity of fCKD according to IRIS Stage; Table S12: Clinical signs reported by the owner as more troublesome for the owner and the cat; Table S13: Descriptive statistics of the biochemical examinations performed in the study population; Table S14: Comparison of analyte concentrations across IRIS stages using ANOVA and Kruskal-Wallis tests; Table S15: Urine examination results; Table S16: Prevalence of Concomitant Disease Categories and 95% Confidence Intervals; Table S17: List of Comorbidities reported in fCKD

cases; Table S18: Results of Permutational Multivariate Analysis of Variance (PERMANOVA) for Concomitant Conditions and combined QoL measures.

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Informed Consent Statement: Written informed consent was obtained from the owners for the participation of their animals in this study.

Data Availability Statement: The data presented in this study are available on request from the corresponding author. The datasets for this article are not publicly available due to concerns regarding owner/practitioner anonymity.

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Conflicts of Interest: T.L.S., A.W. are employees of Zoetis Inc. The remaining authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

ALB	Albumin
AOH	Asclepius One Health
BCS	Body Condition Score
BUN	Blood urea nitrogen
Ca and Ca ⁺⁺	Calcium
CKD	Chronic kidney disease
Cl	Chloride
CREA	Blood creatinine
DSH	Domestic Shorthair
fCKD	Feline chronic kidney disease
HRQL	Health-related quality of life
IQR	Interquartile Range
IRIS	International Renal Interest Society
K	Potassium
MCS	Muscle Condition Score
Na	Sodium
OR	Odds ratios
P	Phosphorus
PERMANOVA	Permutational Multivariate Analysis of Variance
QoL	Quality of life

RWD	Real-world data
RWE	Real-world evidence
r	Pearson correlation coefficient
rpb	Point-biserial correlation coefficient
SDMA	Symmetric dimethylarginine
SG	Specific gravity
sCr	Serum creatinine
TP	Total Proteins
UP/C	Urine protein-to-creatinine

References

1. Reynolds, B.S.; Lefebvre, H.P. Feline CKD: Pathophysiology and Risk Factors—What Do We Know? *J. Feline Med. Surg.* **2013**, *15*, 3–14. [\[CrossRef\]](#)
2. Brown, C.A.; Elliott, J.; Schmiedt, C.W.; Brown, S.A. Chronic Kidney Disease in Aged Cats: Clinical Features, Morphology, and Proposed Pathogenesis. *Vet. Pathol.* **2016**, *53*, 309–326. [\[CrossRef\]](#)
3. Dadousis, C.; Whetton, A.D.; Mwacalimba, K.; Merlo, A.; Wright, A.; Geifman, N. Renal Disease in Cats and Dogs—Lessons Learned from Text-Mined Trends in Humans. *Animals* **2024**, *14*, 3349. [\[CrossRef\]](#)
4. O'Neill, D.G.; Church, D.B.; McGreevy, P.D.; Thomson, P.C.; Brodbelt, D.C. Longevity and Mortality of Cats Attending Primary Care Veterinary Practices in England. *J. Feline Med. Surg.* **2015**, *17*, 125–133. [\[CrossRef\]](#)
5. O'Neill, D.G.; Church, D.B.; McGreevy, P.D.; Thomson, P.C.; Brodbelt, D.C. Prevalence of Disorders Recorded in Cats Attending Primary-Care Veterinary Practices in England. *Vet. J.* **2014**, *202*, 286–291. [\[CrossRef\]](#)
6. Xavier, R. Risk Factors in Dogs and Cats for Development of Chronic Kidney Disease. Available online: <https://www.iris-kidney.com/ckd-risk-factors> (accessed on 10 January 2026).
7. Marino, C.L.; Lascelles, B.D.X.; Vaden, S.L.; Gruen, M.E.; Marks, S.L. Prevalence and Classification of Chronic Kidney Disease in Cats Randomly Selected from Four Age Groups and in Cats Recruited for Degenerative Joint Disease Studies. *J. Feline Med. Surg.* **2014**, *16*, 465–472. [\[CrossRef\]](#)
8. Conroy, M.; Brodbelt, D.C.; O'Neill, D.; Chang, Y.; Elliott, J. Chronic Kidney Disease in Cats Attending Primary Care Practice in the UK: A VetCompass™ Study. *Vet. Rec.* **2019**, *184*, 526. [\[CrossRef\]](#) [\[PubMed\]](#)
9. Bartges, J.W. Chronic Kidney Disease in Dogs and Cats. *Vet. Clin. N. Am. Small Anim. Pract.* **2012**, *42*, 669–692. [\[CrossRef\]](#) [\[PubMed\]](#)
10. Jepson, R.E. Current Understanding of the Pathogenesis of Progressive Chronic Kidney Disease in Cats. *Vet. Clin. N. Am. Small Anim. Pract.* **2016**, *46*, 1015–1048. [\[CrossRef\]](#)
11. DiBartola, S.P.; Rutgers, H.C.; Zack, P.M.; Tarr, M.J. Clinicopathologic Findings Associated with Chronic Renal Disease in Cats: 74 Cases (1973–1984). *J. Am. Vet. Med. Assoc.* **1987**, *190*, 1196–1202. [\[CrossRef\]](#) [\[PubMed\]](#)
12. Elliott, J.; Barber, P.J. Feline Chronic Renal Failure: Clinical Findings in 80 Cases Diagnosed between 1992 and 1995. *J. Small Anim. Pract.* **1998**, *39*, 78–85. [\[CrossRef\]](#)
13. Greene, J.P.; Lefebvre, S.L.; Wang, M.; Yang, M.; Lund, E.M.; Polzin, D.J. Risk Factors Associated with the Development of Chronic Kidney Disease in Cats Evaluated at Primary Care Veterinary Hospitals. *J. Am. Vet. Med. Assoc.* **2014**, *244*, 320–327. [\[CrossRef\]](#) [\[PubMed\]](#)
14. Finch, N.C.; Syme, H.M.; Elliott, J. Risk Factors for Development of Chronic Kidney Disease in Cats. *Vet. Intern. Med.* **2016**, *30*, 602–610. [\[CrossRef\]](#) [\[PubMed\]](#)
15. White, J.D.; Malik, R.; Norris, J.M.; Malikides, N. Association between Naturally Occurring Chronic Kidney Disease and Feline Immunodeficiency Virus Infection Status in Cats. *J. Am. Vet. Med. Assoc.* **2010**, *236*, 424–429. [\[CrossRef\]](#)
16. Kongtasai, T.; Paepe, D.; Meyer, E.; Mortier, F.; Marynissen, S.; Stammeleer, L.; Defauw, P.; Daminet, S. Renal Biomarkers in Cats: A Review of the Current Status in Chronic Kidney Disease. *Vet. Intern. Med.* **2022**, *36*, 379–396. [\[CrossRef\]](#)
17. Paepe, D.; Daminet, S. Feline CKD: Diagnosis, Staging and Screening—What Is Recommended? *J. Feline Med. Surg.* **2013**, *15*, 15–27. [\[CrossRef\]](#) [\[PubMed\]](#)
18. Sparkes, A.H.; Caney, S.; Chalhoub, S.; Elliott, J.; Finch, N.; Gajanayake, I.; Langston, C.; Lefebvre, H.P.; White, J.; Quimby, J. ISFM Consensus Guidelines on the Diagnosis and Management of Feline Chronic Kidney Disease. *J. Feline Med. Surg.* **2016**, *18*, 219–239. [\[CrossRef\]](#)
19. Quimby, J.M. Searching for Biomarkers in Feline Chronic Kidney Disease: A New Frontier. *Vet. J.* **2015**, *206*, 3–4. [\[CrossRef\]](#)
20. Pereira, A.F.; Jota Baptista, C.; Faustino-Rocha, A.; Oliveira, P.A.; Coelho, A.C. Renal Biomarkers in Companion Animals—A Review. *Animals* **2025**, *15*, 818. [\[CrossRef\]](#)

21. Polzin, D.J. Chronic Kidney Disease in Small Animals. *Vet. Clin. N. Am. Small Anim. Pract.* **2011**, *41*, 15–30. [[CrossRef](#)]
22. Bartlett, P.C.; Van Buren, J.W.; Bartlett, A.D.; Zhou, C. Case-Control Study of Risk Factors Associated with Feline and Canine Chronic Kidney Disease. *Vet. Med. Int.* **2010**, *2010*, 957570. [[CrossRef](#)]
23. Bijmans, E.S.; Jepson, R.E.; Syme, H.M.; Elliott, J.; Niessen, S.J.M. Psychometric Validation of a General Health Quality of Life Tool for Cats Used to Compare Healthy Cats and Cats with Chronic Kidney Disease. *Vet. Intern. Med.* **2016**, *30*, 183–191. [[CrossRef](#)]
24. Hays, R.D.; Kallich, J.D.; Mapes, D.L.; Coons, S.J.; Carter, W.B. Development of the Kidney Disease Quality of Life (KDQOLTM) Instrument. *Qual. Life Res.* **1994**, *3*, 329–338. [[CrossRef](#)] [[PubMed](#)]
25. Porter, A.; Fischer, M.J.; Wang, X.; Brooks, D.; Bruce, M.; Charleston, J.; Cleveland, W.H.; Dowie, D.; Faulkner, M.; Gassman, J.; et al. Quality of Life and Outcomes in African Americans with CKD. *J. Am. Soc. Nephrol.* **2014**, *25*, 1849–1855. [[CrossRef](#)]
26. Mujais, S.K.; Story, K.; Brouillette, J.; Takano, T.; Soroka, S.; Franek, C.; Mendelssohn, D.; Finkelstein, F.O. Health-Related Quality of Life in CKD Patients: Correlates and Evolution over Time. *Clin. J. Am. Soc. Nephrol.* **2009**, *4*, 1293–1301. [[CrossRef](#)] [[PubMed](#)]
27. Spitznagel, M.B.; Jacobson, D.M.; Cox, M.D.; Carlson, M.D. Caregiver Burden in Owners of a Sick Companion Animal: A Cross-sectional Observational Study. *Vet. Rec.* **2017**, *181*, 321. [[CrossRef](#)] [[PubMed](#)]
28. Liu, F.; Panagiotakos, D. Real-World Data: A Brief Review of the Methods, Applications, Challenges and Opportunities. *BMC Med. Res. Methodol.* **2022**, *22*, 287. [[CrossRef](#)]
29. U.S. Food and Drug Administration (FDA). Real-World Evidence. Available online: <https://www.fda.gov/science-research/science-and-research-special-topics/real-world-evidence> (accessed on 30 January 2026).
30. Wright, A.; Taylor, D.; Lowe, M.; Barlow, S.; Jackson, J. Replicating the Real-World Evidence Methods Available in Human Health to Assess Burden and Outcomes for Dogs with Chronic Kidney Disease, Their Owners, and the Veterinary Healthcare System in the United States of America. *Front. Vet. Sci.* **2025**, *12*, 1502933. [[CrossRef](#)]
31. Chortara, I.; Tsokana, C.N.; Pavlidou, E.; Sandri, T.L.; Wright, A.; Valiakos, G. Real-World Data on Canine Chronic Kidney Disease in Greece: Clinical and Quality of Life Insights. *Front. Vet. Sci.* **2025**, *12*, 1601044. [[CrossRef](#)]
32. Santos-Araújo, C.; Mendonça, L.; Carvalho, D.S.; Bernardo, F.; Pardal, M.; Couceiro, J.; Martinho, H.; Gavina, C.; Taveira-Gomes, T.; Dinis-Oliveira, R.J. Twenty Years of Real-World Data to Estimate Chronic Kidney Disease Prevalence and Staging in an Unselected Population. *Clin. Kidney J.* **2023**, *16*, 111–124. [[CrossRef](#)]
33. Li, J.; An, J.; Huang, M.; Zhou, M.; Montez-Rath, M.E.; Niu, F.; Sim, J.J.; Pao, A.C.; Charu, V.; Odden, M.C.; et al. Representation of Real-World Adults With Chronic Kidney Disease in Clinical Trials Supporting Blood Pressure Treatment Targets. *J. Am. Heart Assoc.* **2024**, *13*, e031742. [[CrossRef](#)]
34. Rottura, M.; Drago, S.F.A.; Gianguzzo, V.M.; Molonia, A.; Pallio, G.; Scoglio, R.; Marino, S.; Alibrandi, A.; Imbalzano, E.; Squadrito, F.; et al. Chronic Kidney Disease Progression in Diabetic Patients: Real World Data in General Practice. *Heliyon* **2024**, *10*, e30787. [[CrossRef](#)]
35. Lorbach, S.K.; Quimby, J.M.; Nijveldt, E.; Paschall, R.E.; Scott, E.M.; Reid, J. Evaluation of Health-Related Quality of Life in Cats with Chronic Kidney Disease. *J. Feline Med. Surg.* **2025**, *27*, 1098612X251367535. [[CrossRef](#)]
36. Scott, E.M.; Davies, V.; Nolan, A.M.; Noble, C.E.; Dowgray, N.J.; German, A.J.; Wiseman-Orr, M.L.; Reid, J. Validity and Responsiveness of the Generic Health-Related Quality of Life Instrument (VetMetrica™) in Cats With Osteoarthritis. Comparison of Vet and Owner Impressions of Quality of Life Impact. *Front. Vet. Sci.* **2021**, *8*, 733812. [[CrossRef](#)]
37. Davies, V.; Reid, J.; Scott, E.M. Optimisation of Scores Generated by an Online Feline Health-Related Quality of Life (HRQL) Instrument to Assist the Veterinary User Interpret Its Results. *Front. Vet. Sci.* **2021**, *7*, 601304. [[CrossRef](#)]
38. Valiakos, G.; Pavlidou, E.; Zafeiridis, C.; Tsokana, C.N.; Del Rio Vilas, V.J. Antimicrobial Practices among Small Animal Veterinarians in Greece: A Survey. *One Health Outlook* **2020**, *2*, 7. [[CrossRef](#)] [[PubMed](#)]
39. ISO/IEC 17025:2017; General requirements for the competence of testing and calibration laboratories. International Organization for Standardization (ISO): Geneva, Switzerland, 2017.
40. ISO 9001:2015; Quality Management Systems—Requirements. International Organization of Standardization (ISO): Geneva, Switzerland, 2015.
41. Greenberg, N.; Roberts, W.L.; Bachmann, L.M.; Wright, E.C.; Dalton, R.N.; Zakowski, J.J.; Greg Miller, W. Specificity Characteristics of 7 Commercial Creatinine Measurement Procedures by Enzymatic and Jaffe Method Principles. *Clin. Chem.* **2012**, *58*, 391–401. [[CrossRef](#)]
42. Cobbaert, C.M.; Baadenhuijsen, H.; Weykamp, C.W. Prime Time for Enzymatic Creatinine Methods in Pediatrics. *Clin. Chem.* **2009**, *55*, 549–558. [[CrossRef](#)] [[PubMed](#)]
43. International Renal Interest Society. IRIS Staging of Chronic Kidney Disease (CKD) Guidelines. World Small Animal Veterinary Association/IRIS. 2023. Available online: <https://www.iris-kidney.com/iris-staging-system> (accessed on 30 January 2026).

44. WSAVA Nutritional Assessment Guidelines Task Force Members; Freeman, L.; Becvarova, I.; Cave, N.; MacKay, C.; Nguyen, P.; Rama, B.; Takashima, G.; Tiffin, R.; Tsujimoto, H.; et al. WSAVA Nutritional Assessment Guidelines. *J. Small Anim. Pract.* **2011**, *52*, 385–396. [[CrossRef](#)] [[PubMed](#)]
45. World Small Animal Veterinary Association (WSAVA) Global Nutrition Committee. Muscle Condition Score Chart—Cat. WSAVA Global Nutrition Guidelines. 2020. Available online: <https://wsava.org/wp-content/uploads/2020/01/Muscle-Condition-Score-Chart-for-Cats.pdf> (accessed on 30 January 2026).
46. World Small Animal Veterinary Association (WSAVA) Global Nutrition Committee. Body Condition Score—Cat (Updated August 2020). WSAVA Global Nutrition Guidelines. 2020. Available online: <https://wsava.org/wp-content/uploads/2020/08/Body-Condition-Score-cat-updated-August-2020.pdf> (accessed on 30 January 2026).
47. Chalhoub, S.; Langston, C.; Eatroff, A. Anemia of Renal Disease: What It Is, What to Do and What's New. *J. Feline Med. Surg.* **2011**, *13*, 629–640. [[CrossRef](#)]
48. Cowgill, L.D.; James, K.M.; Levy, J.K.; Browne, J.K.; Miller, A.; Lobingier, R.T.; Egrie, J.C. Use of Recombinant Human Erythropoietin for Management of Anemia in Dogs and Cats with Renal Failure. *J. Am. Vet. Med. Assoc.* **1998**, *212*, 521–528. [[CrossRef](#)] [[PubMed](#)]
49. Chakrabarti, S.; Syme, H.M.; Elliott, J. Clinicopathological Variables Predicting Progression of Azotemia in Cats with Chronic Kidney Disease. *Vet. Intern. Med.* **2012**, *26*, 275–281. [[CrossRef](#)]
50. Boyd, L.M.; Langston, C.; Thompson, K.; Zivin, K.; Imanishi, M. Survival in Cats with Naturally Occurring Chronic Kidney Disease (2000–2002). *Vet. Intern. Med.* **2008**, *22*, 1111–1117. [[CrossRef](#)]
51. Trevejo, R.T.; Lefebvre, S.L.; Yang, M.; Rhoads, C.; Goldstein, G.; Lund, E.M. Survival Analysis to Evaluate Associations between Periodontal Disease and the Risk of Development of Chronic Azotemic Kidney Disease in Cats Evaluated at Primary Care Veterinary Hospitals. *J. Am. Vet. Med. Assoc.* **2018**, *252*, 710–720. [[CrossRef](#)]
52. Vetter, C.A.M.; Lourenço, B.N.; Meindl, A.G.; Mees, A.; Stone, A.; Farnsworth, R.; Everett, S.E.; Gonzalez, S.E.; Roth, I.G.; Schmiedt, C. Cross-Sectional Characterization of Renal Function in Cats with Caudal Stomatitis. *J. Feline Med. Surg.* **2023**, *25*, 1098612X231179883. [[CrossRef](#)] [[PubMed](#)]
53. Deschamps-Lenhardt, S.; Martin-Cabezas, R.; Hannedouche, T.; Huck, O. Association between Periodontitis and Chronic Kidney Disease: Systematic Review and Meta-analysis. *Oral Dis.* **2019**, *25*, 385–402. [[CrossRef](#)]
54. Li, S.; Cao, H.; Zhang, Y.; Wang, F.; Huang, G.; Wang, B.; Wei, W.; Wang, G. Periodontal Disease and Chronic Kidney Disease: Mechanistic Insights and Novel Therapeutic Perspectives. *Front. Cell. Infect. Microbiol.* **2025**, *15*, 1611097. [[CrossRef](#)]
55. Hartmann, K.; Pennisi, M.G.; Dorsch, R. Infectious Agents in Feline Chronic Kidney Disease What Is the Evidence? *Adv. Small Anim. Care* **2020**, *1*, 189–206. [[CrossRef](#)]
56. Baxter, K.J.; Levy, J.K.; Edinboro, C.H.; Vaden, S.L.; Tompkins, M.B. Renal Disease in Cats Infected with Feline Immunodeficiency Virus. *Vet. Intern. Med.* **2012**, *26*, 238–243. [[CrossRef](#)] [[PubMed](#)]
57. Little, S.; Levy, J.; Hartmann, K.; Hofmann-Lehmann, R.; Hosie, M.; Olah, G.; Denis, K.S. 2020 AAFP Feline Retrovirus Testing and Management Guidelines. *J. Feline Med. Surg.* **2020**, *22*, 5–30. [[CrossRef](#)] [[PubMed](#)]
58. Freeman, L.M.; Lachaud, M.-P.; Matthews, S.; Rhodes, L.; Zollers, B. Evaluation of Weight Loss Over Time in Cats with Chronic Kidney Disease. *Vet. Intern. Med.* **2016**, *30*, 1661–1666. [[CrossRef](#)]
59. Parker, V.J. Nutritional Management for Dogs and Cats with Chronic Kidney Disease. *Vet. Clin. N. Am. Small Anim. Pract.* **2021**, *51*, 685–710. [[CrossRef](#)]
60. Freeman, L.M. Cachexia and Sarcopenia: Emerging Syndromes of Importance in Dogs and Cats. *J. Vet. Intern. Med.* **2012**, *26*, 3–17. [[CrossRef](#)]
61. Hall, J.A.; Fritsch, D.A.; Jewell, D.E.; Burris, P.A.; Gross, K.L. Cats with IRIS Stage 1 and 2 Chronic Kidney Disease Maintain Body Weight and Lean Muscle Mass When Fed Food Having Increased Caloric Density, and Enhanced Concentrations of Carnitine and Essential Amino Acids. *Vet. Rec.* **2019**, *184*, 190. [[CrossRef](#)]
62. Vogt, A.H.; Rodan, I.; Brown, M.; Brown, S.; Buffington, C.A.T.; Forman, M.J.L.; Neilson, J.; Sparkes, A. AAFP-AAHA: Feline Life Stage Guidelines. *J. Feline Med. Surg.* **2010**, *12*, 43–54. [[CrossRef](#)] [[PubMed](#)]
63. Tarrant, G.; Boyden, L.; Rai, T.; Wright, A.; Cook, A.J.C.; Wells, K. Gaining Insights into Pet Owner Understanding/Lived Experience of Canine Chronic Kidney Disease Using Survey and Social Media Data. *Front. Vet. Sci.* **2025**, *12*, 1506272. [[CrossRef](#)]
64. Elliott, J.; Reyes-Hughes, H.; Hibbert, A.; Blackwell, E.; Finch, N.C. Owners' Experiences of Caring for Cats with Chronic Kidney Disease in the UK. *J. Feline Med. Surg.* **2025**, *27*, 1098612X251314769. [[CrossRef](#)] [[PubMed](#)]
65. Baral, R.M.; Dhand, N.K.; Freeman, K.P.; Krockenberger, M.B.; Govendir, M. Biological Variation and Reference Change Values of Feline Plasma Biochemistry Analytes. *J. Feline Med. Surg.* **2014**, *16*, 317–325. [[CrossRef](#)] [[PubMed](#)]
66. Bellows, J.; Center, S.; Daristotle, L.; Estrada, A.H.; Flickinger, E.A.; Horwitz, D.F.; Lascelles, B.D.X.; Lepine, A.; Perea, S.; Scherk, M.; et al. Evaluating Aging in Cats: How to Determine What Is Healthy and What Is Disease. *J. Feline Med. Surg.* **2016**, *18*, 551–570. [[CrossRef](#)]

67. Dowgray, N.; Pinchbeck, G.; Eyre, K.; Biourge, V.; Comerford, E.; German, A.J. Aging in Cats: Owner Observations and Clinical Finding in 206 Mature Cats at Enrolment to the Cat Prospective Aging and Welfare Study. *Front. Vet. Sci.* **2022**, *9*, 859041. [[CrossRef](#)]
68. Markovich, J.E.; Freeman, L.M.; Labato, M.A.; Heinze, C.R. Survey of Dietary and Medication Practices of Owners of Cats with Chronic Kidney Disease. *J. Feline Med. Surg.* **2015**, *17*, 979–983. [[CrossRef](#)] [[PubMed](#)]

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Contribution of polycystic kidney disease to the development of chronic renal dysfunction in cats

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Abstract

Polycystic kidney disease (PKD) is a hereditary disorder characterized by progressive renal cyst formation, leading to compromised kidney function and an increased risk of chronic kidney disease (CKD). This prospective, 18-month longitudinal study evaluated diagnostic and prognostic markers in 12 PKD-affected cats, utilizing ultrasound imaging alongside biochemical analysis. The study documented a 14.00% increase in cyst volume, a 7.00% enlargement in kidney length, and significant elevations in plasma creatinine, phosphorus, and blood urea nitrogen levels, underscoring their correlation with disease progression. Hematological analysis revealed progressive anemia associated with advanced CKD stages, further outlining the systemic impact of PKD. These findings emphasize the progressive nature of CKD in cats with PKD, despite diligent care provided by pet owners and veterinarians. Also, the importance of routine sonographic evaluation along with hematological and biochemical assessments is highlighted in our study to enhance early detection and improve outcomes in feline PKD. Future longitudinal studies with

expanded cohorts are essential to validate these findings and refine diagnostic criteria for PKD-associated CKD.

Keywords: Cat, Chronic kidney disease, Polycystic kidney disease, Prognosis, Ultrasound

Introduction

Chronic kidney disease (CKD) is the most frequently diagnosed renal disorder in cats, characterized by a structural or functional abnormality in one or both kidneys persisting for more than three months.¹ The International Renal Interest Society staging system is widely used to classify the severity of CKD in dogs and cats. This system primarily relies on fasting blood creatinine concentrations in a well-hydrated and stable patient to assign a stage ranging from one to four. In stage 1 (non-azotemic CKD) creatinine levels are below 1.60 mg dL^{-1} and clinical signs are often absent. In Stage 2, creatinine levels range from 1.60 to 2.80 mg dL^{-1} and early clinical signs such as polyuria and polydipsia may appear. Stage 3 (moderate CKD) is characterized by creatinine levels between 2.90 and 5.00 mg dL^{-1} with clinical signs potentially progressing to anorexia, vomiting and weight loss. Stage 4, known as end-stage kidney disease is identified by creatinine levels exceeding 5.00 mg dL^{-1} and a poor quality of life for the cat.²

Azotemia is characterized by elevated concentrations of serum creatinine and urea, making it a key diagnostic marker. Creatinine serves as a more reliable indicator of glomerular filtration rate because it is minimally influenced by extra renal factors such as diet or hepatic function.³

Polycystic kidney disease (PKD) is an inherited disorder that causes the formation of fluid-filled cysts in the kidneys and, in some cases, other organs such as the liver and pancreas. Diagnostic approaches, including ultrasound imaging and genetic testing, are currently the most reliable tools for diagnosing and monitoring PKD.⁴ Studies on human patients with PKD have demonstrated significant correlations between imaging factors, such as kidney length, volume, and cyst size, and the progression of the disease. These findings suggest that similar markers could be applicable in feline populations.⁵

Ultrasound is a non-invasive, widely used method to assess kidney size, function, and abnormalities. This technique is favored for its accessibility, safety, and effectiveness in clinical settings. While ultrasound is not fully accurate for determining precise kidney and cyst volumes, its consistent measurement accuracy makes it reliable for comparative analysis.⁶ Therefore, the current study focused on relative changes in kidney and cyst volume rather than their exact sizes.

Cats provide a suitable model for studying PKD progression due to similarities in disease manifestation and the feasibility of applying imaging techniques commonly used in human medicine.⁷ By advancing our understanding of the relationship between clinical and imaging markers in feline PKD, this research aims to contribute to both veterinary and comparative medical knowledge. This study investigates the progression of CKD in cats diagnosed with PKD, focusing on the predictive value of blood biomarkers for CKD and their association with imaging parameters, including kidney length, volume, and cyst characteristics.

Methods and Materials

Study design.

This research is a prospective study that was conducted on 198 owned cats at several veterinary clinics and teaching veterinary hospitals of Zabol University and Shahid Bahonar University of Kerman.

Selection criteria.

The cats included on this study had a diagnosis of PKD confirmed through veterinary referrals for routine checkups, the presence of clinical symptoms, or a documented history of PKD in their medical records. Inclusion was not dependent on the cats' sex, breed, or age. To ensure the integrity and feasibility of the study, several exclusion criteria were implemented. Cats diagnosed with end-stage renal disease (Stage 4) with a limited life expectancy were not included. Furthermore, cats which owners were unable to commit to scheduled follow-up appointments or did not provide informed consent for their cat's participation were also excluded. Diagnostic ultrasound studies (UMT-160; Mindray, Shenzhen, China) were performed on referred cats after obtaining written consent from their owners to include in the study. Once PKD was confirmed through ultrasound imaging, blood and urine samples were collected for analysis of biochemical parameters, complete blood count, and urinalysis (on the 1st day of reception). During the initial 6 months of the study, a total of 12 cats with PKD were identified and included in the analysis.

Monitoring process for polycystic kidney disease (PKD) cases.

Affected cats were evaluated at least twice: initially (on the 1st day of reception) and at the end of the study (18 months later) between 2021 and 2023. Each examination included tests for biochemical parameters, complete blood count, ultrasound imaging, urinalysis, and clinical signs,

all of which were meticulously recorded. Ultrasound evaluations were conducted to document changes in cyst size, number, and kidney volume. Moreover, complete blood count, biochemical parameters and urine analysis were monitored again after 18 months in selected cats. In some cases, based on the veterinarian's prescription and clinical findings, additional evaluations (up to four) were conducted at regular intervals. However, due to limited data availability for some cases, conclusions were primarily drawn from the initial and final evaluations. During the ultrasound studies, morphological characteristics of cysts including their location (cortex or medulla), size, and number were meticulously recorded. Changes in kidney and cyst volume over time were determined using ultrasound. All patients were prepared by clipping the hair coat and applying acoustic coupling gel. The kidneys were scanned with the patient in dorsal and lateral recumbency. The left and right kidneys were imaged through the left and right ventral abdominal wall, just caudal to the ribs. A Micro Convex transducer (7.00 - 10.00 MHz) and color Doppler ultrasound machine (Mindray) was used. The treatments administered to affected cats were heterogeneous, including dietary modifications, fluid therapy, antihypertensive medications, and calcium channel blockers guided by referring veterinarians or our internist. Due to the frequent changes in therapeutic and dietary regimens observed in most cats throughout the 18-month study, the research team did not standardize or include these treatments in the analysis. Instead, our focus was on characterizing disease progression using laboratory and sonographic evaluations, independent of therapeutic interventions.

Measuring kidney volume.

Kidney volume was estimated using measurements of length, width, and depth, assuming an ellipsoidal shape. The kidney volume was estimated using the standard ellipsoid formula:

$$V = (\pi \times L \times W \times D) / 6$$

where L , W , and D represent length, width, and depth, respectively. This method is consistent with veterinary ultrasonographic practices for renal volume estimation.⁶

Determining CKD staging.

The International Renal Interest Society staging system was employed to classify CKD severity in PKD-affected cats.¹ The CKD stages were determined based on serum creatinine levels and clinical findings, with Stage 4 not being included.

Biochemical analysis.

Blood samples collected in non-anticoagulant tubes were centrifuged at room temperature for 30 min at 503 g to separate the serum. Biochemical parameters, including creatinine, urea, phosphate, calcium, albumin, total protein, alanine aminotransferase, and gamma-glutamyl transferase, were measured using an automated biochemical analyzer and the following commercial kits were used: creatinine (Jaffé method; Pars Azmoon, Tehran, Iran),⁸ urea/blood urea nitrogen (BUN, urease-glutamate dehydrogenase method; Zist Shimi, Tehran, Iran),⁸ phosphate (acid phosphatase method; Pars Azmoon),⁸ calcium (Arsenazo III method; Pars Azmoon),⁹ albumin (Bromocresol green method; Zist Shimi),¹⁰ total protein (Biuret method; Pars Azmoon),¹¹ alanine amino-transferase (Kinase method; Pars Azmoon), and gamma-glutamyl transferase (Kinase method; Pars Azmoon).¹²

Hematology analysis.

Hematology parameters, including hematocrit, red blood cell count, mean corpuscular volume, lymphocytes, and neutrophils, were measured using an automated hematology analyzer with specific kits for this device (Hema-21; Nsbiotec, Malaga, Spain). Samples were collected in tubes containing EDTA and analyzed within 2 hr of collection.

Urine analysis.

Urine analysis was conducted to assess specific gravity, proteinuria, and the presence of abnormal cells, employing the following procedures. Specific gravity was determined through direct measurement using a handheld refractometer. Proteinuria was evaluated via urine dipsticks, with the results interpreted based on the provided colorimetric scale. For urine sediment microscopy, samples were first centrifuged at 1,500 revolutions *per* minute for 5 min to concentrate the cellular components. The resulting sediment was then examined under a light microscope at a magnification of 400 × to identify any abnormal cellular elements.

Statistical analysis.

The dependent variables measured in this study included serum creatinine, BUN, albumin, phosphate, and calcium levels, hematological parameters, urinalysis results, and ultrasound characteristics such as renal cyst size and number. Changes in the mean values of these parameters from baseline to study completion were analyzed using paired t-tests. Comparisons between cats with PKD and healthy controls were performed using independent t-tests, with Welch's t-test

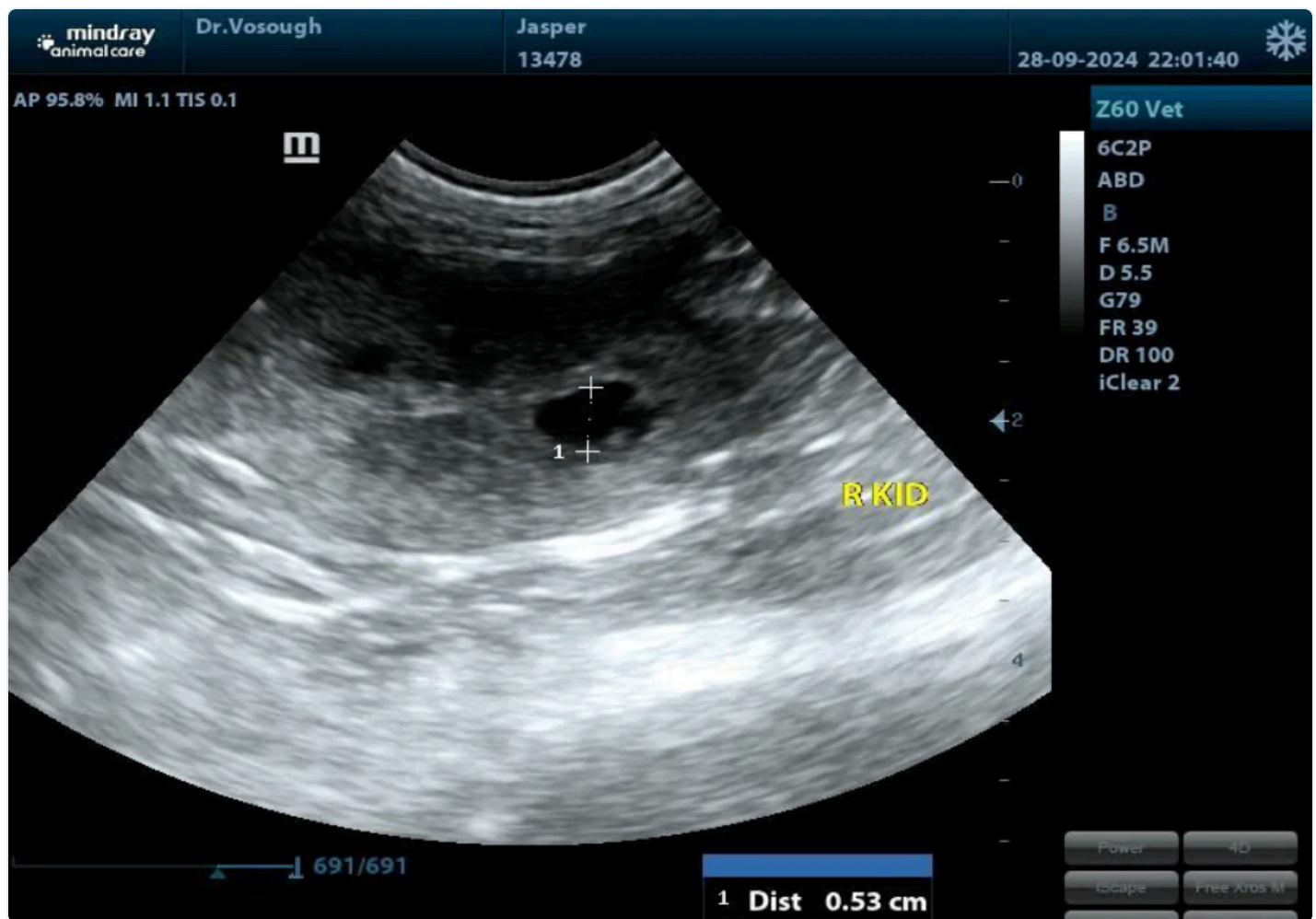
applied for comparisons with unequal variances. All statistical analyses were conducted using IBM SPSS Software (version 29.0; IBM Corp., Armonk, USA).

Results

Population overview.

Out of 198 cats referred for examination, 12 were diagnosed with PKD via ultrasonography ([Fig. 1](#)). The cohort consisted of Persian (n = 92), Himalayan (n = 12), Scottish (n = 8), and domestic shorthair (n = 86) cats, with PKD exclusively identified in Persian (n = 11) and domestic shorthair (n = 1) cats. At baseline, CKD stages were determined using the International Renal Interest Society criteria. Among the PKD-affected cats, 2 (16.66%) were classified as stage 3, 5 (41.66%) as stage 2, and 5 (41.66%) as stage 1. Bilateral kidney involvement was noted in 91.66% (n = 11) of affected cats. All cats participated in follow-up evaluations throughout the study period, except for one 6- year-old cat which entered the study on the first day of reception with stage 3 CKD. This cat progressed to end-stage renal failure by the ninth month, ultimately resulting in death due to end stage kidney disease.

Fig. 1.



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Sagittal plan image of the right kidney in a domestic short hair cat. There is single cyst within the renal cortex. The cyst has anechoic contents, is well defined and the cyst wall blends with the renal cortex.

Ultrasound indicators.

The comparison of cyst dimensions measured at the beginning and end of the study, based on periodic ultrasound assessments, revealed an average increase in cyst volume of 14.00% over the 18-month period. Kidney length measurements revealed a significant increase in PKD-affected cats over the study period. At baseline (day 0), the mean kidney length in PKD-affected cats was $4.20 \pm$

0.70 cm), compared to 3.90 ± 0.60 cm) in healthy controls. By the final day, kidney length in PKD-affected cats had increased by 7.00% (4.50 ± 0.6 cm). In one case, a cyst was exclusively detected in the right kidney, which remained cyst-free throughout the study. In seven cases, cysts were only found in the cortical region of the kidney, while in the remaining cases, cysts were present in both the cortical and medullary regions. An analysis of survival and lifespan within the study cohort revealed that one subject reached the endpoint of kidney survival, ultimately succumbing to kidney failure at month nine. To the authors' knowledge, another subject reached the endpoint of kidney survival—either through euthanasia or death resulting from kidney failure—three months after the study concluded.

Plasma biochemistry factors.

The evaluation of biochemical factors in cases of PKD at baseline (first day of reception) revealed that 16.66% of subjects had concentrations of creatinine (> 1.60 mg dL⁻¹), phosphorus (4.50 mg dL⁻¹), and blood urea nitrogen (35.0 mg dL⁻¹) above standard reference values. Notably, elevated creatinine concentrations alone were observed in 58.33% of cases. Over the course of the study, an increase in plasma levels of creatinine, phosphorus, and BUN was detected, and the mean changes in these variables were statistically significant ($p < 0.05$; [Table 1](#)). This stability suggests that calcium metabolism may not be as directly impacted by PKD progression as other biochemical markers phosphorus or creatinine.

Table 1.

General data and comparison of biochemical factors of healthy group versus cats with polycystic kidney disease (PKD) at two times: Initial and the final day of the study. Data are presented as mean $\pm$ SD.

Variables	Healthy (Initial day)	PKD (Initial day)	PKD (Final day)	<i>p</i> -values		
				Healthy vs. PKD initial day	Healthy vs. PKD final day	PKD initial day vs. final day
Age (months)	44.00 $\pm$ 9.00	52.00 $\pm$ 7.00	70.00 $\pm$ 7.00	–	–	–
Weight (kg)	3.80 $\pm$ 1.20	3.80 $\pm$ 0.80	3.10 $\pm$ 1.20	–	–	–
Creatinine (mg dL⁻¹)	1.48 $\pm$ 0.39	1.82 $\pm$ 0.75	2.81 $\pm$ 1.31	0.109	< 0.001	< 0.001
BUN (mg dL⁻¹)	26.70 $\pm$ 4.49	29.70 $\pm$ 4.93	36.70 $\pm$ 9.30	0.093	< 0.001	0.013
ALP (U L⁻¹)	62.90 $\pm$ 11.54	67.08 $\pm$ 15.29	59.62 $\pm$ 16.19	0.399	0.510	0.202
GGT (U L⁻¹)	2.88 $\pm$ 0.64	3.64 $\pm$ 0.73	3.61 $\pm$ 0.80	0.357	0.003	0.153
Total Protein (g dL⁻¹)	6.71 $\pm$ 0.44	7.10 $\pm$ 0.46	7.50 $\pm$ 0.66	0.025	< 0.001	0.053
Phosphate (mg dL⁻¹)	3.46 $\pm$ 1.09	3.51 $\pm$ 1.23	4.40 $\pm$ 1.22	0.908	0.043	< 0.001
Calcium (mg dL⁻¹)	9.61 $\pm$ 0.29	10.09 $\pm$ 1.33	10.75 $\pm$ 1.5	0.347	0.009	0.179

Variables	Healthy (Initial day)	PKD (Initial day)	PKD (Final day)	<i>p</i> -values		
				Healthy vs. PKD initial day	Healthy vs. PKD final day	PKD initial day vs. final day
Albumin (g dL⁻¹)	3.06 ± 0.46	3.40 ± 0.80	2.98 ± 0.48	0.418	0.615	0.551

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BUN: Blood urea nitrogen; ALP: Alkaline phosphatase; GGT: Gamma-glutamyl transferase.

The lack of significant alterations may also indicate compensatory mechanisms maintaining calcium homeostasis despite declining kidney function. When comparing the PKD-affected cases to the healthy control group ([Table 1](#)), distinct differences emerged. The control group was evaluated at two time points: baseline (1st day of reception) and the final day of the study. A statistical comparison between the baseline results of the control group and the day zero results of the PKD group revealed no significant differences in any of the measured biochemical factors. However, comparisons between the control group and the final study day revealed significant differences in mean levels of creatinine, BUN, phosphorus, gamma-glutamyl transferase enzyme, total protein, and even calcium ($p < 0.05$; [Table 1](#)).

Urinalysis.

Urinalysis of the affected cases included an assessment of urine specific gravity and the presence of proteinuria. On the initial day of the study (day zero), the mean urine specific gravity was 1.020, and proteinuria was detected in four cases, accounting for 33.33% of the total. By the conclusion of the study, the mean specific gravity had increased to 1.025, with proteinuria persisting in the same four cases (33.33%).

Hematology.

Hematologic parameters were compared between healthy cats and those with PKD+. Although some variables showed no significant differences at the beginning of the study (initial day), significant reductions in hematocrit and erythrocyte counts were observed in PKD+ cats by the final day. These findings suggest progressive anemia in affected individuals. Other indices, including mean corpuscular volume, mean corpuscular hemoglobin, mean corpuscular hemoglobin concentration, lymphocyte percentage, and neutrophil band counts, did not differ significantly across groups or time points. Notably, as CKD progressed, erythrocyte counts and hematocrit levels demonstrated a decreasing trend. Detailed values and statistical comparisons are provided in [Table 2](#).

Table 2.

Hematologic parameters comparison between healthy cats and cats with polycystic kidney disease (PKD) at two times: Initial and the final day of the study. Data are presented as mean $\pm$ SD.

Variables	Healthy (Initial day)	PKD (Initial day)	PKD (Final day)	<i>p</i> -values		
				Healthy vs. PKD initial day	Healthy vs. PKD final day	PKD initial day vs. final day
Hematocrit (%)	38.90 $\pm$ 4.80	38.90 $\pm$ 4.80	28.80 $\pm$ 4.70	0.824	< 0.01	0.003
Erythrocytes ($\times 10^6$ per μL)	7.66 $\pm$ 1.23	7.33 $\pm$ 1.38	5.23 $\pm$ 1.71	0.487	< 0.01	0.05
MCV (fL)	40.70 $\pm$ 1.40	40.6 $\pm$ 1.40	40.20 $\pm$ 3.80	0.966	0.829	0.713
MCH (pg)	15.80 $\pm$ 0.90	15.60 $\pm$ 1.10	15.40 $\pm$ 1.30	0.680	0.420	0.680
MCHC (g dL⁻¹)	31.20 $\pm$ 0.80	31.00 $\pm$ 0.90	30.08 $\pm$ 1.10	0.710	0.320	0.710
Lymphocytes (%)	42.30 $\pm$ 1.60	43.30 $\pm$ 1.60	43 $\pm$ 1.40	0.581	0.644	0.252
Band neutrophils (%)	7.40 $\pm$ 1.60	6.90 $\pm$ 1.70	6.90 $\pm$ 2.70	0.926	0.870	0.729
Segmented neutrophils (%)	52.30 $\pm$ 1.40	51.50 $\pm$ 1.40	50.24 $\pm$ 1.10	0.669	0.660	0.201

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MCV: Mean corpuscular volume; MCH: Mean corpuscular hemoglobin; MCHC: Mean corpuscular hemoglobin concentration.

Clinical observations.

During periodic veterinary examinations, clinical symptoms such as anorexia, weakness, vomiting, and diarrhea were documented in a subset of cats affected by PKD. Anorexia and weakness were the most frequently observed symptoms, noted in 33.33% (n = 4) of cases. Vomiting and diarrhea were less common, occurring sporadically in 16.66% (n = 2) of affected cats. Symptoms did not correlate uniformly with CKD stage or severity, and their occurrence varied widely among individuals. No significant associations were identified between clinical signs and biochemical or hematological parameters during the study period.

Discussion

This study evaluated the progression of PKD in cats, focusing on diagnostic tools, biochemical markers, and disease progression indicators over an 18-month period. The study was conducted on 198 cats of various breeds, including Persian, domestic shorthair, Scottish, and Himalayan. The findings revealed that CKD in affected cats progressed rapidly despite diligent follow-up care by pet owners and veterinarians. This progression was closely associated with the monitoring of renal cysts through ultrasonography.

A 14.00% increase in cyst size over 18 months was observed, while the number of detectable cysts remained constant. This contrasts with human studies, which typically report an increase in cyst numbers with age.¹³ However, the findings do not conclusively establish whether the decline in renal function and CKD progression in feline PKD is driven solely by cyst enlargement or an increase in cyst numbers. Further investigation is required to differentiate between these potential mechanisms.

The significant correlation between biochemical parameters and cyst enlargement suggests that monitoring cyst growth could serve as a reliable indicator of PKD progression. However, validating this hypothesis necessitates longitudinal studies with longer observation periods. Interestingly, the findings of Wills *et al.* reported a 66.00% increase in cyst numbers over a 10-month period, potentially due to undetected small cysts during initial assessments or the younger age of the

affected cats in their study. These discrepancies underline the need for standardized methodologies in feline PKD research.¹⁴

Comparisons with human PKD studies further emphasize the value of imaging techniques in disease monitoring. Ultrasound, a cost-effective and accessible method, was identified as an effective tool for tracking kidney enlargement and cyst growth, consistent with findings in human PKD research,^{5,15} demonstrated that ultrasound provides accurate renal volume estimates in cats,⁶ and Wills *et al.* confirmed its repeatability for detecting renal cysts in feline populations.¹⁴ While magnetic resonance imaging has been shown to be highly effective in predicting disease progression and evaluating therapeutic responses, this study advocates for simpler techniques like ultrasound for long-term monitoring and treatment planning in feline PKD.¹⁶

Biochemical markers also played a crucial role in this study. Elevated plasma concentrations of creatinine and phosphorus were significantly correlated with ultrasound findings, including increased cyst size and kidney involvement. Statistical analysis confirmed significant differences in creatinine and phosphorus levels between PKD-affected cats and healthy controls. However, these results contrast with findings from Noori *et al.* which reported no significant correlation between these markers in Persian cats, likely due to the shorter disease duration and younger study population.¹⁷

The significant elevation in plasma creatinine and phosphorus levels reflects declining glomerular filtration rate due to cyst-induced renal parenchymal compression. This aligns with findings in human PKD studies, where cyst expansion correlates with reduced kidney function. Elevated phosphorus levels, observed in advanced CKD stages, further underscore the systemic impact of renal dysfunction.¹⁸

The anemia observed in advanced CKD appears to have multiple underlying causes. A primary factor is erythropoietin deficiency, as declining kidney function reduces production of this essential hormone that stimulates red blood cell production - a well-established phenomenon in human CKD.¹³ Additionally, the accumulation of uremic toxins may contribute by shortening red blood cell survival. Nutritional factors also likely play a role, since decreased appetite in late-stage CKD can lead to deficiencies in iron and vitamins essential for erythropoiesis. While this pattern of normocytic, normochromic anemia mirrors what is seen in human CKD patients, it contrasts with findings from a study,¹⁷ that reported no significant anemia in Persian cats. This discrepancy may reflect differences in either the severity of kidney disease or the duration of observation across studies.

Blood urea nitrogen levels increased over the study period but did not reach statistical significance. This aligns with human studies, where elevated BUN is more indicative of advanced CKD stages.¹⁹ The absence of significant BUN changes in this study may reflect the limited number of cases in advanced PKD stages and the extensive renal function loss observed in the affected population. Similarly, fluctuations in serum creatinine were noted, but these did not correspond to significant changes in CKD staging, which might explain the lack of clinical signs such as uremia.

The stability of plasma calcium levels, despite declining kidney function, suggests compensatory mechanisms such as increased intestinal absorption or reduced renal excretion. This contrasts with human PKD, where hypercalcemia is often observed due to vitamin D dysregulation.²⁰

The lack of observed hypoalbuminemia or proteinuria in our study contrasts with previous findings,²¹ that reported urinary proteomic alterations in PKD-affected cats. Several factors could explain this discrepancy. First, our cohort predominantly consisted of early-stage CKD cases (75.00% in stages 1 - 2), which may not yet exhibit these biochemical changes. Second, standard dipstick urinalysis may not be sensitive enough to detect mild proteinuria. More advanced diagnostic approaches, such as magnetic resonance imaging or novel urinary bio-markers like symmetric dimethylarginine, may be better for identifying early renal damage. Notably, repeated urinalysis performed both at baseline and study completion consistently showed no detectable urinary albumin. These results suggest that serum and urinary albumin measurements may have limited value as markers of progression in feline PKD, although additional studies are needed to verify this conclusion.¹⁷

Advanced imaging techniques such as magnetic resonance imaging or urinary biomarkers like (symmetric dimethylarginine) may be more effective in detecting early renal damage. Urinalysis conducted at baseline and at the study's conclusion also detected no urinary albumin. These results indicate that serum and urinary albumin may not be reliable markers of progression in feline PKD. However, additional research is needed to confirm this.²¹

This study encountered significant limitations. The impact of cyst location (medulla, cortex, or cortex-medulla) on renal function could not be evaluated due to the limited number of cases with medullary cysts. Additionally, the aim of estimating the lifespan of cats with PKD was not achieved, primarily due to the short observation period and variability in disease severity and care practices among the study population. The variability in treatment regimens highlights a study limitation. However, the reliance on objective parameters (e.g., cyst growth, creatinine trends) minimized confounding effects, as these markers are less sensitive to short-term interventions compared to subjective clinical assessments. Due to the challenges of sampling and the limited size of the

statistical population, some correlation analyses (such as the Spearman test) require further validation in larger studies. This research lays the groundwork for future investigations with broader statistical populations to extract findings with greater precision.

This study observed a statistically significant increase in plasma concentrations of creatinine, phosphorus, and BUN over time and at the conclusion of the study compared to baseline levels. In contrast, no significant changes were noted in plasma calcium levels, and variations in calcium concentrations over time were not significant. Periodic ultrasonographic evaluations revealed a 14.00% increase in cyst size from the beginning to the end of the study, with measurements performed in the sagittal plane to ensure accuracy in documenting cyst dimensions.

Additionally, there were statistically significant differences in erythrocyte count and hematocrit levels between the initial and final phases of the study.

Comparative analyses between the affected and control (healthy) groups at baseline and the end of study indicated no significant mean differences in any measured variables at baseline. However, significant changes were observed in creatinine, BUN, phosphorus, calcium, erythrocytes, and hematocrit levels in the affected group by the end of the study.

These findings underscore the progressive nature of CKD in cats with PKD, despite diligent care provided by pet owners and veterinarians. The study highlights the critical role of factors such as creatinine, phosphorus, BUN, erythrocytes, and hematocrit in estimating the progression of PKD. Routine monitoring of these parameters is essential for assessing disease progression and guiding the management of PKD.

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Supplementary Tables

Conflict of interest

The authors declare that there are no conflicts of interest in disclosing this work.

References

1. Bartges JW. Chronic kidney disease in dogs and cats. *Vet Clin North Am Small Anim Pract.* 2012;42(4):669–692. doi: 10.1016/j.cvsm.2012.04.008. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
2. Segev G, Cortellini S, Foster JD, et al. International Renal Interest Society best practice consensus guidelines for the diagnosis and management of acute kidney injury in cats and dogs. *Vet J.* 2024;305:106068. doi: 10.1016/j.tvjl.2024.106068. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
3. Grauer GF. Early detection of renal damage and disease in dogs and cats. *Vet Clin North Am Small Anim Pract.* 2005;35(3):581–596. doi: 10.1016/j.cvsm.2004.12.013. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
4. Gendron K, Owczarek-Lipska M, Lang J, et al. Maine Coon renal screening: ultrasonographical characterisation and preliminary genetic analysis for common genes in cats with renal cysts. *J Feline Med Surg.* 2013;15(12):1079–1085. doi: 10.1177/1098612X13492164. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
5. Grantham JJ, Torres VE, Chapman AB, et al. Volume progression in polycystic kidney disease. *N Engl J Med.* 2006;354(20):2122–2130. doi: 10.1056/NEJMoa054341. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
6. Tyson R, Logsdon SA, Werre SR, et al. Estimation of feline renal volume using computed tomography and ultrasound. *Vet Radiol Ultrasound.* 2013;54(2):127–132. doi: 10.1111/vru.12007. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
7. Yu Y, Shumway KL, Matheson JS, et al. Kidney and cystic volume imaging for disease presentation and progression in the cat autosomal dominant polycystic kidney disease large animal model. *BMC Nephrol.* 2019;20(1):259. doi: 10.1186/s12882-019-1448-1. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
8. Burtis CA, Ashwood ER. *Tietz textbook of clinical chemistry and molecular diagnosis.* 5th ed. St. Louis, USA: Elsevier; 2012. pp. 669–690. [[Google Scholar](#)]
9. Morgan BR, Artiss JD, Zak B. Calcium determination in serum with stable alkaline Arsenazo III and triglyceride clearing. *Clin Chem.* 1993;39(8):1608–1612. [[PubMed](#)] [[Google Scholar](#)]

10. Dumas BT, Watson WA, Biggs HG. Albumin standards and the measurement of serum albumin with bromcresol green. *Clin Chim Acta*. 1997;258(1):21–30. doi: 10.1016/s0009-8981(96)06447-9. [[DOI ↗](#)] [[PubMed](#)] [[Google Scholar ↗](#)]
11. Weichselbaum TE. An accurate and rapid method for the determination of proteins in small amounts of blood serum and plasma. *Am J Clin Pathol*. 1946;10:40–49. [[PubMed](#)] [[Google Scholar ↗](#)]
12. Moss DW. Alkaline phosphatase isoenzymes. *Clin Chem*. 1982;28(10):2007–2016. [[PubMed](#)] [[Google Scholar ↗](#)]
13. Bae KT, Zhou W, Shen C, et al. Growth pattern of kidney cyst number and volume in autosomal dominant polycystic kidney disease. *Clin J Am Soc Nephrol*. 2019;14(6):823–833. doi: 10.2215/CJN.10360818. [[DOI ↗](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar ↗](#)]
14. Wills SJ, Barrett EL, Barr FJ, et al. Evaluation of the repeatability of ultrasound scanning for detection of feline polycystic kidney disease. *J Feline Med Surg*. 2009;11(12):993–996. doi: 10.1016/j.jfms.2009.07.002. [[DOI ↗](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar ↗](#)]
15. Vester U, Kranz B, Hoyer PF. The diagnostic value of ultrasound in cystic kidney diseases. *Pediatr Nephrol*. 2010;25(2):231–240. doi: 10.1007/s00467-008-0981-0. [[DOI ↗](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar ↗](#)]
16. Grantham JJ, Torres VE. The importance of total kidney volume in evaluating progression of polycystic kidney disease. *Nat Rev Nephrol*. 2016;12(11):667–677. doi: 10.1038/nrneph.2016.135. [[DOI ↗](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar ↗](#)]
17. Noori Z, Moosavian H, Esmaeilzadeh H, et al. Prevalence of polycystic kidney disease in Persian and Persian related-cats referred to Small Animal Hospital, University of Tehran, Iran. *Iran J Vet Res*. 2019;20(2):151–154. [[PMC free article](#)] [[PubMed](#)] [[Google Scholar ↗](#)]
18. Chakrabarti S, Syme HM, Elliott J. Clinicopathological variables predicting progression of azotemia in cats with chronic kidney disease. *J Vet Intern Med*. 2012;26(2):275–281. doi: 10.1111/j.1939-1676.2011.00874.x. [[DOI ↗](#)] [[PubMed](#)] [[Google Scholar ↗](#)]
19. Kamal A. Estimation of blood urea (BUN) and serum creatinine level in patients of renal disorder. *Indian J Fundam Appl Life Sci*. 2014;4(4):199–202. [[Google Scholar ↗](#)]
20. Finch NC, Syme HM, Elliott J. Risk factors for development of chronic kidney disease in cats. *J Vet Intern Med*. 2016;30(2):602–610. doi: 10.1111/jvim.13917. [[DOI ↗](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar ↗](#)]

21. Jiwaganont P, Jaturanratsamee K, Thaisakun S, et al. Analysis of serum proteomic in cats with polycystic kidney disease-1 gene mutation. Heliyon. 2024;10(15):e35577. doi: 10.1016/j.heliyon.2024.e35577. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

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Patterns of morbidity, multimorbidity, and mortality in aging cats: findings from seven years of the Cat Prospective Ageing and Welfare Study

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Introduction: Aging is associated with an increased risk of morbidity, multimorbidity and mortality in cats. However, longitudinal cohort studies are somewhat limited in this species. Information about the prevalence and incidence of age-related disease, combinations of morbidities, and factors associated with mortality could provide insights into the impact of aging on the welfare of domestic cats.

Methods: The prevalence and incidence of common age-related diseases, patterns of multimorbidity and causes and risk factors for mortality were investigated in a prospective longitudinal cohort study of 209 aging cats. Cats were enrolled at middle age and attended a veterinary examination every 6 months, until death or censoring, for up to 7 years.

Results: A total of 1,241 examinations were performed (median follow-up 4.4 years; range 0.0–6.9). Dental disease (84%), abnormalities on orthopedic examination (82%) and heart murmurs (59%) were most prevalent within the cohort over the study period. Most cats (96%) developed at least one abnormality on veterinary examination, and multimorbidity was common, with 175 cats (84%) developing two or more morbidities during the study. Death occurred in 59/209 cats (28%) and median life expectancy was 15.2 years. Pedigree cats had approximately double the hazard of mortality (HR 2.62, 95%-CI 1.39, 4.93; $P = 0.003$) compared with mixed-breed cats, with a similar increase in hazard for those with muscle loss at middle age (HR 2.01, 95%-CI 1.13, 3.60, $P = 0.018$) compared with cats without muscle loss.

Discussion: These findings demonstrate that age-related morbidity and multimorbidity are highly prevalent in cats and both should be considered by veterinary professionals examining these older animals. Furthermore, we identified that midlife body condition, muscle condition and pedigree status affected mortality risk, highlighting the importance of assessing these factors in practice and suggesting that targeted strategies to promote healthy aging should start from middle age.

KEYWORDS

chronic disease, cohort, feline, incidence, lifespan, longitudinal, prevalence, survival

1 Introduction

There are an estimated 11 million owned cats in the UK and 61 million owned cats in the USA, with approximately 26% of households in both countries thought to own a cat (1, 2). The average life expectancy for cats attending primary care veterinary practices in the UK is estimated to be 14 years (3). Certain diseases (e.g. chronic kidney disease [CKD], hyperthyroidism, diabetes mellitus, cardiovascular disease, hypertension, degenerative joint disease and neurological disorders) have an increased prevalence in older cats (4–6). These age-related diseases can adversely impact a cat's quality of life, and early diagnosis and interventions could help to reduce this impact (7–9).

The underlying biology of aging is complex, involving many cellular and molecular mechanisms that affect multiple organ systems (10). Aging is associated with the accumulation of multiple chronic conditions within the same individual. Although the terms comorbidity and multimorbidity are sometimes used interchangeably, comorbidity traditionally refers to additional diseases in relation to an index condition, whereas multimorbidity considers co-existing chronic conditions without prioritizing a single disease process (11). Aging cats commonly present with several concurrent age-related disorders (12, 13), which can present a challenge in veterinary practice. Longitudinal studies are important to understand better the development of multimorbidity, including the accumulation of chronic disorders over time, patterns of disease co-occurrence, and their effects on survival.

Mature and senior cats are at increased risk of developing chronic age-related diseases, making regular health assessment an important component of preventive veterinary care (4, 8, 14). Current American Animal Hospital Association (AAHA) and Feline Veterinary Medical Association (Feline VMA) guidelines classify mature (middle aged) cats as between seven and 10 years old, and senior cats as over 10 years old (Quimby et al., 2021). Improved understanding of morbidity and multimorbidity trajectories in aging cats could support earlier diagnosis, monitoring, and management of age-related diseases, leading to improved cat welfare and better support for pet owners.

Existing evidence on aging-related morbidity in cats is limited primarily to cross-sectional studies or investigations of individual diseases. Large scale retrospective studies using electronic health record data from primary care veterinary practices have provided valuable information on prevalence of age-related disease, causes of mortality and breed-related differences in life expectancy in cats (15, 16, 97). However, their cross-sectional design does not allow for incidence rates of disease to be calculated. Longitudinal cohort studies have focused on single disease processes (17–21), have a relatively short follow-up period (9), or have not yet reported on the effects of aging in mature to senior cats (22). There is, therefore, limited evidence describing long-term morbidity, multimorbidity, and mortality outcomes in longitudinal cohorts of aging client-owned cats. Improved knowledge of the development of morbidity and multimorbidity from such a study could inform veterinary wellness plans, improve advice for preventive care and, ultimately, improve understanding of the aging process in this species.

The Cat Prospective Ageing and Welfare Study (CatPAWS) was established in 2017 with the aim of following a cohort of

>200 client-owned pet cats from middle age to end-of-life. The clinical findings at enrolment in 206 cats from this cohort have been previously reported (8), and findings from an initial 2-year follow-up period were described in a doctoral thesis (23). The current study reports longer-term outcomes related to disease incidence, multimorbidity, and mortality in this ongoing cohort, with follow-up periods of up to 7 years.

The aims of this study were: (1) to estimate the prevalence and incidence of common age-related diseases during longitudinal follow-up of a cohort of aging pet cats; (2) to characterize patterns of multimorbidity over time; and (3) to describe mortality outcomes, including causes of death, survival time, and clinical variables associated with increased mortality risk.

2 Materials and methods

2.1 Ethical approval

Ethical approval for this study was granted by the University of Liverpool Veterinary Research Ethics Committee, under the code VREC491 (with amendments a, b, c, d and e), and the Royal Canin Ethical Review Committee. All owners gave their informed consent, in writing, before their cat was enrolled in the study.

2.2 Data collection

Data were collected from cats enrolled in CatPAWS between February 2017 to January 2024; complete study methodology for CatPAWS (including enrolment criteria, study design and clinical information) has been published elsewhere (8, 23), and is also summarized in the supplementary information (Supplementary Figure 1). Briefly, most cats were enrolled between 7 and 10 years of age, with eight cats being marginally outside this age range at enrolment. All cats underwent a veterinary examination every 6 months until death or censoring at the census date (January 2024). At each examination, bodyweight was measured using portable V20 feline scales (Burtons, UK), which were regularly calibrated with a “test” weight, whilst body condition score (BCS) was assessed using a 9-unit scale, as previously described (Laflamme, 1997). Muscle condition score (MCS) was assessed at 10 separate skeletal landmarks: skull, neck, thoracic vertebrae, lumbar vertebrae, left and right scapula, left and right gluteal muscle group and left and right hindlimb muscle groups caudal to the femur (24). Each skeletal landmark was graded from 0 (severe muscle wastage) to 3 (no muscle wastage) using a validated scale (Michel et al., 2011), with a combined MCS score up to a maximum of 30 points created by summing results from all 10 landmarks (24). Systolic blood pressure (SBP) measurements were taken by the Doppler technique with either a Vet BP Doppler (Burtons) or a CAT+ Doppler (Thames medical) blood pressure monitor, and a 2.5 to 3 cm cuff. Measurements of SBP complied with standard guidelines, usually made from the right or left palmar common digital artery, or from the tail coccygeal artery (25). Five readings were obtained and the average of these taken.

A complete physical examination was performed, including cardiac and lung auscultation, abdominal palpation, assessment of eyes, nose, oral examination, aural examination and coat and skin assessment. Cardiac auscultation was completed up to three times per clinical examination, noting the heart rate, the presence of any arrhythmias, and the presence and intensity of any heart murmurs graded from one to six (26–28). Blood samples were collected, using either jugular or cephalic venepuncture, and analyzed for routine hematology and serum biochemistry panels, as per the methods outlined by Dowgray et al. (8). Urine samples were obtained, either by “free catch” or by cystocentesis where practicable, and urinalysis was performed as per the methods outlined by Dowgray et al. (8).

2.3 Criteria for diagnosis of disease

Criteria for diagnosing specific morbidities have been previously described (8, 23) and are summarized below.

Chronic kidney disease (CKD) was diagnosed when plasma creatinine concentration was measured above the reference interval ($>177 \mu\text{mol/L}$) in at least two blood samples, in the absence of identifiable pre-renal causes, ideally with accompanying urinalysis demonstrating impaired urine concentrating ability (urine specific gravity [USG] <1.035 ; (17, 18)). Where contemporaneous urinalysis demonstrated a USG <1.035 together with an increasing plasma creatinine concentration over the study period, progressing to $\geq 140 \mu\text{mol/L}$ in at least two blood samples and in the absence of identifiable pre-renal causes, this was also considered to meet the diagnostic threshold, consistent with International Renal Interest Society (IRIS) classification of at least stage II CKD (98). During the study period, plasma creatinine concentrations were measured by two different laboratory analysers, which had upper reference intervals of 177 and $212 \mu\text{mol/L}$, respectively. To maintain consistency in CKD case definition across the study and to minimize potential bias related to analyser-specific reference intervals, a single upper reference limit of $177 \mu\text{mol/L}$ was applied throughout to ensure uniform classification of CKD status over time.

Hyperthyroidism was diagnosed when the plasma total thyroxine concentration was $\geq 60 \text{ nmol/L}$, or $>50 \text{ nmol/L}$ with an increased free thyroxine concentration confirming hyperthyroidism (29). Hypertension was diagnosed when the SBP was consistently $\geq 160 \text{ mmHg}$ on subsequent SBP measurements (25). Diabetes mellitus was diagnosed when persistent hyperglycaemia and glucosuria were present with consistent clinical signs, and stress-induced hyperglycaemia had been excluded (30).

An oral examination was undertaken with each quadrant of dentition being graded for both gingivitis and calculus using a scoring system of 0 to 4. The presence of feline odontoclastic resorptive lesions (FORLS) was recorded, as was the identification of fractured teeth, oral lesions or stomatitis. Dental disease was diagnosed if a cat had a gingivitis score of ≥ 2 , or any FORLS or stomatitis, equivalent to the point when veterinary advice of a dental treatment under general anesthetic was provided (8, 21). For the current study, the term neoplasia encompassed multiple neoplasia types, including those diagnosed on physical examination

alone (e.g. oral neoplasia, palpable abdominal masses) or after further diagnostic investigations such as imaging and histology e.g., lymphoma, malignant mammary carcinoma, renal carcinoma (20, 31).

A veterinarian (CP or ND) conducted a complete orthopedic examination (OE) in amenable cats using the methodology previously reported (32, 33). For those cats where a full OE was completed, each limb was assigned a score of 0 (no abnormalities in any joint), 1 (either reduced range of movement (ROM) or pain in one or more joints), 2 (reduced ROM and pain detected or palpable thickening of a joint or joint effusion detected) or 3 (reduced ROM and pain and palpable joint thickening in one or more joints). Coxofemoral joints were separately scored as 0 (no abnormalities in either coxofemoral joint), 1 (either pain or reduced ROM in one or both coxofemoral joints) or 2 (pain and reduced ROM in one or both coxofemoral joints). A total OE score from 0 to 14 was assigned for each complete OE, with a larger score indicating greater musculoskeletal impairment (23).

2.4 Multimorbidity Score

As previously determined by Dowgray (23), a multimorbidity score of 0 to 8 was assigned to cats at each examination, whenever diagnostic information on the following conditions was available. The following conditions or clinical examination abnormalities recorded in this score included: CKD, neoplasia, hypertension, hyperthyroidism, heart murmur, diabetes mellitus, dental disease and any degree of orthopedic examination abnormalities.

2.5 Statistical analysis

Unless otherwise stated, all statistical analysis was undertaken using the open-access statistical language and environment, R version 4.4.1 (34). The additional packages “dplyr” version 1.1.4 (35), “ggplot2” version 3.5.1 (36), “sjPlot” version 2.8.16 (37), “gtsummary” version 1.7.2 (38) and “purr” version 1.0.4 (39) were used to examine, manipulate and visualize the data. Additional packages relating to specific statistical analyses are given in the relevant sections below. Continuous data were assessed for normality using histograms, quantile-quantile plots and Shapiro-Wilk testing and described using either mean and standard deviation (SD) or median and interquartile range (IQR), depending on whether or not their distribution was gaussian.

2.5.1 Sample size calculation

A sample size calculation was performed prior to enrolment and estimated a cohort of 194 cats would be required for a cohort study investigating incidence of disease or mortality. This calculation was undertaken using an online tool (<https://epitools.ausvet.com.au/>) and assumed a 95% confidence level, a desired power of 80%, an expected incidence of 10% in the unexposed group and a relative risk of 2.5% (for exposures of 20%; (23)).

2.5.2 Prevalence and incidence rates

Prevalence was reported as the number (percentage) of cats with a particular morbidity, both as prevalence during the entire study period and prevalence in cats by age year group, including pre-existing cases and cases diagnosed at enrolment. Each cat was counted once per age-year group. The prevalence estimates by age-year group and sex were descriptive rather than inferential, with the aim of describing the disease burden across age and sex groups in the cohort. For each disease, 95% confidence intervals (95%-CI) for prevalence were calculated using the Wilson score interval (40).

Binomial mixed-effects logistic regression models testing associations of age, sex and the interaction of age and sex with the diagnosis of each morbidity were explored to account for repeated observations within cats. Models were fit using the *glmer* function of the “lme4” package version 1.1-37 (41) and “splines” package version 4.4.1 (34). In these models, the dependent variable was a binary variable of whether that condition was diagnosed (yes vs. no), fixed effects were age modeled as continuous variable (in decimalised years) with natural (restricted) cubic splines at three degrees of freedom to allow for non-linear age-related changes, sex (binary male vs. female), and the interactions of age (modeled using natural cubic splines at three degrees of freedom) and sex. For several conditions (e.g. diabetes mellitus, CKD, neoplasia and hyperthyroidism), these models failed to converge or produced unstable estimates, due to limited event counts. Conditions with sufficient sample size (hypertension, heart murmurs, OE abnormalities and dental disease) were presented as plots of the predicted probability of disease presence using “sjPlot” and “ggplot2”. Likelihood ratio tests (LRT) were applied to investigate whether age and sex significantly improved model fit compared to models without these separate variables. Significance was set at $P < 0.05$. Models were assessed for convergence, inspection of simulated residuals and evaluation of overdispersion using the “DHARMA” package (42), and posterior predictive check using the “performance” package (43).

Incidence per cat-year at risk was calculated by dividing the number of new cases of disease (excluding cases already diagnosed at enrolment from the numerator and denominator of this calculation) by total cat-time at risk in days (which is the time from enrolment to diagnosis, death or loss to follow up or study end date) and dividing by 365.25 to account for leap years (44).

2.5.3 Age at diagnosis for conditions not present at enrolment

The age at diagnosis was recorded as either the age when the diagnosis was made, or the age at the first study examination following a diagnosis made at another veterinary practice when clinical records from that practice were unavailable. Cats with pre-existing conditions at the time of enrolment to the study were excluded from this specific analysis due to the age of diagnosis being unknown if made prior to enrolment. Therefore, these analyses were examining the age at diagnosis where that condition was not present at the enrolment examination. Information on disease status at enrolment to the study has been previously published (8).

2.5.4 Analysis of multimorbidity

Associations with the following morbidities and clinical findings were assessed: diabetes mellitus, hyperthyroidism, neoplasia, CKD, hypertension, heart murmurs (of any grade), OE abnormalities (of any grade) and dental disease. Firstly, the sample sizes of different combinations of morbidities that occurred in the study cats were calculated. Secondly, pairings of morbidities diagnosed by the end of the study period in the cats were calculated. Thirdly, correlation analyses amongst different morbidities, and also with mortality, were explored using the Phi coefficient (which assesses associations between binary variables), and results were displayed using a heatmap created with the “corrplot” package [version 0.92; (99)]. The Phi coefficient, mathematically equivalent to Pearson’s correlation for binary data, was calculated using Pearson’s method applied to the binary dataset. Corresponding *P*-values were computed and then adjusted for false discovery rate (FDR) using the Benjamini-Hochberg method. Significance was set as FDR adjusted $P < 0.05$. In this correlation analysis the following morbidities were included as binary variables: heart murmur, hypertension, hyperthyroidism, diabetes mellitus, CKD, neoplasia, dental disease, muscle loss of moderate-to-severe (combined MCS 0-19/30), OE score of ≥ 5 . Given the high prevalence of OE abnormalities and muscle loss over the study period, only cats with an OE score of ≥ 5 at any timepoint [classified as “severe”; (23)] and muscle loss of moderate-to-severe (MCS 0-19/30), were included when determining associations with other morbidities to retain meaningful variability.

Finally, a Poisson generalized linear mixed-effects model was created using the *glmer* function of the “lme4” package to assess associations between cat age and changes in multimorbidity scores. In this random intercept model, age in decimalised years was the sole fixed effect and the individual cat was included as a random effect to enable data from multiple visits to be included. Model fit was assessed using the “DHARMA” residual diagnostics package version 0.4.7 (42); no evidence of overdispersion or zero-inflation was found.

2.5.5 Mortality analysis

Survival analyses were performed using the “survival” package version 3.7.0 (45) within R. The primary aim of the survival analyses was to identify baseline risk factors associated with mortality at enrolment to the study in the cats.

Age (in decimal years, calculated as days divided by 365.25 to account for leap years) was used as the underlying time scale. Survival time was defined as the age at death or the age at censoring (i.e. when the cat was still alive at the end of the study or lost to follow-up). Survival status was recorded as a binary variable indicating whether the cat had died (event) or was censored. Statistical significance was set at $P < 0.05$.

To ensure consistency and comparability across models, univariable and multivariable Cox proportional hazards regression analyses were performed, using a complete-case dataset and including only cats with all the following predictor variables

recorded at enrolment. Predictor variables tested were: BCS (categorized as 3–4, 5, 6 and 7–9 with BCS 5 as the reference category; no cat had a BCS <3 at enrolment), MCS (categorized as no muscle loss [combined MCS 30/30] and mild muscle loss [combined MCS 20–29/30; 24]; no cat had a combined MCS <20 at enrolment); bodyweight (kg), sex (male vs. female), breed (pedigree vs. non-pedigree), binary indicators of disease at enrolment including CKD, hypertension, hyperthyroidism and dental disease, and the presence of a heart murmur (categorized as none, grades 1–2, or ≥ 3). Neoplasia and diabetes mellitus at enrolment were not included because only a single cat was affected for each, and these cats were excluded from these analyses.

A separate univariable Cox model, investigating association of multimorbidity score at enrolment with mortality, was applied to a reduced dataset due to missing data. Multimorbidity scores were categorized as 0, 1, 2, 3 or 4–5 morbidities due to only two cats having a multimorbidity score of 5 and none with a multimorbidity score >5 at enrolment.

First, analyses of separate predictor variables at enrolment on risk of death was completed using univariable Cox proportional hazards models. The proportional hazards assumption was assessed using scaled Schoenfeld residuals and Z-tests (using the function *cox.zph* in the “survival” package).

Where violations of the proportional hazards assumption were identified in univariable analyses (as occurred for BCS categories), a time-dependent Cox proportional hazards model was also fit, by splitting survival time using the *survSplit* function of the “survival” package. An initial cutpoint at 10 years of age was considered based on feline mature-senior life stage classifications (Quimby et al., 2021). Examination of Schoenfeld residuals suggested a change in the hazard relationship around 11 years, and models using an 11-year split provided improved proportional hazards diagnostics and more stable estimates. Therefore, the 11-year split model is presented.

Multivariable Cox proportional hazards regression was then used to identify independent predictors of mortality. The multimorbidity score was not included in the multivariable model because it is a composite variable directly derived from the individual morbidities that were evaluated as separate candidate predictors. Exhaustive subset selection was performed as an exploratory approach to evaluate all candidate predictors; all possible combinations of the 10 predictor variables (1,023 models) were evaluated. Model performance was compared using Akaike's Information Criterion (AIC) and Bayesian Information Criterion (BIC); however, model selection was ultimately based on BIC to prioritize parsimony given the relatively limited number of events. The final model had the smallest BIC and met the proportional hazards assumption.

Kaplan Meier curves and Cumulative Hazards curves were assessed visually using “survival”, “ggsurvfit” version 1.1.0 (46) and “survminer” version 0.4.9 (47) packages in R.

Internal validation of the final Cox model was performed using bootstrap resampling (1,000 iterations) to estimate optimism-corrected performance, implemented using the *validate* function of the “rms” package version 8.1–0 (48).

3 Results

3.1 Sample size and summary data

A total of 209 cats, belonging to 149 owners, were enrolled between 7th February 2017 and 5th March 2020. Data from the enrolment examinations of 206 of these cats have been published previously (8), and body condition metrics for these cats have also been recently reported (24). Clinical data from enrolled cats were collected up to 25th January 2024 and included a total of 1,241 visits, with a median of six visits per cat (IQR 3, 9; range 1, 12). Twenty-nine cats (14%) only attended for the initial visit, whilst the remaining 180 cats (86%) attended for ≥ 2 visits. The median follow up time was 4.4 years (IQR 1.8, 5.7; range 0.0, 6.9).

A summary of the signalment and visit data for the 209 cats is included in the supplementary information (Supplementary Tables 1, 2). There were 111 (53%) female cats and 98 (47%) male cats, only two of which were sexually intact (both male cats). One hundred and eighty cats (86%) were non-pedigree mixed breed, with the remaining 29 cats (14%) representing 14 different pedigree breeds. Eight cats were marginally outside the target recruitment age at enrolment (seven cats were marginally younger [aged 6.7–7.0 years] and one cat marginally older [aged 11.4 years]) either because of inaccurate age records held by the primary veterinary practice, or because of inaccurate owner recall at the time of recruitment (23). Therefore, the median age at enrolment was 8.1 years (IQR 7.2, 9.3 years). The median age of the cohort across all study visits was 10.3 years (IQR 8.8, 12.0 years; range 6.7, 16.4 years).

3.2 Loss to follow up

During the study, 59 cats (28%) were lost to follow up, and 59 cats (28%) died, with median time before withdrawal being 443 days (IQR 1.5, 686.5 days; range 1, 2,415 days). The most common reason for withdrawal was lack of owner response to communications from clinical staff (e.g., booking appointments; 28, 48%), followed by owners moving house (9, 15%) and owners withdrawing due to their cat being too stressed by visits (7, 12%; Supplementary Table 3).

3.3 Prevalence and incidence rates of age-related morbidities

Prevalence and incidence rates of certain age-related morbidities in the cats over the course of the study period are shown in Table 1. The most prevalent conditions were dental disease (84%, 95%-CI 79, 89%), OE abnormalities (82%, 95%-CI 77, 87%) and heart murmurs (59%, 95%-CI 53, 66%). The descriptive prevalence of conditions per age-year group by sex is shown in Figure 1, with the prevalence per age-year group over all cats (not stratified by sex) shown in Supplementary Figure 2.

The results of the binomial mixed-effects models investigating predicted probabilities of dental disease, hypertension, heart

TABLE 1 Prevalence, incidence rates and age at diagnosis for various diseases in the 209 study cats.

Disease	Number of cats ¹	Number at enrolment ²	Incident cases ³	Time at risk ⁴	Prevalence ⁵	Incidence rate ⁶	Age at diagnosis ⁷
Dental disease	176	129	47	118	84.2 (78.7, 88.5)	0.398 (0.284, 0.512)	9.3 (8.3, 10.3)
OE abnormalities	172	130	42	85	82.3 (76.6, 86.9)	0.375 (0.245, 0.505)	10.1(9.1, 10.6)
Heart murmur	124	61	63	407	59.3 (52.6, 65.8)	0.155 (0.117, 0.193)	10.4(9.3, 11.6)
Hypertension	41	6	35	668	19.6 (14.8, 25.5)	0.052 (0.035, 0.070)	12.0(10.4, 13.1)
CKD	23	9	14	699	11.0 (7.5, 16.0)	0.020 (0.013, 0.035)	11.6(10.7, 13.5)
Neoplasia	19	1	18	745	9.1 (5.9, 13.8)	0.024 (0.013, 0.035)	11.7(10.8, 13.1)
Hyperthyroidism	18	6	12	713	8.6 (5.5, 13.2)	0.017 (0.007, 0.026)	11.1(10.5, 12.1)
Diabetes Mellitus	4	1	3	741	1.9 (0.7, 4.8)	0.004 (0.000, 0.009)	12.2 (12.2, 12.7)

Cats were enrolled at middle age (between 7–10 years).

Prevalence includes all new and existing cases of each disease, whilst incidence rates exclude pre-existing cases.

¹Number of cats that were diagnosed with each condition over the entire study period.

²Number of cats with each condition at time of enrolment (including those cats with pre-existing diagnosis and those diagnosed at the enrolment examination).

³Incident cases = Total number of cats diagnosed with each condition minus number of cats with the condition present at enrolment.

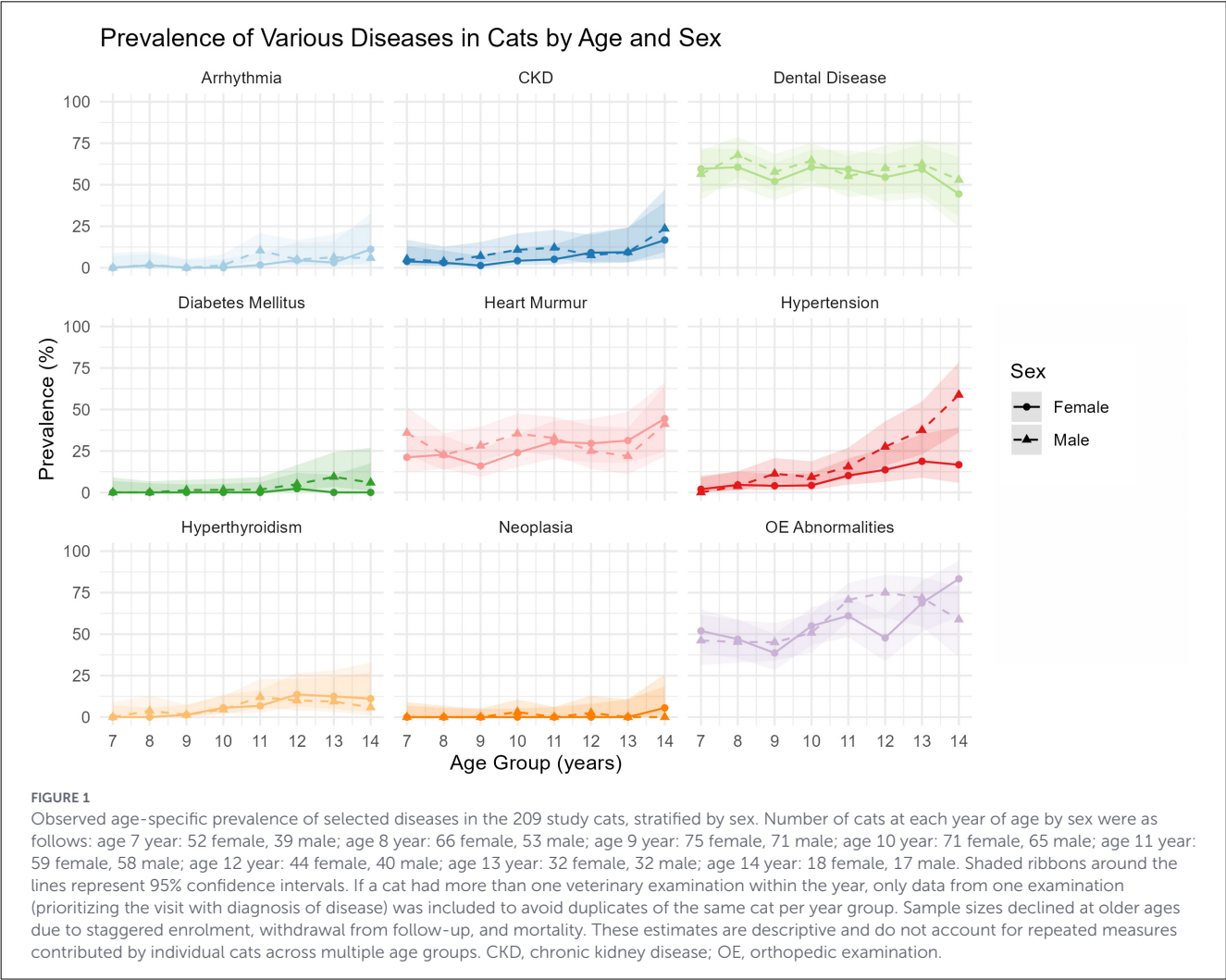
⁴Time at risk equates to the total number of cat-years from enrolment date up to either diagnosis of disease, death, withdrawal from study or the end of the study period.

⁵Prevalence reported as mean percentage, with the 95% confidence interval in brackets.

⁶Incidence rate reported as mean percentage per cat year, with the 95% confidence interval in brackets.

⁷Median age at diagnosis (in years), with inter-quartile range in brackets, which excludes both pre-existing cases and also those diagnosed at initial enrolment.

CKD, chronic kidney disease; OE, orthopedic examination.



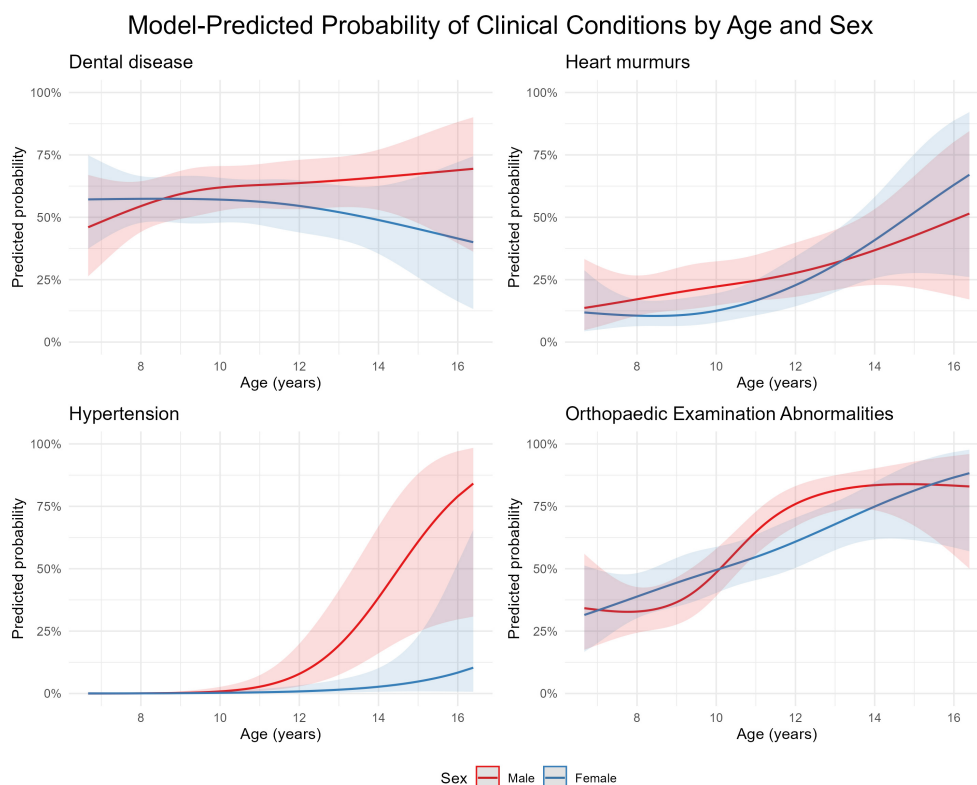


FIGURE 2

Results of binomial mixed-effects models investigating the predicted probabilities of dental disease, heart murmurs, hypertension and orthopedic examination abnormalities in 209 study cats. Each condition was modeled with the fixed effects of age (modeled using natural cubic splines to three degrees of freedom to allow for non-linear changes with age), sex and the interactions of age and sex, with individual cat subjects as random effect to account for repeated measures within cat subjects.

murmurs and OE abnormalities by age and sex are presented in Figure 2, with full model outputs provided in the supplementary information (Supplementary Table 4). Age (modeled using natural cubic splines to three degrees of freedom) was significantly associated with hypertension (LRT: $\chi^2_6 = 161.2$, $P < 0.001$), heart murmurs (LRT: $\chi^2_6 = 28.8$, $P < 0.001$) and OE abnormalities (LRT: $\chi^2_6 = 81.2$, $P < 0.001$), but not dental disease (LRT: $\chi^2_6 = 4.3$, $P = 0.630$). Sex was significantly associated with hypertension (LRT: $\chi^2_4 = 10.8$, $P = 0.028$). The predicted probability of hypertension increased markedly in male cats after approximately 10 years of age (Figure 2).

Cumulative event curves showing the age of development for each morbidity within the cohort are shown in the supplementary information (Supplementary Figure 3).

3.4 Other morbidities

Several other morbidities were either pre-existing or developed in the cohort during the study. Two cats had feline immunodeficiency virus (FIV), and both were euthanased, one cat due to the FIV diagnosis and the other due to concurrent stomatitis and an oral mass. Nine cats had upper respiratory tract infection, based on signs of sneezing or conjunctivitis at one or more visits. A further four cats had chronic gastrointestinal disease, whilst one cat had cerebellar hypoplasia and concurrent chronic rhinitis. Liver disease and pancreatitis were present in three and one cats,

respectively. Idiopathic hypercalcaemia, non-specific dermatitis, otitis externa and cognitive dysfunction syndrome were recorded in one, four, three and one cats, respectively.

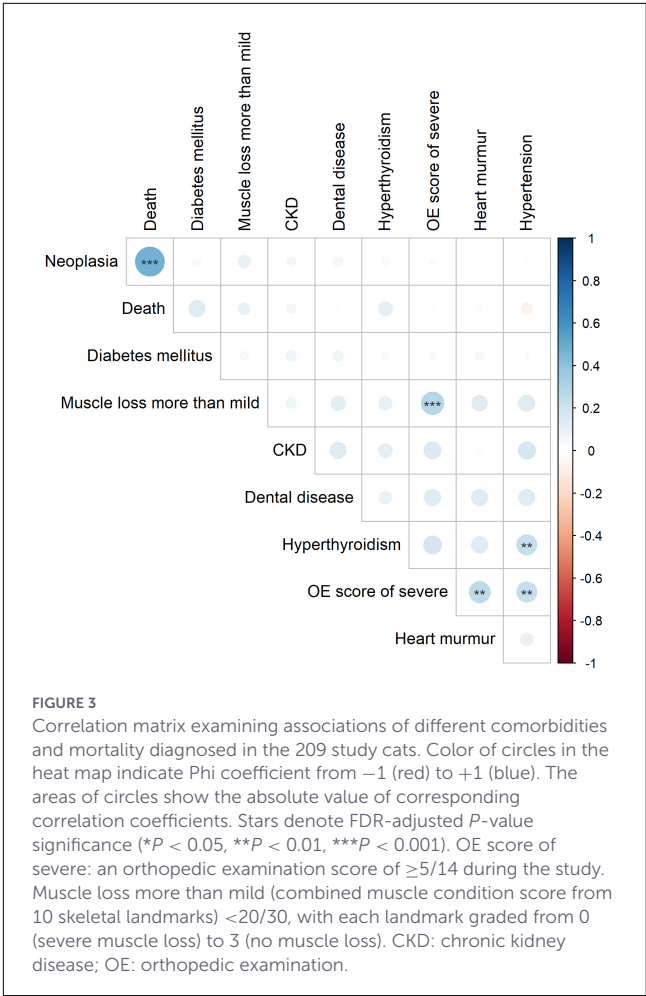
3.5 Multimorbidity in study cats

Associations with the following morbidities and abnormalities on examination were assessed: diabetes mellitus, hyperthyroidism, neoplasia, CKD, hypertension, heart murmurs (of any grade), OE abnormalities (of any grade) and dental disease. Of the 209 cats, only nine cats (4%) did not develop any of these conditions, whilst 25 cats (12%) developed one morbidity, 44 cats (21%) developed two morbidities, 78 cats (37%) developed three morbidities, 40 cats (19%) developed four morbidities, eight cats (4%) developed five morbidities, and five cats (2%) developed six morbidities. The most common grouping of morbidities included dental disease, OE abnormalities and heart murmurs, which occurred in 103 (49%) of the cats, with or without additional morbidities. The different combinations of morbidities seen are shown in the supplementary information (Supplementary Table 5), whilst pairings of different conditions are summarized in Table 2. In the 41 cats that were diagnosed with hypertension, 28 (68%) also had a detectable heart murmur on at least one examination, nine (22%) had concurrent hyperthyroidism, and a further nine (22%) had concurrent CKD. None of the cats that developed hypertension, hyperthyroidism,

TABLE 2 Pairwise co-occurrence of chronic conditions in study cats.

Condition	Heart murmur	Hypertension	Hyperthyroidism	Diabetes mellitus	CKD	Neoplasia	Dental disease	OE abnormalities	No other morbidities
Heart murmur (<i>n</i> = 124)		28 (23)	15 (12)	3 (2)	13 (11)	11 (9)	110 (89)	113 (91)	3 (2)
Hypertension (<i>n</i> = 41)	28 (68)		9 (22)	1 (2)	9 (22)	4 (10)	39 (95)	37 (90)	0 (0)
Hyperthyroidism (<i>n</i> = 18)	15 (83)	9 (50)		0 (0)	4 (22)	1 (6)	17 (94)	16 (89)	0 (0)
Diabetes mellitus (<i>n</i> = 4)	3 (75)	1 (25)	0 (0)		1 (25)	0 (0)	4 (100)	3 (75)	0 (0)
CKD (<i>n</i> = 23)	13 (57)	9 (39)	4 (17)	1 (4)		3 (13)	23 (100)	22 (96)	0 (0)
Neoplasia (<i>n</i> = 19)	11 (58)	4 (21)	1 (5)	0 (0)	3 (16)		17 (90)	17 (90)	1 (5)
Dental disease (<i>n</i> = 176)	110 (63)	39 (22)	17 (10)	4 (2)	23 (13)	17 (10)		153 (87)	13 (7)
OE abnormalities (<i>n</i> = 172)	113 (66)	37 (22)	16 (9)	3 (2)	22 (13)	17 (10)	153 (89)		8 (5)

Values represent the number (and percentage) of cats with the row condition that also had the column condition. Percentages are calculated using the total number of cats with the row condition as the denominator. "No other morbidities" indicates cats diagnosed with the row condition in the absence of any other recorded morbidity. For example, 124 cats had a heart murmur, of which 28 (23%) also had hypertension and three (2%) had no co-occurring morbidities. OE, orthopedic examination; CKD, chronic kidney disease.



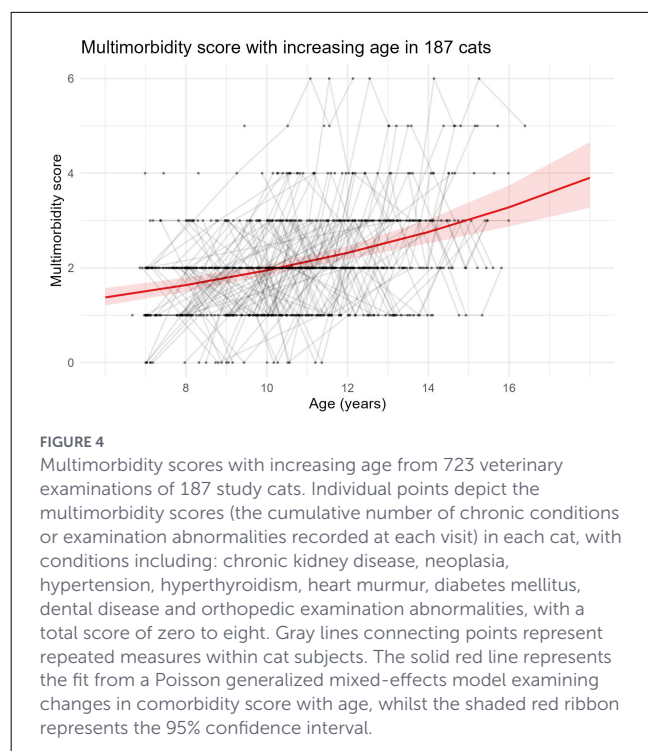
diabetes mellitus or CKD over the course of the study had these conditions in isolation.

Correlation analysis revealed certain pairings of morbidities that developed in the cats over the study period (Figure 3). Hypertension was positively associated both with hyperthyroidism ($\phi = 0.23$, FDR-adjusted $P = 0.007$) and an OE score of $\geq 5/14$ ($\phi = 0.23$, FDR-adjusted $P = 0.007$). Having an OE score of $\geq 5/14$ was also associated with moderate to severe muscle loss ($\phi = 0.28$, FDR-adjusted $P < 0.001$) and the presence of a heart murmur ($\phi = 0.25$, FDR-adjusted $P = 0.004$). Finally, death during the study was positively associated with a diagnosis of neoplasia ($\phi = 0.47$, FDR-adjusted $P < 0.001$).

A multimorbidity score could be calculated for 723 veterinary examinations from 187 cats, where complete diagnostic information was available. Age was associated with an increase in multimorbidity score (incidence rate ratio 1.09, 95% CI 1.07, 1.12; $P < 0.001$; Figure 4). Of these 723 examinations, no cat >12 years had a multimorbidity score of 0 (Figure 4).

3.6 Mortality in study cats

Fifty-nine of the 209 cats (28%) died during the study period, with the mean age at death being 11.6 (SD 2.20) years. Forty-eight cats were euthanised, eight cats died without euthanasia,



and three cats had no record of whether they died by euthanasia or not. During follow-up, 13/29 (45%) pedigree cats and 46/180 (26%) non-pedigree (domestic short, medium or long-haired) cats died. Causes of death, classified by disease grouping, are shown in Table 3. The five most common causes of death were neoplasia (18/59 cats, 31%), cardiovascular (8/59 cats, 14%), endocrine (5/59 cats, 9%), renal (4/59 cats, 7%) and sudden unexplained death (4/59 cats, 7%). Of the 18 cats that died from neoplasia, six were attributed to lymphoma, three had a gastrointestinal tumor, two had an oral neoplasm and two had an unspecified abdominal mass. There was one case each of renal carcinoma, pancreatic mass, cardiac neoplasm, liver neoplasm, and a malignant neoplasia of unknown type. Cardiovascular conditions leading to death included heart failure (five cats) aortic thromboembolic disease (ATE, two cats) and both heart failure and ATE (one cat). The cats that had cause of death classed as endocrine disorders included two cats with diabetes mellitus, and three cats with hyperthyroidism. Three other cats had both heart failure and concurrent hyperthyroidism, recorded separately from those with either cardiovascular disease or hyperthyroidism (Table 3). Finally, the four cats with a renal disorder as cause of death included three with CKD, and one with acute kidney injury.

3.6.1 Life expectancy of study cats

Median survival time, equivalent to the median life expectancy, was 15.2 year (5,557 days), with the lower limit of the 95%-CI being 14.8 year (5,407 days; Figure 5A). It was not possible to estimate the upper 95%-CI due to insufficient deaths during the study period. The overall probability of a study cat being alive at 15 year was 53% (95%-CI: 43%, 66%). The median life expectancy of pedigree cats

TABLE 3 Causes of death in the 59 cats that died during the study period.

Cause of death	Number of cats	Percentage (%)
Neoplasia	18	31
Cardiovascular disease	8	14
Endocrine disorder	5	9
Renal disorder	4	7
Sudden unexplained death	4	7
Cardiovascular disease and endocrine disorder	3	5
Poor quality of life	3	5
Gastrointestinal disorder	2	3
Multiple morbidities	2	3
Neurological	2	3
Unknown	2	3
Anemia	1	2
Hepatopathy	1	2
Infectious disease	1	2
Pancreatic disorder	1	2
Trauma	1	2
Urinary disorder	1	2
Total	59	100

was 12.7 year (4,633 days) compared with 15.8 year (5,787 days) in non-pedigree cats (95%-CI again not reported due to insufficient numbers of events) (Figure 5B).

3.6.2 Factors associated with mortality

Survival analyses were performed on a complete-case dataset including 190 cats with all included predictor variables recorded at enrolment (apart from the univariable Cox proportional hazards model investigating multimorbidity score and mortality, where only 178 cats had a multimorbidity score assigned at enrolment). During the study period, 54 of the 190 cats died and 55 were lost to follow up. The median follow up time was 4.3 years (IQR 1.1, 5.7 years, range 0.0, 6.9 years).

3.6.2.1 Univariable Cox proportional hazards analyses

The results of the univariable Cox proportional hazards analyses investigating the association between clinical factors at enrolment and the hazard of mortality are presented in Table 4.

Sex was not associated with hazard of mortality (HR 0.99; 95% CI 0.58, 1.68; $P = 0.959$). Pedigree cats had a significantly greater hazard of death compared with non-pedigree cats (HR 2.54; 95% CI 1.35, 4.76; $P = 0.004$) (Figure 5B).

None of the cats had muscle loss graded as moderate or severe at enrolment; however, cats with mild muscle loss had nearly twice the hazard of death compared with cats with no muscle loss (HR 1.95; 95% CI 1.09, 3.48; $P = 0.024$) (Figure 5C).

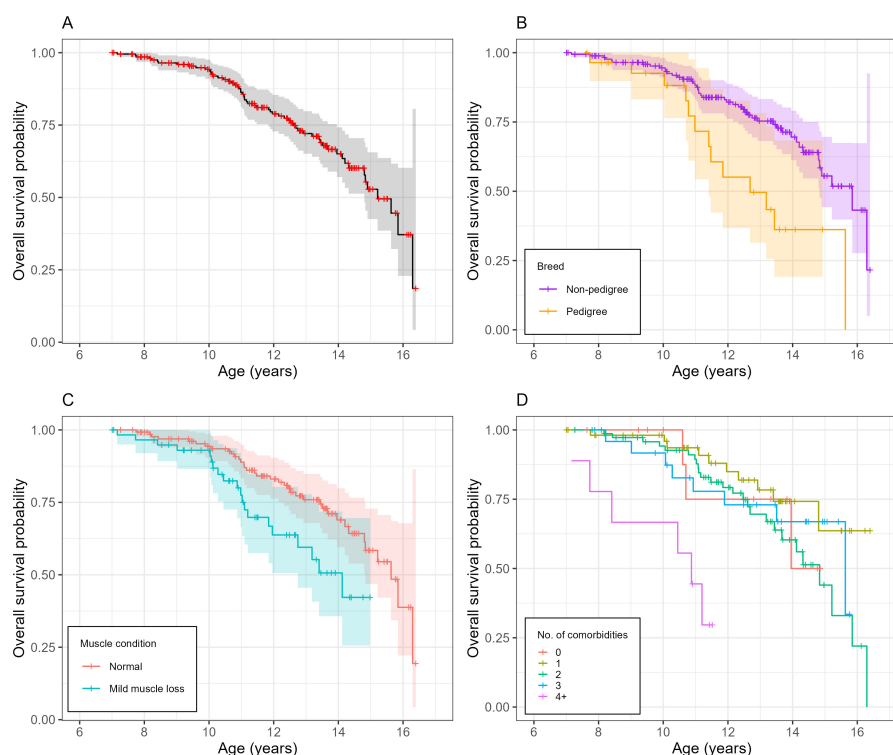


FIGURE 5

Kaplan-Meier curve showing survival probability for (A) all 209 study cats; (B) 209 study cats stratified by breed; (C) 198 cats stratified by muscle condition at enrolment; (D) 178 study cats stratified by number of morbidities at enrolment. The solid lines and shaded ribbon represent survival probability and 95% confidence interval, respectively (not shown in Figure (D) due to the number of survival curves), whilst crosses denote censored data (i.e. age at which cats either dropped out of study or their age at the end of the study period if still alive). Median survival time (i.e. median life expectancy) was 5,557 days (15.2 years).

Cats with a BCS of 3–4/9 had an increased hazard of death compared with those with a BCS of 5/9 (HR 2.48; 95% CI 1.08, 5.67; $P = 0.031$). However, the proportional hazards assumption was violated for this analysis, indicating that the effect of BCS on hazard was not constant over time and, therefore, should be interpreted with caution (Supplementary Figure 4).

In the 178 cats where an enrolment multimorbidity score could be assigned, hazard of death during the study was six times greater in those with ≥ 4 comorbidities (HR 6.30, 95% CI 1.5, 26.1; $P = 0.011$; Figure 5D). Proportional hazards assumptions were met for this model (z-test global $P = 0.577$).

3.6.2.2 Time-dependent Cox proportional hazards model examining changes in baseline hazard with age for enrolment body condition score categories

The results of the univariable Cox proportional hazards model investigating the association of BCS categories at enrolment and survival violated the proportional hazards assumption, suggesting a change in baseline hazard depending on the age of the cat. Therefore, a time-dependent Cox model was also fit, separating survival at ages <11 years and ≥ 11 years, based on examining Schoenfeld residual plots.

BCS group was not associated with survival in cats <11 year, although BCS 3–4 was borderline significant ($P = 0.055$). However, in cats ≥ 11 year, those with BCS 7–9 at enrolment had an increased hazard of death compared to cats with BCS 5 at enrolment (HR

5.04, 95% CI 1.62, 15.69, $P = 0.008$; Table 5). The proportional hazards assumption was satisfied after splitting survival time in this manner (z-test global $P = 0.40$).

3.6.2.3 Multivariable Cox proportional hazards regression model

The five top-performing models and their BIC are provided in the supplementary information (Supplementary Table 6). The best-fit model (LRT $P = 0.002$) had modest discriminatory ability (C-index = 0.611) and contained breed and MCS grade at enrolment as the main predictor variables (Table 6). Age (in decimal years, calculated as days divided by 365.25 to account for leap years) was used as the underlying time scale, thereby inherently adjusting for age and reducing potential confounding by age. In the final model (54 deaths in 190 cats), mortality was positively associated with breed group and pedigree cats had 2.62 times greater hazard of death over the study compared with non-pedigree cats (HR 2.62, 95% CI 1.39, 4.93; $P = 0.003$; Table 6). Cats with mild muscle loss at enrolment had twice the hazard of death compared with cats with no muscle loss (HR 2.02, 95% CI 1.13, 3.60, $P = 0.018$; Table 6).

Internal validation demonstrated modest discrimination (optimism-corrected Dxy = 0.214, corresponding to a C-index of approximately 0.61). The model showed minimal optimism (0.008) and good calibration (slope = 0.995), indicating low risk of overfitting.

TABLE 4 Results of univariable Cox proportional hazards models investigating the association of separate predictor variables at enrolment to the study and mortality in 190 cats.

Characteristic	N	HR	95%-CI	P
Body condition score				
5	53	—	—	—
3–4	31	2.48	1.08, 5.67	0.031
6	48	1.55	0.67, 3.60	0.308
7–9	58	1.99	0.91, 4.35	0.086
Muscle condition score				
No muscle loss	132	—	—	—
Mild muscle loss	58	1.95	1.09, 3.48	0.024
Bodyweight (kg)				
	190	0.99	0.76, 1.28	0.928
Sex				
Female	99	—	—	—
Male	91	0.99	0.58, 1.68	0.959
Breed				
Non-pedigree	162	—	—	—
Pedigree	28	2.54	1.35, 4.76	0.004
Chronic kidney disease				
No	180	—	—	—
Yes	10	1.66	0.60, 4.62	0.332
Hypertension				
No	180	—	—	—
Yes	10	1.13	0.41, 3.17	0.809
Hyperthyroidism				
No	184	—	—	—
Yes	6	3.1	0.95, 10.1	0.062
Dental disease				
No	74	—	—	—
Yes	116	1.49	0.83, 2.68	0.179
Heart murmur grade				
None	132	—	—	—
I–II/VI	47	1.5	0.81, 2.78	0.193
≥III/VI	11	2.26	0.88, 5.81	0.091

Each predictor variable was modeled separately.
Models were based on 190 cats and 54 deaths.
Cats were aged 7–10 years at enrolment and followed for a median of 4.3 years (IQR 1.1 – 5.7 years).
Age was used as the underlying time scale to inherently account for age in the models.
The proportional hazards assumption was violated for body condition score and bodyweight, therefore these results should be interpreted with caution.
HR, hazard ratio; CI, confidence intervals.
P-values shown in bold denote those that met the threshold for significance ($P < 0.05$).

4 Discussion

In this observational cohort study, the most prevalent disease identified was dental disease, which affected 84% of cats. This prevalence is greater than that reported in previous studies, with estimates of 15% in UK cats (16) and 53% in cats >5 years of age from USA veterinary hospitals (49). The prevalence of dental disease was probably greater in this study due to the older average age of the cats studied, its longitudinal nature and that a careful oral

cavity examination was undertaken at each health check, with any dental problems recorded. Overall prevalence of dental disease did not increase with age in this study, which was similar to findings from a recent study of ~18,000 cats attending primary care practice in the UK, where the risk of periodontal disease plateaued after 12 years age (16). The reason for this is not clear but might be because study cats were often treated for their dental disease after it had been identified. The incidence rate for dental disease was estimated 0.398 cases per cat-year, but this might have been an

TABLE 5 Time-dependent Cox model investigating the association between body condition score (BCS) categories at enrolment and survival in study cats with survival split at 11 years.

BCS group	Baseline N	<11 years			≥11 years		
		HR	95% CI	P	HR	95% CI	P
5 (ref)	53	—	—	—	—	—	—
3–4	31	2.75	0.98–7.71	0.055	1.98	0.79–7.98	0.713
6	48	0.74	0.21–2.62	0.638	2.94	0.88–9.85	0.121
7–9	58	0.45	0.11–1.81	0.263	5.04	1.62–15.69	0.008

HR, hazard ratio; CI, confidence interval.

The time-dependent Cox model stratifies survival at 11 years of age, creating 311 cat-intervals; cats surviving past 11 years contribute to both <11-year and ≥11-year strata.

Survival time was split at 11 years of age; HRs are shown separately for <11 years and ≥11 years.

The model was based on 190 cats and 54 deaths.

Baseline N represents cohort size at enrolment. BCS 5/9 was the reference category.

P-values shown in bold met the threshold for significance ($P < 0.05$).

TABLE 6 Multivariable Cox proportional hazards model examining risk factors at enrolment for mortality in 190 study cats.

Characteristic	Number	Hazard ratio	95%-CI	P-value
Breed				
Non-pedigree	162	—	—	—
Pedigree	28	2.62	1.39, 4.93	0.003
Muscle condition				
Normal	132	—	—	—
Mild muscle loss	58	2.01	1.13, 3.60	0.018

95%-CI, 95% confidence interval.

Muscle condition was scored using combined scores from 10 skeletal landmarks each graded from 3 (no muscle wastage) to 0 (severe muscle wastage); mild muscle loss 20–29/30; normal 30/30 (reference category). Age was used as the underlying time scale to inherently account for age in the models. P-values shown in bold met the threshold for significance ($P < 0.05$).

under-estimate due to left censoring; in this respect, dental disease was already present at the time of enrolment in over half the study cats (129/209). A more accurate incidence estimate could be obtained from future studies using cat cohorts recruited earlier in life (50).

Orthopedic examination abnormalities were identified in 82% of the study cats, with an incidence rate of 0.375 new cases per cat-year which included any level of severity of OE abnormalities. The prevalence of radiographic osteoarthritis in the appendicular joints in cats was 61% in one joint and 48% in two or more joints in a study of 100 cats, with prevalence increasing with age (51). However, false positives are possible on orthopedic examination in cats, not least when it is used as the sole method of assessment, for example, when investigating possible pain (32). Conversely, radiographic signs of osteoarthritis in cats do not always correspond with clinical signs of pain or impaired mobility (52). Diagnosing clinical musculoskeletal impairment in aging cats can, therefore, prove somewhat challenging (53). Cat-friendly techniques to reduce stress during veterinary examinations can improve the confidence of pain assessments (54). The use of clinical metrology instruments such as the Feline Musculoskeletal Pain Index can further aid the diagnosis of musculoskeletal disorders in cats, and be useful in monitoring the response to treatment (55).

An OE score of ≥5 was also associated with hypertension, an audible heart murmur on cardiac auscultation and moderate-to-severe muscle loss. Muscle wastage often occurs secondary to chronic pain in cats with degenerative joint disease (56, 57), and can also occur in old age in the absence of other diseases (termed sarcopenia). In humans, sarcopenia leads to impaired mobility

and frailty (58), but decreases in muscle strength or performance have not yet been investigated in aging cats. Osteoarthritis has been linked to an increased risk of cardiovascular disease (59, 60), although the underlying reason for this is complex and still not fully understood. Hypertension is also associated with an increased odds of having osteoarthritis in people, which may not be due solely to chronic pain, but also due to the role the vascular system is thought to play in joint disease (61, 62). The outcomes of orthopedic assessments in the CatPAWS cohort will be investigated in a future study which will include how the severity of OE abnormality changed with age in the cats.

Heart murmurs were identified in 59% of cats in this study, which is similar to previous studies where a prevalence of 60% was reported in cats >9 years of age (27). The prevalence of heart murmurs increased with age but was not associated with the sex of the cats. The incidence rate of newly detected heart murmurs was 0.155 cases per cat-year. Unfortunately, it was not possible to confirm whether the heart murmurs identified were associated with underlying structural cardiac abnormalities because echocardiography was not performed. In a study of 856 cats with audible heart murmurs referred for echocardiography, over half did not subsequently have echocardiographic evidence of heart disease (28). The presence of a heart murmur also had low predictive power for identifying hypertrophic cardiomyopathy (HCM) in a study of 780 cats that underwent echocardiography (27). However, in that study, the presence of a heart murmur (especially grade III/VI and greater), increasing age, male cats and overweight cats were risk factors for HCM in a multivariable model (27).

Hypertension was identified in 20% of the cohort during the study period, and the prevalence increased as age of the cats increased. This prevalence estimate is marginally greater than the 15% prevalence recently reported in a similar cohort of 185 cats aged >10 years (63), but less than the 40% prevalence reported in a large multicentre study of SBP in 8,884 cats >7 years old (64). The estimated incidence rate of new cases of hypertension in the current study was 0.052 per cat-year; unfortunately, it is not possible to make comparisons with previous studies because, to the authors' knowledge, incidence rates have not previously been documented. Hypertension was also associated with cats having muscle loss and hyperthyroidism; hyperthyroidism being known to cause secondary hypertension (25). Further, there was an increased prevalence of hypertension in male cats >10 years old. More detailed longitudinal modeling to assess further associations of various clinical variables with systolic blood pressure measurements and risk of hypertension is currently being undertaken. This future work which will account for possible confounding factors in this apparent sex difference such as body composition and comorbidities.

The estimated incidence rate of CKD diagnosis was low, at 0.020 cases per cat-year, although overall prevalence was 11%, which is greater than in one recent study of electronic medical records where prevalence was 4% in cats >9 year (65), but less than in a second study where prevalence of CKD (with concurrent azotaemia) was 18% (18). Further, a 50% prevalence of CKD was estimated in a study of 86 randomly selected cats aged 6 months to 20 years, whilst the prevalence in 128 cats with degenerative joint disease was 68% (66). The discrepancies amongst studies are likely due to differences in the populations investigated, the study design and the diagnostic procedures. In this respect, the current study used proactive screening veterinary examinations of cats, meaning that cats with early CKD might have been identified earlier than expected, and before the onset of clinical signs. Further, although similar methodology to that of Finch et al. (18) was used to diagnose CKD, the approach used by Marino et al. (66) was different. In that study the abdominal radiographs were taken in all cats to determine kidney size and better identify cases of non-azotaemic CKD (e.g., IRIS stage I), which probably explains the greater prevalences recorded. The approach to CKD diagnosis used in our study may also have underestimated CKD prevalence and incidence by overlooking IRIS stage I and early IRIS stage II CKD. We also recognize that using a threshold of serum creatinine in the current study of >177 $\mu\text{mol/L}$ for laboratory analysers where the upper reference interval was 212 $\mu\text{mol/L}$ could risk overestimating CKD diagnosis; however, it is also noted that the IRIS guidelines do not consider laboratory differences.

Multimorbidity was common in the aging cats. Only 9/209 cats remained free of any abnormality on clinical examination over the study, and 175/209 (84%) developed two or more abnormalities. The most common groupings of conditions were dental disease, OE abnormalities, and heart murmurs (with or without additional morbidities), which affected 103/209 cats (49%). All cats with either hypertension, hyperthyroidism, diabetes mellitus or CKD had developed other comorbidities by the end of the study. However, this analysis did not take into consideration grades of heart murmur or whether the heart murmur was indicative of clinical signs of heart disease. Likewise, any abnormality on OE was

included without accounting for the severity of OE abnormalities and whether the cat had other metrics of musculoskeletal disorders.

Multimorbidity with aging is reported in many species including dogs (67) and humans (68), and poses a challenge for the care of aging patients. The need for polypharmacy to treat different conditions leads to an increase in potential for drug-interactions (69), and trying to manage multiple conditions simultaneously whilst prioritizing overall quality of life can be difficult to balance. In cats, the dilemma of managing cats with both hyperthyroidism and CKD, with or without common musculoskeletal disorders such as osteoarthritis, has been well-documented (12, 13). Hyperthyroidism can mask CKD by various mechanisms, including the increase in glomerular filtration rate caused by thyroxine excess, and treating hyperthyroidism can then lead to an increase in biomarkers such as creatinine associated with CKD (13). Given that common treatments for musculoskeletal pain, such as non-steroidal anti-inflammatory drugs (NSAIDs), can be associated with renal toxicity, there are concerns about their use in cats with CKD (12). However, not providing analgesia to cats with a painful musculoskeletal condition can impact welfare and the safety of long-term NSAID therapy in cats with CKD remains unknown (12).

The process of aging is a risk factor for the development of morbidities, and therapies (both pharmaceutical and non-pharmaceutical) targeting the aging process aim to be a broad-spectrum preventative measure for multiple diseases of aging (70). In experimental studies in mice, drugs such as rapamycin and metformin have been found to prolong healthspan and lifespan (71–75). In dogs, the effect of rapamycin treatment on lifespan and healthspan is currently being investigated (76). Rapamycin has been approved by the FDA in the US for the treatment of subclinical HCM in cats, and reduces maximal thickness of the left ventricle compared to placebo (77). The effect of rapamycin treatment for CKD in cats is currently also under investigation (78). No studies have yet reported whether rapamycin treatment results in an overall increased healthspan or lifespan in cats.

Nutrition also plays a role in healthy aging. Interventions such as caloric restriction (CR) have been known for many decades to impact longevity in short-lived animal models such as mice (Fontana et al., 2010). The long-term impacts of CR have not been fully explored, and longevity outcomes may depend on the age of the subject (79). Older aged rats exposed to CR, for example, had worsening cognitive function and frailty levels (Prvulovic et al., 2022). The effects of dietary supplements and nutraceuticals, such as omega-3 fatty acids and antioxidants including vitamin E have been studied in *C. elegans*, fruit flies and laboratory mice with mixed results (80). Increased intake of oily fish containing omega-3 fatty acids and antioxidants have been found to decrease biological age as measured by epigenetic-clock based biomarkers in humans (81). In cats, 30 cats aged 7 years and above fed a diet supplemented with vitamin E, β -carotene, chicory root (source of prebiotic) and *n*-3 and *n*-6 fatty acids over 5 years lived significantly longer and has less disease incidence than those receiving a control diet (82).

The mortality rate in this study was 28% (59/209 cats), although this might be underestimated given the fact that many (59/209) cats were withdrawn prematurely and some of these might also have died before the end of the study period. The median life expectancy was 15.2 year, which is similar to some previous reports

(14.0 years (3); 15.8 years (83)) but greater than others (11.8 years (84); 11.7 years (85)). Although the present study benefits from its prospective longitudinal cohort design, the sample size was smaller than in the four previous studies that analyzed data from thousands of cats using electronic health records. Furthermore, cats were also only followed from middle age, which might lead to an overestimate of life expectancy since cats dying at a younger age would automatically be excluded. The top causes of death in this current cohort were neoplasia (18 cats, 31% of deaths) and cardiovascular disease (eight cats, 14% of deaths), followed by endocrine disorders (including hyperthyroidism and diabetes mellitus; five cats, 9%), renal disorders (four cats, 7%) and sudden death (four cats, 7%). This contrasts with the O'Neill et al. (3), study where renal disorders were the top cause of death, estimated at 13.6%, with the next most common causes being non-specific illness (12.6%) and neoplasia (12.3%), and cardiac disease being the cause in 4.6% of deaths. However, in that study, mass lesions were included as a separate category (3), which might have resulted in deaths attributed to the neoplasia category being under-estimated. Recently, another prospective cohort study of aging in cats (the "Bristol Cats" study) reported a mortality rate of 362/2,444 cats before the age of 9 years, with the most common cause of death (46% of deaths) being road traffic accidents (22). None of the cats in the present study died from a road traffic accident; it is likely that the risk of road traffic accidents decreases in older cats as they may spend less time outdoors and be less active, as well as having potentially learnt how to avoid road vehicles. Furthermore, before aged 9 years, 16.3% of pedigree cats died whereas 12.6% mixed-breed cats died, possibly showing that pedigrees have a greater mortality rate across life-stages (22).

Risk factors at enrolment for study mortality included being from a pedigree breed, having four or more concurrent morbidities and muscle loss. Age was used as the underlying time-scale, thereby inherently adjusting for age and limiting its potential confounding effect. The associations between pedigree breed, multimorbidity burden and muscle loss with mortality were therefore independent of chronological age at enrolment and unlikely to be explained by age alone. Although, on initial survival analysis using Cox regression, cats that had a BCS of 3–4 at enrolment had an increased hazard of death compared to those cats with BCS 5, the proportional hazards assumption was violated, indicating that the baseline hazard varied with age. After accounting for age-related changes in hazard, BCS at enrolment was no longer associated with survival in cats <11 years, although cats in overweight and obese body condition (BCS 7–9) were at greater risk of death after 11 years of age. This finding may reflect the small number of cats with a BCS of 3 or less, which necessitated grouping underconditioned cats with those of BCS 4. Cats with a BCS of 4 may be at lesser risk of death before 11 years of age than cats with a BCS of 3, potentially diluting the observed effect in younger cats. These results emphasize the need for cats in middle age to maintain an optimal body condition [currently considered to be BCS 5–6/9; (83)], to maximize longevity. Previous studies have also found that BCS of <5/9 or 9/9 (83) or a non-ideal body weight (85) negatively affects longevity in cats. This U-shaped relationship between body condition and mortality risk, where individuals who are either very thin or very fat are at greater risk of mortality, is also well documented in humans (86). Body condition score and MCS were found to change with age in a non-linear fashion in this cohort

(24), and changes in these markers of body composition with age may also impact survival. This study aimed to examine predictors of mortality from enrolment data, but another important question for future analysis would be to examine whether changes in BCS and MCS over the aging process also have an impact on mortality risk.

In our final Cox regression model, the most important predictors of mortality were breed and muscle loss at enrolment. In the same cohort of cats, muscle condition decreased over middle age, and decreased at a greater rate after age 10 years and in those cats that developed age-related morbidities (24). Furthermore, although body condition increased slightly during middle age, there was a subsequent decline after 10 years suggesting that the changes in BCS and MCS do not necessarily correlate with each other over a cat's lifespan (24). Muscle wastage is common in aging cats both as a consequence of age-related morbidities and due to aging related mechanisms (87) and contributes to the frailty phenotype in people and dogs (88–90). Severe epaxial muscle wastage at 12 years (range 9–17 years) is also associated with decreased lifespan in various dog breeds (90). In a recent pilot study, muscle loss was associated with increased odds of a veterinarian classifying a cat as frail and frail cats had an increased risk of 6-month mortality (91). The finding that muscle condition at middle age can be a predictor of future mortality underlines the importance of both recording MCS in clinical examinations and implementing preventative therapies to maintain muscle mass in aging cats. This could include providing adequate dietary protein and, where appropriate, gentle strength-building activities, such as interactive play or low-impact exercise tailored to the pet's ability. Future work to assess the importance of muscle wastage with frailty in cats, and how it is associated with mobility in aging cats is required to gain more knowledge of the effect of muscle wastage on quality of life. Further work to identify targets to prevent muscle wastage, and the effect of dietary protein would also be of use.

The finding of a shorter average lifespan in pedigree cats in the current study is, perhaps, not surprising because pedigree cats have long been hypothesized to have shorter lifespans than mixed-breed cats due to "inbreeding depression" causing an increased likelihood for the expression of deleterious genes (92). The "pedigree" category in the current study had a relatively small sample size and included several cat breeds, though numbers were too small in most individual breeds to permit a separate analysis. In previous studies, pedigree cats had a shorter lifespan (3, 85, 93), although life expectancy varied amongst different breeds (3, 85). In a recent study of life expectancy in 7,936 UK cats, cats of the Sphynx cat breed had the shortest life expectancy (6.7 years), whilst Burmese cats had the longest life expectancy at year zero (14.4 years) which, interestingly, was greater than the mixed-breed cats in that study (85). Variety in longevity between breeds is likely due to different pre-dispositions to diseases. However, in studies relying on data from visits to veterinary clinics, the recorded lifespans may also be biased if the likelihood of attending a veterinary clinic is increased based on the value or perceived value of an animal (94). Pedigree cats are also becoming more popular in the UK; in 2024, more pedigrees were obtained in the preceding 12 months than mixed-breed cats for the first time since data collection began (95). Therefore, it is important to be aware of the potential health risks that certain breeds of cat have.

There were several limitations with this study. Firstly, the sample size was relatively small, and, therefore, some diseases

were uncommon (e.g., diabetes mellitus) meaning that estimates of incidence might be unreliable. Further, stratifying incidence rates by different age ranges, which would have provided additional detail, was explored but was not feasible due to small sample sizes of certain diseases at different ages. Second, the prevalence estimates by age-year group are descriptive statistics and do not take into consideration repeated measures within the same cat subjects and, therefore, may underestimate uncertainty. To overcome this, multi-level models were applied to those diseases with enough events, although a further limitation is the possible effect of survivor bias, with some cats dying or being lost to follow up over the course of the study. Third, some cats had an unknown date of birth, which had been estimated by the owner at the time of acquisition; this might have led to inaccuracies in some estimates although, in fairness, this is likely to be an issue that could affect many such studies. Fourth, orthopedic conditions, such as degenerative joint disease, were mainly identified by OE, but not confirmed definitively with radiography. Likewise, echocardiography was not performed to determine the cause of the heart murmurs in most cats, although such findings were discussed with the owner enabling them to pursue a diagnosis separately (with their primary care veterinarian) if they wished. Determining definitive diagnoses of disease in first opinion clinical veterinary practice can sometimes be challenging either because of owner financial constraints or because doing so would not change the management or treatment plan. Therefore, it was decided that the presence of heart murmurs and not definitively diagnosed cardiac diseases would be discussed in this study. Fifth, some disorders were not analyzed separately including chronic skin conditions, chronic gastrointestinal disease and chronic cat flu, because these diseases occurred in relatively small numbers. Given that this cohort study is ongoing, it might be possible to examine such cases in more detail in the future. Continued data collection will also allow for other diseases to be studied in the same cohort, such as cognitive dysfunction. Sixth, given that cats were enrolled at 7–10 years, there is a risk of left-censoring of the data for estimates of the median age at diagnosis (for specific diseases). Therefore, these estimates are only valid for cats without that pre-existing diagnosis in middle age. Likewise, estimates of median life expectancy may be over-estimated. More accurate estimates will be possible from cohort studies where cats are enrolled from birth. Furthermore, the cats in this study were from one geographic location (the North-West of England) and, therefore, findings might not be generalisable to cats in other locations. Finally, the effect of clustering within owners and veterinary practices the cats were registered with was not accounted for in the analyses.

5 Conclusions and future impact

In this 7-year longitudinal cohort study of aging pet cats, almost all cats (96%) developed at least one abnormality on veterinary examination, and multimorbidity was common, with 84% of cats developing two or more abnormalities during the study period. There are several novel findings including estimates of incidence rates of dental disease and hypertension. Having muscle loss during middle age doubled the hazard of death during subsequent follow up, perhaps suggesting the need for implementing interventions to promote maintenance of an optimal body condition as cats age.

Future studies will need to examine in more depth the associations and risk factors associated with specific morbidities and assess potential interventions that could delay or reduce disease burden. Overall, our findings emphasize the importance of considering multi-morbidity in aging cats and suggest that strategies to improve healthy aging should begin before or during middle age.

Data availability statement

The original contributions presented in the study are included in the article/[Supplementary material](#), further inquiries can be directed to the corresponding author.

Ethics statement

The animal studies were approved by University of Liverpool Veterinary Research Ethics Committee and the Royal Canin Ethical Review Committee. The studies were conducted in accordance with the local legislation and institutional requirements. Written informed consent was obtained from the owners for the participation of their animals in this study.

Author contributions

CP: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Visualization, Writing – original draft, Writing – review & editing. ND: Conceptualization, Data curation, Investigation, Methodology, Project administration, Writing – review & editing. KE: Data curation, Investigation, Project administration, Writing – review & editing. GP: Conceptualization, Methodology, Supervision, Writing – review & editing. DH: Methodology, Writing – review & editing. DM: Methodology, Writing – review & editing. EC: Conceptualization, Methodology, Supervision, Writing – review & editing. AG: Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Writing – review & editing.

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University Veterinary Practice, University of Liverpool. Some of the findings presented in this manuscript were previously published in CRP's doctoral thesis (96).

Conflict of interest

At the time the study was performed, CP was undertaking a post-graduate studentship funded by Royal Canin, a division of Mars Petcare. Since February 2025, CP holds a part-time postdoctoral research position at the University of Liverpool, funded by Royal Canin. ND previously held a post-doctoral research associate position at the University of Liverpool, funded by Royal Canin. KE previously held a research technician position at the University of Liverpool, funded by Royal Canin. AG is an employee of the University of Liverpool whose academic post is funded by Royal Canin. AG and ND have also received financial remuneration and gifts for providing educational material, speaking at conferences, and consultancy work. DM is an employee of Royal Canin.

The remaining author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was used in the creation of this manuscript. Portions of the R code used in this

study were refined with the assistance of ChatGPT (OpenAI) to improve clarity and resolve minor issues. The author(s) reviewed, tested, and validated all AI-assisted code to ensure accuracy and reproducibility. ChatGPT was not involved in the conceptualization, interpretation of results, or writing of the manuscript beyond these code refinements.

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Supplementary material

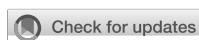
The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fvets.2026.1813450/full#supplementary-material>

References

1. American Veterinary Medical Association. *AVMA 2022 Pet Ownership and Demographic Sourcebook*. Schaumburg, IL: American Veterinary Medical Association. (2022).
2. Cats Protection. Cats and their stats (CATS) report. Report UK. East Sussex: Cats Protection. (2023).
3. O'Neill DG, Church DB, McGreevy PD, Thomson PC, Brodbelt DC. Longevity and mortality of cats attending primary care veterinary practices in England. *J Feline Med Surg*. (2015) 17:125–33. doi: 10.1177/1098612X14536176
4. Bellows J, Center S, Daristotle L, Estrada AH, Flickinger EA, Horwitz DF, et al. Aging in cats: common physical and functional changes. *J Feline Med Surg*. (2016) 18:533–50. doi: 10.1177/1098612X16649523
5. Sordo L, Breheny C, Halls V, Cotter A, Törnqvist-Johnsen C, Caney SMA, et al. Prevalence of disease and age-related behavioural changes in cats: past and present. *Vet Sci*. (2020) 7:85. doi: 10.3390/vetsci7030085
6. Salt C, Saito EK, O'Flynn C, Allaway D. Stratification of companion animal life stages from electronic medical record diagnosis data. *J Gerontol A Biol Sci Med Sci*. (2023) 78, 579–86. doi: 10.1093/gerona/glac220
7. Morris J. Mammary tumours in the cat: size matters, so early intervention saves lives. *J Feline Med Surg*. (2013) 15:391–400. doi: 10.1177/1098612X13483237
8. Dowgray N, Pinchbeck G, Eyre K, Biourge V, Comerford E, German AJ. Aging in cats: owner observations and clinical finding in 206 mature cats at enrolment to the cat prospective aging and welfare study. *Front Vet Sci*. (2022) 9:859041. doi: 10.3389/fvets.2022.859041
9. Mortier F, Daminet S, Marynissen S, Smets P, Paepé D. Value of repeated health screening in 259 apparently healthy mature adult and senior cats followed for 2 years. *J Vet Int Med*. (2024) 38:2089–98. doi: 10.1111/jvim.17138
10. López-Otin C, Blasco MA, Partridge L, Serrano M, Kroemer G. Hallmarks of aging: An expanding universe. *Cell*. (2023) 186:243–78. doi: 10.1016/j.cell.2022.11.001
11. Harrison C, Fortin M, Van Den Akker M, Mair F, Calderon-Larranaga A, Boland F, et al. Comorbidity versus multimorbidity: why it matters. *J Multimorb Comorb*. (2021) 11:2633556521993993. doi: 10.1177/2633556521993993
12. Monteiro B, Steagall PVM, Lascelles BDX, Robertson S, Murrell JC, Kronen PW, et al. Long-term use of non-steroidal anti-inflammatory drugs in cats with chronic kidney disease: from controversy to optimism. *J Small Animal Prac*. (2019) 60:459–62. doi: 10.1111/jsap.13012
13. Geddes R, Aguiar J. Feline Comorbidities: Balancing hyperthyroidism and concurrent chronic kidney disease. *J Feline Med Surg*. (2022) 24:641–50. doi: 10.1177/1098612X221090390
14. Bellows J, Center S, Daristotle L, Estrada AH, Flickinger EA, Horwitz DF, et al. Evaluating aging in cats: how to determine what is healthy and what is disease. *J Feline Med Surg*. (2016) 18:551–70. doi: 10.1177/1098612X16649525
15. O'Neill DG, Church DB, McGreevy PD, Thomson PC, Brodbelt DC. Prevalence of disorders recorded in cats attending primary-care veterinary practices in England. *Vet J*. (2014) 202:286–91. doi: 10.1016/j.tvjl.2014.08.004
16. O'Neill DG, Blenkarn A, Brodbelt DC, Church DB, Freeman A. Periodontal disease in cats under primary veterinary care in the UK: frequency and risk factors. *J Feline Med Surg*. (2023) 25:1098612X231158154. doi: 10.1177/1098612X231158154
17. Jepson RE, Brodbelt D, Vallance C, Syme HM, Elliott J. Evaluation of predictors of the development of azotemia in cats. *J Vet Int Med*. (2009) 23:806–13. doi: 10.1111/j.1939-1676.2009.0339.x
18. Finch NC, Syme HM, Elliott J. Risk factors for development of chronic kidney disease in cats. *J Vet Intern Med*. (2016) 30:602–10. doi: 10.1111/jvim.13917
19. Fox PR, Keene BW, Lamb K, Schober KA, Chetboul V, Luis Fuentes V, et al. International collaborative study to assess cardiovascular risk and evaluate long-term health in cats with preclinical hypertrophic cardiomyopathy and apparently healthy cats: the reveal study. *J Vet Intern Med*. (2018) 32:930–43. doi: 10.1111/jvim.15285

20. Economu L, Stell A, O'Neill DG, Schofield I, Stevens K, Brodbelt D. Incidence and risk factors for feline lymphoma in UK primary-care practice. *J Small Animal Prac.* (2021) 62:97–106. doi: 10.1111/jsap.13266
21. Williams JL, Roberts C, Harley R, Gruffydd-Jones TJ, Murray JK. Prevalence and risk factors for gingivitis in a cohort of UK companion cats aged up to 6 years. *J Small Animal Prac.* (2024) 65:605–14. doi: 10.1111/jsap.13737
22. Taylor AR, McDonald J, Foreman-Worsley R, Hibbert A, Blackwell EJ. Mortality and life table analysis in a young cohort of pet cats in the UK. *J Feline Med Surg.* (2025) 27:1098612X251314689. doi: 10.1177/1098612X251314689
23. Dowgray N. *An Epidemiological, Clinical and Biomechanical study into age-related changes in 206 middle aged cats.* The CatPAW Study. PhD Thesis, University of Liverpool (2020).
24. Pye CR, Dowgray NJ, Eyre K, Pinchbeck G, Biourge V, Moniot D, et al. (2025). Longitudinal changes in bodyweight, body condition, and muscle condition in ageing pet cats: findings from the cat prospective ageing and welfare study. *Front Vet Sci.* (2025) 12:1654002. doi: 10.3389/fvets.2025.1654002
25. Taylor SS, Sparkes AH, Briscoe K, Carter J, Sala SC, Jepson RE, et al. ISFM consensus guidelines on the diagnosis and management of hypertension in cats. *J Feline Med Surg.* (2017) 19:288–303. doi: 10.1177/1098612X17693500
26. Côté E, Manning AM, Emerson D, Laste NJ, Malakoff RL, Harpster NK. Assessment of the prevalence of heart murmurs in overtly healthy cats. *J Am Vet Med Assoc.* (2004) 225:384–8. doi: 10.2460/javma.2004.225.384
27. Payne JR, Brodbelt DC, Luis Fuentes V. Cardiomyopathy prevalence in 780 apparently healthy cats in rehoming centres (the CatScan study). *J Vet Cardiol.* (2015) 17:S244–57. doi: 10.1016/j.jvc.2015.03.008
28. Ferasin L, Ferasin H, Cala A, Creelman N. Prevalence and clinical significance of heart murmurs detected on cardiac auscultation in 856 cats. *Vet Sci.* (2022) 9:564. doi: 10.3390/vetsci9100564
29. Carney HC, Ward CR, Bailey SJ, Bruyette D, Dennis S, Ferguson D, et al. 2016 AAEP guidelines for the management of feline hyperthyroidism. *J Feline Med Surg.* (2016) 18:400–16. doi: 10.1177/1098612X16643252
30. Sparkes AH, Cannon M, Church D, Fleeman L, Harvey A, Hoenig M, et al. ISFM consensus guidelines on the practical management of diabetes mellitus in cats. *J Feline Med Surg.* (2015) 17:235–50. doi: 10.1177/1098612X15571880
31. Dagher E, Abadie J, Loussouarn D, Campone M, Nguyen F. Feline invasive mammary carcinomas: prognostic value of histological grading. *Vet Pathol.* (2019) 56:660–70. doi: 10.1177/0300985819846870
32. Lascelles BDX, Dong Y-H, Marcellin-Little DJ, Thomson A, Wheeler S, Correa M. Relationship of orthopedic examination, goniometric measurements, and radiographic signs of degenerative joint disease in cats. *BMC Vet Res.* (2012) 8:10. doi: 10.1186/1746-6148-8-10
33. Dowgray N, Comerford E, German AJ, Gardiner J, Pinchbeck G, Bates KT. Paw pressure and gait in middle-aged client-owned cats with and without naturally-occurring musculoskeletal disease. *PLoS ONE.* (2024) 19:e0314629. doi: 10.1371/journal.pone.0314629
34. R Core Team. *R: A Language and Environment for Statistical Computing.* Vienna: R Foundation for Statistical Computing (2024).
35. Wickham H, François R, Henry L, Müller K, Vaughan D. dplyr: a grammar of data manipulation. R package version 1.1. 2. *Computer software* (2023).
36. Wickham H. *ggplot2: Elegant Graphics for Data Analysis.* New York, NY: Springer-Verlag New York. (2016).
37. Lüdtke D. *sjPlot: Data Visualization for Statistics in Social Science.* R package version 2.8.16 ed. (2024).
38. Sjöberg DD, Whiting K, Curry M, Lavery JA, Larmarange J. Reproducible summary tables with the gtsummary package. *R J.* (2021) 13:570–80. doi: 10.32614/RJ-2021-053
39. Wickham H, Henry L. purrr: Functional Programming Tools. R package version 1.0.4 ed. (2025).
40. Newcombe RG. Two-sided confidence intervals for the single proportion: comparison of seven methods. *Stat Med.* (1998) 17:857–72. doi: 10.1002/(sici)1097-0258(19980430)17:8<857::aid-sim777>3.0.co;2-e
41. Bates D, Mächler M, Bolker B, Walker S. Fitting linear mixed-effects models using lme4. *J. Stat. Softw.* (2015) 67:1–48. doi: 10.18637/jss.v067.i01
42. Hartig F. DHARMA: residual diagnostics for hierarchical (multi-level/mixed) regression models. CRAN: Contributed Packages. (2016).
43. Lüdtke D, Ben-Shachar MS, Patil I, Waggoner P, Makowski D. Performance: an R package for assessment, comparison and testing of statistical models. *J Open Source Software.* (2021) 6. doi: 10.31234/osf.io/vtq8f
44. Noordzij M, Dekker FW, Zoccali C, Jager KJ. Measures of disease frequency: prevalence and incidence. *Nephron Clin Prac.* (2010) 115:c17–20. doi: 10.1159/000286345
45. Therneau T, Lumley T. R survival package. *R Core Team.* (2013) p. 523.
46. Sjöberg DD, Baillie M, Haesendonckx S, Treis T. ggsvf: flexible time-to-event figures. *R package version.* 0.3: 0. (2023).
47. Kassambara A, Kosinski M, Biecek P, Fabian S. *Survminer: Drawing Survival Curves Using Ggplot2.* R Package Version 0.4. 8. (2020).
48. Harrell Jr FE, Harrell Jr MFE, Hmisc D. Package 'rms'. *Vanderbilt Univ.* (2017) 229.
49. Lund E. Epidemiology of periodontal disease in older cats. *Vet Focus.* (2012) 22:23–4. doi: 10.1055/s-0034-1381881
50. Murray JK, Casey RA, Gale E, Buffington CA, Roberts C, Kinsman RH, et al. Cohort profile: the 'bristol cats study' (BCS)—a birth cohort of kittens owned by UK households. *Int J Epidemiol.* (2017) 46:1749–50e. doi: 10.1093/ije/dyx066
51. Slingerland LI, Hazewinkel HA, Meij BP, Picavet P, Voorhout G. Cross-sectional study of the prevalence and clinical features of osteoarthritis in 100 cats. *Vet J.* (2011) 187:304–9. doi: 10.1016/j.tvjl.2009.12.014
52. Godfrey DR. Osteoarthritis in cats: a retrospective radiological study. *J Small Animal Prac.* (2005) 46:425–9. doi: 10.1111/j.1748-5827.2005.tb00340.x
53. Dowgray N, Comerford E. Feline musculoskeletal ageing: how are we diagnosing and treating musculoskeletal impairment? *J Feline Med Surg.* (2020) 22:1069–83. doi: 10.1177/1098612X20965832
54. Rodan I, Dowgray N, Carney HC, Carozza E, Ellis SL, Heath S, et al. 2022 AAEP/ISFM cat friendly veterinary interaction guidelines: approach and handling techniques. *J Feline Med Surg.* (2022) 24:1093–132. doi: 10.1177/1098612X221101984
55. Enomoto M, Lascelles BDX, Robertson JB, Gruen ME. Refinement of the Feline Musculoskeletal Pain Index (FMPI) and development of the short-form FMPI. *J Feline Med Surg.* (2022) 24:142–51. doi: 10.1177/1098612X211011984
56. Guillot M, Moreau M, Heit M, Martel-Pelletier J, Pelletier JP, Troncy E. Characterization of osteoarthritis in cats and meloxicam efficacy using objective chronic pain evaluation tools. *Vet J.* (2013) 196:360–7. doi: 10.1016/j.tvjl.2013.01.009
57. Gruen ME, Alfaro-Córdoba M, Thomson AE, Worth AC, Staicu AM, Lascelles BD. The use of functional data analysis to evaluate activity in a spontaneous model of degenerative joint disease associated pain in cats. *PLoS ONE.* (2017) 12:e0169576. doi: 10.1371/journal.pone.0169576
58. Cruz-Jentoft AJ, Bahat G, Bauer J, Boirie Y, Bruyère O, Cederholm T, et al. Sarcopenia: revised European consensus on definition and diagnosis. *Age Ageing.* (2019) 48:16–31. doi: 10.1093/ageing/afy169
59. Wang H, Bai J, He B, Hu X, Liu D. Osteoarthritis and the risk of cardiovascular disease: a meta-analysis of observational studies. *Sci Rep.* (2016) 6:39672. doi: 10.1038/srep39672
60. Hall AJ, Stubbs B, Mamas MA, Myint PK, Smith TO. Association between osteoarthritis and cardiovascular disease: systematic review and meta-analysis. *Eur J Prev Cardiol.* (2020) 23:938–46. doi: 10.1177/2047487315610663
61. Ramasamy SK, Kusumbe AP, Schiller M, Zeuschner D, Bixel MG, Milić C, et al. Blood flow controls bone vascular function and osteogenesis. *Nat Commun.* (2016) 7:13601. doi: 10.1038/ncomms13601
62. Lo K, Au M, Ni J, Wen C. Association between hypertension and osteoarthritis: A systematic review and meta-analysis of observational studies. *J Orthopaed Transl.* (2022) 32:12–20. doi: 10.1016/j.jot.2021.05.003
63. Knies M, Kooistra HS, Teske E. Prevalence of persistent hypertension and situational hypertension in a population of elderly cats in The Netherlands. *J Fel Med Surg.* (2023) 25:1098612X231172629. doi: 10.1177/1098612X231172629
64. Sparkes A, Garelli-Paar C, Blondel T, Guillot E. 'The mercury challenge': feline systolic blood pressure in primary care practice—a European survey. *J Feline Med Surg.* (2022) 24:e310–23. doi: 10.1177/1098612X221105844
65. Conroy M, Brodbelt DC, O'Neill D, Chang Y-M, Elliott J. Chronic kidney disease in cats attending primary care practice in the UK: a VetCompass™ study. *Vet Record.* (2019) 184:526. doi: 10.1136/vr.105100
66. Marino CL, Lascelles BDX, Vaden SL, Gruen ME, Marks SL. Prevalence and classification of chronic kidney disease in cats randomly selected from four age groups and in cats recruited for degenerative joint disease studies. *J Feline Med Surg.* (2014) 16:465–72. doi: 10.1177/1098612X13511446
67. Hoffman JM, Creevy KE, Franks A, O'Neill DG, Promislow DEL. The companion dog as a model for human aging and mortality. *Aging Cell.* (2018) 17:e12737. doi: 10.1111/acel.12737
68. Kirchberger I, Meisinger C, Heier M, Zimmermann A-K, Thorand B, Autenrieth CS, et al. Patterns of multimorbidity in the aged population results from the KORA-age study. *PLoS ONE.* (2012) 7:e30556. doi: 10.1371/journal.pone.0030556
69. Pagan F, Wehling M. Polypharmacy in older adults: a narrative review of definitions, epidemiology and consequences. *Eur Geriatr Med.* (2021) 12:443–52. doi: 10.1007/s41999-021-00479-3
70. Campisi J, Kapahi P, Lithgow GJ, Melov S, Newman JC, Verdin E. From discoveries in ageing research to therapeutics for healthy ageing. *Nature.* (2019) 571:183–92. doi: 10.1038/s41586-019-1365-2
71. Anisimov VN, Berstein LM, Egormin PA, Piskunova TS, Popovich IG, Zabezhinski MA, et al. Metformin slows down aging and extends life span of female SHR mice. *Cell Cycle.* (2008) 7:2769–73. doi: 10.4161/cc.7.17.6625

72. Chen C, Liu Y, Liu Y, Zheng P. mTOR regulation and therapeutic rejuvenation of aging hematopoietic stem cells. *Sci Signal.* (2009) 2:ra75. doi: 10.1126/scisignal.2000559
73. Harrison DE, Strong R, Sharp ZD, Nelson JF, Astle CM, Flurkey K, et al. Rapamycin fed late in life extends lifespan in genetically heterogeneous mice. *Nature.* (2009) 460:392–5. doi: 10.1038/nature08221
74. Anisimov VN, Zabezhinski MA, Popovich IG, Piskunova TS, Semchenko AV, Tyndyk ML, et al. Rapamycin extends maximal lifespan in cancer-prone mice. *Am J Pathol.* (2010) 176:2092–7. doi: 10.2353/ajpath.2010.091050
75. Martin-Montalvo A, Mercken EM, Mitchell SJ, Palacios HH, Mote PL, Scheibye-Knudsen M, et al. Metformin improves healthspan and lifespan in mice. *Nat Commun.* (2013) 4:2192. doi: 10.1038/ncomms3192
76. Coleman AE, Creevy KE, Anderson R, Reed MJ, Fajt VR, Aicher KM, et al. Test of Rapamycin in Aging Dogs (TRIAD): study design and rationale for a prospective, parallel-group, double-masked, randomized, placebo-controlled, multicenter trial of rapamycin in healthy middle-aged dogs from the Dog Aging Project. *Geroscience.* (2025) 47:2851–77. doi: 10.1007/s11357-024-01484-7
77. Kaplan JL, Rivas VN, Walker AL, Grubb L, Farrell A, Fitzgerald S, et al. Delayed-release rapamycin halts progression of left ventricular hypertrophy in subclinical feline hypertrophic cardiomyopathy: results of the RAPACAT trial. *J Am Vet Med Assoc.* (2023) 261:1628–37. doi: 10.2460/javma.23.04.0187
78. North Carolina State University College of Veterinary Medicine. *A blinded, randomized, and placebo-controlled clinical trial of a novel veterinary product for the management of chronic kidney disease in cats.* [Online]. Raleigh, NC: NC State University College of Veterinary Medicine. (2024). Available online at: <https://cvm.ncsu.edu/a-blinded-randomized-and-placebo-controlled-clinical-trial-of-a-novel-veterinary-product-for-the-management-of-chronic-kidney-disease-in-cats/> [Accessed November 10, 2025].
79. Dakic T, Jevdjovic T, Vujovic P, Mladenovic A. The less we eat, the longer we live: can caloric restriction help us become centenarians? *Int J Mol Sci.* (2022) 23. doi: 10.3390/ijms23126546
80. Varela-López A, Romero-Márquez JM, Navarro-Hortal MD, Ramírez-Tortosa CL, Battino M, Forbes-Hernández TY, et al. Dietary antioxidants and lifespan: relevance of environmental conditions, diet, and genotype of experimental models. *Exp Gerontol.* (2023) 178:112221. doi: 10.1016/j.exger.2023.112221
81. Fabian P, Blander G, Deehan R, Torkamani A, Nogal B. Causal impact of genetically-determined fish and fish oil intake on epigenetic age acceleration and related serum markers. *Hum Genomics.* (2025) 19:61. doi: 10.1186/s40246-025-00756-3
82. Cupp C, Jean-Philippe C, Kerr W, Patil A, Perez-Camargo G. Effect of nutritional interventions on longevity of senior cats. *Int J Appl Res Vet Med.* (2006) 4:34–50.
83. Teng KT, Mcgreevy PD, Toribio J-aL, Raubenheimer D, Kendall K, Dhand NK. Strong associations of nine-point body condition scoring with survival and lifespan in cats. *J Feline Med Surg.* (2018) 20:1110–8. doi: 10.1177/1098612X17752198
84. Montoya M, Morrison JA, Arrignon F, Spofford N, Charles H, Hours M-A, et al. Life expectancy tables for dogs and cats derived from clinical data. *Frontiers in Veterinary Science.* (2023) 10. doi: 10.3389/fvets.2023.1082102
85. Teng KT-Y, Brodbelt DC, Church DB, O'Neill DG. Life tables of annual life expectancy and risk factors for mortality in cats in the UK. *J Feline Med Surg.* (2024) 26:1098612X241234556. doi: 10.1177/1098612X241234556
86. Aune D, Sen A, Prasad M, Norat T, Janszky I, Tonstad S, et al. BMI and all cause mortality: systematic review and non-linear dose-response meta-analysis of 230 cohort studies with 374 million deaths among 303 million participants. *BMJ.* (2016). 353:2156. doi: 10.1136/bmj.i2156
87. Freeman LM. Cachexia and sarcopenia: emerging syndromes of importance in dogs and cats. *J Vet Intern Med.* (2012) 26:3–17. doi: 10.1111/j.1939-1676.2011.00838.x
88. Hua J, Hoummady S, Muller C, Pouchelon J-L, Blondot M, Gilbert C, et al. Assessment of frailty in aged dogs. *Am J Vet Res.* (2016) 77:1357–65. doi: 10.2460/ajvr.77.12.1357
89. Martin FC, Ranhoff AH. Frailty and sarcopenia, in *Orthogeriatrics: The Management of Older Patients with Fragility Fractures*. eds. P. Falaschi and D. Marsh. Cham: Springer. (2021) p. 53–65.
90. Russell KJ, Mondino A, Fefer G, Griffith E, Saker K, Gruen ME, et al. Establishing a clinically applicable frailty phenotype screening tool for aging dogs. *Front Vet Sci.* (2024) 11:1335463. doi: 10.3389/fvets.2024.1335463
91. Colleran EJ, Delgado MM, Ren Y, German AJ, Gruen ME, Gunn-Moore DA, et al. A non-randomized pilot study to test the feasibility of developing a frailty scale for pet cats. *Front Vet Sci.* (2025) 12:1549566. doi: 10.3389/fvets.2025.1549566
92. Matsumoto Y, Ruamrungsri N, Arahori M, Ukawa H, Ohashi K, Lyons LA, et al. Genetic relationships and inbreeding levels among geographically distant populations of *Felis catus* from Japan and the United States. *Genomics.* (2021) 113:104–10. doi: 10.1016/j.ygeno.2020.11.018
93. Toribio JA, Norris JM, White JD, Dhand NK, Hamilton SA, Malik R. Demographics and husbandry of pet cats living in Sydney, Australia: results of cross-sectional survey of pet ownership. *J Feline Med Surg.* (2009) 11:449–61. doi: 10.1016/j.jfms.2008.06.010
94. Paynter AN, Dunbar MD, Creevy KE, Ruple A. Veterinary big data: when data goes to the dogs. *Animals (Basel).* (2021) 11. doi: 10.3390/ani11071872
95. Cats Protection. *Cats and their stats (CATS) report 2024*. East Sussex: Cats Protection (2024).
96. Pye CR. *Physical, Biochemical, Haematological and Metabolomic Biomarkers of Ageing in Cats: Findings from the Cat Prospective Ageing and Welfare Study.* (thesis). Liverpool: University of Liverpool. (2025).
97. O'Neill DG, Gunn-Moore D, Sorrell S, Mcauslan H, Church DB, Pegram C, Brodbelt DC. Commonly diagnosed disorders in domestic cats in the UK and their associations with sex and age. *J Feline Med Surg.* (2023) 25:1098612X231155016. doi: 10.1177/1098612X231155016
98. International Renal Interest Society. *IRIS Staging of CKD (modified 2019)*. (2019). Available online at: http://www.iris-kidney.com/pdf/IRIS_Staging_of_CKD_modified_2019.pdf (Accessed February 17, 2022).
99. Wei T, Simko V, Levy M, Xie Y, Jin Y, Zemla J. *Corrplot: Visualization of a correlation matrix (0.84) [computer software]*. (2021). Available online at: <https://CRAN.R-project.org/package=corrplot>



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Dogs and cats with acute-on-chronic kidney disease have worse outcomes than those with acute kidney injury following renal replacement therapy

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Objective: To compare survival to discharge and long-term outcomes in dogs and cats with acute-on-chronic kidney disease (ACKD) compared to acute kidney injury (AKI) treated with renal replacement therapy (RRT) and to identify factors associated with outcome.

Design: Retrospective observational design.

Setting: Single referral hospital.

Animals: 136 client-owned animals (67 dogs and 69 cats) treated with RRT between 2017 and 2024.

Measurements and main results: This retrospective observational cohort study reviewed medical records from a referral hospital. Animals were classified as having ACKD ($n = 66$) or AKI ($n = 70$). Short-term survival was evaluated as survival to discharge, restricted mean survival time (RMST) and logistic regression models. Long-term survival was assessed using Kaplan–Meier analysis, RMST and Cox proportional hazards models. Animals with ACKD were older than those with AKI (10.4 ± 3.9 years vs. 6.3 ± 3.3 years; $p < 0.0001$) and had higher serum creatinine concentrations at discharge (3.38 ± 1.16 mg/dL vs. 2.81 ± 2.92 mg/dL; adjusted $p = 0.0073$). Survival to discharge was 53% (35/66) for ACKD and 70% (49/70) for AKI. Short-term survival did not differ significantly between groups (log-rank $p = 0.089$). Among animals survived to discharge, RMST was shorter for ACKD. Median survival time for the ACKD group was 57 days overall and 33 days for dogs.

Conclusion: Although statistically significant differences were not identified, dogs and cats with ACKD had lower survival to discharge and numerically shorter long-term survival than animals with AKI. Nevertheless, 53% of RRT-treated animals with ACKD survived to discharge, demonstrating that short-term survival is achievable in selected ACKD patients.

KEYWORDS

acute kidney injury, acute-on-chronic kidney disease, cat, chronic kidney disease, dog, hemodialysis, renal replacement therapy

Introduction

Acute kidney injury (AKI) and chronic kidney disease (CKD) are major causes of morbidity and mortality in dogs and cats (1, 2). Over the past few decades, advances in veterinary medicine have substantially improved the management of patients with severe kidney disease (3). In particular, the introduction and wider availability of renal replacement therapy (RRT), has expanded the window of treatment as a life-saving option for dogs and cats with severe uremia that do not respond to conventional medical treatment (3, 4).

In clinical practice, veterinarians frequently encounter not only “pure” AKI in previously healthy animals, but also acute-on-chronic kidney disease (ACKD), in which an acute insult is superimposed on pre-existing CKD. Acute kidney injury is potentially reversible, as renal function may recover through regeneration once the underlying cause is treated and appropriate management is instituted on time, whereas CKD is characterized by irreversible structural damage and loss of functional reserve. Consequently, dogs and cats with ACKD, in whom these two pathophysiologic processes coexist, may follow a different clinical course from those with *de novo* AKI (2, 4).

Two recent veterinary studies on ACKD have reported comparable short-term outcome to AKI patients (5, 6). However, most dogs and cats in those studies recovered with medical management without hemodialysis, highlighting the need for data on patients that require hemodialysis. In cases severe enough to necessitate RRT, there is a concern that the exhausted renal reserve in CKD patients may reduce their resilience, together with complications associated with severe illness, may lead to worse prognosis than in patients with *de novo* AKI.

Therefore, the aim of this study was to compare short-term (survival to discharge) and long-term survival (post-survival discharge) between dogs and cats with AKI and those with ACKD treated with RRT. We hypothesized that, among severely azotemic patients requiring RRT, animals with ACKD would have similar survival to discharge compared with those with AKI, but poorer long-term outcomes after discharge. In addition, this study sought to identify risk factors associated with survival, in order to support clinical decision-making and prognostication for patients with ACKD requiring RRT.

Materials and methods

This is a retrospective observational cohort study. Medical records from a single referral veterinary hospital were reviewed for dogs and cats diagnosed with AKI or ACKD 2017 and 2024. From these cases, only patients that received hemodialysis were included. Extracted data comprised signalment (age, sex, body weight, and species), serum creatinine, blood urea nitrogen (BUN) and phosphate concentration at presentation, serum creatinine concentration at discharge, Urine output at presentation, length of hospitalization, total number of dialysis sessions, etiology of AKI status, in-hospital mortality, post-discharge follow-up and survival status. Urine output was categorized as follows: anuria (<0.1 mL/kg/h), oliguria (0.1–1 mL/kg/h), normal urine output

(1–2 mL/kg/h), and polyuria (>2 mL/kg/h) (7). Segev model C score (8) was calculated based on retrospective review of the medical records. Details pertaining to the duration of RRT and the urea reduction ratio for each session, up to the fifth session, were reviewed.

All dogs and cats were eligible for inclusion if they were admitted to the veterinary hospital with an acute onset of clinical signs consistent with AKI, in accordance with the diagnostic criteria of the IRIS AKI guidelines (3). The ACKD group was defined using criteria adapted from two previous studies on ACKD in dogs and cats (5, 6). For dogs, patients were assigned to the ACKD group if they fulfilled at least one of the following indicators of underlying CKD: (a) a documented prior diagnosis of CKD based on persistently increased serum creatinine, with a concurrent rise in serum creatinine of >25% above the previously recorded baseline; (b) historical laboratory evidence of chronic renal parenchymal injury obtained prior to the acute episode (when available), such as persistent renal proteinuria or clinically significant cylindruria, without concurrent baseline azotemia; or (c) abdominal ultrasonographic changes compatible with CKD, defined by the presence of ≥ 2 of the following: increased renal echogenicity, markedly reduced corticomedullary distinction, decreased or asymmetric kidney size, or renal cysts/irregular renal contour. For cats, patients were classified as ACKD if at least one of the following criteria was met: (1) a previous diagnosis of CKD with a concurrent increase in serum creatinine >20% above the most recent measurement; (2) a previous diagnosis of CKD together with urinary findings indicative of acute tubular injury (e.g., glucosuria with normoglycemia, cylindruria); or (3) unequivocal ultrasonographic evidence of CKD, defined by ≥ 2 of the following: increased cortical echogenicity, markedly diminished corticomedullary differentiation, reduced kidney size (<2.8 cm), renal cysts, or irregular renal contour. In both species, animals that did not fulfill these CKD criteria were categorized as belonging to the AKI group. Among animals classified as AKI or ACKD, RRT was performed when extracorporeal support was considered clinically indicated based on the overall clinical assessment. Indications included severe azotemia or uremic signs, fluid overload or oliguria/anuria, acid–base abnormalities, and severe electrolyte derangements when present. Uncontrolled electrolyte imbalance was not the primary indication for RRT in all cases. Dehydration and hypovolemia were assessed and optimized before RRT when clinically feasible. Many animals had already received fluid therapy before referral, whereas some were presented with fluid overload, oliguria, or anuria; therefore, the decision to initiate RRT was based on the overall assessment of volume status and clinical stability.

In general, RRT was delivered using continuous venovenous hemodiafiltration or continuous venovenous hemodialysis modes on CRRT machines (PrisMax, Baxter International Inc.; MultiFiltrate, Fresenius Medical Care) with an extracorporeal circuit incorporating an integrated hemofilter (HF20 set; FMC Paed and Midi) over 4–8 h. The ultrafiltration rate was determined based on hydration status, anticipated tolerance to fluid removal, and the resulting filtration fraction, with the goal of avoiding increases in filtration fraction above 20%. Circuits were primed primarily with 0.9% saline; however, when circuit volume exceeded 20% of the dog’s estimated blood volume, the circuit was primed with whole blood. Anticoagulation was achieved using systemic heparinization in all cases. Anticoagulation was monitored using activated partial thromboplastin time (aPTT) in earlier cases and activated clotting time (ACT) after ACT monitoring became available, with a target of approximately 1.5–2 times the baseline clotting time when feasible. The extracorporeal circuit was also visually

Abbreviations: AKI, Acute kidney injury; CKD, chronic kidney disease; ACKD, acute-on-chronic kidney disease; RRT, renal replacement therapy; OR, Odds ratios; RMST, Restricted mean survival time; ROC, Receiver Operating Characteristic; AUC, area under the curve; CI, confidence interval; HR, hazard ratio.

monitored for clot formation, and patients were monitored for hemorrhage.

Following the etiologic categorization used in previous ACKD studies in dogs and cats, presumptive etiologies were classified as inflammatory, obstructive, ischemic, pyelonephritis, toxic, other renal-related, or unknown (5, 6). Inflammatory causes included pancreatitis, diagnosed based on compatible history, clinical signs, ultrasonographic findings, and increased quantitative serum canine pancreatic lipase concentration; and glomerulopathies, defined by persistent, high-grade proteinuria after exclusion of prerenal and postrenal causes. Ureteral obstruction was diagnosed when patients exhibited compatible clinical signs together with ultrasonographic evidence. Renal ischemia was considered the underlying cause in patients that had received furosemide or nonsteroidal anti-inflammatory drugs (NSAIDs), in those in which the acute episode developed after general anesthesia, in animals with clinical signs of severe dehydration, or in those with a history of traumatic injury. Similar to the previous ACKD studies, furosemide and NSAIDs associated cases were included in the ischemic category when the clinical context suggested reduced renal perfusion or hemodynamic renal injury (9–11). Pyelonephritis was suspected when a positive urine culture and pyuria on urinalysis were present, along with supportive ultrasonographic findings. “Other cause” included cardiorenal syndrome and polycystic kidney disease. A presumptive diagnosis of toxicity was made when there was a clear history of exposure to a specific toxin accompanied by compatible clinical signs, hematologic abnormalities, and ultrasonographic changes. When two or more presumptive causes were identified, the primary cause was selected for statistical analysis. Cases that did not fulfill criteria for any of the defined categories were classified as having an unknown etiology.

Long-term outcome for animals that survived to discharge was defined as the number of days from hospital discharge to death or euthanasia from any cause. Dates of death or euthanasia were obtained from medical records. Animals for which outcome data were unavailable were censored from the analysis. Therefore, survival analyses reflected all-cause mortality rather than kidney disease-specific mortality.

Statistical analysis

All statistical analyses were performed using R software (version 4.3.2; R Foundation for Statistical Computing, Vienna, Austria). For the overall cohort, comparative analyses were conducted between the ACKD and AKI groups. Within each group, categorical variables were summarized as counts (*n*), and most continuous variables were summarized as mean (SD). As exceptions, the number of hemodialysis sessions was summarized as median (Q1, Q3), and length of hospitalization was summarized as both mean (SD) and median (Q1, Q3) (SD = standard deviation; Q1/Q3 = 25th/75th percentiles).

Differences between the ACKD and AKI groups were evaluated for age, body weight, creatinine at presentation, creatinine at discharge, blood urea nitrogen (BUN) at presentation, phosphorus at presentation, Segev model C, length of hospitalization, and number of hemodialysis during hospitalization. The normality assumption for each variable within each group was first assessed using the Shapiro–Wilk test. When the Shapiro–Wilk test yielded a *p*-value <0.05, the normality assumption was considered violated, and a nonparametric two-sample test (Wilcoxon rank-sum test) was used for between-group comparisons. Conversely, when both groups had Shapiro–Wilk

test *p*-values >0.05, a two-sample *t*-test was applied. Because multiple variables were tested, *p*-values were adjusted using the Bonferroni correction to control the family-wise type I error rate. Fisher’s exact test was used to evaluate the association between etiology and disease group in the overall cohort. Heatmaps were used to summarize the number of cases in each group.

Survival to discharge was compared between the ACKD and AKI groups. Kaplan–Meier survival curves and log-rank tests were used to compare short-term outcomes during hospitalization between the ACKD and AKI groups. To examine the association between each predictor variable and survival at discharge, univariable Firth penalized logistic regression was performed, with survival coded as event = 1 in R software. Each model was specified as follows: $\text{logit}\{P(Y=1)\} = \beta_0 + \beta_1 X$ where *Y* denotes survival status and *X* denotes a single predictor variable entered individually. Odds ratios (ORs) were reported to quantify unadjusted associations. Natural log transformations were applied to creatinine and BUN to improve model fit. Variables with a univariable *p*-value <0.20 were selected for inclusion in the subsequent multivariable Firth penalized logistic regression model; sex was excluded because of model instability related to its multiple categorical levels.

Animals that survived to hospital discharge were included in the evaluation of post-discharge long-term survival, with the date of discharge defined as time 0. Death was coded as the event of interest, and animals that were alive at the last follow-up were treated as right-censored. Kaplan–Meier survival curves and log-rank tests were used to compare post-discharge long-term outcomes between the ACKD and AKI. Restricted mean survival time (RMST) was reported for each cohort. For RMST, Wald-type 95% confidence intervals were calculated based on the standard error of the RMST estimate. Median survival time could not be estimated for some cohorts because the survival probability did not fall below 50% in the dataset.

To investigate the unadjusted association between each candidate predictor and long-term survival, univariable Firth-penalized Cox proportional hazards models were first fitted. To reduce overfitting and account for the limited number of events, variables with a univariable *p*-value <0.20 were selected for inclusion in the multivariable model. Adjusted associations were then estimated using penalized (Firth) Cox proportional hazards regression. Likewise, natural log transformations were applied to creatinine and BUN to improve model fit. Assessment of the proportional hazards assumption revealed no major violations.

The clinical utility of the Segev model C score for predicting mortality in dogs and cats undergoing hemodialysis was evaluated. Diagnostic accuracy was assessed using Receiver Operating Characteristic (ROC) analysis (8). The diagnostic accuracy for the entire cohort, the ACKD group, and the AKI group was evaluated based on the area under the curve (AUC). The predictive value was considered inadequate with an AUC of <0.70, acceptable with an AUC of 0.70–0.80, excellent with an AUC of 0.80–0.90, and outstanding with an AUC of >0.90 (12).

Results

Descriptive statistics and grouping details

A total of 136 animals were included, comprising 67 dogs and 69 cats. Based on the predefined criteria, 66 animals were classified into

the ACKD group and 70 into the AKI group. There were no significant differences between two groups in sex or body weight. ACKD group was significantly older than the AKI group (10.39 ± 3.87 years vs. 6.32 ± 3.32 years; $p < 0.0001$). This difference remained significant after Bonferroni correction (adjusted $p < 0.0001$). A significant difference was also observed in serum creatinine (mg/dL) at discharge between two groups (ACKD group, 3.38 ± 1.16 vs. AKI group, 2.81 ± 2.92 ; $p = 0.0008$; adjusted $p = 0.0073$).

The presumptive causes also differed significantly between ACKD and AKI group ($p < 0.001$). Toxicity was more common in the AKI group, whereas cases of unknown origin were relatively more common in the ACKD group (Figure 1). In contrast, no significant differences were identified in creatinine, BUN, phosphorus at presentation, urine output, or Segev model C. In addition, length of hospitalization and the number of hemodialysis did not differ significantly between the groups. Detailed demographic data, laboratory findings, and treatment-related information are presented in Table 1. Urea reduction ratio ($[\text{predialysis urea concentration} - \text{postdialysis urea concentration}] / \text{predialysis urea concentration} \times 100$) from each treatment up to fifth treatments were similar between AKI and ACKD (Table 2).

Short-term survival at discharge

Survival at discharge between groups

The short-term survival rate to discharge among all dogs and cats that underwent hemodialysis was 62% (84/136). Of these, the short-term survival rate was 53% (35/66) in the ACKD group and 70% (49/70) in the AKI group. Within the ACKD group, the short-term survival rate was 48% (15/31) in dogs and 57% (20/35) in cats. Within the AKI group, the corresponding rates were 67% (24/36) in dogs and 74% (25/34) in cats, indicating higher survival rates in cats than in

dogs in both groups. In the ACKD group, the RMST for short-term survival during hospitalization (≤ 27 days) was 15 days (95% CI, 11.7–17.7), which was shorter than that in the AKI group (19 days; 95% CI, 16.6–21.1). Kaplan–Meier analysis showed no statistically significant difference in short-term in-hospital survival between the ACKD and AKI groups (log-rank $p = 0.089$), although the survival curves showed modest separation.

Analysis of factors associated with survival at discharge

To evaluate factors associated with short-term survival, univariable followed by multivariable logistic regression analyses were performed. In the ACKD group, neuter status and length of hospitalization were significantly associated with survival to discharge (Supplementary Table 1). Based on the candidate variables for multivariable analysis ($p < 0.20$), normal urine output also showed a potential association with survival (OR = 2.79; 95% CI, 0.72–13.07; $p = 0.141$). In contrast, age, sex, body weight, species, creatinine at presentation, BUN at presentation, oliguria, polyuria, Segev model C, and number of hemodialysis were not significantly associated with short-term survival.

In the AKI group, Segev model C, length of hospitalization, and number of hemodialysis sessions were significantly associated with survival to discharge. For each one-point increase in Segev model C, the odds of survival decreased by 20%, indicating a lower likelihood of survival with increasing score (OR = 0.80; 95% CI, 0.62–0.96; $p = 0.016$). In addition, each one-day increase in length of hospitalization was associated with a 1.57-fold increase in the odds of survival (OR = 1.57; 95% CI, 1.28–2.03; $p < 0.0001$), and each additional hemodialysis session was associated with a 1.99-fold increase in the odds of survival (OR = 1.99; 95% CI, 1.25–3.60; $p = 0.001$). Candidate variables selected for multivariable analysis included creatinine at

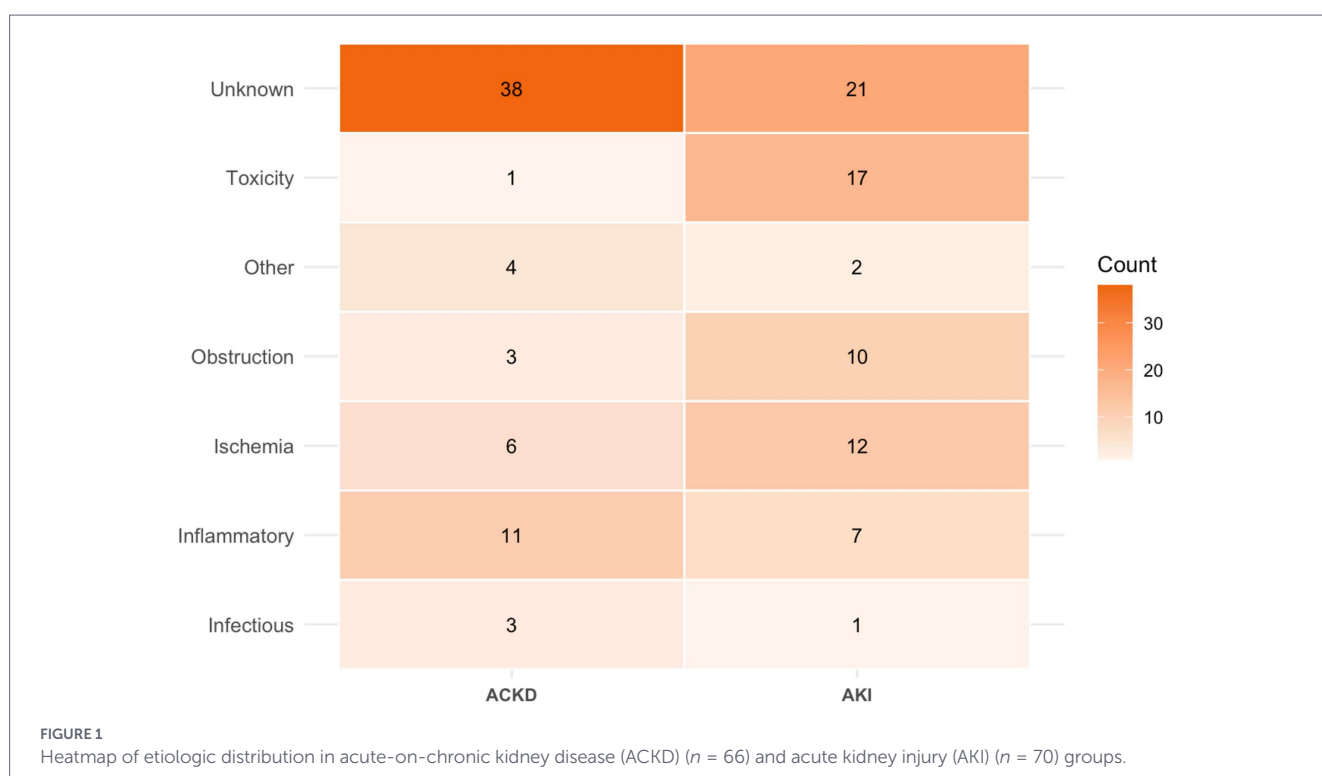


TABLE 1 Baseline characteristics of dogs and cats with acute-on-chronic kidney disease (ACKD) and acute kidney injury (AKI) treated with renal replacement therapy.

Characteristic	ACKD group			AKI group			<i>p</i> value	Adjusted <i>p</i> value
	Survivors	Non-survivors	Total	Survivors	Non-survivors	Total		
Age	10.76 ± 3.40	9.97 ± 4.37	10.39 ± 3.87	6.06 ± 3.37	6.94 ± 3.21	6.32 ± 3.32	<0.0001	<0.0001
Sex	M (0), MN (23), F (1), SF (11)	M (3), MN (16), F (3), SF (9)	M (3), MN (39), F (4), SF (20)	M (3), MN (31), F (1), SF (13)	M (2), MN (8), F (0), SF (11)	M (5), MN (39), F (1), SF (24)		
Weight	5.59 ± 2.47	5.06 ± 2.31	5.34 ± 2.39	6.84 ± 6.63	5.85 ± 2.34	6.55 ± 5.69	0.3041	1.0000
Species	Canine (15), Feline (20)	Canine (16), Feline (15)	Canine (31), Feline (35)	Canine (24), Feline (25)	Canine (12), Feline (9)	Canine (36), Feline (34)		
Creatinine at presentation	10.99 ± 5.30	11.92 ± 8.56	11.43 ± 6.98	11.08 ± 7.59	14.62 ± 10.10	12.14 ± 8.50	0.7572	1.0000
Creatinine at discharge	3.32 ± 1.13	5.10	3.38 ± 1.16	2.81 ± 2.92	NA	2.81 ± 2.92	0.0008	0.0073
BUN at presentation	145.86 ± 59.23	169.81 ± 83.05	157.11 ± 71.86	146.43 ± 65.60	193.43 ± 103.26	160.53 ± 80.96	0.7374	1.0000
Phosphorus at presentation	13.10 ± 7.09	16.81 ± 7.62	14.84 ± 7.52	12.91 ± 5.19	16.66 ± 8.90	14.04 ± 6.69	0.5295	1.0000
Urine output at presentation	A (14), O (7), N (10), P (2), U (2)	A (13), O (9), N (3), P (3), U (3)	A (27), O (16), N (13), P (5), U (5)	A (20), O (10), N (10), P (5), U (4)	A (13), O (5), N (1), P (1), U (1)	A (33), O (15), N (11), P (6), U (5)		
Segev model C	21.44 ± 4.59	22.20 ± 3.35	21.83 ± 3.96	19.89 ± 4.25	23.64 ± 3.60	21.22 ± 4.37	0.5480	1.0000
Etiology	Infectious (2), Inflammatory (7), Ischemia (2), Obstruction (1), Toxicity (1), Other (3), Unknown (19)	Infectious (1), Inflammatory (4), Ischemia (4), Obstruction (2), Toxicity (0), Other (1), Unknown (19)	Infectious (3), Inflammatory (11), Ischemia (6), Obstruction (3), Toxicity (1), Other (4), Unknown (38)	Infectious (1), Inflammatory (4), Ischemia (9), Obstruction (7), Toxicity (11) Other (2), Unknown (15)	Infectious (0), Inflammatory (3), Ischemia (3), Obstruction (3), Toxicity (6), Other (0), Unknown (6)	Infectious (1), Inflammatory (7), Ischemia (12), Obstruction (10), Toxicity (17) Other (2), Unknown (21)	<0.001	
Length of hospitalization (days) [IQR]	10 [7.5–15.5]	4 [2–6]	7.5 [3.25–11]	10 [7–16]	2 [2–4]	8 [3–11]	0.8036	1.0000
Number of hemodialysis [IQR]	3 [2.5–5]	2 [1–4]	3 [2–4.75]	3 [2–4]	2 [1–2]	2 [2–4]	0.1260	1.0000

TABLE 2 Urea reduction ratio (%) during dialysis sessions in dogs and cats with acute-on-chronic kidney disease (ACKD) and acute kidney injury (AKI) treated with renal replacement therapy.

Dialysis session	Total		ACKD		AKI		<i>p</i>
	<i>n</i>	Mean ± SD	<i>n</i>	Mean ± SD	<i>n</i>	Mean ± SD	
1st	130	44 ± 19	66	46 ± 18	64	43 ± 20	0.436
2nd	102	48 ± 14	51	51 ± 9	51	46 ± 17	0.068
3rd	71	50 ± 13	40	52 ± 11	31	47 ± 14	0.114
4th	48	47 ± 18	29	50 ± 14	19	44 ± 23	0.281
5th	29	57 ± 17	18	59 ± 18	11	53 ± 16	0.393

Urea reduction ratio (URR) was calculated as $([\text{predialysis urea concentration} - \text{postdialysis urea concentration}]/\text{predialysis urea concentration}) \times 100$.

ACKD, Acute-on-chronic kidney disease; AKI, acute kidney injury; SD, standard deviation.

presentation (OR = 0.54; 95% CI, 0.24–1.07; $p = 0.078$), BUN at presentation (OR = 0.39; 95% CI, 0.12–1.05; $p = 0.065$), and normal urine output relative to anuria (OR = 4.61; 95% CI, 0.91–46.26; $p = 0.067$).

The remaining variables, including age, sex, neuter status, body weight, species, oliguria, and polyuria, were not significantly associated with short-term survival.

In the subsequent multivariable logistic regression analysis, only length of hospitalization remained independently associated with survival to discharge in the ACKD group (OR, 1.21; 95% CI, 1.09–1.39; $p = 0.0002$), whereas no independent predictor was identified in the AKI group (Supplementary Table 2).

Long-term survival analysis

Comparison of long-term prognosis among animals that survived

Among all dogs and cats that survived to discharge, long-term outcome was evaluated. The median survival time for the ACKD group was 57 days; specifically, the median survival time for dogs was 33 days. The median survival time for cats with ACKD could not be determined because more than 50% were still alive 2,400 days after discharge. The RMST for long-term survival (≤ 1824 days) was 594.9 days (95% CI, 280.7–909.2) in the ACKD group, which was shorter than that in the AKI group (1099.5 days; 95% CI, 710.5–1488.4). Likewise, the Kaplan–Meier survival curves showed that the ACKD group consistently tended to have lower survival probabilities over time than the AKI group (Figure 2). However, despite this trend, log-rank testing indicated that the difference in long-term survival between the ACKD and AKI groups was not statistically significant ($p = 0.21$).

Analysis of factors associated with long-term prognosis

To evaluate factors associated with long-term outcome, univariable Cox proportional hazards analyses were performed, and the results are presented in Supplementary Table 3.

In the ACKD group, age, species, creatinine at discharge, and urine output at presentation were significantly associated with long-term survival. Increasing age was associated with an increased hazard of death (HR = 1.18; 95% CI, 1.00–1.40; $p = 0.044$), and species was also significantly associated with long-term survival (HR = 0.31; 95% CI, 0.11–0.89; $p = 0.030$). In addition, higher log-transformed creatinine at discharge was associated with an increased hazard of death

(HR = 6.63; 95% CI, 1.16–38.00; $p = 0.034$), and urine output at presentation was also significantly associated with poorer long-term outcome (HR = 5.92; 95% CI, 1.42–24.8; $p = 0.015$). Although log-transformed creatinine at presentation did not reach statistical significance, it showed a potential association with long-term outcome and was therefore considered a candidate variable for multivariable analysis (HR = 0.41; 95% CI, 0.15–1.11; $p = 0.080$).

In the AKI group, body weight was the only variable significantly associated with long-term survival. Higher body weight was associated with an increased hazard of death (HR = 1.39; 95% CI, 1.02–1.88; $p = 0.036$). Although not statistically significant, log-transformed creatinine at discharge met the criterion for inclusion as a candidate variable in the multivariable analysis (HR = 80.03; 95% CI, 0.2–32264.91; $p = 0.152$). In contrast, age, neuter status, species, creatinine at presentation, BUN at presentation, urine output at presentation, Segev model C, length of hospitalization, and number of hemodialysis were not significantly associated with long-term survival in the AKI group.

In the multivariable Cox proportional hazards analysis Supplementary Table 4, only urine output at presentation remained independently associated with long-term survival in the ACKD group. Interestingly, relative to anuria, normal urine output was significantly associated with a higher hazard of poor long-term outcome (HR = 5.26; 95% CI, 1.25–33.56; $p = 0.022$). In contrast, no independent predictor was identified in the AKI group.

Application of Segev score to ACKD and AKI group for predicting survival to discharge

When the analysis was performed in the overall cohort ($n = 65$) included in the ROC analysis, the AUC for the Segev score was 0.73 (95% CI, 0.61–0.85), indicating overall acceptable discriminative ability. However, when the cohort was stratified into the two groups, a clear difference in predictive performance was observed. In the AKI group ($n = 33$), the AUC was 0.84 (95% CI, 0.70–0.98), indicating excellent performance for predicting short-term survival. In contrast, in the ACKD group ($n = 32$), the AUC was only 0.59 (95% CI, 0.38–0.81), indicating inadequate mortality prediction.

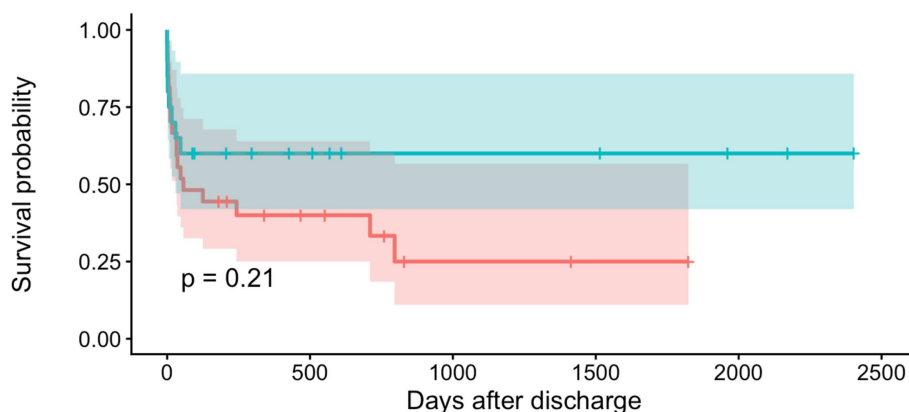


FIGURE 2

Kaplan–Meier survival curves for comparison of long-term survival between acute-on-chronic kidney disease (ACKD) and acute kidney injury (AKI) group. Red: ACKD group; Blue: AKI groups. ACKD: Median: 57d; RMST (≤ 1824 d) = 594.9 (95% CI: 280.7–909.2). AKI: Median: Not reached d; RMST (≤ 1824 d) = 1099.5 (95% CI: 710.5–1488.4).

Discussion

This study compared outcomes between dogs and cats with *de novo* AKI and those with ACKD treated with RRT. Three main findings were identified. First, compared with the AKI group, the ACKD group was significantly older (10.4 ± 3.9 vs. 6.3 ± 3.3 years; $p < 0.0001$) and had a higher serum creatinine at discharge (3.4 ± 1.2 vs. 2.8 ± 2.9 ; adjusted $p = 0.0073$), whereas creatinine, BUN, phosphorus, urine output, and Segev model C score at presentation did not differ significantly between the two groups. These findings suggest that, although the severity of illness at presentation was similar between groups, renal recovery by discharge was less complete as anticipated in the ACKD group. Second, contrary to our hypothesis, survival to discharge was lower in the ACKD group than in the AKI group (53% vs. 70%). Kaplan–Meier analysis during hospitalization likewise showed a trend toward poorer short-term survival in the ACKD group, although the difference did not reach statistical significance ($p = 0.089$). Nevertheless, even among patients requiring hemodialysis, the ACKD group still showed a clinically meaningful short-term survival rate. Third, among patients that survived to discharge, those in the ACKD group had numerically shorter long-term survival and a lower RMST than those in the AKI group, although this difference was not statistically significant ($p = 0.21$). Taken together, these findings describe the clinical outcomes of RRT-treated ACKD patients and suggest that underlying CKD may adversely affect both immediate outcome and the durability of recovery after discharge.

Notably, a significant etiologic difference was observed between ACKD and *de novo* AKI in the present study. To maintain consistency and comparability with previous ACKD studies in dogs and cats, the etiologic categories used in those studies were adopted in the present study. ACKD group was characterized by a markedly higher proportion of cases with unknown etiology (57.6%), compared to AKI group (30%, 21/70). This finding is consistent with previous ACKD reports in dogs (45%) and cats (66%) (5, 6). Taken together, these findings suggest that ACKD reflects a more complex and heterogeneous decompensated state superimposed on chronic renal injury.

In the present study, inflammatory causes (such as pancreatitis or glomerulopathies), toxicities, and infections constituted a major portion of the underlying etiologies. It is important to note that conditions like severe pancreatitis or sepsis act as complex systemic comorbidities rather than isolated renal insults. For instance, pancreatitis induces AKI through a combination of severe hypovolemia and systemic inflammatory response syndrome-mediated renal microvascular endothelial injury. These systemic inflammatory processes, compounded by uremia-induced thrombocytopenia or inflammation-driven coagulopathies, can independently and severely jeopardize patient survival regardless of the efficacy of RRT. Although we attempted to mitigate this confounding effect by adjusting for baseline clinical severity using the Segev model C score in our statistical models, the potential influence of these diverse, non-renal systemic pathologies on the clinical course should be carefully considered when interpreting the survival outcomes.

Previous ACKD studies in dogs and cats found that short-term outcomes were comparable between AKI and ACKD patients (5, 6). In the present study, the discharge survival rate in the ACKD group was 53%, which was lower than the 65% reported in dogs and the 58% reported in cats in previous ACKD studies. An important distinction, however, is that those earlier studies evaluated broader

canine and feline populations, including less severe cases that recovered with medical management alone. By contrast, the present study was restricted to patients with kidney injury severe enough to require RRT. In this setting, the lower discharge survival rate in ACKD patients suggests that the burden of underlying CKD has a more direct influence on prognosis. Importantly, however, 53% of ACKD patients still survived to discharge, indicating that RRT should not be regarded as futile for patients with ACKD. Rather, ACKD should be viewed as a condition associated with a more guarded short-term prognosis than AKI, while still retaining a meaningful likelihood of survival to discharge. Meanwhile, 70% of the AKI cohort of the present study survived to discharge which was higher than that reported in previous study of dogs (53%) and cats (50%) (12, 13). Although direct comparison is limited, this difference is likely attributable to differences in study period, case selection, and treatment modality (PIRRT vs. IHD).

With regard to short-term prognostic factors, after multivariable analysis, length of hospitalization was the only independent variable significantly associated with survival to discharge in the ACKD group, whereas no independent predictor remained significant in the AKI group. These findings should be interpreted cautiously. Length of hospitalization is unlikely to represent a true prognostic factor; rather, it more likely reflects survivor bias, as patients survived long enough to receive ongoing supportive care and demonstrate signs of renal recovery. In other words, length of hospitalization should be viewed not as a prognostic predictor, but as a course-related variable that tends to be longer among survivors. The number of hemodialysis, which was associated with short-term survival in the univariable analysis in the AKI group but lost significance after multivariable analysis, may be interpreted similarly. Accordingly, these variables should not be interpreted as indicating that dialysis treatment or prolonged hospitalization itself causes survival, but rather that patients who survived the early critical period and continued to receive hemodialysis had a greater opportunity for recovery.

In the present study, median survival time after discharge was 57 days in dogs and cats with ACKD, specifically 33 days in dogs. Kaplan–Meier curves and RMST suggested persistently lower survival probabilities in ACKD group over time. Even without statistical significance, this trend is clinically important as it suggests that the effect of underlying CKD extends beyond the initial hospitalization. The lack of statistical significance was likely influenced by the limited sample size and censoring. Although direct comparison is limited, this trend is broadly consistent with previous reports showing median survival times of 105 days in canine ACKD and 66 days in feline ACKD, in contrast to 1,322 days in dogs recovering from *de novo* AKI (5, 6, 14). Taken together, these findings indicate that some ACKD patients treated with RRT survived the acute crisis to discharge, whereas the long-term trajectory after discharge remains strongly affected by the progression of underlying CKD, incomplete renal recovery from acute injury, and subsequent post-discharge complications.

In the ACKD group, only urine output remained significant in the multivariable model, with normuria being associated with a higher risk of death than anuria. This finding is not physiologically intuitive and contrasts with previous studies that have considered urine production a potential marker of recovery (13, 15, 16). Given the small sample size, limited number of events, and wide confidence intervals in the Cox analysis, this result is best viewed as an exploratory signal. One possible explanation is that, in CKD patients, apparently normal urine output does not necessarily indicate preserved renal reserve, but

may instead reflect impaired concentrating ability, chronic tubulointerstitial damage, or non-oliguric decompensation. In addition, urine output at presentation may have been influenced by prior fluid therapy and diuretic exposure. Because these treatments were inconsistently documented in referred cases, adjustment for these variables was not feasible. One possible explanation is that urine output in ACKD may reflect the relative contribution of acute versus chronic injury. Patients with a more prominent acute component may be more likely to present with oliguria or anuria, whereas patients with a greater chronic component may maintain apparently normal urine output despite substantial loss of renal functional reserve. Overall, these findings suggest that the prognostic value of urine output in ACKD may differ from its interpretation in *de novo* AKI.

The relationship between creatinine at discharge and long-term survival is also noteworthy. In the ACKD group, higher discharge creatinine was associated with poorer long-term outcome in the univariable analysis and showed a strong directional effect in the multivariable model, although it did not reach statistical significance. This finding is consistent with the feline ACKD study by Chen et al., in which creatinine at discharge was also identified as a factor associated with long-term outcome (6). This result is clinically intuitive, as creatinine at discharge may serve as a surrogate marker for the degree of residual renal dysfunction after the acute episode. In other words, among ACKD patients that survived hospitalization, the extent of renal recovery achieved by the time of discharge may have a greater influence on subsequent survival than the severity of creatinine at presentation. In contrast, no stable independent predictor of long-term outcome was identified among AKI survivors, which may reflect the smaller number of past injury events.

Another clinically relevant finding was the predictive performance of the Segev model C (8). Overall, the model showed acceptable discrimination in entire cohorts and excellent in AKI group, but its discriminative ability was inadequate in ACKD group. Consistent with this, univariable analysis showed that higher Segev model C scores were associated with lower short-term survival only in the AKI group (OR = 0.80; 95% CI, 0.62–0.96; $p = 0.016$), not in the ACKD group. This suggests that a prognostic scoring system developed for AKI may retain clinical utility in dogs and cats with *de novo* AKI requiring RRT, but may perform less well when applied to those with AKI and concurrent underlying CKD. This is not unexpected as prognosis in ACKD is determined not only by the severity of the acute insult, but also by chronic nephron loss, baseline renal reserve, and the ability to recover to a functionally sustainable state after the crisis (17, 18). Accordingly, our findings suggest that ACKD-specific prognostic tools may be needed. Ideally, future models should incorporate variables reflecting both acute severity and chronic renal burden, such as baseline creatinine, CKD stage, diagnostic imaging, and biomarkers of renal recovery at discharge.

Because all animals in this cohort received RRT, the present study cannot determine whether RRT improved survival compared with conventional medical management. A previous proportional meta-analysis found no statistically significant difference in mortality between animals treated with dialysis and those managed conservatively (19). Moreover, in a single-center retrospective cohort, identifying an untreated comparison group with equivalent indications for RRT and comparable baseline disease severity is challenging, because the decision not to pursue RRT may be influenced by concurrent disease, clinical instability, owner preference, financial limitations, or transition to palliative care. Such a comparison would therefore be highly susceptible to being confounded by indication. Future

multicenter studies using prospectively defined eligibility criteria and severity-matched or appropriately adjusted comparison groups are needed to evaluate the clinical efficacy of RRT compared to medical management alone.

This study has several limitations. First, there are inherent limitations associated with the retrospective, single-center study design. Due to the retrospective nature, comprehensive screening for non-azotemic underlying CKD relied entirely on available historical records. Because urinalysis data or specific tubular injury biomarkers (e.g., urinary Cystatin B) were not uniformly evaluated across all patients prior to referral, a degree of misclassification bias cannot be completely excluded; some early-stage, non-azotemic CKD patients may have been inadvertently categorized into the *de novo* AKI group. Second, the small sample size within subgroups and the substantial amount of censored data in the long-term analyses reduced statistical power and widened confidence intervals. Third, RRT outcomes are influenced by non-biological factors such as owner finances and euthanasia decisions which affect survival independently of disease severity.

Conclusion

To conclude, this is the first retrospective study comparing outcomes and prognostic factors between ACKD and AKI in dogs and cats with kidney injury severe enough to require hemodialysis. In this study, ACKD was associated with older age, a different etiologic distribution, lower survival to discharge, and shorter long-term survival than AKI group. Still, more than half of the ACKD patients survived to discharge, suggesting that RRT may remain a reasonable supportive option in selected ACKD dogs and cats. These findings suggest that RRT in ACKD should not be dismissed as futile, but rather approached with careful consideration. Larger prospective multicenter studies are needed to improve prognostic assessment and risk stratification.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

The animal studies were approved by the Signature Animal Medical Center Ethics committee. The studies were conducted in accordance with the local legislation and institutional requirements. Written informed consent was obtained from the owners for the participation of their animals in this study.

Author contributions

WA: Investigation, Data curation, Writing – review & editing, Writing – original draft. KK: Methodology, Writing – original draft, Writing – review & editing, Formal analysis, Investigation. JK: Data curation, Writing – review & editing, Writing – original draft. YL:

Writing – original draft, Writing – review & editing. TL: Writing – original draft, Writing – review & editing. JH: Resources, Writing – original draft, Writing – review & editing, Formal analysis, Conceptualization, Supervision, Project administration, Validation, Methodology, Investigation.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

- Rimer D, Chen H, Bar-Nathan M, Segev G. Acute kidney injury in dogs: etiology, clinical and clinicopathologic findings, prognostic markers, and outcome. *J Vet Intern Med.* (2022) 36:609–18. doi: 10.1111/jvim.16375
- Dadousis C, Whetton AD, Mwacalimba K, Merlo A, Wright A, Geifman N. Renal disease in cats and dogs—lessons learned from text-mined trends in humans. *Animals (Basel).* (2024) 14:3349. doi: 10.3390/ani14233349
- Segev G, Cortellini S, Foster JD, Francey T, Langston C, Londoño L, et al. International renal interest society best practice consensus guidelines for the diagnosis and management of acute kidney injury in cats and dogs. *Vet J.* (2024) 305:106068. doi: 10.1016/j.tvjl.2024.106068
- Chen H, Cowgill LD, Francey T, Jepson RE, Langston C, Schweighauser A, et al. International renal interest society best practice consensus guidelines on the use of continuous renal replacement therapy in dogs and cats. *Vet J.* (2026) 315:106548. doi: 10.1016/j.tvjl.2026.106548
- Dunaevich A, Chen H, Musseri D, Kuzi S, Mazaki-Tovi M, Aroch I, et al. Acute on chronic kidney disease in dogs: etiology, clinical and clinicopathologic findings, prognostic markers, and survival. *J Vet Intern Med.* (2020) 34:2507–15. doi: 10.1111/jvim.15931
- Chen H, Dunaevich A, Apfelbaum N, Kuzi S, Mazaki-Tovi M, Aroch I, et al. Acute on chronic kidney disease in cats: etiology, clinical and clinicopathologic findings, prognostic markers, and outcome. *J Vet Intern Med.* (2020) 34:1496–506. doi: 10.1111/jvim.15808
- Vaden SL, Levine J, Breitschwerdt EB. A retrospective case-control of acute renal failure in 99 dogs. *J Vet Intern Med.* (1997) 11:58–64. doi: 10.1111/j.1939-1676.1997.tb00074.x
- Segev G, Langston C, Takada K, Kass PH, Cowgill LD. Validation of a clinical scoring system for outcome prediction in dogs with acute kidney injury managed by hemodialysis. *J Vet Intern Med.* (2016) 30:803–7. doi: 10.1111/jvim.13930
- Giorgi ME, Mochel JP, Yuan L, Adin DB, Ward JL. Retrospective evaluation of risk factors for development of kidney injury after parenteral furosemide treatment of left-sided congestive heart failure in dogs. *J Vet Intern Med.* (2022) 36:2042–52. doi: 10.1111/jvim.16571
- Lomas AL, Grauer GF. The renal effects of NSAIDs in dogs. *J Am Anim Hosp Assoc.* (2015) 51:197–203. doi: 10.5326/JAAHA-MS-6239
- Whelton A. Nephrotoxicity of nonsteroidal anti-inflammatory drugs: physiologic foundations and clinical implications. *Am J Med.* (1999) 106:13S–24S. doi: 10.1016/S0002-9343(99)00113-8
- Mandrekar JN. Receiver operating characteristic curve in diagnostic test assessment. *J Thorac Oncol.* (2010) 5:1315–6. doi: 10.1097/JTO.0b013e3181ec173d
- Eatroff A, Langston C, Chalhoub S, Poeppel K, Mitelberg E. Long-term outcome of cats and dogs with acute kidney injury treated with intermittent hemodialysis: 135 cases (1997–2010). *J Am Vet Med Assoc.* (2012) 241:1471–8. doi: 10.2460/javma.241.11.1471
- Bar-Nathan M, Chen H, Rimer D, Segev G. Long-term outcome of dogs recovering from acute kidney injury: 132 cases. *J Vet Intern Med.* (2022) 36:1024–31. doi: 10.1111/jvim.16435
- Segev G, Nivy R, Kass PH, Cowgill LD. A retrospective study of acute kidney injury in cats and development of a novel clinical scoring system for predicting outcome for cats managed by hemodialysis. *J Vet Intern Med.* (2013) 27:830–9. doi: 10.1111/jvim.12108
- Thoen ME, Kerl ME. Characterization of acute kidney injury in hospitalized dogs and evaluation of a veterinary acute kidney injury staging system. *J Vet Emerg Crit Care.* (2011) 21:648–57. doi: 10.1111/j.1476-4431.2011.00689.x
- Rudinsky AJ, Harjes LM, Byron J, Chew DJ, Toribio RE, Langston C, et al. Factors associated with survival in dogs with chronic kidney disease. *J Vet Intern Med.* (2018) 32:1977–82. doi: 10.1111/jvim.15322
- Cowgill LD, Polzin DJ, Elliott J, Nabity MB, Segev G, Grauer GF, et al. Is progressive chronic kidney disease a slow acute kidney injury? *Vet Clin N Am Small Anim Pract.* (2016) 46:995–1013. doi: 10.1016/j.cvsm.2016.06.001
- Legatti SAM, El Dib R, Legatti E, Botan AG, Camargo SEA, Agarwal A, et al. Acute kidney injury in cats and dogs: a proportional meta-analysis of case series studies. *PLoS One.* (2018) 13:e0190772. doi: 10.1371/journal.pone.0190772

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Article

Iron-Handling, Lipid-Oxygenation, and Hypoxia-Response Gene Expression in the Renal Cortex of Cats with Chronic Kidney Disease: An Analysis-Plan-Guided Secondary Analysis

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Simple Summary

Chronic kidney disease is common in older cats and often worsens over time, but the tissue changes behind the disease are not fully understood. This study reanalyzed public gene expression data from the kidney cortex, the outer part of the kidney, from cats without kidney disease and cats with mild or more advanced chronic kidney disease. We focused on genes related to low oxygen, iron handling, and fat-related injury signals because these processes may help explain how kidney tissue changes during disease. Several genes changed as disease severity increased, but they did not move together as one simple pathway. Instead, the results suggested mixed inflammation, low oxygen response, iron handling, and tissue remodeling changes. The findings do not prove a specific form of cell death or identify the exact cell types involved. They do provide candidate patterns that can be tested in future studies using tissue imaging, protein measurements, and cell level methods. This work helps narrow down which kidney pathways deserve deeper study in cats with chronic kidney disease.

Abstract

Chronic kidney disease (CKD) is common in older cats, but cortical transcript-level relationships among hypoxia response, iron handling, and lipid oxygenation are poorly defined. This analysis-plan-guided, hypothesis-generating secondary analysis used public feline renal RNA-seq data (GSE303653). The internal plan fixed the gene panel, composite construction, primary inferential test, and quality-control thresholds before the review of the present expression results, but was not publicly registered. After technical quality control, 21 renal cortex samples from control, CKD 1/2, and CKD 3/4 cats were analyzed. A 23-gene panel and whole-transcriptome differential expression were evaluated using likelihood ratio testing as the primary panel-level screen, with pairwise DESeq2 contrasts, Spearman summaries, enrichment, medulla, composite, and marker-set analyses as secondary or exploratory context. VEGFA, FTL, and NCOA4 decreased with ordinal disease group, whereas ALOX5 and HIF1A increased; eight panel genes were stage-associated by likelihood ratio testing. The equal-weight composite was nonmonotonic. Advanced CKD enrichment was dominated by immune and inflammatory terms, while GPX4 and ferroptosis-pathway enrichment were not stage-significant. The findings support heterogeneous transcript-level remodeling, including ALOX5-associated inflammatory/lipid-oxygenation signal and HIF1A-VEGFA divergence, rather than evidence of ferroptotic cell death, pathway activation, or cell-specific mechanism.



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Keywords: feline chronic kidney disease; iron metabolism; lipid oxygenation; hypoxia signaling; renal cortex; RNA sequencing

1. Introduction

Chronic kidney disease (CKD) is common in older cats [1]. Longitudinal work in geriatric cats has identified risk factors for CKD development [2]. Clinical staging in cats is commonly based on International Renal Interest Society (IRIS) criteria [3]. In feline renal tissue, naturally occurring CKD has been associated with increased HIF1A transcription and decreased VEGFA transcription [4]. Integrated multi-omics has since shown broader cortical and medullary remodeling with hypoxia-related, metabolic, and inflammatory signals in feline CKD [5]. Hypoxia-related mechanisms and their relevance to cats have been reviewed [6]. What remains unresolved is narrower than a general CKD transcriptomics gap: in the publicly available feline kidney RNA-seq dataset, it is unclear whether the renal cortex shows consistent stage-associated relationships among hypoxia-response, iron-handling, and lipid-oxygenation genes, or whether bulk-tissue changes are better explained as heterogeneous remodeling across inflammatory, tubular, endothelial, and fibrotic compartments.

These gene families are related, but they are not equivalent biological readouts. Ferroptosis is a regulated cell-death process most convincingly supported by convergent evidence for iron-dependent lipid peroxidation, loss of glutathione peroxidase 4 activity, and compatible injury phenotypes [7–10]. By contrast, HIF1A, VEGFA, HMOX1, ALOX5, ferritin genes, and iron-transport genes are also broadly involved in oxidative stress, inflammation, tissue remodeling, hypoxia response, and renal injury. Renal hypoxia signaling is not a uniform transcript-level output: HIF-1alpha and HIF-2alpha occupy different renal cell compartments [11], progressive human kidney disease can show hypoxia-response signaling despite reduced VEGF-A expression [12], and renal oxygen sensing is governed largely by HIF hydroxylation and protein stabilization rather than by HIF transcript abundance alone [13]. Ferritinophagy can mobilize ferritin-bound iron through NCOA4 [14], 5-lipoxygenase signaling can contribute to renal fibrosis and CKD progression [15], and heme oxygenase-1 induction can be cytoprotective in renal injury models [16]. These links justify examining the genes together, but they do not justify treating bulk renal cortex transcript abundance as direct evidence of ferroptotic cell death, pathway activation, or cell-specific mechanisms.

Our objective was to determine whether the expression of genes related to iron handling, lipid oxygenation, and hypoxia response differs across ordinal disease groups in the feline renal cortex. We used a 23-gene analysis-plan-defined panel together with a whole-transcriptome context to evaluate whether these genes changed coherently or showed divergent transcript-level patterns. We hypothesized that at least some panel genes would differ by ordinal disease group. We treated the equal-weight composite as an analysis-plan-defined descriptive summary, not as a primary biologic endpoint or as a test of ferroptotic cell death.

2. Materials and Methods

This was an analysis-plan-guided cross-sectional secondary analysis of publicly available feline renal RNA-sequencing data from GSE303653 [5]. Reporting followed STROBE guidance [17]. The quality-control exclusion thresholds, 23-gene cortical panel, equal-weight composite definition, primary inferential test, and planned sensitivity analyses were documented in an internal statistical analysis plan before the review of the present

expression results (Supplementary Document S1). The plan was not deposited in a public registry; therefore, the term analysis-plan-guided is used to indicate protection against ad hoc analytic drift within this project, not externally verifiable preregistration. The primary analysis used the renal cortex because CKD 3/4 medulla samples were limited; medulla results were summarized descriptively as exploratory only.

The analytical hierarchy was defined as follows. The primary inferential analysis was the DESeq2 likelihood ratio test (LRT) for disease-group association within the 23-gene cortical panel. Secondary analyses were pairwise DESeq2 Wald contrasts and Spearman ordinal trend summaries for panel genes. Exploratory analyses included whole-transcriptome enrichment, medulla summaries, equal-weight composite sensitivity analyses, and bulk marker-set summaries used to contextualize possible cell-composition effects. This hierarchy was added to make clear that the study was not designed to infer ferroptotic cell death, pathway activation, disease progression trajectories, or cell-specific mechanisms from bulk RNA-seq.

Two samples were excluded before hypothesis testing because they met technical quality-control failure criteria fixed in the analysis plan (mapping rate < 35% OR duplication > 70% OR robust within-tissue principal-component-analysis outlier with Mahalanobis distance $p < 0.01$). These cutoffs were chosen as conservative bounds that nf-core/rnaseq runs on moderate-depth tissue RNA-seq are expected to meet by a wide margin; samples falling outside them typically reflect technical failures such as degradation, library bias, or mislabeling rather than interpretable biological variation. PCA was used only as a technical outlier screen within tissue, not as evidence for downstream biological interpretation. The exclusions comprised SRR34712210, a CKD 1/2 cortex sample, and SRR34712180, a CKD 3/4 medulla sample. After applying these criteria, the primary analytic set contained 21 cortex samples from 21 cats: 6 controls, 8 CKD 1/2, and 7 CKD 3/4. Stage assignments were taken from the source study, which classified cats by IRIS stage [3,5].

The cortical panel included 23 genes chosen to represent intersecting, but non-equivalent, biology: iron storage and transport, ferritin handling/ferritinophagy, lipid oxygenation and lipid-peroxidation defense, heme/redox handling, and hypoxia-response/oxygen-sensing transcripts. The panel was intended to test whether these transcript groups behaved coherently in bulk cortex, not to define a validated ferroptosis signature. The original statistical analysis plan defined an equal-weight composite of the two gene arms; this earlier terminology is retained only when referring to the analysis plan or figures that reproduce those analyses. The composite itself is interpreted descriptively because the iron-handling/lipid-oxygenation and hypoxia-response arms moved in different directions across ordinal disease groups, making gene-level results more informative than the averaged summary.

Read processing and transcript quantification followed the nf-core/rnaseq workflow (v3.16.1) [18] with Salmon (v1.10.1) pseudo-alignment against the *Felis catus* 9.0 reference (Ensembl release 113), as detailed in Appendix A [19]. Transcript-level abundances were imported into R with tximeta v1.20.1, length-scaled (countsFromAbundance = lengthScaledTPM), and aggregated to gene level using the GTF gene_id attribute; DESeq2 v1.44.0 independent filtering was applied to remove low-count genes before pairwise Wald tests and the omnibus LRT. We used DESeq2 because it is a widely used negative-binomial framework for count-based RNA-seq differential expression and because the goal was a transparent secondary analysis rather than comparison of statistical engines [20]. Alternative frameworks such as edgeR or limma-voom could affect borderline genes and are therefore treated as a limitation rather than as a separate set of confirmatory analyses.

Gene-level stage associations were assessed with Spearman correlations and bootstrap confidence intervals. Spearman correlations with a 3-level ordinal predictor produce

substantial tied ranks and should be interpreted as descriptive group-separation summaries rather than continuous progression models. We refer to this predictor as ordinal disease group (control, CKD 1/2, CKD 3/4) rather than as CKD stage, because controls are not stage 0 and IRIS stages 1/2 and 3/4 are collapsed in the source classification. Whole-transcriptome differential expression used DESeq2 with 3 pairwise Wald contrasts (default unshrunk maximum-likelihood log₂ fold change estimates) and an LRT for stage-dependent expression [20]. Pairwise differential expression required Benjamini–Hochberg-adjusted $p < 0.05$ and $|\log_2 \text{fold change}| > 1$ within each contrast. The fold-change filter was used as an effect-size threshold for reporting, not as evidence that genes below that threshold lacked biological relevance.

These choices differ from the source-study report by Li et al. [5] in three respects: Li used edgeR's glmQLFTest rather than DESeq2, applied a Benjamini–Hochberg threshold of 0.10 rather than 0.05, and did not document use of a fold-change filter. Under our DESeq2 plus Benjamini–Hochberg $p < 0.05$ criterion without the $|\log_2 \text{fold change}| > 1$ reporting threshold, the cortex DEG count for CKD 3/4 versus controls was 4560, differing by 56 genes (about 1.2%) from Li's reported 4616 and consistent with the two-sample technical exclusion described above. The CKD 1/2 versus controls discrepancy (539 here vs. 6 in Li) is larger and reflects, at minimum, the different statistical engine and threshold; the smaller early-stage contrast is therefore interpreted more cautiously.

Functional enrichment used g:Profiler with the g:SCS multiple-testing correction [21]. Because cat-specific GO and pathway databases are sparse, gene sets were mapped to their human orthologs (HGNC symbols) and queried against the Homo sapiens reference, with the 16,959 cortex-tested genes used as the custom background to avoid genome-wide-default bias. Ortholog/query retention was high for the DEG sets used for enrichment: 535 of 539 CKD 1/2 versus control DEGs (99.3%), 2678 of 2686 CKD 3/4 versus control DEGs (99.7%), and 249 of 250 CKD 3/4 versus CKD 1/2 DEGs (99.6%) were retained for g:Profiler queries (Table S2). To contextualize possible cell-composition shifts, mean z-scored log₂(counts per million + 1) expression of conserved cell-type marker sets (macrophage/monocyte, proximal tubule, endothelial, fibroblast/myofibroblast, and cytokine/NF-kappaB) was computed per cortex sample and summarized by ordinal disease group (Table S7). These marker-set analyses were exploratory context only and were not used as formal cell-type deconvolution. Composite summaries are reported as medians, interquartile ranges, means, standard deviations, and Hedges' g with bootstrap confidence intervals as descriptive secondary analyses. A formal LRT-based power calculation was not performed because LRT power for transcriptome-wide RNA-seq is dispersion-dependent and not summarized by a single effect-size threshold; the closest available power statement applies to the panel-level Spearman analyses (see Discussion). Additional processing details, sensitivity analyses, exploratory medulla summaries, and the cell-composition marker analysis are provided in Appendix A and Supplementary Tables S1–S7.

3. Results

3.1. Sample Characteristics and Quality Control

Source-study cohort 2 comprised renal tissues from 23 cats [5]. Review of the public GSE303653 deposit identified 40 renal tissue samples, of which 38 samples from 22 cats passed technical quality-control criteria. Two samples were excluded before hypothesis testing because they met the analysis-plan-defined failure thresholds described in Materials and Methods: one medulla sample with mapping rate 21.6% and duplication 80.6%, and one cortex sample with mapping rate 32.2%, duplication 73.5%, and principal-component-analysis outlier status. The retained set comprised 21 cortex samples for the primary analysis and 17 medulla samples for exploratory summaries; the GSE303653 deposit listed

22 cortex-labeled and 18 medulla-labeled RNA-seq records (40 total), and the source publication's report of 21 cortical samples corresponds to the analytic set after the technical exclusion described above. Source-cohort demographic information reported by Li et al. is shown in Table 1 [5].

Table 1. Source-cohort demographics and analytic-sample distribution.

Characteristic	Overall Cohort	Control	CKD 1/2	CKD 3/4
Age, years (mean)	–	13.7	14.3	12.8
Sex, F/M	14/9	–	–	–
Breed, DSH/DLH	19/4	–	–	–
Cortex samples (primary)	–	6	8	7
Medulla samples (exploratory)	–	6	7	4
Excluded samples	–	0	1 cortex	1 medulla

Demographic data are from the full 23-cat source cohort reported by Li et al. and are not stratified to the retained analytic subset used here. Individual-level age, sex, breed, serum creatinine, and symmetric dimethylarginine values were not available for the retained subset from the public deposit. The excluded samples were SRR34712210 (CKD 1/2 cortex) and SRR34712180 (CKD 3/4 medulla). Dashes (–) indicate values not reported for that subgroup.

3.2. Gene-Level Stage Associations Within the Cortical Panel

Gene-level analyses were more informative than the original equal-weight composite. In the cortical panel, the strongest decreasing stage associations were VEGFA (correlation coefficient, -0.81 ; 95% confidence interval [CI], -0.92 to -0.59), FTL (-0.78 ; 95% CI, -0.93 to -0.46), and NCOA4 (-0.72 ; 95% CI, -0.88 to -0.46). The strongest increasing associations were ALOX5 (0.77 ; 95% CI, 0.50 to 0.93), HMOX1 (0.56 ; 95% CI, 0.17 to 0.80), HIF1A (0.54 ; 95% CI, 0.07 to 0.85), and STEAP3 (0.53 ; 95% CI, 0.05 to 0.83 ; Table 2; Figure 1A; full panel in Table S1). After Benjamini–Hochberg correction across the 23 panel-level Spearman tests, 10 of 11 nominally significant correlations were retained at $q < 0.05$; BNIP3 (raw $p = 0.040$, $q \approx 0.082$) was the gene that fell below the BH threshold (Table S1). Eight of 23 panel genes were also stage-dependent by likelihood ratio testing using the same Benjamini–Hochberg adjustment within the LRT family: VEGFA, HIF1A, FTL, NCOA4, STEAP3, EGLN1, ALOX5, and EGLN3. C1SD1 and HMOX1 reached significance only on the secondary Spearman screen and not on the primary likelihood ratio test, and were therefore treated as weaker, exploratory candidates. By contrast, GPX4 and EPAS1 showed little stage association (correlation coefficients, -0.36 and -0.09 , respectively; Table S1). GPX4 did not meet panel-level Spearman, likelihood ratio test, or pairwise differential-expression thresholds in any contrast. ALOX5 was the only panel gene meeting the differential-expression threshold in CKD 1/2 versus controls, whereas ALOX5, HIF1A, FTL, NCOA4, and VEGFA met that threshold in CKD 3/4 versus controls. HIF1A and FTL differed between CKD 3/4 and CKD 1/2 (Figure 1B).

Table 2. Stage associations within the cortical panel (top genes).

Gene	Functional Group	Spearman ρ	Spearman q	LRT Adj. p
VEGFA	Hypoxia response	-0.81	<0.001	1.22×10^{-7}
FTL	Iron handling	-0.78	<0.001	3.39×10^{-4}
ALOX5	Lipid oxygenation/inflammation	0.77	<0.001	5.05×10^{-3}
NCOA4	Ferritin handling	-0.72	0.001	6.12×10^{-4}
C1SD1	Iron handling	-0.63	0.010	0.054
EGLN3	Hypoxia response	-0.59	0.020	0.020
HMOX1	Redox/heme handling	0.56	0.028	0.077

Table 2. Cont.

Gene	Functional Group	Spearman ρ	Spearman q	LRT Adj. p
<i>HIF1A</i>	Hypoxia response	0.54	0.035	7.10×10^{-6}
<i>STEAP3</i>	Iron handling	0.53	0.037	1.23×10^{-3}
<i>EGLN1</i>	Hypoxia response	−0.50	0.047	4.32×10^{-3}

Top stage-associated genes from the 23-gene cortical panel; full 23-gene results are provided in Supplementary Table S1, and pairwise DESeq2 log₂ fold changes with adjusted p values are provided in Supplementary Table S2a. Spearman ρ = correlation coefficient with ordinal disease group; Spearman q = Benjamini–Hochberg-adjusted p across all 23 per-gene panel-level correlations. LRT adj. p = panel-adjusted likelihood ratio test for any disease-group effect.

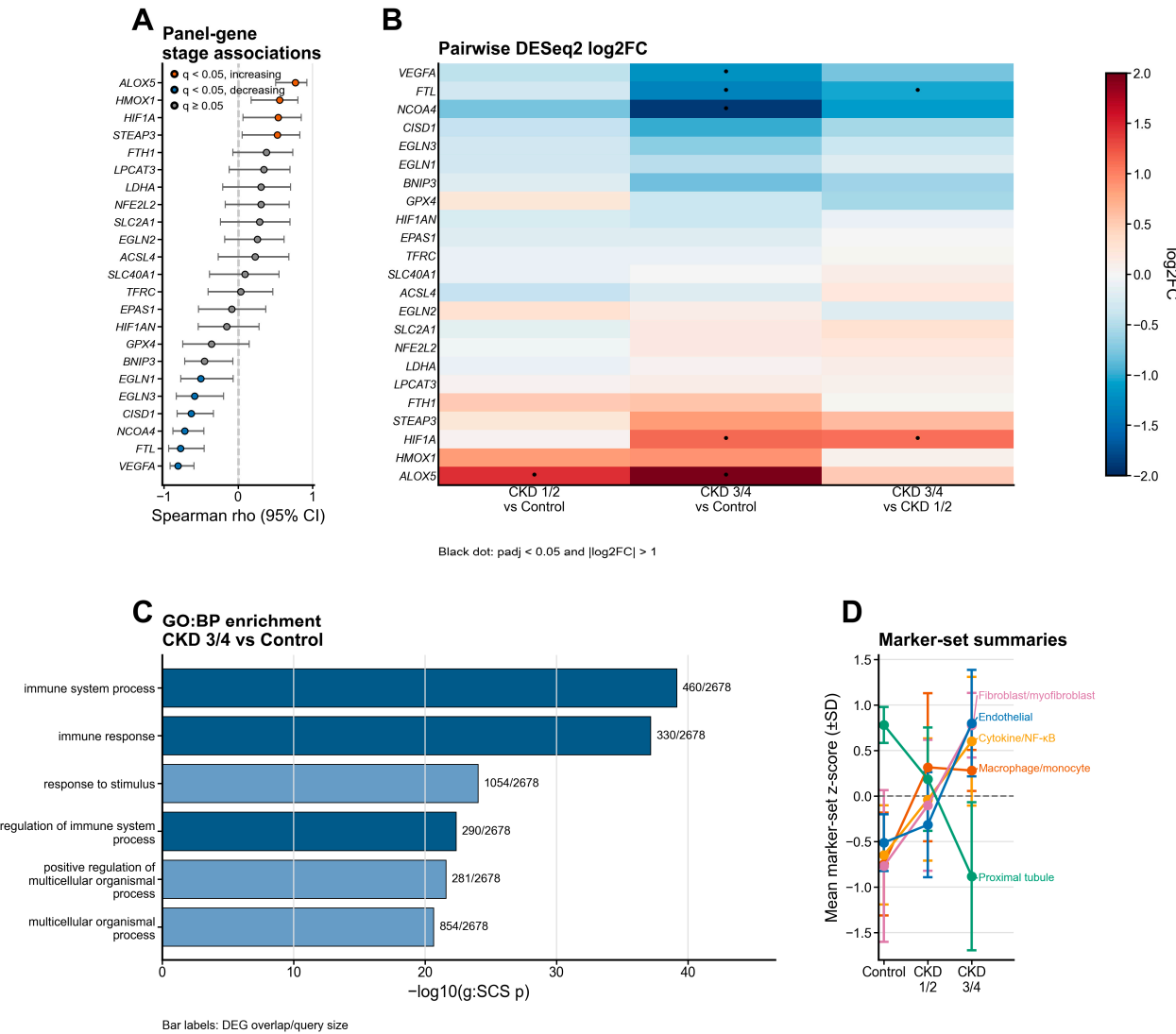


Figure 1. Summary of gene-level and whole-transcriptome context by ordinal disease group in feline renal cortex. **(A)** Forest plot of Spearman rho values with 95% confidence intervals for the 23-gene cortical panel; orange and blue indicate genes increasing or decreasing with ordinal disease group at Spearman $q < 0.05$, respectively, and gray indicates $q \geq 0.05$. **(B)** Heatmap of pairwise DESeq2 log₂ fold changes for panel genes; black dots indicate adjusted $p < 0.05$ and |log₂ fold change| > 1. **(C)** Top enriched GO:BP terms among CKD 3/4 versus control differentially expressed genes; bars show $-\log_{10}(\text{g:SCS } p)$, and labels show DEG overlap/query size. **(D)** Exploratory marker-set summaries from conserved marker groups; points show mean z-scored log₂(CPM+1) marker-set score by ordinal disease group, and error bars show standard deviation. Panels are generated from Supplementary Tables S1, S2a, S5, and S7.

3.3. Whole-Transcriptome Differential Expression and Pathway Context

Whole-transcriptome differential expression increased with disease severity. CKD 1/2 versus control yielded 539 differentially expressed genes (DEGs), CKD 3/4 versus control yielded 2686, and CKD 3/4 versus CKD 1/2 yielded 250 (Table S2). Functional enrichment of CKD 3/4 versus control DEGs was dominated by immune and inflammatory biology, including immune system process and immune response, with reactive oxygen species metabolic process appearing among less dominant significant terms (Figure 1C; Table S5). A ferroptosis-specific pathway was not significantly enriched. Exploratory bulk marker-set summaries were used to contextualize possible cell-composition effects (Figure 1D; Table S7): macrophage/monocyte markers were elevated in both CKD 1/2 and CKD 3/4 relative to controls, fibroblast/myofibroblast and cytokine/NF- κ B marker sets increased with ordinal disease group, and proximal-tubule markers decreased sharply in CKD 3/4. These summaries were exploratory and were not used as formal deconvolution.

3.4. Analysis-Plan-Defined Composite Summary

The analysis-plan-defined equal-weight cortical composite was a less informative summary than the underlying gene-level results (Table 3; Figure S1). The composite did not change monotonically with stage (median values 0.20 for controls, -0.09 for CKD 1/2, 0.26 for CKD 3/4), per-gene stage associations were heterogeneous within both panel arms (Figure S2), and all three pairwise composite effect sizes had bootstrap 95% confidence intervals crossing zero (Table 4). Sensitivity analyses, including leave-one-out exclusion of a single technically retained sample with an extreme score, did not change this central interpretation; composite sensitivity summaries are reported in Tables S3 and S6.

Table 3. Descriptive statistics for the equal-weight cortical composite.

Stage	n	Median Composite	IQR	Mean Composite	SD
Control	6	0.20	0.47	0.16	0.29
CKD 1/2	8	-0.09	0.51	-0.15	0.40
CKD 3/4	7	0.26	0.12	0.03	0.73

The equal-weight composite retains the earlier terminology used in the statistical analysis plan and is presented here as a descriptive secondary summary. Rounded values are shown to 2 decimal places in the manuscript text. Composite sensitivity summaries are provided in Supplementary Table S3, and within-arm heterogeneity in gene-level stage associations is shown in Figure S2.

Table 4. Pairwise effect sizes for the equal-weight cortical composite.

Comparison	Hedges' g	95% CI
Control vs. CKD 1/2	0.80	$-0.13, 2.13$
Control vs. CKD 3/4	0.21	$-1.65, 1.05$
CKD 1/2 vs. CKD 3/4	-0.29	$-2.44, 0.65$

All bootstrap 95% confidence intervals cross zero, consistent with an unstable, low-information composite summary. These estimates are descriptive and not used for confirmatory inference.

3.5. Exploratory Medulla Summary

Exploratory medulla summaries did not mirror the cortex pattern exactly, although interpretation was limited by the small number of CKD 3/4 medulla samples ($n = 4$). Median medulla composite values were -0.04 for controls, 0.20 for CKD 1/2, and 0.01 for CKD 3/4 (Table S4).

4. Discussion

The central finding of this cortical RNA-sequencing analysis was heterogeneous transcript-level remodeling rather than uniform activation of a single ferroptosis-related

program. ALOX5, HIF1A, HMOX1, and STEAP3 increased with ordinal disease group, whereas VEGFA, FTL, NCOA4, EGLN1, and EGLN3 decreased. Because the original equal-weight composite averaged genes moving in opposite directions, it performed poorly as a summary statistic. The most informative result is therefore the split behavior of individual genes within the 23-gene panel, not the composite itself.

This pattern is compatible with, but more restricted than, the existing feline literature. Lourenço and colleagues reported higher HIF1A and lower VEGFA transcript levels in kidneys from cats with naturally occurring CKD than in controls [4]. Li and colleagues then documented broader cortical and medullary remodeling with hypoxia-related, metabolic, and inflammatory signals in feline CKD by integrated multi-omics [5]. The present analysis differs from Li et al. in three main ways: it emphasizes renal cortex, applies an analysis-plan-defined 23-gene panel focused on iron handling, lipid oxygenation, and hypoxia response, and evaluates broad ordinal disease groups rather than CKD versus control alone. Methodological differences also help explain why differential-expression counts differed between studies, especially for the CKD 1/2 comparison. For that reason, the present results should be read as a cautious secondary analysis of the public dataset rather than as a replacement for the source-study transcriptomic interpretation.

The ferroptosis interpretation should be deliberately narrow. Experimental kidney disease models support ferroptosis most convincingly when glutathione peroxidase 4 is suppressed, lipid-oxygenation injury is demonstrated directly, and ferritinophagy-linked iron mobilization is supported [8–10,22]. In the present dataset, GPX4 was not stage-associated, ferroptosis-pathway enrichment was not significant, and NCOA4 decreased rather than increased. Reduced NCOA4 and FTL expression may reflect altered ferritin handling, loss or remodeling of ferritin-rich tubular populations, inflammatory replacement, or a combination of these processes; bulk transcript abundance alone cannot distinguish among these explanations. Recent feline biomarker work reporting increased urine GPX4 in cats with CKD reflects ongoing interest in this pathway [23], but the present tissue-level transcriptomic results do not provide evidence sufficient to claim ferroptotic cell death or a classical ferroptosis-driven model.

ALOX5 is better interpreted as an inflammatory/lipid-oxygenation signal than as a specific ferroptosis readout. In renal disease models, 5-lipoxygenase contributes to fibrosis and CKD progression, and macrophages can be an important renal source of 5-lipoxygenase expression [15]. In this dataset, ALOX5 was higher in both CKD 1/2 and CKD 3/4 than in controls, while whole-transcriptome enrichment in CKD 3/4 was dominated by immune and inflammatory terms. However, bulk RNA-seq cannot determine whether higher ALOX5 reflects inflammatory-cell abundance, transcriptional activation within infiltrating or resident cells, or both. HMOX1 should be interpreted with similar caution because heme oxygenase-1 can be cytoprotective and antifibrotic in renal injury models [16]. These findings are most appropriately framed as transcript-level evidence of inflammatory/lipid-oxygenation and redox remodeling, not as proof of a cell-specific mechanism.

The hypoxia-related result is best described as transcript-level divergence between HIF1A and VEGFA. HIF1A transcript abundance increased with ordinal disease group, whereas VEGFA decreased sharply and EPAS1 showed little stage association. Because renal oxygen sensing is governed largely by HIF hydroxylation and protein stabilization, rising HIF1A messenger RNA should not be equated with uniform HIF pathway activation [13]. HIF-1 α and HIF-2 α also occupy different renal cell compartments [11], and progressive human kidney disease can show hypoxia-response signaling despite reduced VEGF-A expression [12]. Recent feline work reported that plasma HIF-1 α protein concentrations increased in early CKD but declined in advanced stages [24], whereas HIF1A transcript abundance increased across ordinal disease groups in the present analysis. This cross-study

difference is compatible with several explanations, including post-transcriptional regulation, tubular remodeling, fibrosis, inflammatory remodeling, altered vascular or endothelial transcriptional state, or differences between plasma and tissue measurements. The present data do not distinguish among these possibilities.

Cell composition is the main interpretive limitation of this study. Because this analysis used bulk cortex RNA sequencing, apparent expression changes may reflect both within-cell transcriptional shifts and changing proportions of tubular, inflammatory, endothelial, and fibrotic populations. Loss of transcript-rich tubular epithelium, expansion of inflammatory infiltrates, fibrosis, endothelial rarefaction, or altered endothelial transcriptional state could influence genes such as VEGFA, HMOX1, FTL, NCOA4, and ALOX5. This limitation constrains the interpretation to bulk-tissue transcript-level associations.

The exploratory marker-set summaries provide context for this limitation but do not replace formal deconvolution. Mean z-scored $\log_2(\text{CPM}+1)$ expression of conserved marker sets showed that macrophage/monocyte markers (CD68, CD163, and LYZ) were elevated in both CKD 1/2 and CKD 3/4 relative to controls, cytokine/NF- κ B and fibroblast/myofibroblast marker sets increased with ordinal disease group, and proximal-tubule markers (SLC34A1, LRP2, and MIOX) decreased sharply in CKD 3/4 (Table S7). These patterns are consistent with inflammatory and fibrotic cortical remodeling, but they cannot quantify cell fractions or definitively assign ALOX5, VEGFA, FTL, or NCOA4 changes to specific cell populations. Cell-resolved or spatial transcriptomic studies will be needed to separate cellular abundance from within-cell transcriptional regulation.

The whole-transcriptome enrichment results are consistent with this calibrated interpretation. If a coherent ferroptosis program were the dominant cortical signal captured by bulk RNA sequencing, stronger agreement between the panel findings and ferroptosis-specific enrichment would be expected. Instead, immune and inflammatory processes dominated the advanced-stage transcriptome, while the 23-gene panel showed internally divergent stage-associated signals. This supports a model of overlapping inflammatory, redox, iron-handling, and hypoxia-response remodeling rather than a single defining mechanism.

This study has additional limitations. It was cross-sectional and therefore cannot establish temporality. Sample size was modest. With $n = 21$, the panel-level Spearman analyses were powered (80%, $\alpha = 0.05$, two-sided) to detect only strong monotone associations ($|\rho| \gtrsim 0.58$); moderate associations may have been missed, confidence intervals for several stage associations were wide, and the relative ranking of candidate genes should be treated as provisional pending replication. A formal power analysis was not performed for the DESeq2 LRT or pairwise differential-expression analyses, whose power depends on dispersion, mean expression, filtering, model specification, and multiple-testing burden. The CKD groups were collapsed into CKD 1/2 and CKD 3/4 because the available sample size did not support stable inference across individual IRIS stages, so the results should not be interpreted as precise stage-specific trajectories. Descriptive composite estimates in CKD 3/4 were fragile to single-sample exclusion: leaving out one technically retained sample shifted the CKD 3/4 mean composite from 0.03 to 0.30 (Table S3), so late-stage descriptive estimates should be interpreted as provisional rather than as stable summary statistics. Individual-level demographic, treatment, comorbidity, serum creatinine, and symmetric dimethylarginine data were not available from the public deposit for the analytic subset, precluding covariate-adjusted analyses and representing systematic missing data that could introduce unmeasured confounding; the age comparison ($p = 0.85$) reported by Li et al. applies to the 23-cat source cohort, not the 21-cat analytic subset [5]. The statistical analysis plan was internally fixed before the review of the present expression results but was not externally time-stamped in a public registry. Finally, messenger RNA abundance does not establish protein stabilization, enzyme activity, cellular localization, iron phe-

notype, lipid-oxygenation injury, pathway activation, or cell death. Those questions will require orthogonal validation by tissue protein assays, iron mapping, lipid-oxygenation measurements, and cell-resolved transcriptomic or spatial approaches.

The strengths are the analysis-plan-guided panel, whole-transcriptome context, transparent reporting of significant and nonsignificant results, explicit handling of the unstable composite, and direct integration of exploratory marker-set and enrichment findings with the limitations of bulk RNA-seq. In practical terms, these data identify ALOX5-associated inflammatory/lipid-oxygenation signal and transcript-level HIF1A–VEGFA divergence as candidate patterns for prospective study in feline CKD. Although pharmacologic inhibition of 5-lipoxygenase reduces fibrosis and CKD progression in experimental models, the present bulk RNA-sequencing data do not justify therapeutic inference [15]. If validated, these transcript patterns may help define biologically distinct cortical response states rather than a single generic oxidative-injury mechanism.

5. Conclusions

In the renal cortex of cats with CKD, genes related to iron handling, lipid oxygenation, and hypoxia response showed bidirectional, gene-level change rather than coordinated activation of a single pathway. Increased ALOX5 and HIF1A together with decreased VEGFA, FTL, and NCOA4 were consistent with heterogeneous inflammatory/lipid-oxygenation, iron-handling, and hypoxia-response remodeling in bulk cortex. The absence of stage-associated GPX4 expression and ferroptosis-pathway enrichment means that the present transcriptomic data do not provide evidence sufficient to claim ferroptotic cell death, pathway activation, or cell-specific mechanism. Cell-composition shifts may contribute materially to these associations and should be resolved with cell-resolved approaches. These results identify ALOX5-associated inflammatory/lipid-oxygenation signal and transcript-level HIF1A–VEGFA divergence as candidate patterns for prospective validation in feline CKD.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/vetsci13060604/s1>, Supplementary Document S2. STROBE checklist for cross-sectional studies; Figure S1. Composite summary and high-influence-sample sensitivity; and Figure S2. Within-arm heterogeneity of gene-level stage associations. The statistical analysis plan, supplementary tables, figure source tables, and figure-generation code are deposited in the Figshare source package at <https://doi.org/10.6084/m9.figshare.32621199> (accessed on 29 April 2026). Supplementary Document S1. Statistical Analysis Plan v1.2 (original analysis plan). Table S1. Full per-gene Spearman correlations for all 23 panel genes with ordinal disease group. Table S2. Whole-transcriptome differential-expression and likelihood ratio test summary, including enrichment query sizes and unmapped counts. Table S2a. Panel gene differential-expression results for all three pairwise contrasts. Table S2b. Panel gene likelihood ratio test results. Table S3. Sensitivity analysis for the equal-weight cortical composite with and without the technically retained high-influence sample. Table S4. Exploratory medulla descriptive statistics for the equal-weight composite. Table S5. g:Profiler enrichment results for CKD 3/4 versus control differentially expressed genes. Table S6. Sensitivity analyses including SLC7A11 inclusion, trimmed means, and weight perturbation. Table S7. Exploratory cell-composition marker analysis: mean z-scored log₂(CPM+1) expression of conserved macrophage/monocyte, proximal-tubule, endothelial, fibroblast/myofibroblast, and cytokine/NF-kappaB marker sets by ordinal disease group in renal cortex.

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Informed Consent Statement: Not applicable.

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Conflicts of Interest: The author declares no conflicts of interest.

Appendix A. Detailed Materials and Methods

Appendix A.1. Data Source and Analytical Cohort

This study used Cohort 2 from GSE303653, generated by Li et al. from cats with spontaneous CKD and healthy controls [5]. The public deposit contained 40 renal samples (22 cortex and 18 medulla samples) from 23 cats. Two samples were excluded before hypothesis testing because they met the analysis-plan-defined technical quality-control failure criteria stated in Materials and Methods (mapping rate < 35% OR duplication > 70% OR robust within-tissue principal-component-analysis outlier with Mahalanobis distance $p < 0.01$): one medulla sample (mapping rate, 21.6%; duplication, 80.6%) and one cortex sample (mapping rate, 32.2%; duplication, 73.5%) that also was a principal-component outlier. The retained dataset therefore comprised 38 samples from 22 cats: 21 cortex samples for the primary analysis and 17 medulla samples for exploratory summaries. Cortex group sizes were 6 controls, 8 CKD 1/2, and 7 CKD 3/4. Medulla group sizes were 6, 7, and 4, respectively. Stage assignments followed the source study classification, which was based on IRIS stages [3,5]. Because the public deposit did not include a complete individual-level covariate table for the analytic subset, demographic variables were reported from the source cohort description rather than re-estimated for the retained subset [5].

Appendix A.2. RNA-Seq Processing and Panel Definition

Raw reads were quality-filtered with fastp and assessed with FastQC. Transcript quantification used Salmon against the *Felis catus* 9.0 reference genome (Ensembl release 113) [19]. The broader workflow followed nf-core/rnaseq conventions [18]. The 23-gene cortical panel was chosen before hypothesis testing and comprised 13 genes in an iron-handling/lipid-oxygenation arm (GPX4, ACSL4, FTH1, FTL, TFRC, HMOX1, NCOA4, SLC40A1, NFE2L2, C1SD1, LPCAT3, STEAP3, and ALOX5) and 10 hypoxia-response genes (HIF1A, EPAS1, EGLN1, EGLN2, EGLN3, VEGFA, BNIP3, LDHA, SLC2A1, and HIF1AN). All 23 genes were detected in the cortex. The original statistical analysis plan referred to the equal-weight composite as the ferroptosis–HIF axis (FHA) score; that historical term is retained in Supplementary Document S1 because it reflects the analytical record, whereas the manuscript uses more specific biologic language when interpreting the results.

Appendix A.3. Composite Computation and Cortex Analyses

For descriptive composite analyses, gene-level counts were converted to counts per million, transformed as $\log_2(\text{counts per million} + 1)$, and standardized within tissue. The composite was calculated as 0.5 multiplied by the mean z-score of the 13-gene iron-handling/lipid-oxygenation arm plus 0.5 multiplied by the mean z-score of the 10-gene

hypoxia-response arm. Arm scores were also summarized separately. Because the observed cortical composite pattern was nonmonotonic and sample size was limited, composite results were treated as descriptive rather than confirmatory and are reported as medians, interquartile ranges, means, standard deviations, and Hedges' g with bootstrap 95% confidence intervals. All bootstrap confidence intervals used 10,000 nonparametric case-level resamples with random seed 20260301 and percentile confidence intervals. Gene-level stage associations were estimated with Spearman correlations and bootstrap confidence intervals. Whole-transcriptome differential expression used DESeq2 with the design formula $\sim$ disease_stage in cortex [20]. Pairwise contrasts were CKD 1/2 versus control, CKD 3/4 versus control, and CKD 3/4 versus CKD 1/2, computed as default DESeq2 Wald tests using the unshrunk maximum-likelihood log₂ fold change estimates (no lfcShrink was applied; $|\log_2 \text{fold change}| > 1$ was used as an effect-size threshold (low-count behavior is handled by DESeq2 independent filtering)). Stage-dependent expression across all three groups was evaluated with a likelihood ratio test against the reduced model ~ 1 , with Benjamini–Hochberg adjustment within the LRT family. Significant genes for pairwise contrasts were defined as Benjamini–Hochberg-adjusted $p < 0.05$ within the contrast AND $|\log_2 \text{fold change}| > 1$. Functional enrichment of differentially expressed genes used g:Profiler with the g:SCS multiple-testing correction [21].

Appendix A.4. Sensitivity Analyses and Exploratory Medulla Summary

The technically retained cortex sample with the most negative composite score was evaluated in a leave-one-out sensitivity analysis. Because that sample passed technical quality-control thresholds, it was retained in the primary analysis and reported separately only to show its effect on late-stage variance. Additional sensitivity analyses from the statistical analysis plan included SLC7A11 inclusion, trimmed means, and weight perturbation; those results are summarized in Supplementary Tables S3 and S6. Medulla analyses were limited to descriptive summaries because the CKD 3/4 medulla sample size was four and no inferential claims were planned for that tissue. All analyses were performed in R v4.4.3 using DESeq2 v1.44.0, Salmon v1.10.1, and g:Profiler (accessed on February 2026).

References

1. Reynolds, B.S.; Lefebvre, H.P. Feline CKD: Pathophysiology and risk factors—What do we know? *J. Feline Med. Surg.* **2013**, *15*, 3–14. [CrossRef] [PubMed]
2. Finch, N.C.; Syme, H.M.; Elliott, J. Risk factors for development of chronic kidney disease in cats. *J. Vet. Intern. Med.* **2016**, *30*, 602–610. [CrossRef] [PubMed]
3. International Renal Interest Society (IRIS). IRIS Staging System. Available online: <https://www.iris-kidney.com/iris-staging-system> (accessed on 25 March 2026).
4. Lourenço, B.N.; Coleman, A.E.; Tarigo, J.L.; Berghaus, R.D.; Brown, C.A.; Rissi, D.R.; Stanton, J.B.; Brown, S.A.; Schmiedt, C.W. Evaluation of profibrotic gene transcription in renal tissues from cats with naturally occurring chronic kidney disease. *J. Vet. Intern. Med.* **2020**, *34*, 1476–1487. [CrossRef] [PubMed]
5. Li, Q.; Cominetti, O.; Holzwarth, J.A.; Summers, S.; Wang, X.; Dayon, L. Integrated multi-omics analysis of renal metabolism in domestic cats with spontaneous chronic kidney disease. *Commun. Biol.* **2025**, *8*, 1794. [CrossRef] [PubMed]
6. Spencer, S.; Wheeler-Jones, C.; Elliott, J. Hypoxia and chronic kidney disease: Possible mechanisms, therapeutic targets, and relevance to cats. *Vet. J.* **2021**, *274*, 105714. [CrossRef] [PubMed]
7. Dixon, S.J.; Lemberg, K.M.; Lamprecht, M.R.; Skouta, R.; Zaitsev, E.M.; Gleason, C.E.; Patel, D.N.; Bauer, A.J.; Cantley, A.M.; Yang, W.S.; et al. Ferroptosis: An iron-dependent form of nonapoptotic cell death. *Cell* **2012**, *149*, 1060–1072. [CrossRef] [PubMed]
8. Stockwell, B.R.; Friedmann Angeli, J.P.; Bayir, H.; Bush, A.I.; Conrad, M.; Dixon, S.J.; Fulda, S.; Gascón, S.; Hatzios, S.K.; Kagan, V.E.; et al. Ferroptosis: A regulated cell death nexus linking metabolism, redox biology, and disease. *Cell* **2017**, *171*, 273–285. [CrossRef] [PubMed]
9. Belavgeni, A.; Meyer, C.; Stumpf, J.; Hugo, C.; Linkermann, A. Ferroptosis and necroptosis in the kidney. *Cell Chem. Biol.* **2020**, *27*, 448–462. [CrossRef] [PubMed]

10. Friedmann Angeli, J.P.; Schneider, M.; Proneth, B.; Tyurina, Y.Y.; Tyurin, V.A.; Hammond, V.J.; Herbach, N.; Aichler, M.; Walch, A.; Eggenhofer, E.; et al. Inactivation of the ferroptosis regulator Gpx4 triggers acute renal failure in mice. *Nat. Cell Biol.* **2014**, *16*, 1180–1191. [[CrossRef](#)] [[PubMed](#)]
11. Rosenberger, C.; Mandriota, S.; Jürgensen, J.S.; Wiesener, M.S.; Hörstrup, J.H.; Frei, U.; Ratcliffe, P.J.; Maxwell, P.H.; Bachmann, S.; Eckardt, K.-U. Expression of hypoxia-inducible factor-1 α and -2 α in hypoxic and ischemic rat kidneys. *J. Am. Soc. Nephrol.* **2002**, *13*, 1721–1732. [[CrossRef](#)] [[PubMed](#)]
12. Rudnicki, M.; Perco, P.; Enrich, J.; Eder, S.; Heininger, D.; Bernthaler, A.; Wiesinger, M.; Sarközi, R.; Noppert, S.-J.; Schramek, H.; et al. Hypoxia response and VEGF-A expression in human proximal tubular epithelial cells in stable and progressive renal disease. *Lab. Investig.* **2009**, *89*, 337–346. [[CrossRef](#)] [[PubMed](#)]
13. Haase, V.H. Oxygen sensors as therapeutic targets in kidney disease. *Nephrol. Ther.* **2017**, *13*, S29–S34. [[CrossRef](#)] [[PubMed](#)]
14. Mancias, J.D.; Wang, X.; Gygi, S.P.; Harper, J.W.; Kimmelman, A.C. Quantitative proteomics identifies NCOA4 as the cargo receptor mediating ferritinophagy. *Nature* **2014**, *509*, 105–109. [[CrossRef](#)] [[PubMed](#)]
15. Montford, J.R.; Bauer, C.; Dobrinskikh, E.; Hopp, K.; Levi, M.; Weiser-Evans, M.; Nemenoff, R.; Furgeson, S.B. Inhibition of 5-lipoxygenase decreases renal fibrosis and progression of chronic kidney disease. *Am. J. Physiol.-Ren. Physiol.* **2019**, *316*, F732–F742. [[CrossRef](#)] [[PubMed](#)]
16. Correa-Costa, M.; Semedo, P.; Monteiro, A.P.F.S.; Silva, R.C.; Pereira, R.L.; Gonçalves, G.M.; Marques, G.D.M.; A Cenedeze, M.; Faleiros, A.C.G.; Keller, A.C.; et al. Induction of heme oxygenase-1 can halt and even reverse renal tubule-interstitial fibrosis. *PLoS ONE* **2010**, *5*, e14298. [[CrossRef](#)] [[PubMed](#)]
17. von Elm, E.; Altman, D.G.; Egger, M.; Pocock, S.J.; Gøtzsche, P.C.; Vandenbroucke, J.P.; Initiative, S. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: Guidelines for reporting observational studies. *Lancet* **2007**, *370*, 1453–1457. [[CrossRef](#)] [[PubMed](#)]
18. Ewels, P.A.; Peltzer, A.; Fillinger, S.; Patel, H.; Alneberg, J.; Wilm, A.; Garcia, M.U.; Di Tommaso, P.; Nahnsen, S. The nf-core framework for community-curated bioinformatics pipelines. *Nat. Biotechnol.* **2020**, *38*, 276–278. [[CrossRef](#)] [[PubMed](#)]
19. Patro, R.; Duggal, G.; Love, M.I.; Irizarry, R.A.; Kingsford, C. Salmon provides fast and bias-aware quantification of transcript expression. *Nat. Methods* **2017**, *14*, 417–419. [[CrossRef](#)] [[PubMed](#)]
20. Love, M.I.; Huber, W.; Anders, S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol.* **2014**, *15*, 550. [[CrossRef](#)] [[PubMed](#)]
21. Raudvere, U.; Kolberg, L.; Kuzmin, I.; Arak, T.; Adler, P.; Peterson, H.; Vilo, J. g:Profiler: A web server for functional enrichment analysis and conversions of gene lists (2019 update). *Nucleic Acids Res.* **2019**, *47*, W191–W198. [[CrossRef](#)] [[PubMed](#)]
22. Jin, L.; Yu, B.; Wang, H.; Shi, L.; Yang, J.; Wu, L.; Gao, C.; Pan, H.; Han, F.; Lin, W.; et al. STING promotes ferroptosis through NCOA4-dependent ferritinophagy in acute kidney injury. *Free Radic. Biol. Med.* **2023**, *208*, 348–360. [[CrossRef](#)] [[PubMed](#)]
23. Hsu, W.L.; Huang, S.H.; Hseu, Z.Y.; Wu, V.-C.; Chen, Y.-H.; Lee, Y.-J. Evaluation of urine glutathione peroxidase 4 in cats with chronic kidney disease. *Front. Vet. Sci.* **2026**, *12*, 1756038. [[CrossRef](#)] [[PubMed](#)]
24. Chen, C.H.; Hsu, W.L.; Tsai, P.J.; Lai, C.F.; Wu, M.T.; Lee, Y.J. Evaluation of hypoxia-inducible factor-1 α and urine non-transferrin-bound iron concentrations in cats with chronic kidney disease. *Front. Vet. Sci.* **2024**, *11*, 1482998. [[CrossRef](#)] [[PubMed](#)]

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